Introductory Physics Experiments for Undergraduates

付浩 Esmond Agurgo Balfour 著

科学出版社

（O-6738.31）

Introductory Physics Experiments for Undergraduates

科学出版社互联网入口
高教数理出版分社：(010)64015178 销售：(010)64031535
E-mail：mph@mail.sciencep.com
销售分类建议：物理教育

www.sciencep.com
ISBN 978-7-03-051407-3
9 787030 514073 >
定 价：45.00元

Introductory Physics Experiments for Undergraduates

付　浩　Esmond Agurgo Balfour　著

科学出版社

北京

内 容 简 介

本书是为电子科技大学中外合作办学机构格拉斯哥学院编写的基础物理实验英文教材. 主要内容包括实验前的一些相关基础知识与要求，测量与不确定度的评定、数据处理方法，涵盖电学、力学、光学、磁学、声学的 15 个基本物理实验. 附录提供了英文物理实验报告的撰写要点，基本物理常数，以及配套本书的考卷样本和参考答案.

本书可作为中外合作办学项目理工科专业的基础物理实验教材或参考书，也可作为全国各高校理工科专业物理实验双语课程的教材或参考书.

图书在版编目(CIP)数据

大学物理实验=Introductory Physics Experiments for Undergraduates：英文 / 付浩，(加纳) 李志雄著. —北京：科学出版社，2017.1

ISBN 978-7-03-051407-3

Ⅰ. ①大… Ⅱ. ①付… ②李… Ⅲ. ①物理学-实验-高等学校-教材-英文 Ⅳ. ①O4-33

中国版本图书馆 CIP 数据核字(2017)第 000606 号

责任编辑：窦京涛 /责任校对：邹慧卿

责任印制：徐晓晨 /封面设计：华路天然工作室

科 学 出 版 社出版

北京东黄城根北街 16 号

邮政编码：100717

http://www.sciencep.com

北京中石油彩色印刷有限责任公司 印刷

科学出版社发行 各地新华书店经销

*

2017 年 1 月第 一 版 开本：720×1000 1/16

2017 年 11 月第二次印刷 印张：15

字数：302 000

定价：45.00 元

(如有印装质量问题，我社负责调换)

Preface

Physical laboratory experiments are an essential complement to the study of physics and engineering. Most theoretical principles are confirmed or otherwise disputed by experimental work. On the other hand, experimental results are important basis for understanding the theoretical model of a physical system. It is therefore very important to take accurate measurements and to properly carry out experiments to avoid misleading results. This book is thus focused on the fundamentals of good laboratory practices necessary for personal safety and minimizing experimental error to the bearers minimum.

The book has three chapters with a total of fifteen experiments. Chapter one is the introduction which presents in detail the safety measures to be taken in the laboratory, equipment maintenance and care and how to concisely present and discuss the obtained results in the form of a standard laboratory report. In Chapter two, how to accurately read various measurements during an experiment and the processes of uncertainty estimations are well covered. The final chapter has fifteen experiments with specific instructions to guide students through their practical execution in the laboratory.

It is very important to read through and understand Chapters one and two before attempting any experiments in Chapter three.

Hao Fu
Professor, Glasgow College and School of Physical Electronics, UESTC

Esmond Agurgo Balfour
Ph.D candidate, School of Physical Electronics, UESTC

Contents

Preface

1. INTRODUCTION 1

1.1 Safety 1

1.2 Laboratory Objectives 2

1.3 Development of Character and Sense of Responsibility 2

1.4 Preparation for the Actual Laboratory Work 3

1.5 Equipment Care 4

1.6 Materials to Be Provided by the Student 5

1.7 Performance of the Experiment 5

1.8 Experimental Report 5

1.9 Proficiency in the Laboratory 6

1.10 Use of the Origin Software 7

2. EXPERIMENTAL UNCERTAINTY AND DATA ANALYSIS 8

2.1 Measurement 8

2.2 Types of Errors 11

2.3 Statistical Analysis of Random Errors 15

2.4 Accuracy and Precision 20

2.5 Uncertainty 21

2.6 Instrument Error 26

2.7 Propagation of Uncertainties 30

2.8 Significant Figures 34

2.9 Computations with Measured Values 36

2.10 Graphical Representation of Data 38

2.11 Least-Squares Fitting 42

Problems for Chapter 2 48

3. EXPERIMENTS 50

3.1 Measurement of Resistance by Ammeter-Voltmeter Method 50

3.2 Use Hall Effect to Measure Magnetic Field 64

3.3 The Oscilloscope 75

3.4 Measurement of the Young's Modulus of Wire by Elongating 84

3.5 Laser Holography ····· 97
3.6 Newton's Rings ····· 106
3.7 Polarized Light ····· 115
3.8 Measuring Laser Wavelength and Index of Refraction of Air by Michelson Interferometer ····· 126
3.9 Measurement of the Apex Angle of a Prism and the Wavelengths of Mercury Lights Using a Spectrometer ····· 138
3.10 The Franck-Hertz Experiment ····· 152
3.11 The Millikan Oil Drop Experiment ····· 163
3.12 The Wheatstone Bridge and the Prototype of Electric Balance ····· 172
3.13 Measuring the Ultrasonic Speed ····· 184
3.14 The Photoelectric Effect ····· 196
3.15 The Potentiometer ····· 205
APPENDIX I Laboratory Report Writing Format ····· 215
A1-1 Abstract ····· 215
A1-2 Introduction ····· 215
A1-3 Experimental Procedure ····· 215
A1-4 Results ····· 216
A1-5 Discussion ····· 216
A1-6 Conclusion ····· 216
A1-7 References ····· 216
A1-8 Appendices ····· 217
APPENDIX II Materials Properties and Physical Constants ····· 218
APPENDIX III Physical Experiments I Sample Examination Paper ····· 221
Part A Multiple Choice ····· 221
Part B Long Questions ····· 225
Answers to Sample Questions ····· 229
References ····· 232

1. INTRODUCTION

You are advised to read the entire introduction very carefully. References to it will be made in many sections of this book.

1.1 Safety

The most important thing in the laboratory is your safety and that of others. Experiments are designed to be done safely, and as such proper caution must always be exercised. A major potential danger arises from the lack of knowledge of the equipment used and the required procedure. Upon entering the physics laboratory at the first instance, you will probably find the equipment for the experiment on the laboratory bench. Restrain your curiosity and do not play with the equipment. You may hurt yourself and/or damage the equipment. A good general rule is:

> *Do not touch or turn on laboratory equipment until its use and/or setup has been properly explained and understood and authorized by the instructor to do so.*

Certain items used in various experiments can be particularly dangerous, for example, hot objects, electricity, mercury lamps, and radioactive sources. In some instances, such as with hot objects and electricity, basic common sense and knowledge are required.

However, in other instances, such as with mercury lamps and radioactive sources, you may not be aware of the possible dangers. Mercury lamps may emit ultraviolet radiation that can be harmful to your eyes. Consequently, some sources need to be properly shielded. Some radioactive sources are solids and are encapsulated to prevent direct contact. Others are in liquid form and are transferred during an experiment, so care must be taken to avoid spillage or to reduce the danger of spillage to the barest minimum. Proper handling is therefore important.

In general, necessary precautions will be given in the experiment descriptions. *Note them well.* As pointed out earlier, experiments are designed to be carried out safely. Yet a common kitchen match can be dangerous if used improperly. You should take extra care to follow the procedure carefully and adhere to the precautions prescribed. Another good rule for the laboratory is:

If you have any questions about the safety of a procedure, ask your instructor for clarification before proceeding with the experiment.

The physics laboratory is a place to learn and practice safety.

1.2 Laboratory Objectives

The laboratory is a workshop for students and the place where they get firsthand knowledge of physical principles and experimental methods through the handling of apparatus designed to demonstrate the meaning and application of specific theories. Some of the more specific objectives are:

a. To acquire training in scientific methods of observation and recording of data.

b. To acquire techniques in the handling and adjustment of equipment.

c. To gain an understanding of the limitations and strengths of experimentation.

d. To obtain experience in the use of graphical representation.

e. To collect data and to develop confidence in one's ability to compute reliable answers or to determine valid relation.

By developing the skill of computing the required results from experimental data for comparison with known values of the desired quantities, the student acquires the necessary confidence needed to perform an experiment and to determine some quantity or relationship which was previously unknown.

1.3 Development of Character and Sense of Responsibility

Prospective employers and placement offices frequently request information from physics instructors on the character, attitude, honesty, and dependability of

students. The instructor makes an evaluation of these traits from observations of the student's performance in class and in the laboratory. It is therefore important to take laboratory experiment seriously. The laboratory is a place for serious thought and investigation, and the following suggestions should help you to develop the above-mentioned traits:

a. Be prompt in arriving at your work station and be well prepared concerning the principles of the experiment. If, for some good reason, you are late or absent, report the matter to the instructor.

b. Work quietly and attempt to make the most careful observations possible by adjusting the equipment to perform at its best.

c. Be honest in making and recording observations. Record data as indicated by your equipment and not as you thought they were supposed to be, if they differ. Do not copy data, conclusions, or computations from any other source. If your results seem to be outside the limits predicted by the experimental uncertainties, recheck your measurements and computations. If this does not give the answer, provide the best possible explanation for the discrepancy.

d. Have the entire procedure well in mind and perform the various steps in the order that will make the best use of your time. Cooperate with your partner, if allowed, so that each of you gets experience in manipulating the equipment. However, you should compute your results independently and compare to check on the accuracy of your work.

e. Always remain at your assigned station and do not disturb other people in the laboratory concerning any part of the experiment. Do not fidget with other equipment in the laboratory that is not part of your assigned experiment.

f. Always abide by any precautions that your instructor may have given you regarding the proper handling of the equipment. Delicate equipment may be easily damaged.

1.4 Preparation for the Actual Laboratory Work

The efficiency of performance in the laboratory depends largely on the preparation made before the experimental work begins. The entire experiment should be

read in detail before any measurements are made. It is also advisable for the student to review sections in the class textbooks that deal with the principles under investigation.

Laboratory experiments are usually intended to give students the opportunity to discover for themselves what may not be obvious at the start of the experiment, so they should not be concerned if they do not know the expected results at the beginning. At the start of the laboratory period the instructor will discuss any required special instructions needed for the apparatus being used, including precautions and perhaps some special techniques which should be used to get the best results. The instructor may also choose to discuss the underlying theory at the start of the laboratory period and/or discuss the results at the end of the laboratory period. The details of how to perform the experiment can be found in the—Experimental Procedure—section of each experiment.

1.5 Equipment Care

The equipment provided for the laboratory experiments are often expensive and in some cases quite delicate. If used improperly, certain pieces of apparatus can be damaged. The general rules given above concerning personal safety also apply to equipment care.

Even after familiarizing oneself with the equipment, it is often advisable or required to have an experimental setup checked and approved by the instructor before putting it into operation. This is particularly true for electrical experiments. Applying power to improperly wired circuits can cause serious damage to meters and other apparatus.

If a piece of equipment is broken or does not function properly, it should be reported to the laboratory instructor. Disassemble the experimental setup after you complete an experiment and leave the bench neatly as found, unless you are otherwise instructed.

If you accidentally break some equipment or the equipment stops working properly during an experiment, *report it to your instructor.* Otherwise, the next

time the equipment is used, a great deal of time may be wasted trying to get good results.

1.6 Materials to Be Provided by the Student

Tools and other materials which are not considered as general laboratory apparatus will be needed at various times. These items include graph paper, straight edge, protractor, data entry sheet and a hand calculator. You should always have your related textbook available for reference.

1.7 Performance of the Experiment

Before beginning the experimental work always read the entire procedure to get a general idea of what is to be done. You should always arrange and adjust the apparatus to give the best performance possible and then make and record readings as precisely as the apparatus will permit. Always estimate one significant figure beyond the smallest graduation on the instrument being read.

Data should never be recorded on scrap paper and then transferred to your record form. If, after you have recorded a reading, you decide that it is in error and should be discarded, mark through it and record the corrected reading below it. Always record the proper unit beside the measured value or at the heading of a column when a whole column of readings use the same unit.

Do not hesitate to discuss any details of the experiment with the laboratory instructor during the laboratory session. You may want to question certain procedures or suggest improvements in the method. A good question may be more important than a good answer.

1.8 Experimental Report

The form of the report required will be designated by the instructor in the course. In any case, the original data should be presented in a neat form, such

as that suggested for each experiment in this book. The data should be followed by sample calculations showing the method of obtaining the results. If the experiment requires several computations of the same type, only one of each type need be shown in the report.

Each experiment has a stated "purpose". Use this as a guide to your investigations. Your report should include a separate section in which you clearly and concisely state your results and conclusions. This section should respond to the question: "To what extent was the experiment's purpose accomplished?" This discussion should stick to the facts, and all conclusions should be supported by reference to your data and observations. Avoid idle, unsupported speculation.

Many of the questions at the end of each experiment are intended to stimulate thought and to guide the student in drawing conclusions concerning the results. These questions are to be answered in a discussion style and the answers so worded that the reader can ascertain the question from the answer. The sheet containing the questions may be removed from a lab manual if the instructor prefers that both questions and answers be a part of the report and has instructed so.

1.9 Proficiency in the Laboratory

The factors that will be used to measure your proficiency in the laboratory fall into two general categories as defined below.

General laboratory performance including:

a. Conduct in the laboratory.

b. Care and technique in operating the equipment.

c. Ability to grasp the fundamental principles demonstrated.

Presentation of experimental results including:

a. Neatness and orderly arrangement of recorded data and computations.

b. Interpretation of experimental data and conclusions drawn from it.

c. Answers to the assigned questions.

d. Answers to quiz questions if given.

1.10 Use of the Origin Software

In laboratory experiments you will often need to plot graphs of your results. There are many software applications for data processing, such as Excel, Matlab, etc. However, in this book, the *Origin* software is recommended. This software also provides a multitude of data analysis tools. Here you will need to use a relatively small number of its features but it is advised that you familiarize yourself with its features as many will be required in your lab work during the year.

2. EXPERIMENTAL UNCERTAINTY AND DATA ANALYSIS

2.1 Measurement

Laboratory work in physics involves the use of apparatus to make measurement which can be used for one of two purposes; either to make a definite determination, such as the elementary electric charge, the wavelengths of lights, the refractive index of air, etc. or to investigate the validity of a law. In both cases, the observations must be faithfully recorded and should be as reliable as possible.

In the classical definition, *measurement* is the determination or estimation of ratios of quantities. It is the assignment of a number to a characteristic of an object or an event, which can be compared with other objects or events. A measurement tells us about a property of something. It might tell us how heavy an object is, or how hot, or how long it is. It gives a number to that property. Measurements are always made using an instrument of some kind. Rulers, stopwatches, weighing scales, and thermometers are all measuring instruments. The result of a measurement is normally in two parts: a number and a unit of measurement, e. g. "How long is it? ··· 2 metres. "Quantities frequently measured include time, length, area, volume, pressure, mass, force, and energy.

There are some processes that might seem to be measurements, but are not. For example, comparing two pieces of string to see which is longer is not really a measurement. Counting is not normally viewed as a measurement. Often, a test is not a measurement: tests normally lead to a "yes/no" answer or a "pass/fail" result.

Measurements are classified into ***direct*** and ***indirect***. Direct measurement means measuring exactly the quantity of interest with a given instrument without need for further manipulations. In physics, there are some quantities that are

very easy to measure directly. These include mass, distance, temperature, time and electric current. So to measure how long a piece of wood is one would just measure its length using a tape measure. To determine the temperature of a cup of water, all one needs is just a thermometer. Indirect measurement refers to determining the value of the quantity in question by measuring some other quantities for manipulation. For example, the most obvious way to find the velocity v of an object is to measure the distance traveled, d, and the time taken, t, and then to calculate v as $v = d/t$. It is worth noting that you can measure one quantity by using direct or indirect measurement. For example, you can measure the magnetic field by measuring the voltages and currents based on the Hall effect. It is indirect measurement. However, if you have a teslameter you can read the magnetic field directly.

When doing a direct measurement, we must read an instrument. There are three parts that must be reported. The first is the reading itself, called the nominal value of the measurement. The second is the unit of the measurement. The third part is to estimate the uncertainty of the reading.

Important points to remember:

a. A measurement without units is meaningless.

b. A measurement without an estimated range of uncertainty is also meaningless.

Let us consider an example that illustrates more clearly the crucial importance of knowing how big the uncertainties are.

Suppose we are faced with a problem like the one said to have been solved by Archimedes. We are asked to find out whether a crown is made of pure gold, as claimed, or a cheaper alloy. Following Archimedes, we decide to test the crown's density ρ knowing that the densities of 18-karat gold and the suspected alloy are

$$\rho_{gold} = 19.3 \text{ g/cm}^3$$

and

$$\rho_{alloy} = 17.8 \text{ g/cm}^3$$

If we can measure the density of the crown, we should be able to decide whether the crown is really gold by comparing ρ with the known densities ρ_{gold} and ρ_{alloy}.

Suppose we summon two experts in the measurement of density. The first expert, Muller, might make a quick measurement of ρ and report that his best estimate for ρ is 19.0 g/cm^3 and that it almost certainly lies between 17.5 and 20.5 g/cm^3. Our second expert, Halberstam, might take a little longer and then report a best estimate of 17.9 g/cm^3 and a probable range from 17.7 to 18.1 g/cm^3. The findings of our two experts are summarized in Fig. 2-1.

The first point to notice about these results is that although Halberstam's measurement is much more precise, Muller's measurement is probably also correct. Each expert states a range within which he is confident ρ lies, and these ranges overlap; so it is perfectly possible (and even probable) that both statements are correct.

Note next that the uncertainty in Muller's measurement is so large that his results are of no use. The densities of pure gold and of the alloy both lie within his range, from 17.5 to 19.5 g/cm^3. So no conclusion can be drawn from Muller's measurements. On the other hand, Halberstam's measurements indicate clearly that the crown is not genuine; the density of the suspected alloy, 17.9 g/cm^3, lies comfortably inside Halberstam's estimated range of 17.7 to 18.1 g/cm^3, but that of pure gold, 19.3 g/cm^3, is far outside it. Evidently, if the measurements are to allow a conclusion, the experimental uncertainties must not be too large. The uncertainties do not need to be extremely small, however. In this respect, our example is typical of many scientific measurements, for which uncertainties have to be reasonably small (perhaps a few percent of the measured value) but for which extreme precision is often unnecessary.

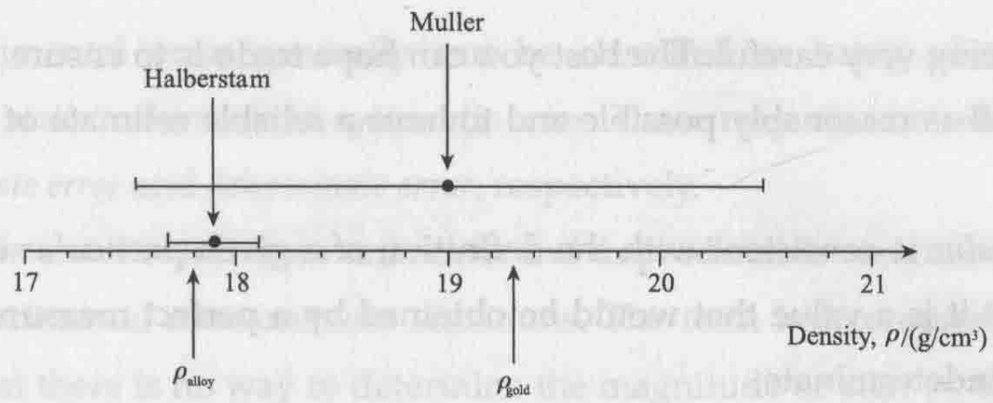


Fig.2-1 Two measurements of the density of a supposedly gold crown. The two dots show Muller's and Halberstam's best estimates for the density. The two horizontal error bars show their margins of error, the ranges within which they believe the density probably lies. Muller's uncertainty is so large that both gold and the suspected alloy fall within his margins of error. Therefore, his measurement does not determine which metal was used. Halberstam's uncertainty is appreciably smaller, and his measurement shows clearly that the crown is not made of gold

The most important point about the two experts' measurements is that, like most scientific measurements, they would both have been useless if they had not included reliable statements of their uncertainties. In fact, if we knew only the two best estimates (19.0 g/cm^3 for Muller and 17.9 g/cm^3 for Halberstam), not only would we have been unable to draw a valid conclusion, but we could actually have been misled, because Muller's result (19.0 g/cm^3) seems to suggest the crown is genuine.

2.2 Types of Errors

Error is defined as the result of a measurement minus the true value of the measurand. It is sometimes called *absolute error*. Measurand is a particular quantity subject to measurement.

$$error = result\ of\ measurement - true\ value \tag{2.1}$$

It should be noted that *absolute error* can be positive or negative and should not be confused with *absolute value of error*, which is the modulus of the error.

In science, the word error does not carry the usual connotations of the terms mistake or blunder. Errors are not mistakes and you cannot eliminate

them by being very careful. The best you can hope to do is to ensure that errors are as small as reasonably possible and to have a reliable estimate of how large they are.

True value is consistent with the definition of a given particular quantity. It means that it is a value that would be obtained by a perfect measurement and by nature indeterminate.

Relative error is the error of measurement divided by a true value of the measurand. It is commonly expressed as a percentage and is also called ***percent error*** of an experimental value.

$$relative\ error = \frac{absolute\ error}{true\ value} \times 100\% \tag{2.2}$$

In both (2.1) and (2.2) a conventional true value or accepted value is used since a true value cannot be determined.

Example 2.1 A cylindrical object is measured to have a diameter d of 20.24 mm and a circumference c of 63.85 mm. What is the value of π determined from the measurements? What are the absolute error and the relative error, respectively, if the accepted value of π with two decimal places is 3.14?

Solution With, $d = 20.24$ mm and $c = 63.85$ mm

$$c = \pi d \quad \text{or} \quad \pi = \frac{c}{d} = \frac{63.85\ \text{mm}}{20.24\ \text{mm}} = 3.15$$

then,

$$\begin{aligned} \text{absolute error} &= \text{result of measurement} - \text{true value} \\ &= 3.15 - 3.14 \\ &= 0.01 \end{aligned}$$

and,

$$\begin{aligned} \text{relative error} &= \frac{\text{absolute error}}{\text{true value}} \times 100\% \\ &= \frac{0.01}{3.14} \times 100\% \\ &= 0.3\% \end{aligned}$$

Experimental error generally can be classified as being of two types, *random* or *statistical error* and *systematic error*. These may also be referred to as *indeterminate error* and *determinate error*, respectively.

Random errors result from unknown and unpredictable variations that arise in all experimental measurement situations. The term indeterminate refers to the fact that there is no way to determine the magnitude or sign (+, too large; –, too small) of the error in any individual measurement. Conditions leading to random errors include:

a. Unpredictable fluctuations in temperature or line voltage.

b. Mechanical vibrations of an experimental setup.

c. Specimen variations. If the measurement trials are being made on a number of presumably identical samples, minor differences in chemistry, physical structure, optical properties, etc. , between one measurement specimen and another, will give rise to random errors.

The effect of random errors may be reduced and minimized by improving and refining experimental techniques such as optimizing the experimental setup or averaging large number of repeated measurements.

Systematic errors are associated with particular measurement instruments or techniques, such as an improperly calibrated instrument or bias on the part of the observer. The term *systematic* implies that the same magnitude and sign of experimental uncertainty are obtained when the measurement is repeated several times. *Determinate* means that the magnitude and sign of the uncertainty can be determined if the error is identified. Conditions that may cause systematic errors include:

a. An improperly "zeroed" instrument.

b. A faulty instrument, such as a thermometer that reads 101℃ when immersed in boiling water at standard atmospheric pressure. Such a thermometer is faulty because the reading should be 100℃.

c. Personal error, such as using a wrong constant in calculation or always taking a high or low reading of a scale division. Reading a value from a measurement scale generally involves aligning a mark on the scale. The

alignment—and hence the value of the reading—can depend on the position of the eye (parallax). Examples of such personal systematic error are shown in Fig. 2-2.

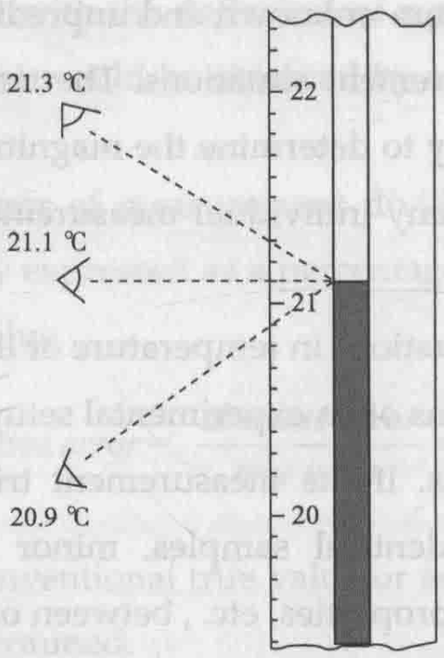


Fig. 2-2 Example of personal error due to parallax in reading

Avoiding systematic errors depends on the skill of the observer to recognize the sources of such errors and to prevent or correct them.

Example 2.2 A 30-meter steel tape measure is designed for use at a temperature of 20 ℃. Suppose that we need to use the tape outdoors when the temperature is only −9.0 ℃. Since the steel tape will contract in the cold, a systematic error is expected to occur. Since the scale divisions are closer together, the measured results will be greater than the actual value. How much contraction is there?

Solution It is known that steel contracts by 11.4 millionths of its length for each 1 ℃ drop in temperature. We have a total temperature drop of 29 ℃, so the total fractional contraction is

$$29 \times 11.4 \times 10^{-6} = 331 \times 10^{-6}$$

So, the total contraction in centimeters is

$$331 \times 10^{-6} \times 30 \times 10^{2} = 0.99 \text{ (cm)}$$

This would be the systematic error for each 30 meters measurement if we neglected the contraction due to temperature.

2.3 Statistical Analysis of Random Errors

2.3.1 The Mean and the Standard Deviation

Suppose we need to measure some quantity x and we have identified all sources of systematic error and reduced them to a negligible level. Because all remaining sources of errors are random, we should be able to detect them by repeating the measurement several times. We might, for example, make the measurement five times to obtain the results

$$71, 72, 72, 73, 71$$

(where, for convenience, we have omitted any units).

Given the five measured values, the question arises as to what should we take for our best estimate x_{best} of the quantity x? Reasonably, our best estimate would seem to be the *average* or *mean* x of the five values found. Thus,

$$\begin{aligned} x_{best} &= \bar{x} \\ &= \frac{71+72+72+73+71}{5} \\ &= 71.8 \end{aligned}$$

More generally, suppose we make N measurements of the quantity x (all using the same equipment and procedure) and find the N values to be,

$$x_1,\ x_2,\ \cdots,\ x_N$$

Once again, the best estimate for x is usually the average of $x_1, x_2, \cdots, x_N$. That is,

$$x_{best} = \bar{x} \tag{2.3}$$

where,

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_N}{N} = \frac{\sum_{i=1}^{N} x_i}{N}$$

In the last line, the useful sigma notation has been introduced according to which

$$\sum_{i=1}^{N} x_i = \sum_{i} x_i = \sum x_i = x_1 + x_2 + \cdots + x_N \tag{2.4}$$

The second and the third notations of the expression here are common abbreviations, which will be used hereafter when there is no danger of confusion.

The concept of the average or mean is almost certainly familiar to most readers. Our next concept, which is the *standard deviation*, is probably less so. The standard deviation of the measurements x_1, x_2, $\cdots$, x_N is an estimate of the *average error of the measurements* $x_1, x_2, \cdots, x_N$ and is determined as follows.

Given that the mean $\bar{x}$ is our best estimate of the quantity x, it is natural to consider the difference $x_i - \bar{x} = d_i$. This difference, often called the *deviation* (or residual) of x_i from x, tells us how much the i th measurement x_i differs from the average of x. If the deviations $d_i = x_i - \bar{x}$ are all very small, our measurements are all close together and presumably very precise. If some of the deviations are large, our measurements are obviously not so precise.

To be sure that the idea of the deviation is understood, let us calculate the deviations for the set of five measurements above with mean of 71.8. These deviations can be listed as shown in Table 2.1. Notice that the deviations are not (of course) all the same size; $|d_i|$ is small if the i th measurement, x_i, happens to be close to $\bar{x}$, but $|d_i|$ is large if x_i is far from $\bar{x}$. Notice that some of the d_i values are positive and others negative because some of the x_i are bound to be higher than the average x and some are bound to be lower.

Table 2.1 Calculations of deviations

Trial number i	Measured value x_i	Deviation $d_i = x_i - \bar{x}$
1	71	−0.8
2	72	0.2
3	72	0.2
4	73	1.2
5	71	−0.8
		$\sum d_i = 0$

To estimate the average reliability of the measurements x_1, x_2, $\cdots$, x_5, we might naturally try averaging the deviations d_i. Unfortunately, a glance at Table 2.1 shows, the average of the deviations is zero. In fact, this average will be zero for any set of measurements x_1, x_2, $\cdots$, x_N because the definition of the average $\bar{x}$ ensures that $d_i = x_i - \bar{x}$ is sometimes positive and sometimes negative in just such a way that $\bar{d}$ is zero. Obviously, then, the average of the deviations is not a useful way to characterize the reliability of the measurements x_1, x_2, $\cdots$, x_N.

The best way to avoid this annoyance is to square all the deviations, which will create a set of positive numbers, and then average these numbers. ① If we then take the square root of the result, we obtain a quantity with the same units as x itself. This number is called the *standard deviation* of $x_1, x_2, \cdots, x_N$, and is denoted σ_x

$$\sigma_x = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(d_i)^2} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i - \bar{x})^2} \tag{2.5}$$

With this definition, the standard deviation can be described as the *root mean square* (or RMS) deviation of the measurements x_1, x_2, $\cdots$, x_N. It proves

① Another possibility would be to take the absolute values $|d_i|$ and average them, but the average of the d_i^2 proves more useful. The average of the $|d_i|$ is sometimes (misleadingly) called the average deviation.

to be a useful way to characterize the reliability of the measurements. As will be discussed shortly, the Eq. (2.5) is sometimes modified by replacing the denominator N by $N-1$.

To calculate the standard deviation σ_x as defined by Eq. (2.5), we must first compute the deviations d_i, square them, average these squares, and then take the square root of the result. For the data in Table 2.1, we start this calculation in Table 2.2.

Table 2.2 Calculations of the standard deviations

Trial number i	Measured value x_i	Deviation $d_i = x_i - \bar{x}$	Deviation squared d_i^2
1	71	−0.8	0.64
2	72	0.2	0.04
3	72	0.2	0.04
4	73	1.2	1.44
5	71	−0.8	0.64
	$\bar{x} = \frac{359}{5} = 71.8$	$\sum d_i = 0$	$\sum d_i^2 = 2.80$

Summing the numbers d_i^2 in the fourth column of Table 2.2 and dividing by 5, we obtain the quantity σ_x^2 (often called the *variance* of the measurements),

$$\sigma_x^2 = \frac{1}{N}\sum d_i^2 = \frac{2.80}{5} = 0.56$$

Taking the square root, we find the standard deviation

$$\sigma_x \approx 0.7$$

Thus the average error of the five measurements 71, 72, 72, 73, 71 is approximately 0.7.

Unfortunately, the standard deviation has an alternative definition. There are theoretical arguments for replacing the factor N in Eq. (2.5) by $(N-1)$ and defining the standard deviation σ_x of x_1, x_2, $\cdots$, x_N as

$$\sigma_x = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(d_i)^2} = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(x_i - \bar{x})^2} \qquad (2.6)$$

The new "improved" definition is obviously a little larger than that of Eq. (2.5) and also to say that Eq. (2.6) corrects a tendency for Eq. (2.5) to understate the error in the measurements x_1, x_2, $\cdots$, x_N, especially if the number of measurements N is small. This tendency can be understood by considering the extreme (and absurd) case that $N = 1$ (that is, only one measurement). Here, the average of x is equal to our one reading x_1 and the one deviation is automatically zero. Therefore, the definition by Eq. (2.5) gives the absurd result $\sigma_x = 0$. On the other hand, the definition by Eq. (2.6) gives 0/0; that is, with definition by Eq. (2.6), σ_x is undefined, which correctly reflects our total ignorance of the error after just one measurement. The definition by Eq. (2.6) is sometimes called the *population standard deviation* and Eq. (2.5) the *sample standard deviation.*

The difference between the two Eqs. (2.5) and (2.6) is almost always numerically insignificant. You should always repeat a measurement many times (at least five, and preferably many more). Even if you make only five measurements ($N = 5$), the difference between $\sqrt{N} = 2.2$ and $\sqrt{N-1} = 2$ is, for most purposes, insignificant. Nevertheless, you need to be aware of both definitions. In the physics laboratory, using the more conservative (that is, larger) definition in Eq. (2.6) is almost always best, but in any case, your laboratory report should state clearly which definition you are using so that your readers can check the calculations for themselves.

2.3.2 The Standard Deviationof the Mean

If x_1, x_2, $\cdots$, x_N are the results of N measurements of the same quantity x, then, as we have seen, our best estimate for the quantity x is their mean $\bar{x}$. We have also seen that the standard deviation (σx characterizes the average error of the separate measurements x_1, x_2, $\cdots$, x_N. Our answer, $x_{best} = \bar{x}$, however, represents a judicious combination of all N measurements, and we have every reason to think it will be more reliable than anyone of the measurements taken

alone. It has been proven that the uncertainty in the final answer $x_{best} = \bar{x}$ is given by the standard deviation σ_x divided by $\sqrt{N}$. This quantity is called the *standard deviation of the mean*, or SDOM, and is denoted $\sigma_{\bar{x}}$

$$\sigma_{\bar{x}} = \sigma_x / \sqrt{N} = \sqrt{\frac{1}{N(N-1)} \sum_{i=1}^{N} (x_i - \bar{x})^2} \tag{2.7}$$

Its other common names are *standard error* and *standard error of the mean*.

2.4 Accuracy and Precision

Accuracy and *precision* are commonly used synonymously in daily life, but in experimental measurements there is an important distinction. The *accuracy* of a measurement signifies how close it comes to the true (or accepted) value—that is, how nearly correct it is.

Precision refers to the agreement among repeated measurements—that is, the "spread" of the measurements or how close they are together. The more precise a group of measurements, the closer together they are. However, a large degree of precision does not necessarily imply accuracy as illustrated in Fig. 2-3.

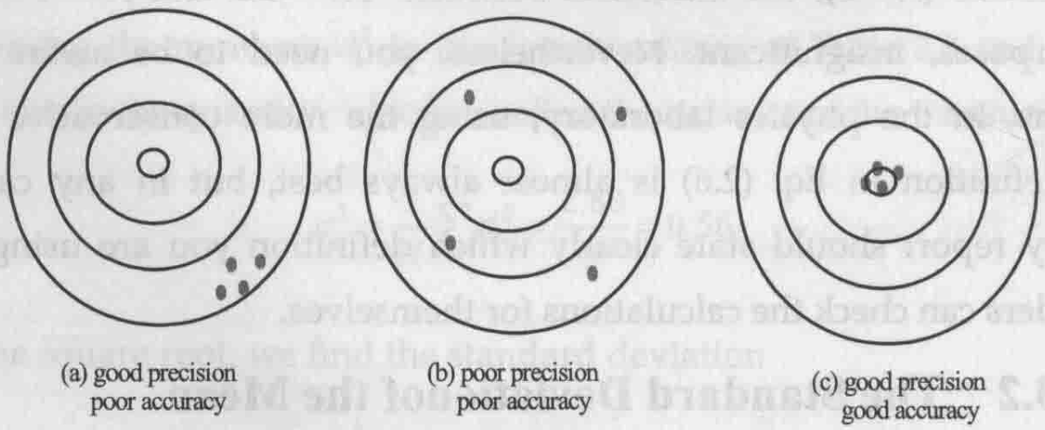


Fig. 2-3 Accuracy and precision. The true value in this analogy is the bull's eye. The degree of scattering is an indication of precision—the closer together the shots, the greater the precision. A group (or symmetric grouping with an average) close to the true value represents accuracy

Obtaining greater accuracy for an experimental value depends in general on *minimizing systematic errors*. Obtaining greater precision for an experimental

value depends on *minimizing random errors.*

2.5 Uncertainty

Uncertainty of a measurement is the doubt that exists about the result of the measurement. You might think that well-made rulers, clocks and thermometers should be trustworthy, and give the right answers. But for every measurement — even the most careful—there is always a margin of doubt. In everyday speech, this might be expressed as "give or take" . . . e. g. a stick might be two meters long "give or take a centimeter".

2.5.1 Uncertainty of a Direct Measurement

Uncertainty of a measurement comprises, in general, many components. Some of these components may be evaluated from the statistical distribution of the results of series of measurements and can be characterized by experimental standard deviations. The other components, which also can be characterized by standard deviations, are evaluated from assumed probability distributions based on experience or other information. In this book, only two components, the Type A and the Type B evaluation of uncertainties are considered.

2.5.1.1 Type A Evaluation of Uncertainty

This component, known as the Type A standard uncertainty or simply Type A uncertainty, can be found from statistical distribution of measurement results and can be characterized by experimental standard deviation. It is denoted μ_A and associated basically with random errors in classical approach.

$$\mu_A = \sigma_x / \sqrt{N} = \sqrt{\frac{1}{N(N-1)} \sum_{i=1}^{N} (x_i - \bar{x})^2} \tag{2.8}$$

It is found by statistical analysis of results of repeated measurements. Its value decreases with number of measurements. Especially, if N=1, the readings of direct measurement can be considered the same during the process of repeat

measurements. In this case, $\mu_A = 0$ for N=1.

2.5.1.2 Type B Evaluation of Uncertainty

In physical experiments, many of the experiments are conducted only once and the Type A uncertainty is zero. Here, we introduce the Type B standard uncertainty, or simply Type B uncertainty, to evaluate the uncertainty of the measurement. The uncertainties which are not in accordance with the statistic principles are called the Type B uncertainty, symbolized μ_B. There are many components resulting in the Type B uncertainty. In this course, we only consider the *instrument error* and the *reading error*, which are denoted by $\Delta_{\text{Instru.}}$ and Δ_{Read}, respectively.

The Type B uncertainty resulting from the instrument error is shown here

$$\mu_{\Delta_{\text{Instru.}}} = \frac{\Delta_{\text{Instru.}}}{C} \tag{2.9}$$

where, C is the confidence probability. In this book, $\sqrt{3}$ is adopted for C. Therefore, we use the following formula to calculate Type B uncertainty resulting from the instrument error.

$$\mu_{\Delta_{\text{Instru.}}} = \frac{\Delta_{\text{Instru.}}}{\sqrt{3}} \tag{2.10}$$

The magnitude of $\Delta_{\text{Instru.}}$ for different apparatus used in the laboratory experiments will be illustrated in the next section.

For the scale-marked measuring tools or instruments, if the scale is clear and with good lighting, we should estimate fractions of a division, which are the measured value's reading error marked Δ_{Read}.

Likewise, the Type B uncertainty resulting from the reading error is given as,

$$\mu_{\Delta_{\text{Read}}} = \frac{\Delta_{\text{Read}}}{\sqrt{3}} \tag{2.11}$$

In our courses the magnitude of Δ_{Read} is determined to be 0.2 of the smallest division of the tools or instruments. For example, the smallest division on a meter stick is 1 mm and the Δ_{Read} is 0.2 × 1 mm = 0.2 mm. It is worth noting that

reading error occurs in most of the results measured by using analog instruments. One exception is that the vernier calipers or instruments with the vernier don't have reading error. However, there is no reading error in the results registered on the digital device. Therefore, when using a digital instrument we will not consider the component from the reading error in computing the Type B uncertainty.

Generally speaking, the measured value's evaluating error Δ_{Read} is far smaller than the instrument's greatest possible error $\Delta_{\text{Instru.}}$ So, we use $\Delta_{\text{Instru.}}$ to define Type B uncertainty of the measured results. However, in some circumstances, reading error is far bigger than the instrument error, for example, when we use electronic second watch to time (Why?).

The errors, $\Delta_{\text{Instru.}}$ and Δ_{Read}, are both independent and neither of them satisfies the statistic principles. We usually use geometric adding to evaluate the Type B uncertainty.

$$\mu_{\text{B}} = \sqrt{\mu_{\Delta_{\text{Instru.}}}^2 + \mu_{\Delta_{\text{Read}}}^2} \tag{2.12}$$

2.5.1.3 Combined Standard Uncertainty

Combining the Type A standard uncertainty μ_{A} and the Type B standard uncertainty μ_{B}, the uncertainty of a direct measurement σ is found as

$$\sigma = \sqrt{\mu_{\text{A}}^2 + \mu_{\text{B}}^2} \tag{2.13}$$

2.5.2 Expression of the Measurement Result

The measurement result for one direct measurement quantity or one indirect measurement quantity can be reported in the form:

$$x = (\bar{x} \pm \sigma)\ \text{unit} \quad (P = 68.3\%) \tag{2.14}$$

Where, $\bar{x}$ is the best estimate value, which also refers to the mean or average of the measurement, and σ is the uncertainty of the measurement. Normally, the probability of $P = 68.3\%$ in the above expression can be omitted.

The physical meaning of the final expression of the measurement is that the

probability of the true value lying in the range from $\bar{x}+\sigma$ to $\bar{x}-\sigma$ is 68.3% (see Fig. 2-4).

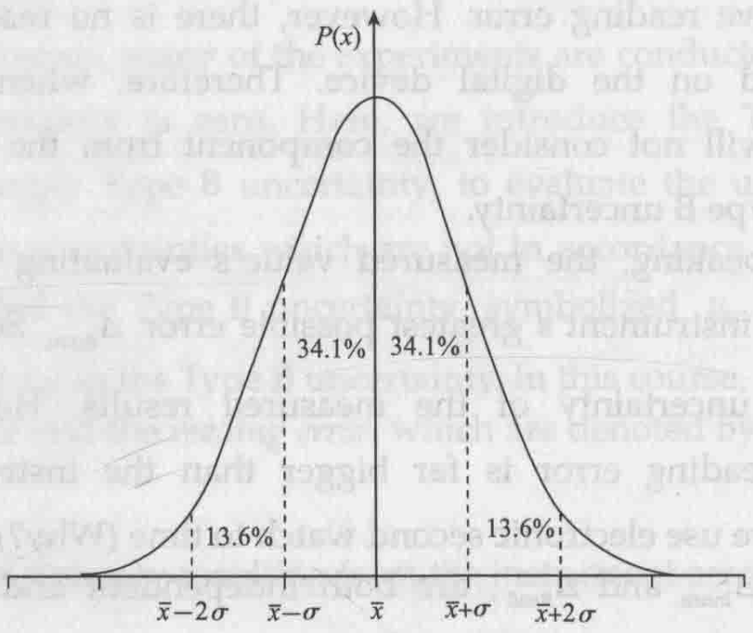


Fig. 2-4 The normal distribution. The area between $\bar{x} \pm n\sigma (n = 1 \text{ and } 2)$ is the probability of a measurement within $n (n = 1 \text{ and } 2)$ standard deviation of $\bar{x}$

The limiting distribution $f(x)$ for measurement of some quantity x tells us the probability of obtaining any given value of x. Specifically, the integral

$$\int_a^b f(x)\,\mathrm{d}x$$

is the probability that anyone measurement gives an answer in the range $a \leqslant x \leqslant b$. When a very large number of measurements are made the spread of results about the mean value, if determined by random fluctuations, will be represented by a Normal Distribution (also called the Gaussian distribution) and the integral can be evaluated. In particular, we can calculate the probability that a measurement will fall within one standard deviation σ of the average $\bar{x}$. This probability is

$$\frac{1}{\sigma\sqrt{2\pi}}\int_{\bar{x}-\sigma}^{\bar{x}+\sigma} \mathrm{e}^{-(x-\bar{x})^2/2\sigma^2}\,\mathrm{d}x$$

It can be easily computed on a computer and is given by 68.3%. In addition, the probability that a measurement will fall inside 2σ is 95.4%; that for 3σ is 99.7% (see Fig. 2-4).

There are two major rules for stating the uncertainty.

Rule 2.1a *Experimental uncertainty should almost always be rounded to one significant figure* (see 2.8 for more on significant figures).

Uncertainties are usually only known roughly, so quote the uncertainties to no more than two significant figures. One significant figure is usually sufficient. Thus, if some calculation yields acceleration of gravity $g = 9.82$ m/s^2 and its uncertainty $\sigma_g = 0.043\,85$ m/s^2, the uncertainty should be rounded to $\sigma_g = 0.04$ m/s^2, and the conclusion should be rewritten as

$$g = (9.82 \pm 0.04)\,\text{m/s}^2$$

If the leading digit in the uncertainty σ is a 1, then keeping two significant figures in σ may be better. For example, suppose that some calculation gave the uncertainty $\sigma = 0.14$. Rounding this number to $\sigma = 0.1$ would be a substantial proportionate reduction, so we could argue that retaining two figures might be less misleading, and quote $\sigma = 0.14$. The same argument could perhaps be applied if the leading digit is a 2 but certainly not if it is any larger.

Rule 2.1b *The last significant figure in any stated answer (the best estimated value or the mean) should usually be of the same order of magnitude (in the same decimal position) as the uncertainty. (Note: It hints that both the best estimated value and the experimental uncertainty must be in the same unit.)*

For example, the answer 92.81 mm with an uncertainty of 0.3 mm should be rounded as

$$L = (92.8 \pm 0.3)\ \text{mm}$$

If its uncertainty is 3, then the answer should be given as

$$L = (93 \pm 3)\ \text{mm}$$

An important qualification to rules 1 and 2 is as follows: To reduce inaccuracies caused by rounding, *any numbers to be used in subsequent calculations*

should normally retain at least one more significant figure than is required for the final result. At the end of the calculations, the final answer should be rounded to remove these extra, insignificant figures. An electronic calculator will happily carry numbers with far more digits than are likely to be significant in any calculation you make in a laboratory. Obviously, these numbers do not need to be rounded in the middle of a calculation but certainly must be rounded appropriately for the final answers.

Note that the uncertainty in any measured quantity has the same dimensions as the measured quantity itself. Therefore, writing the units (m/s^2, cm^3, etc.) after both the answer *and* the uncertainty is clearer and more economical. By the same token, if a measured number is so large or small that it calls for scientific notation (the use of the form 3×10^3 instead of 3 000, for example), then it is simpler and clearer to put the answer and uncertainty in the same form. For example, the result of

$$\text{measured charge} = (1.61 \pm 0.05) \times 10^{-19} \text{ coulombs}$$

is much easier to read and understand in this form than it would be in the form

$$\text{measured charge} = (1.61\times10^{-19} \pm 5\times10^{-21}) \text{ coulombs}$$

2.6 Instrument Error

In order to evaluate the Type B uncertainty the instrument error $\Delta_{\text{Instru.}}$ should be determined. In this section, we will discuss how to determine the instrument error $\Delta_{\text{Instru.}}$.

Instrument error refers to the combined accuracy and precision of a measuring instrument, or the difference between the actual value and the value indicated by the instrument (error). It is also named the greatest possible error of the instrument. All instruments, no matter how sophisticated, have a limit to their precision. The errors may be due to wrong construction, calibration of the measuring instruments. Usually, the manufacturer of the instrument will specify the error. It is always written on the back of the instrument. However, if

neither you nor your laboratory instructor can find this information, the following sections provide the rule of thumb for some common measuring instruments.

2.6.1 Instruments for Measuring Length

General tools used to measure length include rule, caliper, micrometer screw gauge, tape, and so on. Main parameters and the greatest possible errors of the tools are shown in Table 2.3.

Table 2.3 Greatest possible errors of some length measuring tools

Instruments	Division /mm	Greatest possible error $\Delta_{Instru.}$ /mm
Steel rule	1	0.1
Plastic meter stick	1	0.5
Micrometer screw gauge	0.01	0.005
Tape measure	1	0.2×L‰+0.3 (L = measured value, mm)
Vernier caliper	0.02	0.02
	0.05	0.05

2.6.2 Instruments for Measuring Time

Mechanical stop watch, quartz electric watch and digit millisecond watch are often used to record time. In physics experiment, the greatest possible errors of the stop watch and quartz electric watch are always 0.01 s. The greatest possible errors of the digit millisecond watch is 1 ms.

2.6.3 Analog Instruments for Measuring Electric Quantities

According to the national standard, the accuracy class of the electric instruments can be classified into 0.1, 0.2, 0.5, 1.0, 1.5, 2.5, and 5.0. They correspond to the percentages of ±0.1%, ±0.2%, ±0.5%, ±1.0%, ±1.5% , ±2.5%, and ±5.0%, respectively. Generally, these accuracy classes have been marked on the apparatus. The instruments' limit error (or the greatest possible error) can be

obtained through the accuracy class.

$$\Delta_{\text{Instru.}} = a\% \times N_{\text{m}} \tag{2.15}$$

where, $\Delta_{\text{Instru.}}$ is the instrument's limit error (or the greatest possible error), N_{m} is the full scale reading for the chosen measurement range, a is the accuracy class.

In addition, a resistance box is often used in the physics laboratory. Its greatest possible error is the smallest division on the panel.

2.6.4 Digital Instruments for Measuring Electric Quantities

Nowadays, many modern laboratory instruments use digital displays. For example, you can measure weights using an electronic scale. It has the advantage that you don't have to judge what the central value of your measurement is. It gives the specific reading automatically.

The greatest possible error, $\Delta_{\text{Instru.}}$, of a digital device can be obtained by

$$\Delta_{\text{Instru.}} = a\% \times N_x + a \text{ digit}(s) \tag{2.16}$$

where, a is the accuracy class, N_x is the reading registered on the display, and α is the product of a specified integer and the unit of the last digit of the reading. For example, a digital voltmeter with accuracy class of 0.02 and measurement range of 2 V is used to measure a voltage. The measurement result is 1.4786 V. The manufacturer specifies that the instrument error of the voltmeter is 0.02% × N_x + 2 digits. In this case, the unit of the last digit is 0.0001 V and the specified integer is 2. So, $\alpha = 0.0001 \times 2 = 0.0002$ V and the instrument error is given as 0.02% × 1.4786 + 0.0002 = 0.0005 V.

Example 2.3 There are 10 readings for the length (L) of a string taken by a tape measure with division of 1 mm: 53.27, 53.25, 53.23, 53.29, 53.24, 53.28, 53.25, 53.20, 53.24, 53.21 (unit: cm). Please calculate the uncertainty in the length.

Solution

Step 1: Calculate the mean of the readings.

$$\bar{L} = \frac{1}{N}\sum_{i=1}^{10} L_i$$
$$= \frac{53.27 + 53.25 + \cdots + 53.21}{10}$$
$$= \frac{532.47}{10} = 53.247(\text{cm})$$

Step 2: Using Eq. (2.8) to calculate the Type A standard uncertainty.

$$\mu_{AL} = \sqrt{\frac{\sum_{i=1}^{N}(L_i - \bar{L})^2}{N \times (N-1)}}$$
$$= \sqrt{\frac{(53.27 - 53.247)^2 + (53.25 - 53.247)^2 + \cdots + (53.21 - 53.247)^2}{10 \times (10-1)}}$$
$$= \sqrt{\frac{7.61 \times 10^{-3}}{90}}$$
$$= 9.2 \times 10^{-3}(\text{cm})$$

Step 3: Calculate the Type B uncertainty.

According to Table 2.3, the greatest possible error of tape measure is

$$\Delta_{\text{Instru.}} = 0.2 \times L‰ + 0.3 = 0.2 \times 532.47‰ + 0.3 = 0.41\ (\text{mm}) = 0.041\ (\text{cm})$$

$$\mu_{\Delta_{\text{Instru.}}} = \frac{\Delta_{\text{Instru.}}}{\sqrt{3}} = \frac{0.041}{\sqrt{3}} = 0.024(\text{cm})$$

The reading error and its Type B uncertainty component are

$$\Delta_{\text{Read}} = 0.2 \times 1 = 0.2(\text{mm}) = 0.02(\text{cm})$$

$$\mu_{\Delta_{\text{Read}}} = \frac{\Delta_{\text{Read}}}{\sqrt{3}} = \frac{0.02}{\sqrt{3}} = 0.01(\text{cm})$$

So, the combined Type B uncertainty is

$$\mu_{BL} = \sqrt{\mu_{\Delta_{\text{Instru.}}}^2 + \mu_{\Delta_{\text{Read}}}^2}$$
$$= \sqrt{0.024^2 + 0.01^2}$$
$$= 0.026(\text{cm})$$

Step 4: Calculate the combined uncertainty.

$$\sigma_L = \sqrt{\mu_{AL}^2 + \mu_{BL}^2}$$
$$= \sqrt{0.0092^2 + 0.026^2}$$
$$= 0.03(\text{cm})$$

Step 5: Express the final result.

$$L = L \pm \sigma_L$$
$$L = (53.34 \pm 0.03)\,\text{cm}$$

2.7 Propagation of Uncertainties

Most physical quantities usually cannot be measured by a single direct measurement but are instead found in two distinct steps. First, we measure one or more quantity(ies) by direct measurement. Second, we use the so measured values of these quantities to calculate the quantity of interest itself. For example, to find the area of a rectangle, you actually measure its length l and breadth h and then calculate its area A as,

$$A = lh$$

Any reader with experience in introductory laboratory can easily think of more examples. In fact, a little thought will show that almost all interesting measurements involve these two distinct steps of direct measurement followed by calculation.

When a measurement involves these two steps, the estimation of uncertainties also involves two steps. We must first estimate the uncertainties in the quantities measured directly and then determine how these uncertainties "propagate" through the calculations to produce an uncertainty in the final answer. This propagation of errors is the main subject of this section.

Suppose that x, $\cdots$, z are measured with uncertainties σ_x, $\cdots$, σ_z and the measured values are used to compute the function $f(x, \cdots, z)$. If the uncertainties in x, $\cdots$, z are independent and random, then the general rule for the uncertainty in f is

$$\sigma_f = \sqrt{\left(\frac{\partial f}{\partial x}\sigma_x\right)^2 + \cdots + \left(\frac{\partial f}{\partial z}\sigma_z\right)^2} \tag{2.17}$$

The term $\frac{\partial f}{\partial x}\sigma_x$ by itself is the uncertainty, or partial uncertainty, in f caused by the uncertainty in x alone. In the same way, $\frac{\partial f}{\partial y}\sigma_y$ is the partial uncertainty in f due to σ_y alone, and so on. Referring to Eq. (2.17), we see that the total uncertainty in f is the square root of the sum of the squares of the partial uncertainties due to each of the separate uncertainties $\sigma_x, \sigma_y, \cdots, \sigma_z$ (provided the latter are independent). This is a good way to think about the result in Eq. (2.17) and it suggests the simplest way to use Eq. (2.17) to calculate the total uncertainty in f. In using the Eq. (2.17), first calculate the partial uncertainties in f due to $\sigma_x, \sigma_y, \cdots, \sigma_z$ separately, then simply combine these separate uncertainties in quadrature to give the total uncertainty.

Table 2.4 shows the combined formula to evaluate the uncertainty of the functions used frequently. In Table 2.4, $\frac{\sigma_f}{|f|}$ is defined as relative uncertainty (or fractional uncertainty). It is often used when the function is products or quotients. In other words, we can first calculate the relative uncertainty $\frac{\sigma_f}{|f|}$ in products or quotients and then calculate the uncertainty of the function using

$$\sigma_f = \frac{\sigma_f}{|f|} \times |f| \tag{2.18}$$

Table 2.4 Formula to evaluate the uncertainty of the functions used frequently

Functions	the formula to evaluate the uncertainty	
$f = ax + \cdots + bz - (cu + \cdots + dw)$	$\sigma_f = \sqrt{(a\sigma_x)^2 + \cdots + (b\sigma_z)^2 + (c\sigma_u)^2 + \cdots + (d\sigma_w)^2}$	(2.19)
$f = \frac{x \times \cdots \times z}{u \times \cdots \times w}$	$\frac{\sigma_f}{\lvert f\rvert} = \sqrt{\left(\frac{\sigma_x}{x}\right)^2 + \cdots + \left(\frac{\sigma_z}{z}\right)^2 + \left(\frac{\sigma_u}{u}\right)^2 + \cdots + \left(\frac{\sigma_w}{w}\right)^2}$	(2.20)
$f = \frac{x^k \times y^n}{z^m}$	$\frac{\sigma_f}{\lvert f\rvert} = \sqrt{\left(k\frac{\sigma_x}{x}\right)^2 + \left(n\frac{\sigma_y}{y}\right)^2 + \left(m\frac{\sigma_z}{z}\right)^2}$	(2.21)

Example 2.4 Suppose you measure the volumes of water in two beakers as

$$V_1 = (130 \pm 3)\ \text{ml}$$

$$V_2 = (65 \pm 2)\ \text{ml}$$

and then carefully pour the contents of the first into the second. What is your prediction for the total volume $V = V_1 + V_2$ with its uncertainty, σ_V ?

Solution The best estimate for V is easily found as

$$V = V_1 + V_2 = 130 + 65 = 195\ (\text{ml})$$

It gives V as the sum of two factors, V_1, and V_2. According to Eq. (2.19) the uncertainty in V is

$$\sigma_v = \sqrt{(\sigma_{v_1})^2 + (\sigma_{v_2})^2}$$
$$= \sqrt{(3)^2 + (2)^2}$$
$$= 3\ \text{ml}$$

Thus, our final answer is,

$$V = (195 \pm 3)\text{ml}$$

Example 2.5 Suppose that we measure g, the acceleration of gravity, using a simple pendulum. The period of such a pendulum is well known to be $T = 2\pi\sqrt{l/g}$, where l is the length of the pendulum. Thus, if l and T are measured, we can find g as $g = 4\pi^2 l/T^2$. Suppose we measure the period T for one value of the length l and get the results. If the measured period $T = (1.936 \pm 0.004)$ s for the measured length $l = (0.930 \pm 0.001)$ m, what is the result for g?

Solution This result gives g as the product or quotient of three factors, $4\pi^2$, l, and T^2. If the various uncertainties are independent and random, the fractional uncertainty in our answer is just the quadratic sum of the fractional uncertainties in these factors. The factor $4\pi^2$ has no uncertainty, and according to Eq. (2.21) the fractional uncertainty in g is

$$\frac{\sigma_g}{|g|} = \sqrt{\left(\frac{\sigma_l}{l}\right)^2 + \left(2\frac{\sigma_T}{T}\right)^2}$$

$$= \sqrt{\left(\frac{0.001}{0.930}\right)^2 + \left(2\times\frac{0.004}{1.936}\right)^2}$$

$$= 0.0043$$

The best estimate for g is easily found from as

$$g = \frac{4\pi^2 l}{T^2}$$

$$= \frac{4\times 3.142^2 \times 0.930}{1.936^2}$$

$$= 9.80\ (\mathrm{m/s^2})$$

According to Eq. (2.18)

$$\sigma_g = \frac{\sigma_g}{|g|}\times|g|$$

$$= 0.0043\times 9.80$$

$$= 0.04\ (\mathrm{m/s^2})$$

Thus, based on these measurements, our final answer is

$$g = (9.80 \pm 0.04)\ \mathrm{m/s^2}$$

Having found the measured value of g and its uncertainty, we would naturally compare these values with the accepted value of g. If the latter has its usual value of 9.81 m/s^2, the present value is entirely satisfactory.

NOTE

When we calculate the best estimate for g with the obtained value, the significant figures of constant π must be the same as or more than measured values. It is in order to reduce the error and uncertainty caused by mathematical operation. In the above example, $\pi = 3.142$ is adopted, whose significant figures are the same as that of T and one more than that of l.

2.8 Significant Figures

In general, there are *exact* numbers and *measured* numbers (or quantities). Factors such as the 100 used in calculating percentage and the 2 in $2\pi r$ are exact numbers. Measured numbers, as the name implies, are those obtained from measurement instruments and generally involve some error or uncertainty.

In reporting experimentally measured values, it is important to read instruments correctly. The degree of uncertainty of a number read from a measurement instrument depends on the quality of the instrument and the fineness of its measuring scale. When reading the value from a calibrated scale, only a certain number of figures or digits can properly be obtained or read. That is, only acertain number of figures are *significant*. This depends on the *least count* of the instrument scale, which is the smallest subdivision on the measurement scale. It is the unit of the smallest reading that can be made without estimating. For example, the least count of a meter stick is usually the millimeter (mm). We commonly say "the meter stick is calibrated in centimeters (numbered major divisions) with a millimeter least count. " (See Fig. 2-5)

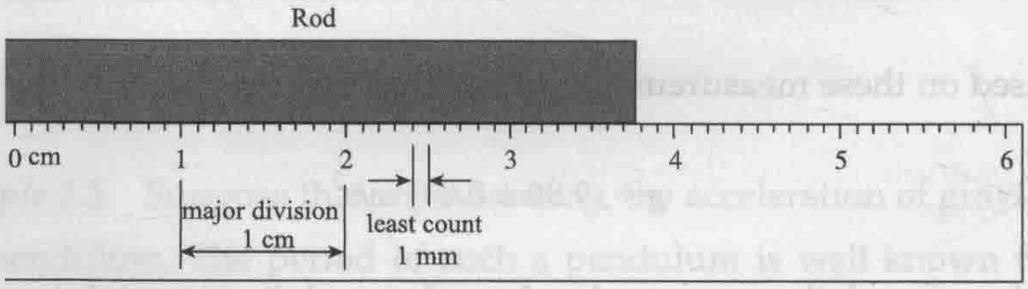


Fig. 2-5 Least count. Meter sticks are commonly calibrated in centimeters (cm), the numbered major divisions, with a least count or smallest subdivision of millimeters (mm)

The *significant figures*(sometimes called *significant digits*) of a measured value include all the numbers that can be read directly from the instrument scale, *plus* one doubtful or estimated number-the fractional part of the least count smallest division. For example, the length of the rod in Fig. 2-5 is 37.5 mm. The rod's length is known to be between 37 mm and 38 mm. The estimated fraction is taken to be 5/10 of the least count (mm), so the doubtful figure is 5, giving 37.5 mm with three significant figures.

Thus, measured values contain inherent uncertainty or doubtfulness because of the estimated figure. However, the greater the number of significant figures the greater the reliability of the measurement that the number represents. For example, the length of an object may be read as 36.5 mm(three significant figures) on one instrument scale and as 36.505 mm (five significant figures) on another. The latter reading is from an instrument with a finer scale (why?) and gives more information and reliability.

Zeros and the decimal point must be properly dealt with in determining the number of significant figures in a result. For example, how many significant figures does 0.0543 m have? What about 209.4 m and 2705.0 m? In such cases, the following rules are generally used to determine significance:

Rule 2.2a *Zeros at the beginning of a number are not significant. They merely locate the decimal point.*

For example, 0.0543 m has three significant figures (5, 4, and 3).

Rule 2.2b *Zeros within a number are significant.*

For example, 209.4 m has four significant figures (2, 0, 9, and 4).

Rule 2.2c *Zeros at the end of a number after the decimal point are significant.*

For example, 2705.0 has five significant figures (2, 7, 0, 5, and 0).

Rule 2.2d *The exact number in the formula is without uncertainty and has infinite significant figures.*

For example, the 2 in the formula $2\pi r$ is an exact number and has infinite significant figures.

Some confusion may arise with whole numbers that have one or more zeros at the end without a decimal point. Consider, for example, 300 kg, where the zeros (called trailing zeros) may or may not be significant. In such cases, it is not clear which zeros serve only to locate the decimal point and which are actually part of the measurement (and hence significant). That is, if the first zero from the left (300 kg) is the estimated digit in the measurement, then only two digits are reliably known, and there are only two significant figures. Similarly, if the last zero is the estimated digit (300 kg), then there are three significant figures. This ambiguity can be removed by using *scientific (powers of 10) notation*:

3.0×10^2 kg has two significant figures.

3.00×10^2 kg has three significant figures.

This notation is also helpful in expressing the significant figures in large numbers. For example, suppose that the average distance from Earth to the Sun, 93 000 000 miles, is known to have only four significant figures. This is easily expressed in scientific notation as 9.300×10^7 mi.

Except for the scientific notation, there are various conventions to address this issue. A bar may be placed over the last significant figure and any trailing zeros following this are insignificant; for example, "13$\bar{0}$0" has three significant figures. The last significant figure of a number may be underlined; for example, "2$\underline{0}$00" has two significant figures. A decimal point may be placed after the number; for example "100." indicates specifically that three significant figures are meant.

2.9 Computations with Measured Values

Calculations are often performed with measured values and uncertainties are "propagated" by the mathematical operations—for example, multiplication or division. That is, uncertainties are carried through to the results by the mathematical operations.

The error can be better expressed by statistical methods. However, a widely used procedure for estimating the uncertainty of a mathematical result involves the use of significant figures.

The number of significant figures in a measured value gives an indication of the uncertainty or reliability of a measurement. Hence, you might expect that the result of a mathematical operation can be no more reliable than the quantity with the least reliability, or smallest number of significant figures, used in the calculation. That is, *reliability cannot be gained through a mathematical operation.*

It is important to report the results of mathematical operations with the proper number of significant figures. This is accomplished by using rules for multiplication, division, addition, and subtraction. To obtain the proper number of significant figures, one rounds the results off. The general rules used for

mathematical operations are as follows.

Rule 2.3a *When multiplying and dividing quantities, leave as many significant figures in the answer as there are in the quantity with the least number of significant figures.*

Rule 2.3b *When adding or subtracting quantities, leave the same number of decimal places (rounded) in the answer as there are in the quantity with the least number of decimal places.*

Rule 2.3c *When doing logarithm, leave the same number of decimal places (rounded) in the answer as there are in the quantity with the least significant figures of the antilogarithm.*

The general rules used for rounding are as follows.

Rule 2.4a *If the first digit to be dropped is less than 5, leave the preceding digit as is.*

Rule 2.4b *If the first digit to be dropped is 5 or greater, increase the preceding digit by one.*

Notice that in the method of rounding, five digits (0, 1, 2, 3, and 4) are rounded down and five digits (5, 6, 7, 8, and 9) are rounded up.

What the rules for determining significant figures mean is that the result of a calculation can be no more accurate than the least accurate quantity used. In other words, you cannot gain accuracy in performing mathematical operations.

Example 2.6 Calculate the following by applying the rules of arithmetic operations (*s. f.* stands for significant figures): (1) Multiply 2.5 m by 1.308 m; (2) Divide 882.0 s2 by 0.245 s; (3) Add up the figures, 46.4, 1.37, 0.505; (4) Subtract 4.5 from 163; (5) Find (a) log 1.983 and (b) log 1983.

Solution (1)Multiplication:

$$\underset{(2\text{ s. f.})}{2.5\text{ m}} \times \underset{(4\text{ s. f.})}{1.308\text{ m}} = 3.27\text{ m}^2 \approx \underset{(2\text{ s. f.})}{3.3\text{ m}^2}$$

(rounding off)

(2)Division:

$$\frac{\underset{(4\text{ s. f.})}{882.0\text{ s}^2}}{\underset{(3\text{ s. f.})}{0.245\text{ s}}} = 3600\text{ s} = \underset{(3\text{ s. f.})}{3.60\times10^3\text{ s}}$$

(3)Addition:

$$46.4 + 1.37 + 0.505 = 48.275 \approx 48.3$$

(46.4 has the least number of decimal places)

(4)Subtraction:

$$163 - 4.5 = 158.5 \approx 159$$

(rounding off)

(163 has the least number of decimal places)

(5)Logarithm:

(a) $\log 1.983 = 0.2973227\cdots \approx 0.2973$

(4 s. f.) (rounding off) (4 decimal places)

(b) $\log 1983 = 3.29732271\cdots \approx 3.2973$

(4 s. f.) (rounding off) (4 decimal places)

2.10 Graphical Representation of Data

It is often convenient to represent experimental data in graphical form, not only for reporting but also to obtain information.

Quantities are commonly plotted using rectangular Cartesian axes (X and Y). The horizontal axis (X) is called the *abscissa*, and the vertical axis (Y), the *ordinate*. The location of a point on the graph is defined by its coordinates x and y, written as (x, y) with reference to the origin O, the intersection of the X and Y axes.

When plotting data, choose axis scales that are easy to plot and read. The graph in Fig. 2-6 shows an example of scales that are too small. This "bunches up" the data, making the graph too small, and the major horizontal scale values make it difficult to read intermediate values. Also, the dots or data points should not be connected. Choose scales so that most of the graph paper is used. The graph in Fig. 2-7 shows data plotted with more appropriate scales. As a

general rule, it is convenient to choose the unit of the first major scale division to the right or above the origin or zero point as 1, 2, or 5(or multiples or submultiples thereof, for example, 10 or 0.1) so that the minor (intermediate) scale divisions can be easily interpolated and read.

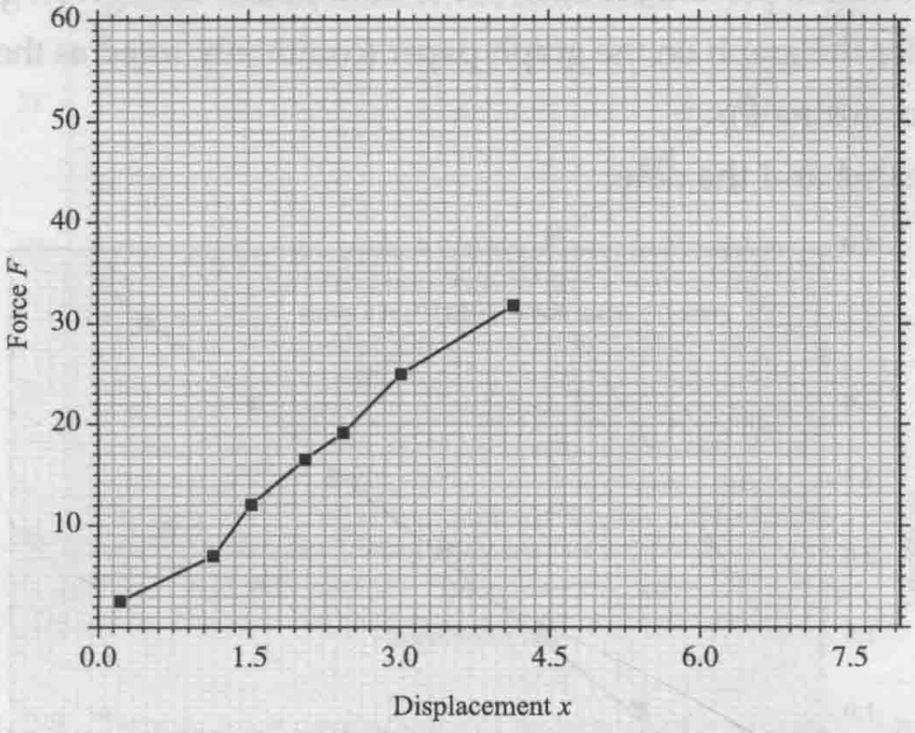


Fig. 2-6 An example of an improperly labeled and plotted graph

Also note in Fig. 2-6 that *scale units* on the axes are not given. For example, you don't know whether the units of displacement are feet, meters, kilometers, or whatever. *Scale units* should always be included, as in Fig. 2-7. It is also acceptable, and saves time, to use standard unit abbreviations, such as *N* for newton and *m* for meter. This will be done on subsequent graphs.

With the data points plotted, draw a smooth line described by the data points. *Smooth* means that the line does not have to pass exactly through each point but connects the general areas of significance of the data points (*not* connecting the data points as in Fig. 2-6). The graphin Fig. 2-7 with an approximately equal number of points on each side of the line gives a "line of best fit".

Graphs should have the following elements (see Fig. 2-7):

a. Each axis labeled with the major and least count divisions indicated.

b. Each axis labeled with the quantity symbol.

c. Each axis must have units of the quantities indicated.

d. The symbols for the data points of each curve must be different from each other if there are several curves on a given graph. The different symbols of data points must all be defined.

e. A smooth line plotted for each set of data points on a given graph.

f. A title for the graph on the graph paper (commonly listed as the *y*-coordinate versus the *x*-coordinate).

g. Your name and the date.

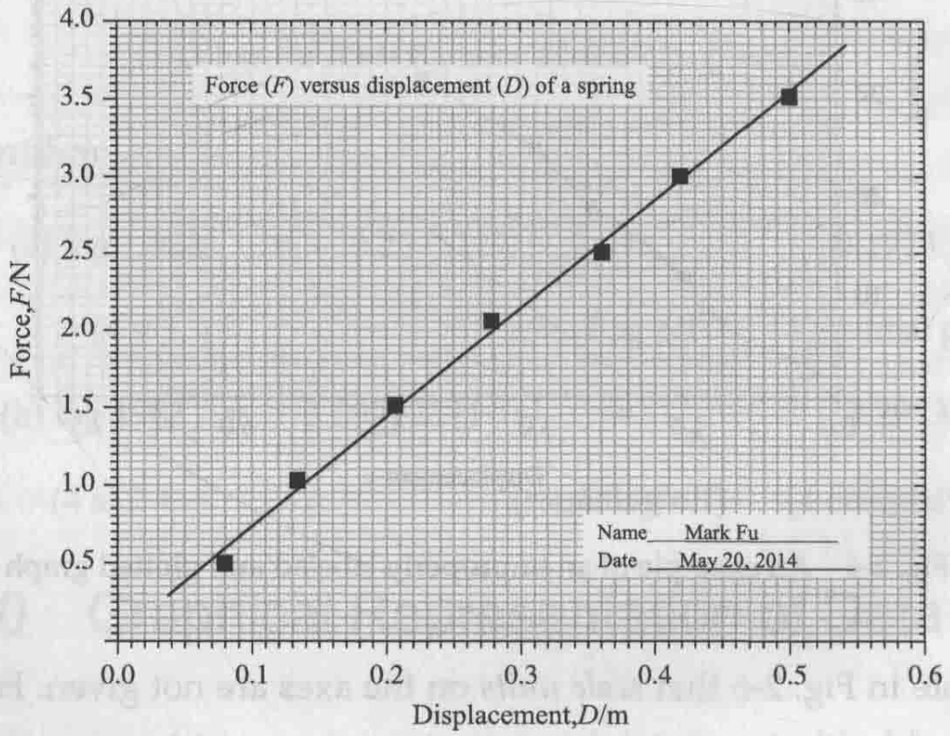


Fig. 2-7 An example of a properly labeled and plotted graph

The two quantities (*x* and *y*) are often linearly related (at this level). That is to say, there is an algebraic relationship of the form $y = ax + b$, where *a* and *b* are constants. When the values of such quantities are plotted, the graph is a straight line, as shown in Fig. 2-7. The *a* in the algebraic relationship is called the *slope* of the line and is equal to the ratio of the intervals $\Delta y/\Delta x$. Any set of intervals may be used to determine the slope of a straight-line graph. The *b* in the algebraic relationship is called the *y-intercept* and is equal to the value of the *y*-coordinate where the graph line intercepts the *Y*-axis. Notice from the relationship that $y = ax + b$, so that when $x = 0$, then $y = b$. If the intercept is at the origin (0, 0), then $b = 0$.

It should be noted that when we use the coordinates of the points to

calculate a and b the points should be chosen *relatively far apart on the line*. For best results, *points corresponding to data points should not be chosen*, even if they appear to lie on the line (see Fig. 2-8).

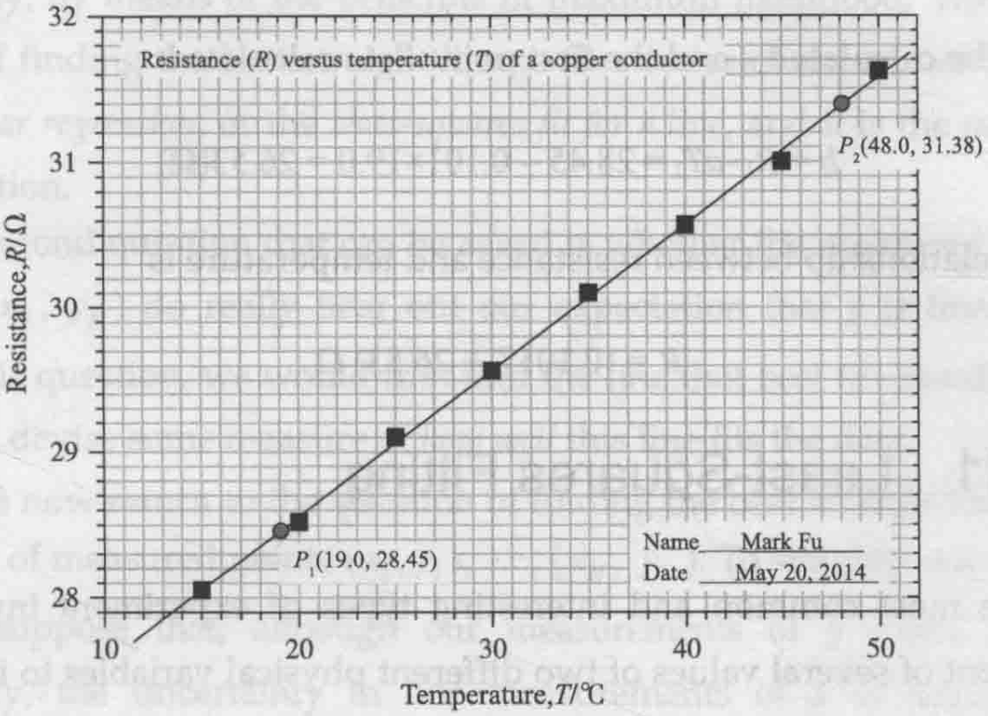


Fig. 2-8 Temperature dependence of resistance of a copper conductor

***Example* 2.7** It is well known that the resistance of a copper conductor has nearly a linear algebraic relationship with its temperature. Table E2.1 shows the measured values of the temperature and the corresponding resistance of a copper conductor. Use these data to draw a graph and find out the relationship between the resistance and the temperature for a copper conductor.

Table E2.1 The experimental results of temperature (T) and the resistance (R) for a copper conductor

T/℃	15.0	20.0	25.0	30.0	35.0	40.0	45.0	50.0
R/Ω	28.05	28.52	29.10	29.56	30.10	30.57	31.00	31.62

Solution

The graph of temperature dependence of resistance of a copper conductor and the best fitted line has been shown in Fig. 2-8. Suppose that the resistance and temperature have the relationship, $R = aT + b$. We choose two widely

separated points, $P_1(19.0, 28.45)$ and $P_2(48.0, 31.38)$ to calculate a and b.

$$a = \frac{R_2 - R_1}{T_2 - T_1} = \frac{31.38 - 28.45}{48.0 - 19.0} = 0.101\,(\Omega/^{\circ}\mathrm{C})$$

Then, use the calculated a and the first point P_1 to calculate b

$$b = R_1 - aT_1 = 28.45 - 0.101 \times 19.0 = 26.53(\Omega)$$

Thus, the relationship between resistance and temperature is

$$R = (0.101T + 26.53)\ \Omega$$

2.11 Least-Squares Fitting

One of the most common and interesting types of experiment involves the measurement of several values of two different physical variables to investigate the mathematical relationship between the two variables. For instance, an experimenter might drop a stone from various different heights $h_1, \cdots, h_N$ and measure the corresponding times of fall $t_1, \cdots, t_N$ to see if the heights and times are connected by the expected relation $h = gt^2/2$.

Probably the most important experiments of this type are those for which the expected relation is *linear*. For instance, if we believe that a body is falling with constant acceleration g, then its velocity v should be a linear function of the time t,

$$v = v_0 + gt$$

More generally, we will consider any two physical variables x and y that we suspect are connected by a linear relation of the form

$$y = a + bx \tag{2.22}$$

where a and b are constants.

If we were to measure N different values $x_1, \cdots, x_N$ and the corresponding values $y_1, \cdots, y_N$, we can ask two questions. First, if we take for granted that y and x *are* linearly related, then the interesting problem is to find the straight line

$y = a + bx$ that best fits the measurements, that is, to find the best estimates for the constants a and b based on the data (x_1, y_1), $\cdots$, (x_N, y_N). This problem can be approached graphically, as discussed in Section 2.10. It can also be treated analytically, by means of the principle of maximum likelihood. This analytical method of finding the best straight line to fit a series of experimental points is called *linear regression*, or the *least-squares fit for a line*, and it is the main subject of this section.

The second question that can be asked is whether the measured values (x_1, y_1), $\cdots$, (x_N, y_N) do really bear out our expectation that y is linear in x. To answer this question, we would first find the line that best fits the data, but we must then devise some measure of *how well* this line fits the data.

Let us now return to the question of finding the best straight line $y = a + bx$ to fit a set of measured points (x_1, y_1), $\cdots$, (x_N, y_N). To simplify our discussion, we will suppose that, although our measurements of y suffer appreciable uncertainty, the uncertainty in our measurements of x is negligible. This assumption is often reasonable, because the uncertainties in one variable often are much larger than those in the other, which we can therefore safely ignore. We will further assume that the uncertainties in y all have the same magnitude.

It is understandable that the measured values y_i is not equal to $a + bx_i$ due to the fact that errors are not avoidable in measurements. Therefore, we can get

$$\begin{aligned}
\varepsilon_1 &= y_1 - (a + bx_1) \\
\varepsilon_2 &= y_2 - (a + bx_2) \\
&\vdots \\
\varepsilon_i &= y_i - (a + bx_i) \\
&\vdots \\
\varepsilon_k &= y_k - (a + bx_k)
\end{aligned}$$

where ε is the error in y. The best estimated values of a and b should satisfy

$$\sum_i^k \varepsilon_i^2 = \sum_i^k (y_i - a - bx_i)^2 = \min \tag{2.23}$$

Eq. (2.23) means that the sum of the error squares ε_i^2 is a minimum (This is why the method is known as least-squares fitting) and then

$$\frac{\partial}{\partial a}\left(\sum_{i=1}^{k}\varepsilon_i^2\right)=-2\sum_{i=1}^{k}(y_i-a-bx_i)=0$$

$$\frac{\partial}{\partial b}\left(\sum_{i=1}^{k}\varepsilon_i^2\right)=-2\sum_{i=1}^{k}(y_i-a-bx_i)x_i=0$$

So,

$$\begin{cases}\sum_{i=1}^{k}y_i-ka-b\sum_{i=1}^{k}x_i=0\\ \sum_{i=1}^{k}x_iy_i-a\sum_{i=1}^{k}x_i-b\sum_{i=1}^{k}x_i^2=0\end{cases} \tag{2.24}$$

Suppose that,

$$\bar{x}=\frac{1}{k}\sum_{i=1}^{k}x_i$$

$$\bar{y}=\frac{1}{k}\sum_{i=1}^{k}y_i$$

$$\overline{x^2}=\frac{1}{k}\sum_{i=1}^{k}x_i^2$$

$$\overline{xy}=\frac{1}{k}\sum_{i=1}^{k}x_iy_i$$

Here, $\bar{x},\bar{y}$, $\overline{x^2}$, $\overline{xy}$ are the means of x_i, y_i, x_i^2, and $x_i y_i$, respectively. Eq. (2.24) can be changed to

$$\begin{cases}\bar{y}-a-b\bar{x}=0\\ \overline{xy}-a\bar{x}-b\overline{x^2}=0\end{cases} \tag{2.25}$$

Solving Eq. (2.25) the least-squares estimates for the constants a and b are determined.

$$a=\frac{\bar{x}\cdot\overline{xy}-\bar{y}\cdot\overline{x^2}}{\bar{x}^2-\overline{x^2}} \tag{2.26}$$

$$b = \frac{\bar{x} \cdot \bar{y} - \overline{xy}}{\bar{x}^2 - \overline{x^2}} \tag{2.27}$$

The results of Eqs. (2.26) and (2.27) give the best estimates for the constants a and b of the straight line $y = a + bx$, based on the k measured points $(x_1, y_1), \cdots, (x_k, y_k)$. The resulting line is called the *least-squares fit* to the data, or the *line of regression* of y on x.

To answer how well the line fits the data, we define a relation coefficient as

$$\gamma = \frac{\sum_{i=1}^{k}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i}^{k}(x_i - \bar{x})^2 (y_i - \bar{y})^2} \tag{2.28}$$

The relation coefficient gives a measure of the degree to which each data fits the line. The closer $|\gamma|$ is to 1, the more the data fits the straight line.

Example 2.8 A student makes a scale to measure masses with a spring. She attaches its top end to a rigid support, hangs a pan from its bottom, and places a meter stick behind the arrangement to read the length of the spring. Before she can use the scale, she must calibrate it; that is, she must find the relationship between the mass in the pan and the length of the spring. To do this calibration, she gets five accurate 2-kg masses, which she adds to the pan one by one, recording the corresponding lengths l_i as shown in the first three columns of Table E2.2. Assuming the spring obeys Hooke's law, she anticipates that l should be a linear function of m,

$$l = A + Bm \tag{E2.1}$$

(Here, the constant A is the unloaded length of the spring, and B is g/k, where k is the usual spring constant.) The calibration equation (E2.1) will let her find any unknown mass m from the corresponding length l, once she knows the constants A and B. To find these constants, she uses the method of least squares. What are her answers for A and B? Plot her calibration data and the line given by her best fit for Eq. (E2.1). If she puts an unknown mass m in the pan and observes the spring's length to be $l = 53.2$ cm, what is m?

Table E2.2 Masses m_i (in kg) and lengths l_i (in cm) for a spring balance. The "x" and "y" in quotes indicate which variables play the roles of x and y in this example

Trial number	"x" Load, m_i	"y" Length, l_i	m_i^2 /kg^2	$m_i l_i$ /kg · cm
1	2	42.0	4	84.0
2	4	48.4	16	193.6
3	6	51.3	36	307.8
4	8	56.3	64	450.4
5	10	58.6	100	586.0
k=5	$\overline{m_i}$ =6	$\overline{l_i}$ =51.32	$\overline{m_i^2}$ = 44	$\overline{m_i l_i}$ = 324.36

Solution As often happens in such problems, the two variables are not called x and y, and one must be careful to identify which is which. Comparing Eq. (E2.1) with the standard form, $y = a + bx$, we see that the length l plays the role of the dependent variable y, while the mass m plays the role of the independent variable x. The constants A and B are given by Eqs. (2.26) and (2.27), with the replacements

$$x_i \equiv m_i, \quad y_i \equiv l_i$$

(This correspondence is indicated by the headings "x" and "y" in Table E2.2) To find A and B, we need to find the means $\overline{m_1}$, $\overline{l_1}$, $\overline{m_1^2}$, $\overline{m_1 l_1}$.

$$\overline{m} = \frac{1}{k}\sum_{i=1}^{k} m_i = \frac{2+4+\cdots+10}{5} = 6\,(\text{kg})$$

$$\overline{l} = \frac{1}{k}\sum_{i=1}^{k} l_i = \frac{42.0+48.4+\cdots+58.6}{5} = 51.32\,(\text{cm})$$

$$\overline{m^2} = \frac{1}{k}\sum_{i=1}^{k} m_i^2 = \frac{4+16+\cdots+100}{5} = 44\,(\text{kg}^2)$$

$$\overline{ml} = \frac{1}{k}\sum_{i=1}^{k} m_i l_i = \frac{84.0+\cdots+586.0}{5} = 324.36\,(\text{kg}\cdot\text{cm})$$

Therefore, the last two columns of Table E2.2 show the quantities m_i^2 and $m_i l_i$ and the corresponding means are shown at the bottom of each column.

Using Eq. (2.26) we find the intercept (the unstretched length, A)

$$A=\frac{\overline{m}\cdot\overline{ml}-\overline{l}\cdot\overline{m^2}}{\overline{m}^2-\overline{m^2}}=\frac{6\times324.32-51.32\times44}{6^2-44}=39.0\,(\text{cm})$$

Finally, from Eq. (2.27) we find the slope

$$B=\frac{\overline{m}\cdot\overline{l}-\overline{ml}}{\overline{m}^2-\overline{m^2}}=\frac{6\times51.32-324.36}{6^2-44}=2.1\,(\text{cm/kg})$$

A plot of the data and the line using these values of A and B is shown in Fig. E2-1. If the mass m stretches the spring to 53.2 cm, then according to Eq. (E2.1) the mass is

$$m=\frac{l-A}{B}=\frac{53.2-39.0}{2.1}=6.8\,(\text{kg})$$

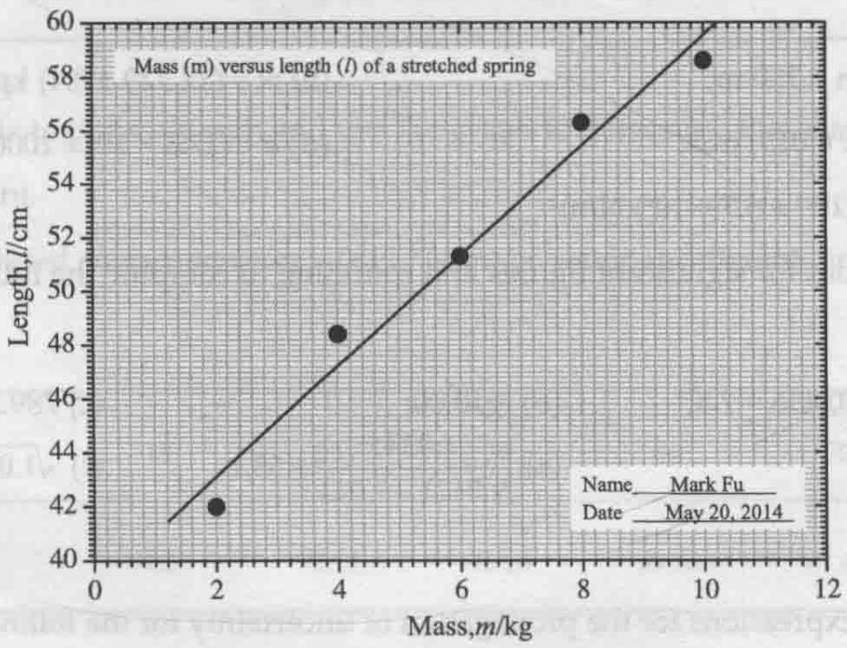


Fig. E2-1 A plot of the data from Table E2.2 and the best-fit line

NOTE

The means $\overline{x}$, $\overline{y}$, $\overline{x^2}$, $\overline{xy}$ are intermediate quantities to calculate the intercept and the slope. Their results should have as many digits as the calculator shows us in order to reduce the error induced by the mathematical operation. In addition, the last significant figure in a and b should usually be of the same order of magnitude (in the same decimal position) as the quantity y.

Problems for Chapter 2

1. Rewrite the following values in unit of mm (keep the significant figures consistent).

(a) 0.050 m; (b) 1.37 m; (c) 4.0 cm; (d) 6.0 μm

2. Rewrite the following equations in their clearest and most appropriate forms.

(a) $x = (3.323 \pm 1.4)$ mm; (b) $t = (1\,234\,567 \pm 54\,321)$ s ;

(c) $\lambda = (5.33 \times 10^{-7} \pm 3.21 \times 10^{-9})$ m ; (d) $r = (0.000\,000\,538 \pm 0.000\,000\,03)$mm

3. State the number of significant figures that there is for each of the following quantities, respectively.

(a) $U = 1.000$ kV; (b) $L = 1.001$ cm; (c) $T = 0.0010$ s;

(d) $V = 3.900 \times 10^3$ m/s; (e) $m = 10.010$ kg; (f) Natural number 10

4. Emend the incorrect expressions according to the rules for the final measurement results.

(a) $L = 2.0$ km $\pm$ 100 m; (b) $m = (72.320 \pm 0.4)$ kg;

(c) $v = (1.23 \pm 0.015)$ m/s; (d) $h = (27.3 \times 10^4 \pm 2000)$ km;

(e) $E = (1.93 \times 10^{11} \pm 6.79 \times 10^9)$ N/m^2

5. Use the rules for significant figures and rounding to compute the following arithmetic expressions.

(a) $123.98 - 40.456 + 7.8$; (b) lg10.00; (c) $789.30 \times 50 \div 0.100$;

(d) 1.00^2; (e) $\dfrac{4.0345}{9.0121 - 9.011} + 58.1$; (f) $\sqrt{1.00}$;

(g) 100.0^2

6. Write the expressions for the propagation of uncertainty for the following functions.

(a) $N = x + y - 2z$; (b) $N = \dfrac{x - y}{x + y}$;

(c) $N = \dfrac{v}{\sqrt{1 + at}}$ (a is constant) ; (d) $V = \dfrac{\pi d^2 h}{4}$

7. For the following instruments state which reading(s) is/are possibly correct.

(a)A vernier caliper has accuracy of 0.05 mm. Which reading(s) is/are possibly correct?

32.50 mm; 32.48 mm; 33.26 mm; 32.5 mm; 32.500 mm

(b)A vernier caliper has accuracy of 0.02 mm. Which reading(s) is/are possibly correct?

45.22 mm; 52.78 mm; 64.05 mm; 84 mm; 73.464 mm

(c)A micrometer caliper has accuracy of 0.01 mm. Which reading(s) is/are possibly correct?

0.50 mm; 0.5 mm; 0.500 mm; 0.5000 mm

8. A wheatstone bridge method is used to measure resistance, R. The instrument error and the accuracy of the apparatus are 2 Ω and 1Ω, respectively. Ten readings are taken as follows; R_i=142.3; 142.7; 141.9; 142.4; 142.8; 142.1; 142.0; 141.8; 142.2; 142.3 (unit: Ω). Compute the mean value of R_i, the Type A uncertainty of R, the Type B uncertainty of R, and the combined uncertainty of R, respectively. Write the final measurement result.

9. Ammeter-Voltmeter method was used to measure resistance. The data are listed in the table below.

Voltage, U/V	0.00	1.00	2.01	3.05	4.00	5.01	5.99	6.98	8.00	9.00	9.99	11.00
Current, I/mA	0.00	2.00	4.00	6.00	8.00	10.00	12.00	14.00	16.00	18.00	20.00	22.00

Use these data to draw a graph and use the graph to determine the relationship between voltage and current.

10. The measured surface tension γ ($\times 10^{-3}$ N/m) of water at different temperatures are as following.

Temperature, t/℃	10	20	30	40	50	60
Surface tension, γ/ ($\times 10^{-3}$ N/m)	74.22	72.75	71.18	69.56	67.91	66.18

Suppose that $\gamma = aT - b$ (T is in the unit of Kelvin, K). Use the above data and least-squares fitting method to find a and b. Compute the relation coefficient and interpret the result in terms of how well the line fits the data.

3. EXPERIMENTS

3.1 Measurement of Resistance by Ammeter-Voltmeter Method

Prelab Assignment 3.1

(1)In the following diagram (Fig. P3.1-1), a current goes through an ammeter. The range of the ammeter and its accuracy grade are 50 mA and 1, respectively. What's the magnitude of the current?

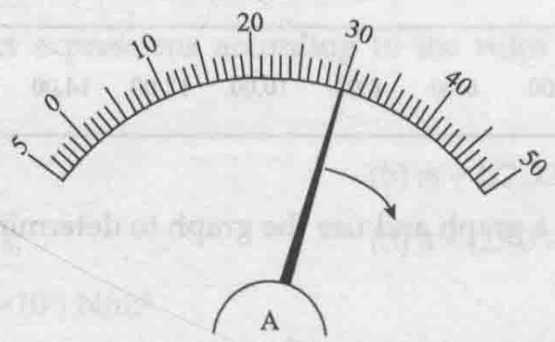


Fig. P3.1-1 Ammeter reading scale

(2)Prove that the true resistance R is given approximately by

$$R = R'\left(1 + \frac{R'}{R_v}\right)$$

where $R' = V/I$ is the measured resistance as given by the voltmeter and ammeter readings for measurements taken using the arrangement in Fig. 3.1-2, R_v is the voltmeter resistance. Is the true resistance larger or smaller than the apparent resistance? Explain.

(3)Prove that the true resistance R is given by

$$R = R'\left(1 - \frac{R_a}{R'}\right)$$

where $R' = V/I$ is the measured resistance as given by the voltmeter and

ammeter readings for measurements done by the arrangement in Fig. 3.1-3, R_a is the ammeter resistance. Is the true resistance larger or smaller than the apparent resistance? Explain.

(4)The resistance R can be determined from Ohm's law, $R = V/I$, by measuring the voltage across the resistor and the current through the resistor. Consider that the voltage and the current are 1.002 V and 23.1 mA respectively with the corresponding uncertainties of 0.005 V and 0.1 mA. Compute the uncertainty in R and present the final measurement result for R.

3.1.1 Introduction and Objectives

The magnitude of a resistance can be measured by several methods. One common method is to measure the voltage drop V across a resistance in a circuit with a voltmeter and the current I through the resistance with an ammeter. By Ohm's law, then, $R = V/I$. However, the ratio of the measured voltage and current does not give an exact value of the resistance because of the resistances of the meters.

After performing this experiment and analyzing the data, you should be able to:

(1)Describe the two ways to measure resistance with an ammeter and voltmeter, and explain how they differ.

(2)Correctly read the analog electric instruments according to their accuracy class and measurement range.

(3)Understand the performance of a diode.

3.1.2 Required Equipment

- Ammeter (with two scales, 0~15 mA and 0~30 mA)
- Voltmeter (with three scales, 0~1.5 V, 0~3 V, and 0~7.5 V)
- Rheostat
- Two linear resistors (with accepted values of 51 Ω and 1.00×10^3 Ω with power ratings of 1 W)
- A diode (non-linear resistor with power rating of 1 W)
- Power supply

• Connecting wires(Fig. 3.1-1)

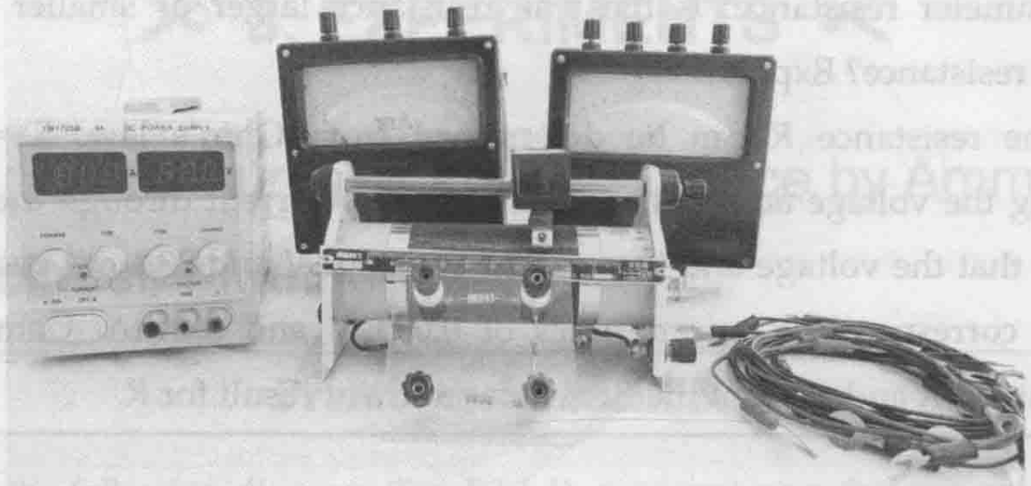


Fig. 3.1-1 Experimental setup

3.1.3 Theory

There are two basic arrangements by which resistance is measured with an ammeter and a voltmeter. One circuit is shown in Fig. 3.1-2. The total current I through the resistor, R_{x1}, is measured with an ammeter, and the potential difference or voltage drop V across the resistor is measured with a voltmeter. Then, by Ohm's law, $R_{x1} = V/I$.

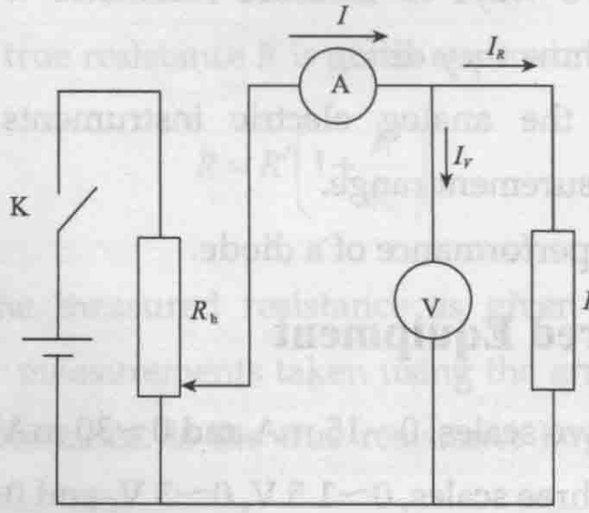


Fig. 3.1-2 Ammeter-outside circuit diagram. This is one of the basic arrangements for measuring resistance with an ammeter and a voltmeter

Strictly speaking, however, this value of the resistance is not altogether correct, since the current registered on the ammeter divides between the

resistance R_{x1} and the voltmeter in parallel. A voltmeter is a high-resistance instrument and draws relatively little current, provided that voltmeter resistance, R_v, is much greater than R_{x1}. Hence, it is more appropriate to write

$$R_{x1} \cong \frac{V}{I} \quad \text{if} \quad R_v \gg R_{x1} \tag{3.1-1}$$

where R_{x1} is the approximate resistance.

For more accurate resistance measurement, one must take the resistance of the voltmeter into account. The current drawn by the voltmeter is $I_v = V/R_v$. Since the total current I divides between the resistance and the voltmeter in the parallel branch

$$I = I_R + I_v$$

or

$$I_R = I - I_v \tag{3.1-2}$$

where I_R is the true current through the resistance. Then, by Ohm's law,

$$R'_{x1} = \frac{V}{I} = \frac{V}{I - I_v} = \frac{V}{I - \dfrac{V}{R_v}} \tag{3.1-3}$$

where R'_{x1} is the corrected resistance.

In Fig. 3.1-3, the ammeter measures the current through R_{x2}, but the voltmeter is across R_{x2} and the ammeter. Therefore, the true value of R_{x2} is *less* than the measured value, if the measured value is taken to be V / I.

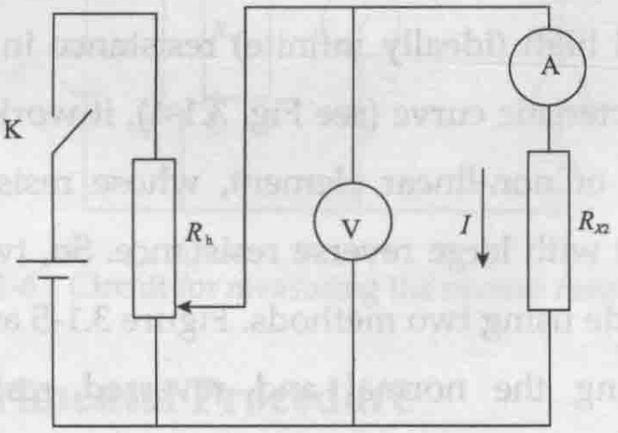


Fig. 3.1-3 Ammeter-inside circuit diagram. This is another basic arrangement for measuring resistance with an ammeter and a voltmeter

In this case, the ammeter measures the current through R_{x2} alone, but now the voltmeter reads the voltage drop across *both* the ammeter *and* the resistance. Since the ammeter is a low resistance instrument, to a good approximation,

$$R_{x2} \cong \frac{V}{I} \quad \text{if} \quad R_a \ll R_{x2} \tag{3.1-4}$$

where R_a is the resistance of the ammeter. When $R_a \ll R_{x2}$, the voltage drop across R_a—that is, $V_a = IR_a$—is small compared to that across R_{x2}, which is $V_R = IR_{x2}$.

Taking the voltage drop or the resistance of the ammeter into account,

$$V = V_R + V_a = IR'_{x2} + IR_a = I(R'_{x2} + R_a) = IR'$$

and

$$R' = (R'_{x2} + R_a) \tag{3.1-5}$$

Solving for R'_{x2} and substituting for R'

$$R'_{x2} = \frac{V}{I} - R_a \tag{3.1-6}$$

where R'_{x2} is the corrected resistance.

In electronics, a diode is a two-terminal electronic component with asymmetric conductance, and it has low (ideally zero) resistance to current flow in one direction, and high (ideally infinite) resistance in the other. Diode has abrupt reverse characteristic curve (see Fig. 3.1-4), it works in the reverse state. The diode is a type of non-linear element, whose resistance in the forward direction is small but with large reverse resistance. So, two resistances may be measured for the diode using two methods. Figure 3.1-5 and Fig. 3.1-6 show the circuits for measuring the normal and reversed resistances of a diode, respectively.

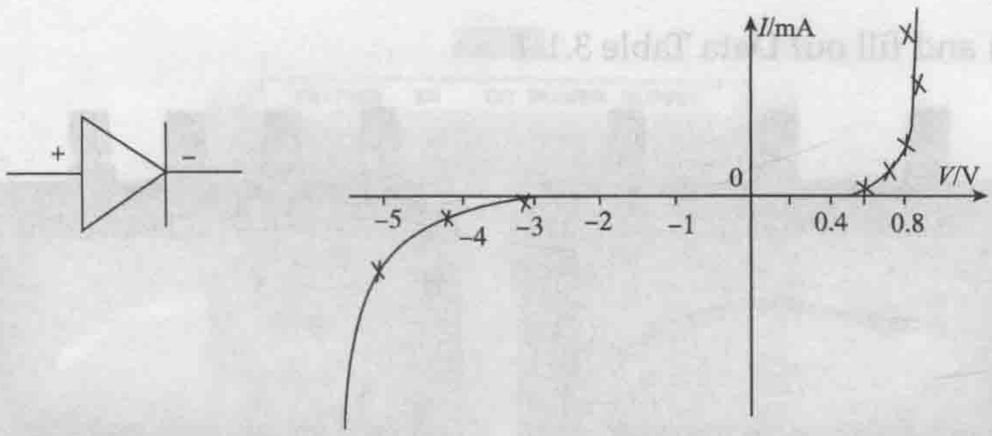


Fig. 3.1-4 A symbol of a diode (left) and its current (I)–voltage (V) characteristics

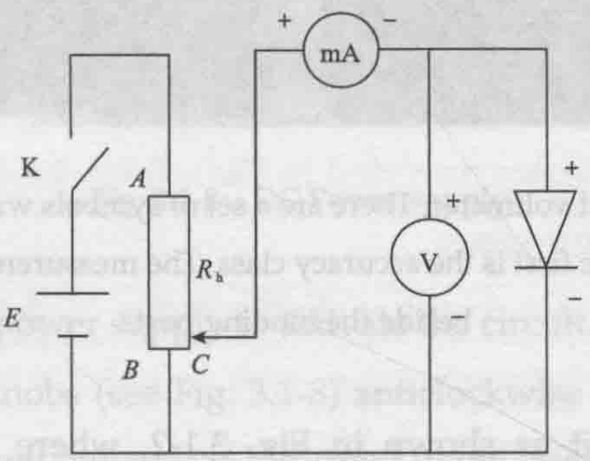


Fig. 3.1-5 Circuit for measuring the normal resistance

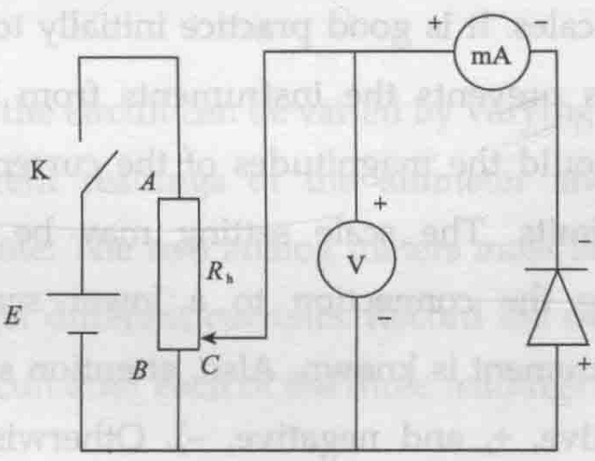


Fig. 3.1-6 Circuit for measuring the reverse resistance

3.1.4 Experimental Procedure

(1)Find the information such as the accuracy class, the measurement ranges (also called *scales*) and the corresponding resistances on the instruments

(Fig. 3.1-7) and fill out Data Table 3.1-1.

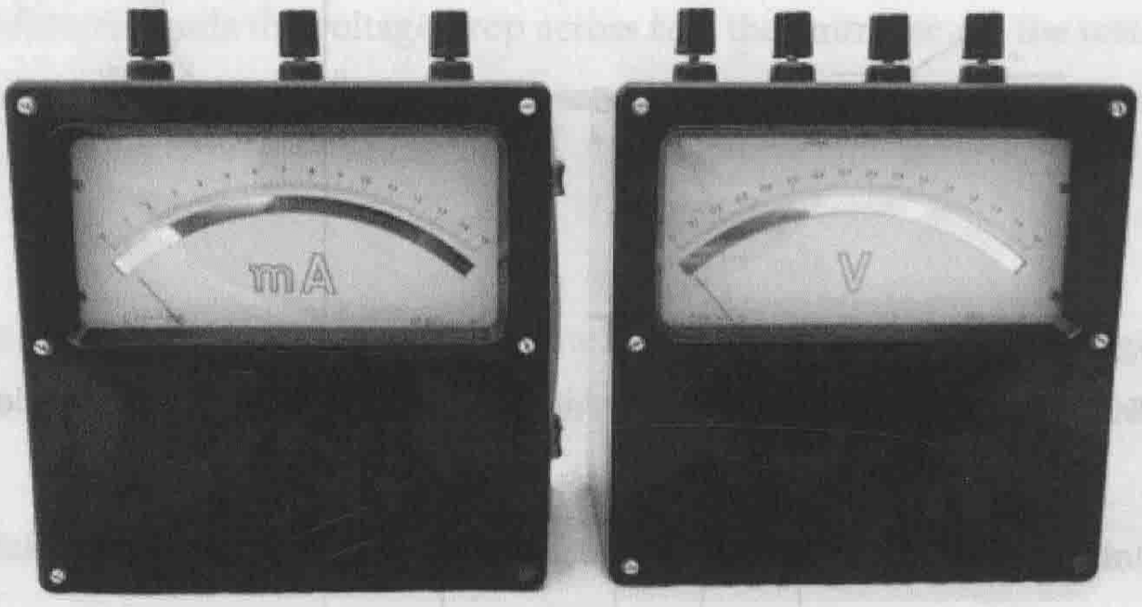


Fig. 3.1-7 Ammeter and voltmeter. There are a set of symbols written at the right bottom corner of the display. The first is the accuracy class. The measurement ranges can be found beside the binding posts

(2)Set up a circuit as shown in Fig. 3.1-2, where R_{x1} is a small known resistance (~51Ω) and R_h is the rheostat. Record the value of R_{x1} in the first of the spaces provided for this purpose at the top of Data Table 3.1-1. Most common meters have more than one scale connections, with a binding post common to all other scales. It is good practice initially to make connections to the largest scale. This prevents the instruments from being "pegged" (and possibly damaged) should the magnitudes of the current and voltage exceed the smaller scales' limits. The scale setting may be changed for greater sensitivity by moving the connection to a lower scale after the general magnitude of a measurement is known. Also, attention should be given to the proper polarity (positive, +, and negative, –). Otherwise, the meter will be "pegged" in the opposite direction. Connect "+" to "+" and "–" to "–". However, *do not activate the circuit until you check and ensure the correctness of the connection.* (Caution: the power rating for both linear resistors is 1 W. Before any measurements, you should do some computation to decide which measurement ranges of the meters you will use.)

Fig. 3.1-8 DC Power supply

(3)Setting up the power supply to activate the circuit.

i. Turn the four knobs (see Fig. 3.1-8) anticlockwise fully. These knobs are for adjusting current and voltage, as indicated in the figure.

ii. Push the power button in.

iii. Tune the coarse knob of current clockwise by a small angle.

iv. Tune the coarse and/or fine knobs of voltage clockwise to set the voltage to a desired value.

(4)The current in the circuit can be varied by varying the rheostat resistance R_h. Take three different readings of the ammeter and voltmeter for three different currents. (Note: The two analog meters must be laid down when you read.) Adjust R_h to get different currents. Record the data in Data Table 3.1-2, and deactivate the circuit after each of the three readings until the rheostat is set for the next reading.

(5)Using Eq. (3.1-3), compute the corrected value of R for each current setting and find the average value. Compare this with the accepted value by finding the percent error. Enter your results in Data Table 3.1-2.

(6)Set up a circuit as shown in Fig. 3.1-3. This is accomplished by changing only one wire in the previous circuit. Repeat the measurements as in Procedure

2 for this circuit, recording your findings in Data Table 3.1-3.

(7)

i. Compute the resistance $R_{x1} = V/I$ directly from each set of current and voltage measurements, and find the average value.

ii. When one is not taking into account the ammeter resistance, R_{x1} is taken to be the value of the resistance R. Compare the average experimental value of R_{x1} with the accepted value of R by finding the percent error.

iii. Using the values of R_a and R_{x1}, compute the corrected resistance R'_{x1} [Eq. (3.1-6)]. Enter your results in Data Table 3.1-3.

(8)Repeat the previous steps with a large known resistance. Record its accepted value in the space provided in Data Table 3.1-4, and use Data Tables 3.1-4 and 3.1-5 for your findings.

(9)(Optional) Set up a circuit as shown in Fig. 3.1-5 to measure the normal resistance of a diode. (Note: select the scales of 0～1.5 V for a voltmeter, and 0～30 mA for an ammeter, respectively). Record the data in Data Table 3.1-6.

(10)Set up a circuit as shown in Fig. 3.1-6 to measure the reverse resistance of a diode. (Note: select the scales of 0～7.5 V for a voltmeter, and 0～30 mA for an ammeter, respectively). Record the data in Data Table 3.1-7.

3.1.5 Experimental Data

Data Table 3.1-1 ***Purpose*** **: To compute the instrument error and reading error for further calculations of uncertainty and to learn to read the voltmeter and ammeter correctly**

Apparatus	Voltmeter			Ammeter	
Accuracy class					
Range (unit)					
Resistance (unit)					
Instrument Error (unit)					
Reading Error (unit)					

(*Don't forget UNITS*)

Data Table 3.1-2 ***Purpose*****: To use ammeter outside method to measure resistance with a small value**

Accepted value R_{x1}:_____; Voltmeter resistance R_V=_____.

Rheostat setting	Current, *I*/mA		Voltage, *V*/V		$R_{x1}=V/I$ /Ω	$R'_{x1}=\frac{V}{1-\frac{V}{R_V}}$ /Ω
	Range	Instrument error	Range	Instrument error		
1						
2						
3						
				Average		
				Relative error		

Data Table 3.1-3 ***Purpose*****: To use ammeter inside method to measure resistance with a small value**

Accepted value R_{x1}_____; Ammeter resistance R_a=_____

Rheostat setting	Current, *I*/mA		Voltage, *V*/V		$R_{x1}=V/I$ /Ω	$R'_{x1}=\frac{V}{I}-R_a$ /Ω
	Range	Instrument error	Range	Instrument error		
1						
2						
3						
				Average		
				Relative error		

Your name and student number:_____ Instructor's initial:_____

Data Table 3.1-4 ***Purpose*: To use ammeter outside method to measure resistance with a large value**

Accepted value R_{x2}:_____; Voltmeter resistance R_V=_____

Rheostat setting	Current, I/mA		Voltage, V/V		$R_{x1}=V/I$ /Ω	$R'_{x2}=\frac{V}{1-\frac{V}{R_V}}$ /Ω
	Range	Instrument error	Range	Instrument error		
1						
2						
3						
				Average		
				Relative error		

Data Table 3.1-5 ***Purpose*: To use ammeter inside method to measure resistance with a large value**

Accepted value R_{x2}_____; Ammeter resistance R_a=_____

Rheostat setting	Current, I/mA		Voltage, V/V		$R_{x2}=V/I$ /Ω	$R'_{x2}=\frac{V}{I}-R_a$ /Ω
	Range	Instrument error	Range	Instrument error		
1						
2						
3						
				Average		
				Relative error		

NOTE: The following measurements are optional.

Data Table 3.1-6 ***Purpose* : To measure the normal resistance of a diode**

V/V	0.000	0.100	0.200	0.300	0.400						
I/mA						5.0	10.0	15.0	20.0	25.0	30.0

Data Table 3.1-7 ***Purpose* : To measure the reverse resistance of a diode**

V/V	0.00	1.00	2.00	2.50	3.00						
I/mA						5.0	10.0	15.0	20.0	25.0	30.0

Your name and student number:_____ Instructor's initial:_____

3.1.6 Calculations

(1)Show one sample calculation of the instrument error by using the data in Data Table 3.1-1.

(2)Use the last set of data in Data Table 3.1-2 to show the sample calculation of the approximate resistance and the corrected resistance.

(3)Follow the steps below to find the uncertainty for the first set of measurement in Data Table 3.1-2.

a. Use the measured voltage and current to calculate the magnitude of resistance.

$$R_{x1} = \frac{V}{I}$$

b. Find the Type A uncertainty in current I, μ_{A-I}.

$$\mu_{A-I} = \sqrt{\frac{1}{N(N-1)}\sum_{i=1}^{N}(I_i - \bar{I})^2}$$

c. Find the Type B uncertainty in current I, μ_{B-I}.

$$\mu_{\Delta_{\text{Instru.}}-I} = \frac{\Delta_{\text{Instru.}-I}}{\sqrt{3}}$$

$$\mu_{\Delta_{\text{Read.}}-I} = \frac{\Delta_{\text{Read.}-I}}{\sqrt{3}}$$

$$\mu_{B-I} = \sqrt{\mu^2_{\Delta_{\text{Instru.}}-I} + \mu^2_{\Delta_{\text{Read}}-I}}$$

d. Find the combined uncertainty in current I, σ_I.

$$\sigma_I = \sqrt{\mu^2_{A-I} + \mu^2_{B-I}}$$

e. Find the Type A uncertainty in voltage V, μ_{A-V}.

$$\mu_{A-V} = \sqrt{\frac{1}{N(N-1)}\sum_{i=1}^{N}(V_i - \bar{V})^2}$$

f. Find the Type B uncertainty in voltage V, μ_{B-V}.

$$\mu_{\Delta_{\text{Instru.}}-V} = \frac{\Delta_{\text{Instru.}-V}}{\sqrt{3}}$$

$$\mu_{\Delta_{\text{Read.}}-V} = \frac{\Delta_{\text{Read}-V}}{\sqrt{3}}$$

$$\mu_{B-V} = \sqrt{\mu^2_{\Delta_{\text{Instru.}}-V} + \mu^2_{\Delta_{\text{Read}}-V}}$$

g. Find the combined uncertainty in voltage V, σ_V.

$$\sigma_V = \sqrt{\mu^2_{A-V} + \mu^2_{B-V}}$$

h. Compute the relative uncertainty in resistance R_{x1} and hence $\sigma_{R_{x1}}$.

$$\frac{\sigma_{R_{x1}}}{|R_{x1}|} = \sqrt{\left(\frac{\sigma_V}{V}\right)^2 + \left(\frac{\sigma_I}{I}\right)^2}$$

$$\sigma_{R_{x1}} = |R_{x1}| \times \frac{\sigma_{R_{x1}}}{|R_{x1}|}$$

3.1.7 Graphing (Optional)

Use the data in Data Tables 3.1-6 and 3.1-7 to draw a graph to show the *I-V* characteristics of a diode.

3.1.8 Post Lab Questions

(1)After completing Data Tables 3.1-2 and 3.1-3, what conclusions can you draw?

(2)After completing Data Tables 3.1-4 and 3.1-5, what conclusions can you draw?

(3)If in general R were calculated as $R = V/I$, which circuit arrangement in Fig. 3.1-2 and Fig. 3.1-3 would have the smallest error? Explain.

(4)In the calculation section, you use the equation $R = V/I$ and the

measurements to compute the magnitude of the resistance and its uncertainty. Does the accepted resistance lie in the range determined by the magnitude of the resistance and its uncertainty? Why?

(5)An ideal ammeter would have zero resistance, and an ideal voltmeter would have an infinite resistance. Explain why we would desire these ideal cases when using the meters.

(6)Why can a diode be applied as the core component of a manostat? (Optional)

3.2 Use Hall Effect to Measure Magnetic Field

Prelab Assignment 3.2

(1)In Fig. P3.2-1, there are n charge carriers per cubic meter in a strip of semiconductor with a strip shape. Each carrier with negative charge q passes through an area A with a speed v. The applied magnetic field B is perpendicular to the upper surface of the strip.

i. What is the magnitude of the current I_H?

ii. When the magnetic field B is switched on, what force will be exerted on one moving charge carrier? What is the magnitude of the force? Mark the direction of the force on the diagram.

iii. After some time there will be higher density of negative charge near the front edge than near the rear edge. Why?

iv. After the accumulation of the negative charge carriers at the edge of the strip, what force will be exerted on one moving charge? How big must it be so that there is no further change in the number of negatively charged particles near the edge?

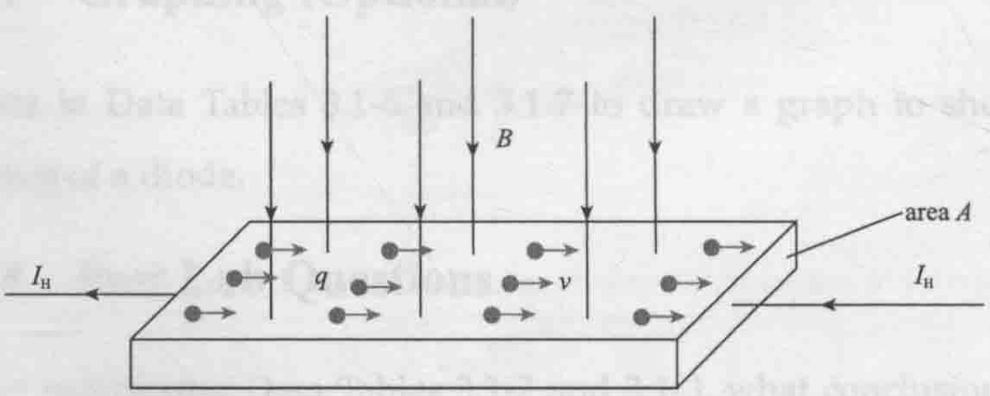


Fig. P3.2-1 A charged semiconductor in a magnetic field

(2)Was Edwin Hall awarded the Noble Prize in history? Who won the Noble Prize in physics for the contribution in connection with Hall effect? Give some details on the winner's name, the year, and the specific contribution.

3.2.1 Introduction and Objectives

The Hall effect is basic to solid-state physics and an important diagnostic tool

for the characterization of materials, particularly semi-conductors. In this experiment, the Hall effect will be used to measure magnetic field induced by solenoid.

In 1879 Edwin Hall observed that when an electrical current passes through a sample placed in a magnetic field, a potential proportional to the current and to the magnetic field is developed across the material in a direction perpendicular to both the current and to the magnetic field. This effect is known as the Hall effect, and is the basis of many practical applications and devices such as magnetic field measurements, and position and motion detectors.

With the measurements he made, Hall was able to determine for the first time the sign of charge carriers in a conductor. Even today, Hall effect measurements continue to be a useful technique for characterizing the electrical transport properties of metals and semiconductors.

After successfully completing this laboratory workshop, you should be able to:

(1) Explain the response of the charge carriers in a material to a magnetic field.

(2) Calculate the magnitude of magnetic field from the Hall voltage and the known experimental variables (Hall coefficient and Hall current).

3.2.2 Required Equipment

- Magnetic field measuring system(Fig. 3.2-1)

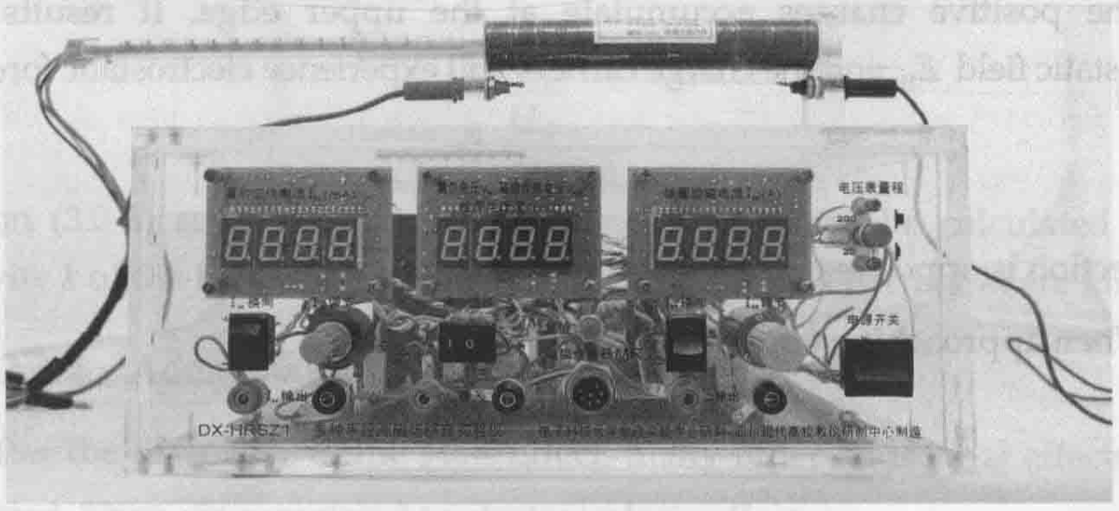


Fig. 3.2-1 Magnetic field measuring system

3.2.3 Theory

3.2.3.1 Hall Effect

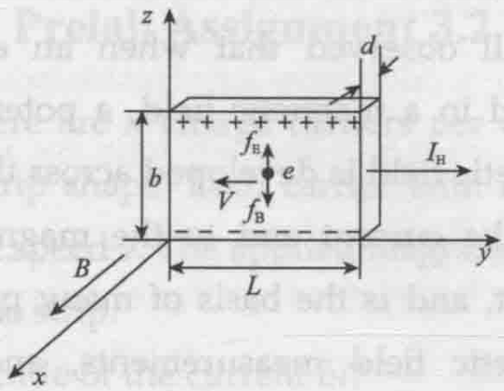


Fig. 3.2-2 Schematic diagram of Hall effect

Shown in Fig. 3.2-2 is an electrified semiconductor with current of I_H. The semiconductor is in the form of the plate of length L, width b and thickness d. In terms of the set of coordinate axes in Fig. 3.2-2, the magnetic field is in the x-direction, and denoted as B. The current I_H is called Hall current and is in the y-direction. We see that the actual motion of carriers is in the y-direction with the average velocity v.

The carriers will experience Lorentz force f_B

$$f_B = evB$$

where e is the charge of a carrier and B is the applied magnetic field. Therefore, along the z-direction, negative carriers accumulate at the lower edge of the plate and the positive charges accumulate at the upper edge. It results in an electrostatic field E_H and the charge carriers will experience electrostatic force f_E

$$f_E = eE_H$$

Its direction is opposite to the direction of the Lorentz forc f_B.

When approaching equilibrium,

$$f_B = f_E$$

thus,

$$evB = eE_{\mathrm{H}} = e\frac{V_{\mathrm{H}}}{b} \tag{3.2-1}$$

where V_{H} is called Hall voltage and

$$V_{\mathrm{H}} = vBb \tag{3.2-2}$$

Hall current I_{H} can be expressed as

$$I_{\mathrm{H}} = -nevbd \tag{3.2-3}$$

where n is the density of current-carrying charges in the material.

Then from Eq. (3.2-3) we find that

$$v = -I_{\mathrm{H}} / nebd \tag{3.2-4}$$

Using Eqs. (3.2-2) and Eq. (3.2-4) we obtain

$$V_{\mathrm{H}} = -\frac{I_{\mathrm{H}}B}{ned}$$

or,

$$V_{\mathrm{H}} = R\frac{I_{\mathrm{H}}B}{d} = kI_{\mathrm{H}}B \tag{3.2-5}$$

where $R = -\frac{1}{ne}$ is known as Hall coefficient of the semiconductor material and $k = \frac{R}{d}$ is called the sensitivity of the Hall sensor.

From Eq. (3.2-5) we find that

$$B = \frac{V_{\mathrm{H}}}{kI_{\mathrm{H}}} \tag{3.2-6}$$

Equation (3.2-6) suggests that the magnetic field B can be calculated if the sensitivity k of the Hall sensor, Hall voltage V_{H}, and Hall current I_{H} are known.

3.2.3.2 Additional Effects of the Hall Effect

Soon after the discovery of the Hall Effect, many other additional effects were discovered. They are associated to electro-caloric effect.

1)Etinghausen Effect

In 1887, Etinghausen found that the different carriers have different drift speeds and so will experience different magnetic forces. This phenomenon leads to a transverse temperature gradient which in turn results in a potential difference, V_E Its direction is decided by I_H, and magnetic field B.

2)Nernst Effect

The resistances of the two welding points through which current flows are not the same and have different temperatures when the Hall sensor is working. So, a thermal diffusion current exists between the terminals due to the Seebeck effect. When a magnetic field is applied perpendicular to the plate, a transverse electric field appears between opposite points on the edges of the plate. The resulting voltage, V_N, is B-dependent.

3)Righi-Leduc Effect

The drift speeds of carriers of the thermal diffusion current aren't the same, a transverse temperature gradient appears. It leads to an additional electric potential difference, V_{RL}.

4)Unequal Electric Potential Difference

Actually, the materials for the Hall sensors are not homogenous; there will appear an unequal electric potential difference, V_O. The direction of V_O is I_H-dependent.

In accordance with the characteristics of the above additional effects, we can eliminate them through the method of anti-direction.

When $+B, +I_H$

$$V_1 = V_H + V_E + V_N + V_{RL} + V_O$$

When $+B, -I_H$

$$V_2 = -V_H - V_E + V_N + V_{RL} - V_O$$

When $-B, -I_H$

$$V_3 = V_H + V_E - V_N - V_{RL} - V_O$$

When $-B, +I_H$

$$V_4 = -V_H - V_E - V_N - V_{RL} + V_O$$

Hence,

$$V_H = \frac{1}{4}(V_1 - V_2 + V_3 - V_4) - V_E$$

Because,

$$V_E \ll V_H$$

We can ignore it, so

$$V_H = \frac{1}{4}(V_1 - V_2 + V_3 - V_4) \qquad (3.2\text{-}7)$$

3.2.4 Experimental Procedure

(1)Record the parameters of the instrument you used and enter them in the Data Table 3.2-1. Turn on the magnetic field measurement system. Set the Hall current I_H = 5.00 mA and the excitation current I_M = 0 mA.

(2)At this time, the magnetic field in the solenoid is zero. According to Eq. (3.2-2), the Hall voltage must be zero. However, the instrument shows some non-zero readings. They are called "zero readings" (systematic errors). You can record four zero readings, V_{01} ($+I_M$, $+I_H$), V_{02} ($+I_M$, $-I_H$), V_{03} ($-I_M$, $-I_H$), V_{04} ($-I_M$, $+I_H$), when you change the directions of both the excitation current I_M and the Hall current I_H. It is worth noting that the position of the Hall sensor can be anywhere.

(3)Study the relationship between the magnetic induction B and the position for the solenoid. Keep the Hall current I_H = 5.00 mA, set the excitation current I_M = 0.500 A, and choose the scale of 200 mV for the voltmeter (you can find it on the upper right). Move the Hall sensor into the solenoid. When the Hall sensor is located at the position of x = –30 mm, change the directions of both excitation current I_M and Hall current I_H to record four voltages, V_1 ($+I_M$, $+I_H$), V_2 ($+I_M$, $-I_H$), V_3 ($-I_M$, $-I_H$), and V_4 ($-I_M$, $+I_H$). Write down these voltages in Data Table 3.2-2. Then change the Hall sensor's position to record the different Hall voltages in Data Table 3.2-2 for the corresponding x value.

(4)Study the relationship between the magnetic induction B in the center

and the excitation current I_M of the solenoid. Repeat Step 2 to record the zero readings again in the space below the heading of Data Table 3.2-3. Hold the Hall sensor at the center of a solenoid ($x = 75$ mm). Set the Hall current $I_H = 5.00$ mA and the excitation current $I_M = 0.050$ A, change the directions of both the excitation current I_M and the Hall current I_H to record four Hall voltages, V_1, V_2, V_3, V_4. Record these readings in Data Table 3.2-3. Then change the excitation currents with intervals of 0.050 A to complete Data Table 3.2-3.

3.2.5 Experimental Data

Data Table 3.2-1 ***Purpose*: To record the instrument parameters for further calculations**

Hall Sensitivity, k /[mV (mA·T)]	Turns, N	Length, l /mm	Diameter, D /mm	Hall Current, I_H /mA
245				

Data Table 3.2-2 ***Purpose*: To study the relationship between the magnetic induction B and the positions**

"Zero Readings"(I_M=0.0000 A): V_{01}=_____, V_{02}=_____, V_{03}=_____, V_{04}=_____, Excitation Current, I_M=_____

Voltage/mV \ x / mm		−30.0	−20.0	−12.0	−7.0	−3.0	0.0	3.0	7.0	12.0	20.0	40.0	75.0
V_1 ($+I_M$, $+I_H$)	*Readings*: V_1'												
	$V_1 = V_1' - V_{01}$												
V_2 ($+I_M$, $-I_H$)	*Readings*: V_2'												
	$V_2 = V_2' - V_{02}$												
V_3 ($-I_M$, $-I_H$)	*Readings*: V_3'												
	$V_3 = V_3' - V_{03}$												
V_4 ($-I_M$, $+I_H$)	*Readings*: V_4'												
	$V_4 = V_4' - V_{04}$												
$V_H = \frac{1}{4}(V_1 - V_2 + V_3 - V_4)$													
$B = V_H / kI_H$ /mT													
B_{theory} / mT (Eqs. (3.2-8) and (3.2-9))													
Relative error / %													

Data Table 3.2-3 ***Purpose*: To study the relationship between the magnetic induction B in the center and the excitation current I_M**

"Zero Readings"(I_M=0.0000A): V_{01}=_____, V_{02}=_____, V_{03}=_____, V_{04}=_____

Voltage/mV \ I_M / A		0.050	0.100	0.150	0.200	0.250	0.300	0.350	0.400	0.450	0.500	0.550
V_1 ($+I_M$, $+I_H$)	*Readings*: V_1'											
	$V_1 = V_1' - V_{01}$											
V_2 ($+I_M$, $-I_H$)	*Readings*: V_2'											
	$V_2 = V_2' - V_{02}$											
V_3 ($-I_M$, $-I_H$)	*Readings*: V_3'											
	$V_3 = V_3' - V_{03}$											
V_4 ($-I_M$, $+I_H$)	*Readings*: V_4'											
	$V_4 = V_4' - V_{04}$											
$V_H = \frac{1}{4}(V_1 - V_2 + V_3 - V_4)$												
$B = V_H / kI_H$ /mT												

Your name and student number:_____ Instructor's initial:_____

3.2.6 Calculations

(1)In Data Table 3.2-2, show the sample calculation of the magnetic induction B at the position $x = -30$ mm.

(2)In Data Table 3.2-3, show the sample calculation of the magnetic induction B with the excitation current $I_M = 0.050$ mA.

(3)Theoretically, the magnetic induction B at the edge of a solenoid can be computed using,

$$B = \frac{\mu_0 N I_M}{2\sqrt{l^2 + (D/2)^2}} \tag{3.2-8}$$

and the magnetic induction B in the center of a solenoid can be computed using,

$$B = \frac{\mu_0 N I_M}{\sqrt{l^2 + D^2}} \tag{3.2-9}$$

where, μ_0 is the permeability of vacuum and $\mu_0 = 4\pi \times 10^{-7}$ N/A^2, N is the turns of a solenoid, I_M is the excitation current, l is the length of a solenoid, and D is the diameter of a solenoid.

Compute the magnetic induction B at the edge and in the center of solenoid by using Eqs. (3.2-8) and (3.2-9), respectively. Show the sample calculations and enter the results in Data Tables 3.2-2 and 3.2-3. (NOTE: The smaller metric units, such as mm and mA, must be converted to standard units. As such, the calculated magnetic induction B is in tesla.)

(4)Assume that the magnetic induction B in the center is a linear function of the excitation current I_M through the solenoid,

$$B = a + bI_M \tag{3.2-10}$$

Use the least-squares fitting method and the data in Data Table 3.2-3 to find the constants a and b. Write down the final relationship.

3.2.7 Graphing

(1)Using the data in Data Table 3.2-2, draw a graph of magnetic induction dependence on position of Hall sensor in a solenoid.

(2)Using the data in Data Table 3.2-3, draw a graph of magnetic induction in the center of a solenoid dependence on the excitation current.

3.2.8 Post Lab Questions

(1)How would Hall sensitivity be increased according to its definition?

(2)How do you utilize the Hall effect to measure a magnetic field? (NOTE: You should talk about the type of the material as the sensor, how to put the sensor in the field, what happens in the sensor, how to eliminate the additional effects, and how to calculate the Hall voltage as well as the magnetic induction.)

3.3 The Oscilloscope

Prelab Assignment 3.3

(1)The core of an analog oscilloscope is a special type of vacuum tube known as a cathode ray tube (CRT). What goes on inside the tube to produce waveform displays on the screen?

(2)When measuring AC voltage signals, what do the two axes (horizontal and vertical) of the oscilloscope screen represent?

(3)Explain how Lissajous patterns are displayed?

3.3.1 Introduction and Objectives

An oscilloscope is a type of electronic test instrument that allows observation of constantly varying signal voltages, usually as a two-dimensional plot of one or more signals as a function of time. The observed waveform can be analyzed for such properties as amplitude, frequency, rise time, time interval, distortion and others. Modern digital instruments may calculate and display these properties directly.

Oscilloscopes are used in the sciences, medicine, engineering, and telecommunications industry. General-purpose oscilloscopes are used for maintenance of electronic equipment and laboratory work. Special-purpose oscilloscopes may be used for such purposes as analyzing an automotive ignition system or to display the waveform of the heartbeat as an electrocardiogram.

The purpose of this experiment is to familiarize the student with the oscilloscope and basic types of measurement that can be made with it.

3.3.2 Required Equipment

- Dual trace oscilloscope(Fig. 3.3-1)
- Two function generators
- Connecting cables

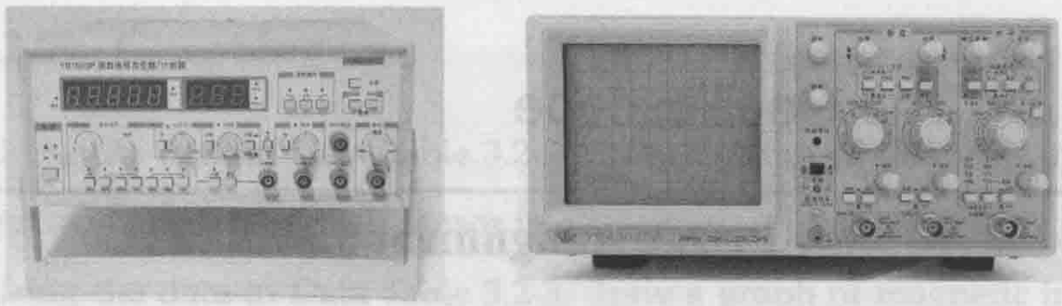


Fig. 3.3-1 Function generator and dual trace oscilloscope

3.3.3 Theory

One of the more versatile laboratory measuring tools is the cathode ray oscilloscope, frequently just called an oscilloscope. The display part of an oscilloscope is the cathode ray tube (CRT), the general construction of which is shown in Fig. 3.3-2. The electron gun consists of a number of components including a source of electrons, high-voltage accelerating electrodes, and electron focusing devices. The basic purpose of the electron gun is to produce a very well defined beam of constant velocity electrons traveling down the center of the CRT.

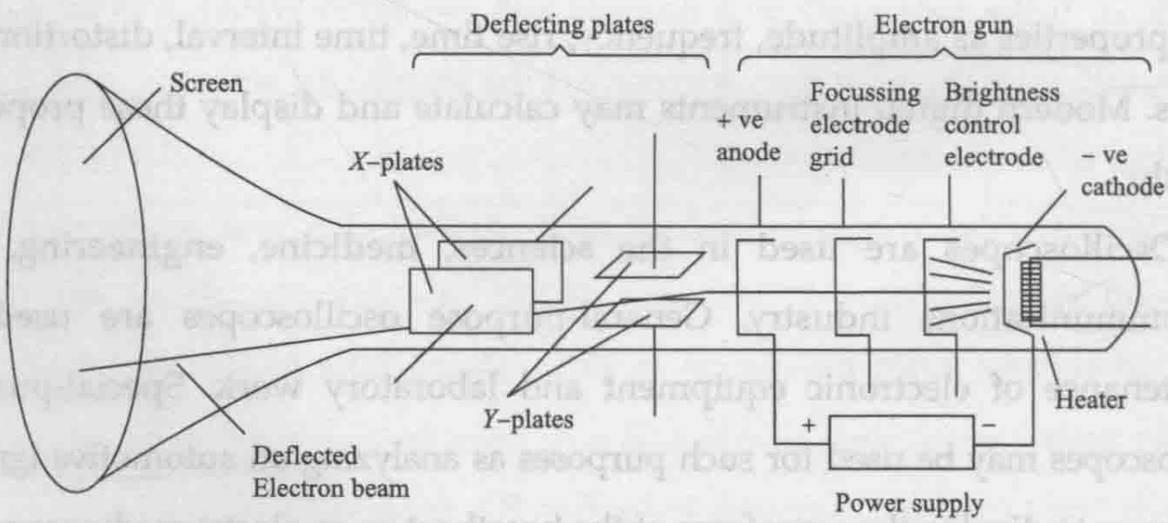


Fig. 3.3-2 Cathode ray tube (CRT)

The key to the versatility of the CRT are two sets of electron deflection plates which deflect the electron beam in the vertical and horizontal directions as a function of the voltage applied to each set of plates. By selecting the appropriate applied voltages to these plates it is possible to deflect the beam so that it strikes any point of the screen. In practice the applied voltage on the plates is usually varied as a function of time causing the beam to sweep in some

pattern across the screen. The most common technique is to cyclically apply a linearly increasing voltage, *i. e.* , the sawtooth voltage, to the horizontal plates causing the beam to sweep from left to right across the screen. Without any vertical signal this appears as a continuous line across the screen, because of the slow decay of the fluorescent screen and the persistence of vision of the human eye. When a signal is applied to the vertical plates the beam will be deflected vertically as it sweeps left to right causing a wave pattern to form.

Figure 3.3-3 shows a block diagram of an oscilloscope. The functions of the sub-blocks are given as follows:

(1)The power supply provides the voltages necessary to produce the electron beam.

(2)The CRT is shown in detail in Fig. 3.3-2.

(3)The vertical amplifier amplifies the input signal to the vertical deflection plates so that small amplitude input signals produce an observable deflection of the electron beam.

(4)The horizontal amplifier amplifies the input signal or the sawtooth voltage which is then applied to the horizontal deflection plates.

(5)The sawtooth voltage generator provides the sawtooth voltage that causes the electron beam spot to move from left to right at a constant (but adjustable) speed. The sawtooth voltage may be synchronized with a respective vertical input voltage.

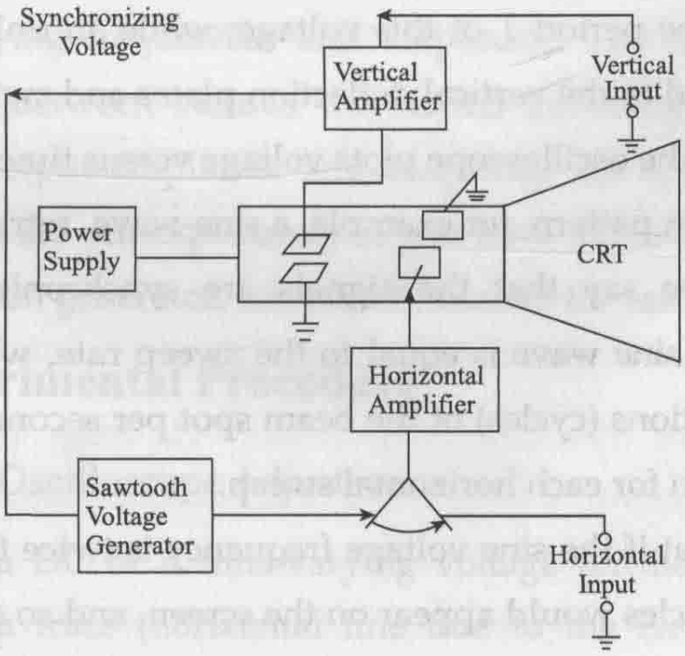


Fig. 3.3-3 Block diagram of an oscilloscope

The sawtooth or "time-base" voltage U_t increases linearly with time as shown in Fig. 3.3-4. The period T of the sawtooth voltage is adjustable. Fig. 3.3-4 shows the horizontal motion of the electron beam spot. It will be shown below that changing the period of T of the sawtooth, changes the horizontal speed of the beam spot.

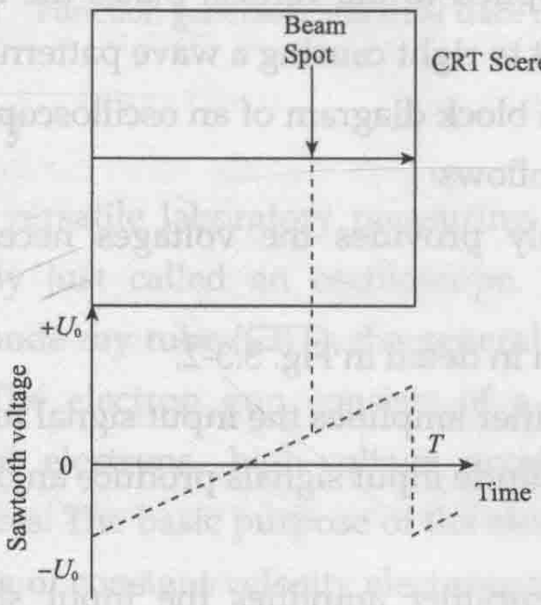


Fig. 3.3-4 Relationship between the sawtooth voltage and the horizontal motion of the electron beam spot

During the operation of an oscilloscope, a sawtooth voltage is applied to the horizontal plates to move the beam spot from left to right at constant speed, determined by the period T of this voltage; while an unknown voltage (study voltage) is applied to the vertical deflection plates and swings the beam spot up and down. Thus the oscilloscope plots voltage versus time on the CRT screen.

If a waveform pattern, for example, a sine-wave, retraces itself and appears to stand still, we say that the signals are synchronized. In this case the frequency of the sine wave is equal to the sweep rate, which is the number of horizontal repetitions (cycles) of the beam spot per second. That is, there is one vertical oscillation for each horizontal sweep.

It follows that if the sine voltage frequency is twice that of the sweep rate, two sine-wave cycles would appear on the screen, and so on. For example, if the sweep rate frequency is 100 Hz and two cycles of a sine wave appear on the screen, the frequency input is 200 Hz. In general, a stationary pattern results

when the ratio of the sweep-rate frequency and the frequency of the input-voltage signal is an integral.

Perhaps the simplest measurement one can make with an oscilloscope is ac or dc voltage. The oscilloscope becomes a high-resistance voltmeter simply by calibrating the vertical deflections to some known volts per division of deflection and then connecting the unknown voltage to the vertical (Y) input. The voltage is then easily determined by reading the deflection on the CRT. Note that the oscilloscope measures instantaneous voltage. Thus, the maximum voltage is from zero to a peak or peak-to-peak divided by 2.

The frequency of an AC source can be determined by calibrating the horizontal sweep rate and then connecting the unknown source to the Y input. The periodic sweep of the beam causes the input pattern to retrace itself and the pattern appears to stand still. By determining the number of x divisions required for one complete cycle of the waveform, the period is easily determined. For example, if a sine wave covers two divisions for one complete cycle and the sweep rate is 2 msec per division then the period would be 4 msec; so the frequency would be 250 Hz. The frequency is the reciprocal of the period.

In addition to the oscilloscope this experiment requires a function generator. This instrument is a variable oscillator designed to output various waveforms with varying frequencies and amplitude. The controls generally allow one to choose between various waveforms including sine waves and square waves. The output amplitude and frequency are usually selected by several controls which are self-explanatory. If you have difficulty understanding how to set your function generator, ask your instructor for assistance.

3.3.4 Experimental Procedure

3.3.4.1 Initial Oscilloscope Adjustments

In order to observe a DC or a time-varying voltage on the scope it is first necessary to obtain a trace (horizontal line due to the electron beam spot motion) on the screen. The control knobs of an oscilloscope are shown in Fig. 3.3-5. The procedure to establish the trace is as follows.

(1)Turn on the POWER.

(2)Turn the INTENSITY (or BRILLIANCE) knob fully clockwise. (This makes the beam spot as bright as possible. This shortens the lifespan of the CRT. For longer life, the brightness should be kept low.)

(3)Set the TIME/DIV knob to 0.5 ms (or 0.2 ms). (This sets the period T of the sawtooth voltage and hence the constant horizontal speed of the beam spot.) The CAL (for VARIABLE) knob varies speed of the spot between calibration. The TIME/DIV calibration is correct only when the CAL knob is in the calibrate position (pushed in or turned fully clockwise).

(4)Adjust the vertical POSITION (or Y-SHIFT) and the horizontal POSITION (or X-SHIFT) knobs until the trace is centered on the screen.

(5)Focus the trace by using the FOCUS knob.

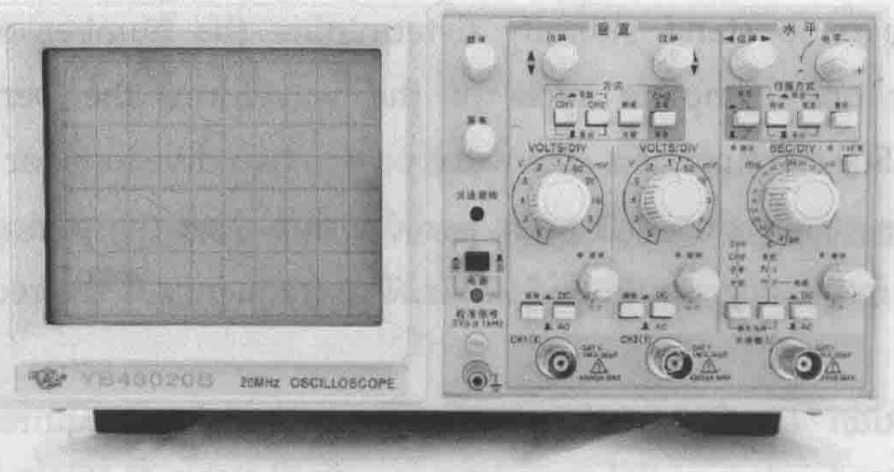


Fig. 3.3-5 Panel of an oscilloscope

The oscilloscope is now ready to be used. The oscilloscope in the laboratory is a dual trace oscilloscope, i. e. , it has two vertical inputs (CHl and CH2) for the simultaneous display of two waveforms.

3.3.4.2 Peak-to-Peak Voltage and Period Measurements

Select a sine wave with frequency of 50.00 Hz from the function generator. Connect this waveform to CH2 of the oscilloscope. Adjust the display knobs to make the sine wave stand still (Do you recognize which knobs to adjust?). Read the Y divisions, H (in cm), between the two peaks of a cycle and the vertical sensitivity, V(in V/cm). The peak-to-peak voltage input is equal to $H \times V$ (in V). Record the data in Data Table 3.3-1.

Meaure the input period. Read the X divisions, L (in cm), of a complete cycle for the sine wave and the horizontal sensitivity, t (in ms/cm). The period input is equal to $L \times t$ (in ms). Record the data in Data Table 3.3-1.

3.3.4.3 Using Lissajous Patterns to Measure the Frequency

So far we have used the oscilloscope to plot voltage versus time. The horizontal deflection was produced by a sawtooth voltage.

On the oscilloscope, the voltage source connected to CH1 or CH2 can be applied to the horizontal deflection by using *X-Y* working mode, which disconnects the sawtooth voltage generator. For two sine waves, if their oscillatory directions are vertical reciprocally and the frequencies of the two signals have an integral rate, the synthetic figure will be a stable closed figure. In other words, if two sine waves with integral frequency ratio are connected to CH1 and CH2, respectively, we can observe a stable closed pattern on the screen by using *X-Y* mode. The closed curves are Lissajous patterns (see Data Table 3.3-2).

For the Lissajous patterns, if there are N_x tangential points in *X*-direction and N_y tangential points in *Y*-direction, we have

$$\frac{N_x}{N_y} = \frac{f_y}{f_x} \tag{3.3-1}$$

where f_x and f_y are the frequencies of the waveforms applied to the horizontal deflection and vertical deflection, respectively.

We can use Eq. (3.3-1) to meaure the frequency using the following steps.

(1)Turn the TIME/DIV knob fully anticlockwise to *X-Y*. Apply a 50.00 Hz sine wave to CH2.

(2)Use the function generator to supply a sine wave. Connect this signal to CH1.

(3)Adjust the function generator frequency to about 50 Hz and then make fine adjustment to obtain a stationary cycle (or oval) pattern on the screen. Record the frequency on the generator in Data Table 3.3-2.

(4)Adjust the generator frequency to obtain different stationary patterns. Record the frequecies in Data Table 3.3-2.

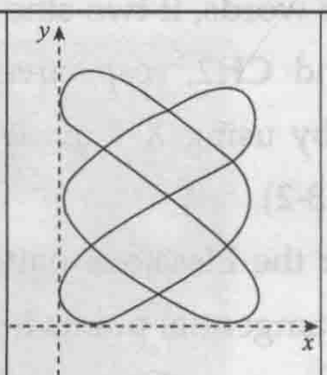
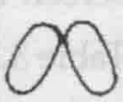
3.3.5 Experimental Data

Data Table 3.3-1 ***Purpose*: To measure the peak-to peak voltage and period of a sine wave**

Peak-to-peak voltage, V_{pp}		Period, T	
Vertical sensitivity, V		Horizontal sensitivity, t	
Height, H		Length, L	
$V_{pp} = V \cdot H$		$T = t \cdot L$	

(*Don't forget UNITS*)

Data Table 3.3-2 ***Purpose*: To measure the frequency using Lissajous pattern**

Known frequency f_y / Hz	Lissajous Patterns					
	N_x	1	1	1	2	2
	N_y	1	2	3	3	1
	$f_y : f_x$	1 : 1	1 : 2	1 : 3	2 : 3	2 : 1
50.00 (CH2)	f_x' / Hz (read from generator)					
	$f_x = \frac{N_y}{N_x} f_y$ / Hz (calculate)					

Your name and student number:_____ Instructor's initial:_____

3.3.6 Calculations

(1)Show the sample calculation of the peak-to-peak voltage of the sine wave.

(2)Show the sample calculation of the period of the sine wave.

(3)Show one sample calculation of the unknown frequency f_x in Data Table 3.3-2.

3.3.7 Post Lab Questions

(1)One of the most useful controls for an oscilloscope is the triggering control. Without proper "triggering", a waveform will scroll horizontally across the screen rather than staying "locked" in place. Describe how the triggering control is able to "lock" an AC waveform on the screen so that it appears stable to the human eye. What, exactly, is the triggering function doing that makes an AC waveform appear to stand still?

(2)What is the input frequency of a signal that covers 4.2 divisions (or cm) when the horizontal sweep rate is set to a 4 ms?

(3)An oscilloscope is used to display an AC voltage signal as shown in Fig. Q3.3-1.

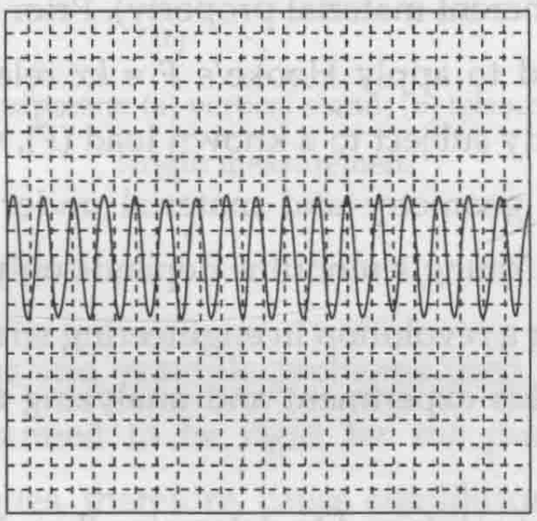


Fig. Q3.3-1 A waveform on an oscilloscope

What display control(s) on the oscilloscope need to be adjusted in order to show fewer cycles with a greater height of this signal on the screen?

3.4 Measurement of the Young's Modulus of Wire by Elongating

Prelab Assignment 3.4

(1)What's the instrument error for a vernier caliper with the smallest division of 0.02 mm? What's its reading error?

(2)What's the definition of Young's modulus? In this experiment, which quantities will be measured in order to determine the Young's modulus? In Eq. (3.4-5), which quantity is the most critical? Why?

(3)Describe the main functions of an optical lever in this experiment. What's the magnification of the optical lever in Fig. 3.4-3?

3.4.1 Introduction and Objectives

Young's modulus, also known as the tensile modulus or elastic modulus, is a quantity used to characterize materials. It is named after Thomas Young, the 19th century British scientist. It is an inherent material property (the term modulus refers to an inherent material property). Prior to Young's contribution, engineers were required to apply Hooke's $F = kx$ relationship to identify the deformation (x) of a body subject to a known load (F), where the constant (k) is a function of both the geometry and material under consideration. Thomas Young pointed out that Young's modulus depends only on the material, not its geometry, thus allowing a revolution in engineering strategies.

After performing this experiment and analyzing the data, you should be able to:

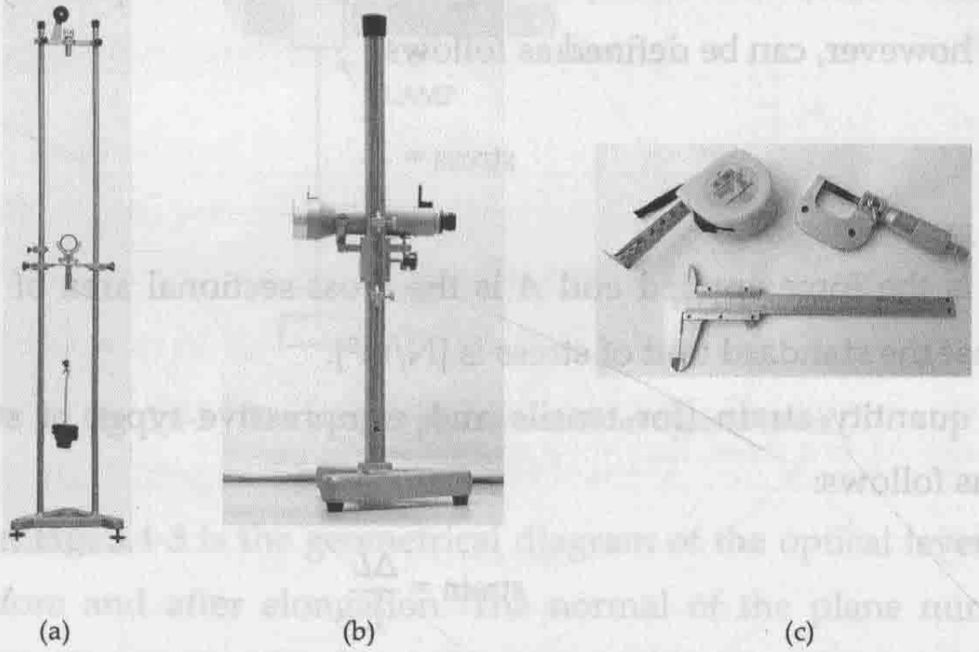
(1)Study elasticity of solids and to determine their Young's modulus.

(2)Use accurate measurement instruments correctly to measure the small lengths of objects.

(3)Understand the methods employed to eliminate the systematic error in some specific situations.

3.4.2 Required Equipment

- Vertical stand to support the wire(Fig. 3.4-1)
- Optical lever
- Set of kilogram weights
- Vernier caliper
- Telescope
- Micrometer screw gauge
- Steel tape



(a) (b) (c)

Fig. 3.4-1 Experimental equipment: (a) vertical stand; (b) telescope; and (c) instruments for length measurement

3.4.3 Theory

In most applications and example problems used in introductory physics courses, objects are assumed to be rigid for the purpose of simplification. Applied forces and torques, therefore, cause equilibrium, translational motion, and/or rotational motion to occur without deformation. Personal experience, however, tells us otherwise. When supposedly rigid materials are subject to great forces, permanent deformation is a definite possibility. Car crashes are an unfortunately common example. Stories of buildings and bridges collapsing under duress are less common, but they do occur.

Engineers need to be extremely cognizant of the properties of the materials

that they use in their designs. When subject to a particular stress, or force per unit area, materials will respond with a particular strain, or deformation. If the stress is small enough, the material will return to its original shape after the stress is removed, exhibiting its elasticity. If the stress is greater, the material may be incapable of returning to its original shape, causing it to be permanently deformed. At some even greater value of stress, the material will break or fracture.

There are different types of stress: tension or tensile stress, compression or compressive stress, shear stress, and hydraulic stress. The quantity for all types of stress, however, can be defined as follows:

$$\text{stress} = \frac{F}{A} \tag{3.4-1}$$

where F is the force applied and A is the cross-sectional area of the material. Notice that the standard unit of stress is [N/m^2].

The quantity strain (for tensile and compressive types of stress) can be defined as follows:

$$\text{strain} = \frac{\Delta L}{L} \tag{3.4-2}$$

where L is the original length of the material, and ΔL is the change in length that results after the stress is applied. Notice that strain is a dimensionless quantity.

Young's modulus, E, is a constant that describes the ratio of stress to strain for a material experiencing either tensile or compressive stress in elastic deformation.

$$E = \frac{\text{Stess}}{\text{Strain}} = \frac{F \times L}{A \times \Delta L} \tag{3.4-3}$$

Given the fact that ΔL is generally a very small length for most materials, measuring Young's modulus accurately can be a difficult task. In this experiment we use an optical lever (see Fig. 3.4-2) to measure the small elongations. Optical lever is made up of one small circular plane mirror and

three metal toes which form one isosceles triangle. The two front metal toes must be placed in one of the grooves in the flat and the back toe must be on the surface of the clamp, as shown in Fig. 3.4-2. The length of the back metal toe can be adjusted if necessary. The distance b from the back metal toe to the line between two front metal toes is called the length of optical lever.

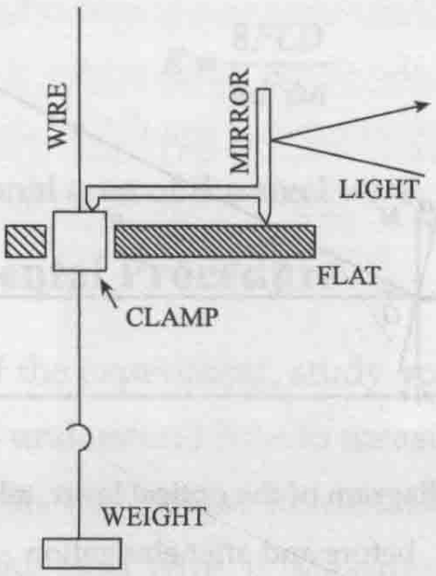


Fig. 3.4-2 The proper placement of an optical lever

Shown in Fig. 3.4-3 is the geometrical diagram of the optical lever and the steel rule before and after elongation. The normal of the plane mirror M is accordant with the axes (BO) of the telescope when there is no force on the wire. At this point, the reading on the measuring scale from the telescope is n_0. When weight is added, the wire is stretched by ΔL. The back metal toe of the optical lever is dropped so that the angle α of the normal of the plane mirror is formed. At this time, the reading on the measuring scale from the telescope is n_1. According to the reference law of light, $\angle n_0 OB' = \angle B'O n_1 = \alpha$ and $\angle n_0 OB'' = 2\alpha$, where OB' is the normal of the plane mirror after the tilt. The difference in readings of measuring scale between the initial and final situations is,

$$\Delta n = |n_1 - n_0|$$

From the geometry, we know

$$\tan\alpha = \frac{\Delta L}{b}$$

$$\tan 2\alpha = \frac{\Delta n}{D}$$

where, D is the distance between the plane mirror and the measuring scale (meter stick near the telescope).

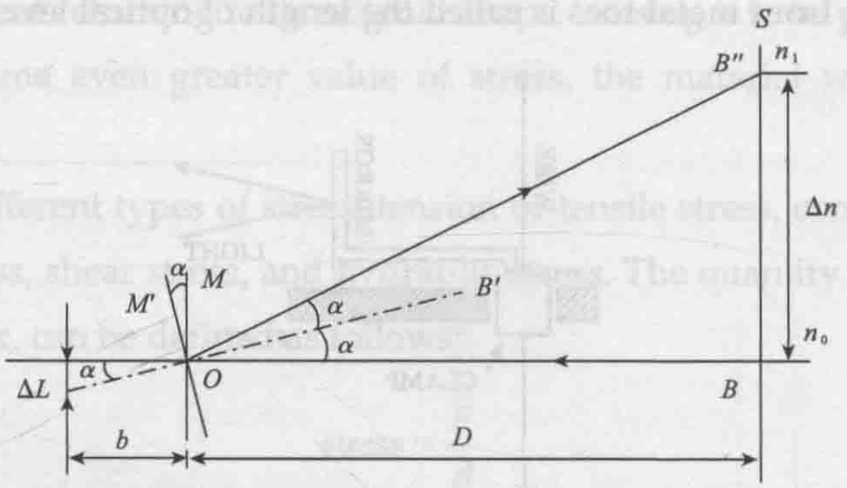


Fig. 3.4-3 The geometrical diagram of the optical lever, telescope, and the steel rule (S) before and after elongation

Because ΔL is a small quantity and $\Delta L \ll b$,

$$\tan\alpha \cong \alpha \cong \frac{\Delta L}{b}$$

Similarly, $\Delta n \ll D$,

$$\tan 2\alpha \cong 2\alpha = \frac{\Delta n}{D}$$

Then,

$$\frac{\Delta L}{b} = \frac{\Delta n}{2D}$$

Thus,

$$\Delta n = \frac{2D}{b} \cdot \Delta L = k \cdot \Delta L \qquad (3.4\text{-}4)$$

where, k is the magnification for the optical lever, $k = 2D/b$.

In this experiment, b is 4~8 cm, D is 1~2 m, so the magnification for the

optical lever could be 25~100. The elongation of the wire ΔL is a small length quantity, which is difficult to measure directly. However, using the optical lever, the small quantity is magnified into a quantity that can be read easily.

Finally, using Eq. (3.4-4) ΔL may be substituted into Eq. (3.4-3) considering that $A = \pi d^2/4$, the Young's modulus can then be written as

$$E = \frac{8FLD}{\pi d^2 \Delta n} \tag{3.4-5}$$

where, A is the cross sectional area of the steel wire.

3.4.4 Experimental Procedure

(1)At the beginning of the experiment, study your apparatus very carefully and work with it until you understand how to measure elongation.

(2)Add two weights on the weight holder. Use a micrometer caliper to measure the diameter of the steel wire, d. Measurement is performed in three directions at each of three points (upper end, middle, and lower end) along its length. It is worth noting that before the first reading, you should note the "zero reading" of the micrometer. Enter your measurements in Data Table 3.4-1. (Hint: Read the next section to learn to use a micrometer.)

(3)Use the steel tape to measure the original length of the steel wire, L_0. Enter your measurements in Data Table 3.4-1.

(4)Place the optical lever such that the two front metal toes are put in one of the transverse grooves on the flat while the back metal toe is placed on the surface of the clamp (see Fig. 3.4-2). Adjust the height of the telescope to the same height as the optical lever. Attention: do not touch the wire and do not put the back toe into the cracks between the clamp and the flat (see Fig. 3.4-2).

(5)Adjust the telescope and the optical lever to see the image of the meter stick by following the steps below.

i. Move the telescope to point to the plane mirror by aligning the gap (or circle) and post on the telescope and the plane mirror in one line.

ii. Put the eye outside the telescope in the direction of gap (or circle) -post; looking for the image of the steel rule from the outside of the telescope. If you

cannot see the meter stick in the mirror, adjust the height of the telescope or adjust the screw bolt in front of the plane mirror to alter the tilting of the plane mirror (see Fig. 3.4-3).

iii. Put your eye on the telescope and adjust the eye piece to make the cross clear. Rotate the bolt on the right side of the telescope (to adjust the image distance) to see the image of the measuring scale clearly (see Fig. 3.4-4) so that the horizontal line is near the 0 cm mark while the two weights are on the weight holder. Attention: if you can not see the image of the meter stick or part of the vision is blurry, you can adjust the screw under the telescope.

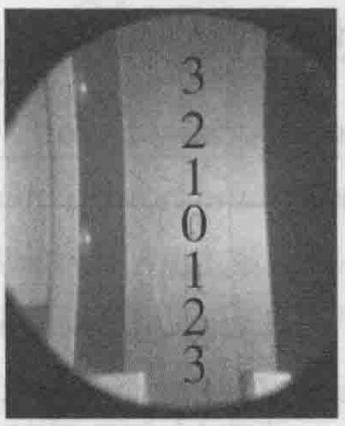


Fig. 3.4-4 The image of the scale marks of the meter stick

iv. Add weights to the weight holder. The telescope will give readings of the measuring scale. Load the measured wire successively in steps of one kilogram weight up to nine kilograms, taking the telescope readings after each addition. Attention: Only read the scale when the system (the loaded wire) is stationary. Enter the readings in Data Table 3.4-2.

v. Add one more weight up to ten kilograms without reading the scale. Unload one weight per time and read the telescope at each step. Enter the readings in Data Table 3.4-2.

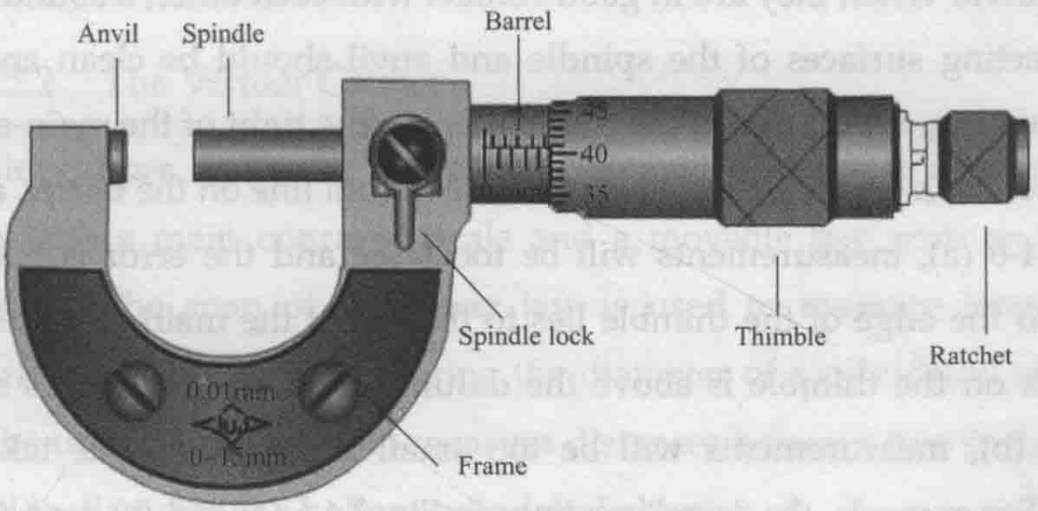
(6)Use the steel tape to measure the distance between the optical lever and the steel rule, D_0. Enter it in Data Table 3.4-1.

(7)Measure the length of the optical lever, b_0, by using the vernier caliper. Put the optical lever on a piece of white paper and press. You can see three dots corresponding to the three toes. The distance b is from the back metal toe to the line across the two front metal toes.

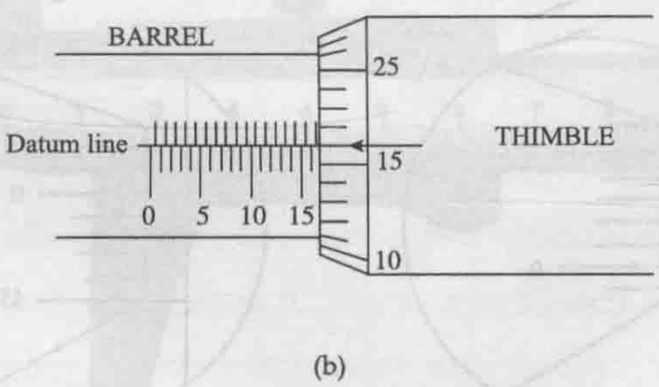
3.4.5 Accurate Instruments for Measuring Lengths

3.4.5.1 The Micrometer Caliper (Micrometer screw Gauge)

The micrometer caliper [see Fig. 3.4-5(a)] provides for accurate measurements of small lengths. A micrometer is particularly convenient in measuring the diameters of thin wires and the thicknesses of thin sheets. The axial main scale on the barrel (or "sleeve") is calibrated in the 1.0 mm and 0.5 mm scale divisions below and above the datum line. The thimble scale is calibrated in 0.01 mm. One complete rotation of the thimble (50 divisions) moves it through 0.5 mm, which is the minor division calibrated on the barrel.



(a)



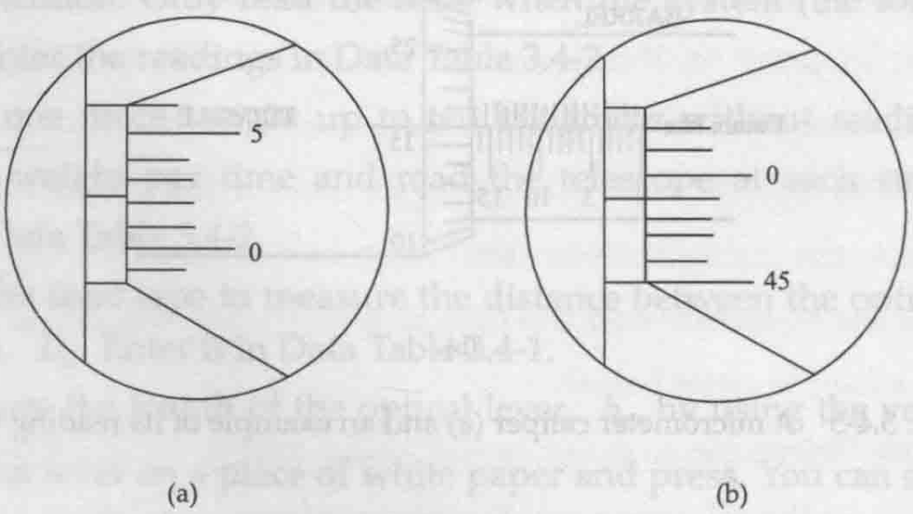
(b)

Fig. 3.4-5 A micrometer caliper (a) and an example of its reading (b)

Measurements are taken by noting the position of the edge of the thimble on the main scale and the position of the datum (or "reading") line on the thimble scale. For example, for the drawing in Fig. 3.4-5(b), the micrometer has

a reading of 16.661 mm. The edge of the thimble is past the half millimeter mark after the 16 mm major division. Therefore, on the main scale is a reading of 16.000 mm plus one 0.500 mm division (scale above datum line), giving 16.500 mm. In addition, the datum line is past the 0.16 mm scale mark on the thimble. So, the reading on the thimble scale is 0.161 mm, where the last digit 1 is the estimated or doubtful figure. Thus, the final measurement is 16.500 + 0.161 = 16.661 (mm).

As with most instruments, a zero check should be made and a zero correction applied to each reading if necessary. As for a micrometer caliper, a zero reading is made by rotating the ratchet until the spindle comes into contact with the anvil. When they are in good contact with each other, a sound is heard. The contacting surfaces of the spindle and anvil should be clean and free of dust. In zeroing, if the edge of the thimble lies to the right of the main-scale zero or the zero mark on the thimble is below the datum line on the barrel, as shown in Fig. 3.4-6 (a), measurements will be too large and the error is taken to be positive. If the edge of the thimble lies to the left of the main-scale zero or the zero mark on the thimble is above the datum line on the barrel, as shown in Fig. 3.4-6 (b), measurements will be too small and the error is taken to be negative. For example, the "zero" readings in Fig. 3.4-6 (a) and (b) are + 0.023 mm and – 0.010 mm, respectively. The zero correction is made by subtracting the zero reading from the measurement (or "actual") reading.



(a) (b)

Fig. 3.4-6 Zero reading of a micrometer

(a) positive error; (b) negative error

Summarizing the corrections in equation form,

Corrected reading = actual reading – zero reading

For example, for a positive zero reading of +0.023 mm as in Fig. 3.4-6 (a),

Corrected reading = actual reading – 0.023 mm

For a negative zero reading of – 0.010 mm as in Fig. 3.4-6 (b), then

Corrected reading = actual reading – (– 0.010) mm

= actual reading + 0.010 mm

3.4.5.2 The Vernier Caliper

The vernier caliper, as shown in Fig. 3.4-7, commonly called a vernier, consists of a rule with a main engraved scale and a movable jaw with an engraved vernier scale. The span of the lower jaw is used to measure length and is particularly convenient for measuring the diameter of a cylindrical object. The span of the upper jaw is used to measure distances between two surfaces, such as the inside diameter of a hollow cylindrical object.

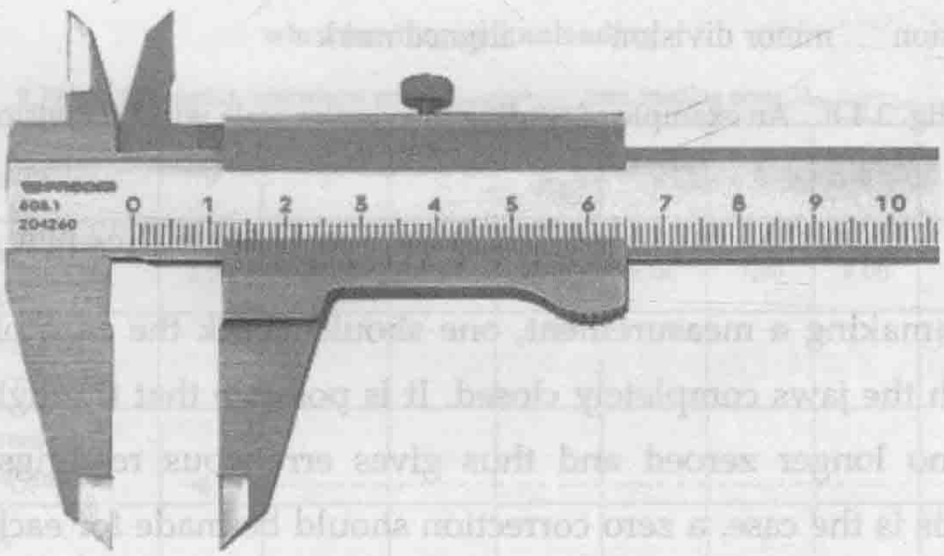
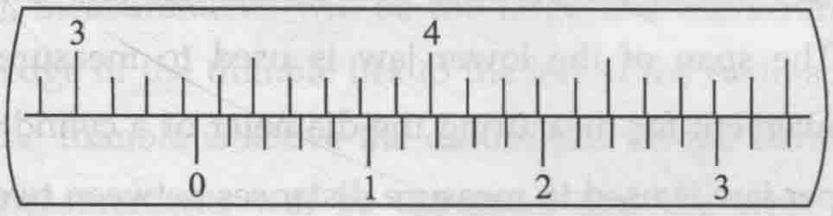


Fig. 3.4-7 A vernier caliper

The main scale is calibrated in centimeters with a millimeter least count.

The movable vernier scale has 50 divisions and one division is 1/50 of 1 mm, or 0.02 mm. If a vernier has 20 divisions on the vernier scale, one division is 0.05 mm. A measurement is made by closing the jaws on the object to be measured and reading where the zero mark on the vernier scale falls on the main scale.

In Fig. 3.4-8, the first two significant figures are read directly from the main scale. The vernier zero mark is past the 3 mm line after the 3 cm major division mark, so there is a reading of 3.3 cm. The next significant figures are obtained by referring to the vernier scale markings below the main scale. There is always a vernier mark that coincides with a mark on the main scale and this vernier mark number is the last two digits. In the figure, the sixteenth mark (or the second mark after 3 on the vernier scale) to the right of the vernier zero coincides with a mark on the main scale, so the last two significant figures are 32 (0.032 cm or 0.32 mm). Therefore, the measurement is



3 cm + 0.3 cm + (0.002 × 16) cm = 3.332 cm = 33.32 mm

major division minor division aligned mark

Fig. 3.4-8 An example of reading the vernier scale with 50 divisions

$$3\ \text{cm} + 0.3\ \text{cm} + (0.002 \times 16)\ \text{cm} = 3.332\ \text{cm} = 33.32\ \text{mm}$$

Before making a measurement, one should check the zero of the vernier caliper with the jaws completely closed. It is possible that through misuse the caliper is no longer zeroed and thus gives erroneous readings (systematic error). If this is the case, a zero correction should be made for each reading. In zeroing, if the vernier zero lies to the right of the main-scale zero, the error is taken to be positive. Similarly, if the vernier zero lies to the left of the main-scale zero the error is negative. In both cases, the zero correction is made by subtracting the zero reading from the measurement reading.

3.4.6 Experimental Data

Data Table 3.4-1 *Purpose* : To record the instrument parameters

(Unit: mm)

Diameter, d_i										
	Zero reading, d_0	0.000	Instrument error, $\Delta_{instr.}$		0.005	Reading error, Δ_{Read}			0.002	
	Trials	Point 1			Point 2			Point 3		
		1	2	3	4	5	6	7	8	9
	Actual reading, d_i'	0.638	0.620	0.621	0.640	0.602	0.628	0.618	0.698	0.612
	Corrected reading $d_i = d_i' - d_0$	0.638	0.620	0.621	0.640	0.602	0.628	0.618	0.698	0.612
Instrument error of tape, Δ_{tape}=(0.2×reading/1000+0.3)						Length of the optical lever				
L_0 (original reading)		660.0	D_0 (original reading)		1378.0	b_0(original reading)			63.80	
Instrument error, $\Delta_{L\text{-tape}}$		0.4	Instrument error, $\Delta_{D\text{-tape}}$		0.6	Instrument error, $\Delta_{b\text{-vernier}}$			0.02	
Estimated error, $\Delta_{L\text{-eval.}}$		2	Estimated error, $\Delta_{D\text{-eval.}}$		5	Estimated error, $\Delta_{b\text{-eval.}}$			0.2	
L (after rounding to an integer)		660	D (after rounding to an integer)		1378	b (after rounding to one decimal place)			63.8	

Data Table 3.4-2 *Purpose* : To record the meter stick readings observed from the telescope when loading and unloading

g_{local}= 9.79m/s^2 (Chengdu), Instrument error, $\Delta_{\Delta n\text{-stick}}$= 0.5 mm, reading error, $\Delta_{\Delta n\text{-read}}$= 0.2 mm

Trial	0	1	2	3	4	5	6	7	8
Total mass of the weights, m/kg	2.00	3.00	4.00	5.00	6.00	7.00	8.00	9.00	10.00
Meter stick reading, loading n_i'/mm	85.0	91.0	98.0	103.1	105.3	107.2	106.3	112.1	
Meter stick reading, unloading n_i''/mm	73.2	81.3	87.0	91.2	99.0	105.1	109.1	114.4	
Mean, $n_i = \frac{1}{2}(n_i' + n_i'')$ /mm	79.1	86.2	92.5	97.2	102.2	106.2	107.7	113.3	

Your name and student number:_____ Instructor's initial: Jing wu

3.4.7 Calculations

(1)Compute the diameter of the wire.

$$\bar{d} = \frac{1}{9}(d_1 + d_2 + \cdots + d_9)$$

(2)Use the following equation to compute the change in length after loading 4 kg weights.

$$\overline{\Delta n} = \frac{1}{4}\left[|n_7 - n_3| + |n_6 - n_2| + |n_5 - n_1| + |n_4 - n_0|\right]$$

(3)Use the following equation to compute the Young's modulus of the measured wire.

$$\bar{E} = \frac{8FLD}{\pi \bar{d}^2 b \overline{\Delta n}}$$

where $F = mg = 4 \times g$ (N).

(4)Use the propagation of uncertainties to compute the uncertainty of the measured Young's modulus.

$$\sigma_E = \bar{E} \times \sqrt{\left(\frac{\sigma_L}{L}\right)^2 + \left(\frac{\sigma_D}{D}\right)^2 + \left(2\frac{\sigma_d}{d}\right)^2 + \left(\frac{\sigma_b}{b}\right)^2 + \left(\frac{\sigma_{\overline{\Delta n}}}{\overline{\Delta n}}\right)^2}$$

Firstly, you must determine the uncertainties, σ_L, σ_D, σ_d, σ_b, and $\sigma_{\overline{\Delta n}}$ of the direct measurements.

(5)Express the final measurement result in the following form.

$$\bar{E} = \bar{E} \pm \sigma_E (\mathrm{N/m^2})$$

3.4.8 Post Lab Questions

(1)Which measurement is more critical, the length of the wire or its elongation? Why? Explain why the length of the wire is measured with a meter stick while a very sensitive length measuring device is required for the elongation?

(2)The initial load on the hanger was not included in our calculation. Why not? Could we have used a different value as our initial load? Explain.

3.5 Laser Holography

Prelab Assignment 3.5

(1)What is coherence as applied to optics. State the required conditions for observing coherence pattern?

(2)In the double-slit experiment, what method was utilized to obtain two beams with same frequency for coherence? In laser holography experiment, what method was used to obtain two beams in order to observe coherence pattern?

(3)What are the differences between normal photography and holography?

(4)Can laser used in holography be replaced by other light sources? Why?

3.5.1 Introduction and Objectives

Holography is an extraordinary example of the interaction that occurs between two light beams of the same coherent source that is optically recorded and then reconstructed by varying either the phase or amplitude. The discovery of holography was an unexpected result of research into improving electron microscopes at the British Thomson-Houston Company in Rugby, England in 1947. Discovered by Hungarian physicist Gabor Denes, holography is a technique that allows the coherent light scattered from an object to be recorded and later reconstructed so that it appears as if the object is in the same position relative to the recording medium as it was when recorded. Remarkably, the image changes as the position and orientation of the viewing system changes in exactly the same way as an observer would see a real life object. The basic idea behind Gabor Denes' theory was that in order to obtain perfect optical imaging, the total of all the information has to be used; not just the amplitude, as in usual optical imaging, but also the phase. In this manner a complete spatial picture could be obtained displaying length, width and depth. However, at the time Gabor developed holography, coherent light sources were not available and the field could not advance until the development of Dr. Townes' maser in 1953. In

this experiment, we will explore the variety of optical arrangements (and their limitations) necessary in order to produce hologram.

The goal of this experiment is to learn the basics of holography by making a two-beam transmission hologram.

3.5.2 Required Equipment

- He-Ne laser source(Fig. 3.5-1)
- Optical shutter
- Plane mirrors
- Beam splitter
- Beam expanders
- Film plate

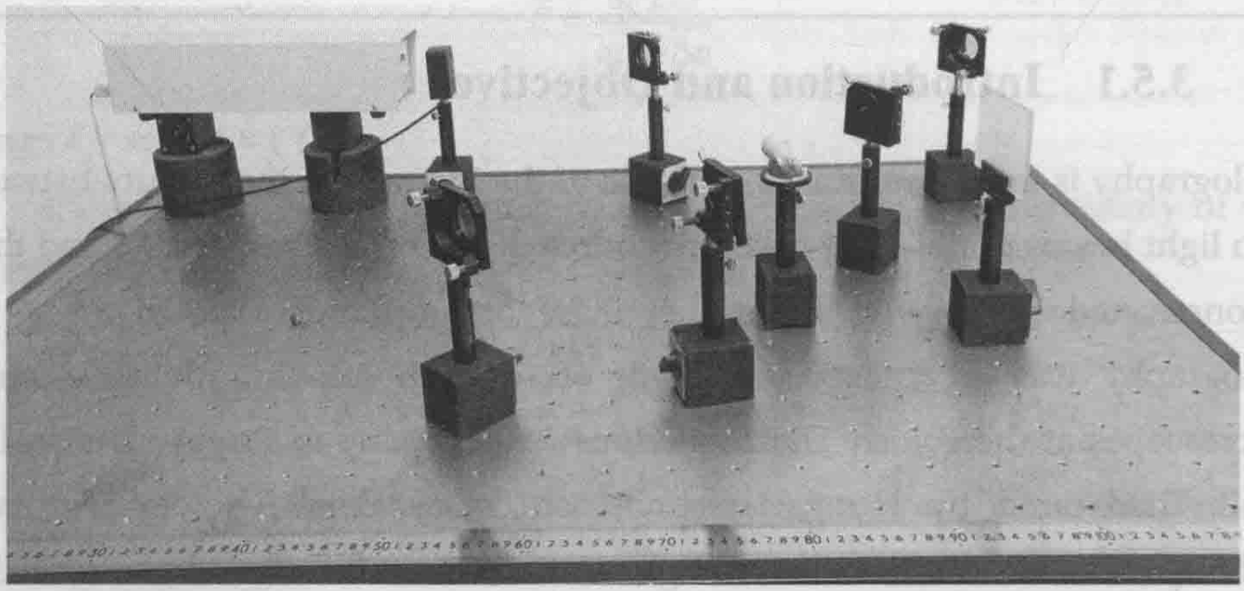


Fig. 3.5-1 Instruments for holography and their arrangements

3.5.3 Theory

In conventional photography, the intensity and the wave-length of the light is made use of, where the image is recorded by the intensity distribution of the light reflected by the object on a photographic emulsion by optical means: lenses and mirrors. With holography, the phase of the light as well as the intensity and wavelength are used to produce the three dimensional image. Thus, unlike in conventional photography, we make use of the phase content of the light in order to record all of the optical information of the object.

In order for any hologram to be created, a source of coherent radiation is required. Coherent radiation is characterized by two components, spatial and temporal coherence. The concepts of spatial and temporal coherence are important in describing the phase characteristics of any imaging system. In general, the concept of coherence is related to the stability of phase. Two waves are said to be coherent if they maintain the same phase. This implies they have the same frequency. If a wave is temporally coherent, then the phase is unchanged throughout time. Spatially coherent waves maintain the same phase as the beam is projected out along an area. As an analogy, we can imagine a successive line of buoys spaced evenly apart all rising and falling with each passing wave. If all the buoys peaked on the crests of the waves simultaneously, the waves could be considered temporally coherent and then if the same thing would occur for a grid of buoys (spatially shifted copies), we could crudely assume that the waves were spatially coherent. Coherent radiation is key in this experiment in order to construct the necessary fringe patterns inherent in the hologram. The creation of these fringes begins at our radiation source, the He-Ne laser. Once the light leaves the source, it is split and directed along two separate paths (see Fig. 3.5-2). One path leads to an object where the light scatters off in all directions in the form of circular waves en route to a recording medium. The other path directs the light onto the same recording medium but from a different angle and leaves the beam virtually unchanged in its planar form. However, before either direction of radiation reaches the medium, they interfere with one another constructively and destructively due to the minor path differences produced by variations in the spatial depth of the object. This interference of the two light waves creates a pattern of concentric rings that form lines of exposure onto the recording medium called a Fresnel zone plate. Because each pattern of rings is produced by a point source from the illuminated object, we can expect a complex array of concentric rings to form the necessary fringe patterns in which to accurately describe the object upon reconstruction. Moreover, it is the concentric ring spacings that encode the depth of the object onto the medium. If one part *a* of an object is closer to the medium than another part *b*, then the path difference between those two parts

result in differently spaced sets of rings which creates the appearance of depth when viewing the hologram.

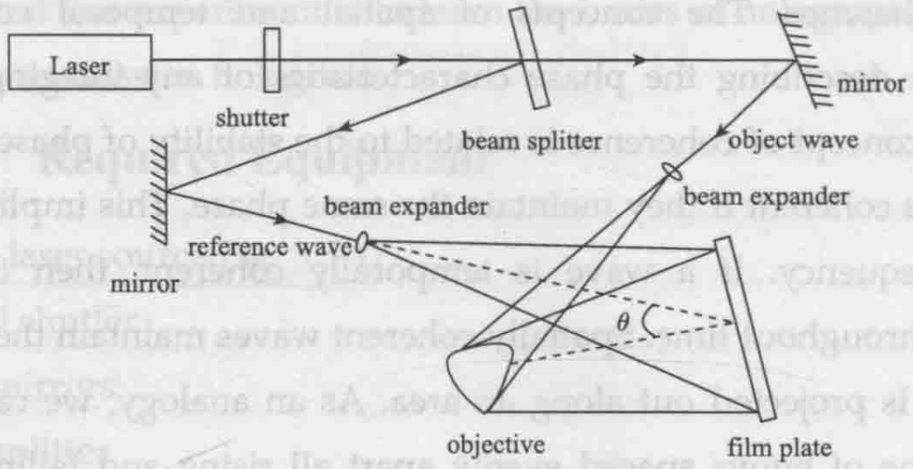


Fig. 3.5-2 Arrangement of the transmission hologram

Once exposed, the holographic film is developed through a series of a chemical rinses and then air dried. The reconstruction of the recorded image now only employs unaltered reference beam and not the beam scattered off the object. With the plate back in its original recording position, the reference illumination interacts with the developed fringes which, remembering, dictate the position of each object point as it relates to the reference. The outcome is the reconstruction of the original light wave by the reference which creates a virtual image of the object in a process of wave-front reconstruction. In optics, a virtual image is an image in which the outgoing rays from a point on the object never actually intersect at a point and are said to diverge. The virtual image is created from the reference beam diverging from the plate as it diffracts around the fringes. Furthermore, an observer traces the divergent light back through the plate along an imaginary path to where the object appears as an orthoscopic (correctly oriented) virtual image (see Fig. 3.5-3). In practice, it is better for an observer to view the hologram along the first order object wave as opposed to the zeroth order reference wave which may damage the eyes. Another phenomenon which occurs from the diffraction of the reference beam results in the formation of the complex conjugate of the virtual image whose angle of projection is opposite to that of the orthoscopic image. This event appears on the observer's side of the plate and is known as the pseudoscopic image. Even

though the pseudoscopic (or real) image is predicted, it was not however observed in this experiment.

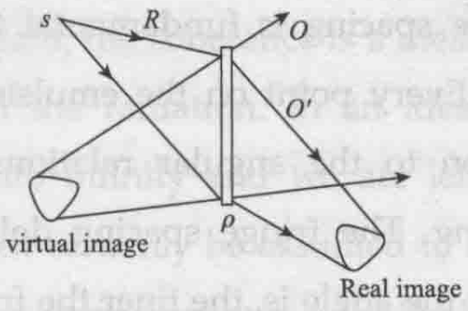


Fig. 3.5-3 Schematic of the diffraction that occurs between the reference light and the holographic plate fringes and the resultant images that occur

3.5.3.1 Plane Hologram

There exist two main categories of holograms: plane and volume holograms. This experiment is focus on plane hologram. As the angle difference between the object and reference beam changes, so does the spacing of the patterns in the emulsion. If the angle difference remains much less than 90 degrees, the hologram is called a plane hologram and the holographic information is largely contained in the two dimensional plane of the emulsion. Therefore, we can idealize the fringe patterns as being encoded into a plane. Likewise, if the difference in angle approaches 180 degrees, the hologram is considered a volume hologram. It is important to note that the emulsion surface of the volume holograms cannot be idealized to two dimensions due to the almost parallel direction of the fringe pattern to the emulsion.

In the formation of the holographic fringes, the angle between the reference and object is significant. Each point on the object bears a distinct relationship to the point reference dictated by the angle formed between the light reflected off the object and reference as they meet on the surface of the emulsion. The spacing of the fringes can be shown by simple geometrical construction to follow the equation known as Bragg's law

$$d = \frac{\lambda}{\sin\theta} \tag{3.5-1}$$

where, θ is the angle between the reference wave and the object wave (see Fig. 3.5-2); d is the spacing of the fringes in the film plate. The relationship between angle and fringe spacing is fundamental to the reconstruction and recording of holograms. Every point on the emulsion and on the object will provide a slight variation to the angular relationship and consequently a variation in fringe spacing. The fringe spacing delineates the object during reconstruction. The larger the angle is, the finer the fringes. If the angle is small the farther apart the fringes are. But we must remember that this is relative to the thickness of the emulsion. As the fringes become finer the requirement for a fairly precise alignment of the hologram with the illuminating beam to affect constructive refraction/diffraction by the fringe pattern becomes exceedingly important whereas the slightest vibrations can prohibit the creation of these fringes.

In plane absorption holograms, the spacing between fringes is normal to the surface of the plate and is capable of absorbing light. Once reconstructed, the absorption hologram depicts the contrast and depth of the object through diffraction of light around the Fresnel zone rings. This modulates the amplitude while keeping the phase constant. Another type of plane hologram is a phase hologram. A phase hologram is exposed in a similar way to an absorption hologram, but the difference arises not only in the exposure time, but during the developing process. In order to ensure the recording of all the optical information of the object onto the plate, it is necessary to double the exposure time of the emulsion. Then, after the plate is developed and fixed, the silver halide is bleached away effectively removing the silver and leaving a translucent film with a different index of refraction to that of the unexposed portion of the plate. Now, when light is passed through the plate the amplitude is held constant from the lack of dark Fresnel zone rings, but the waves are refracted by the varying translucent film thus modulating the phase.

3.5.3.2 Holography Experiment Pre-requisites

As discussed earlier, the theory of holography relies heavily on a coherent source of illumination. Again, the coherence is a measurement of temporal and spatial correlation within the radiation. In an idealized environment, these correlations extend off into infinity and we are left with a perfect laser. In practice, the laser can most certainly be assumed to be coherent, but just as in all experiments; we must concern ourselves with the limitations inherent within the system. Therefore, if temporal coherence tells us how monochromatic the source is, and given that we can expect at least a certain number of spontaneous instabilities to affect our system, that is after a time t, the light will momentarily fail to be coherent. Thus, a coherence length of the light can be measured to give us a kind of timescale over which we can expect our system to be stable and coherent. This idea is fundamental when attempting to create any of the holograms, in that, it gives us an upper bound on the time at which we expose our plates. Furthermore, if, for example, we have a setup which requires an exposure time greater than the coherence time, we can expect that setup to fail from introducing those random instabilities briefly mentioned.

We can explore the stability and thus the coherence length of the hologram recording system by implementing what corresponds to a long-legged Michelson interferometer on the same surface as the one to be used to record the hologram. In this setup, the beam is directed first through a splitter and then each subsequent beam is reflected back onto itself by the addition of mirrors where the path length of each beam is comparable except to a wavelength. The beams are superimposed at the splitter and then projected onto a viewing surface by a divergent lens. Thus, the superimposed beams interfere to create fringe patterns which we can measure. By placing a power meter between the viewing surface and the expanded beam, we are able to measure the random instabilities through an oscilloscope and thus predict a stable time interval. As a result we concluded that after a stable time length of approximately 11 seconds a significant source of noise would be spontaneously introduced. We then

explored our own possible contributions to noise by talking, yelling, stomping and blowing into the beam path. It would be found that while talking has very little effect on the coherence, the latter of the three did at least introduce some source of noise. In addition, interference is also noticed in the beam from the interaction with dust particles in the air which could contribute as a source of noise. In conclusion, the surface was found to be extremely stable.

Because the He-Ne laser is spatially coherent for approximately 34 cm, it was satisfactory for the difference in beam paths to be a few centimeters. However, all attempts were made to achieve a nearly zero beam path difference to ensure the coherence.

As described by Bragg's law, there exists a maximum angle allowed between the reference and object beam given the color of light and maximum fringe number. In our experiment the recommended angle is about 30°.

The ratio of the intensity of the reference beam to the object beam should be between 4 : 1 and 10 : 1, but is recommended to start with a ratio of 4 : 1. The absolute power, which is total intensity projected onto the emulsion multiplied by the exposure time, should be 100 microjoules cm^{-2} and thus gives us a reasonable method of predicting necessary exposure times given a particular setup.

3.5.4 Experimental Procedure

(1)Refer to the setup as shown in Fig. 3.5-2 and suitably arrange the light path. Set up the optical components according to the already designed light path. Adjust the light path and take the holograph of the given still object.

Specifically the following are required:

i. The optical path difference between the object wave and reference wave should be close to zero.

ii. Adjust the beam splitter mirror in order to make sure that the intensity ratio between the reference wave and object wave is between 4 : 1 and 10 : 1.

iii. The angle between the object light and reference light should be about 30°.

iv. The object and the plate should be illuminated evenly.

v. During the course of exposure, keep the system as stable as possible. Avoid the influence of parasitic light, vibration, sound and airflow.

(2)Observing the hologram playback image of the object.

i. Refer to the light path shown in Fig. 3.5-3. Light the emulsion surface of the hologram with playback light wave that is consistent with the reference light wave. Find the virtual image of the playback object through the hologram formed at the back.

ii. Change observing direction and observe the change of image carefully.

iii. Observe the image by nearer and farer scenery.

3.5.5 Calculations

No calculation for this experiment. You are not required to write lab report for this experiment.

3.6 Newton's Rings

Prelab Assignment 3.6

(1)Why are Newton's rings circular?

(2)In modern physics, light has been demonstrated to have wave-particle duality. Which nature is demonstrated in Newton's rings experiment?

(3)Why are the dark and bright fringes observed in newton's rings?

(4)The sodium lamp in this experiment has two yellow components. Why can it be used as the light source for the experiment?

3.6.1 Introduction and Objectives

This experiment is one of many classic demonstrations of the wave-like properties of light. Sir Isaac Newton first reported on the ring-shaped interference pattern in 1675 (see Fig. 3.6-1), but he used the results to argue for a particulate (what he called "corpuscular") nature of light in his famous treatise on the nature of light, "Opticks" (1705). Robert Hooke famously disagreed with Newton's corpuscular theory, and in fact he had observed the rings ten years before Sir Isaac himself, in 1665. Unfortunately for Hooke, he died in 1703 just before Newton published "Opticks".

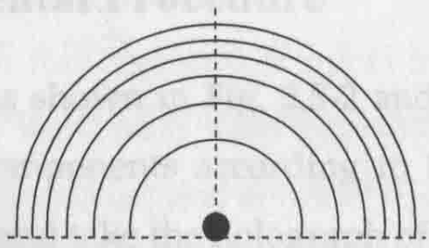


Fig. 3.6-1 Newton's own drawing of the pattern of rings from his book *Opticks*

When two waves of equal intensity combine, the intensity of the resulting wave can be anywhere between zero and four times the intensity of the individual waves, depending upon the phase difference between them. When they are in phase (the maxima coincide with maxima and the minima coincide

with minima), maximum intensity results. When they are exactly out of phase (the maxima of one wave coincide with the minima of the other wave), the result is zero intensity.

In thin film interference, a beam of light is separated into two parts that travel along different paths and then recombine. If the difference in path lengths is exactly an integral number of wavelengths, the waves recombine in phase, and constructive interference occurs. If the path lengths differ by exactly one half-wavelength, the waves are out of phase when they recombine and destructive interference occurs.

After performing this experiment and analyzing the data, you should be able to:

(1)Understand the concept of *interference*.

(2)Demonstrate thin-film interference and determine how the interference pattern depends on the wavelength of incident monochromatic light.

(3)Quantify experimental uncertainty.

3.6.2 Required Equipment

- Traveling microscope(Fig. 3.6-2)
- Sodium lamp
- Newton's rings setup

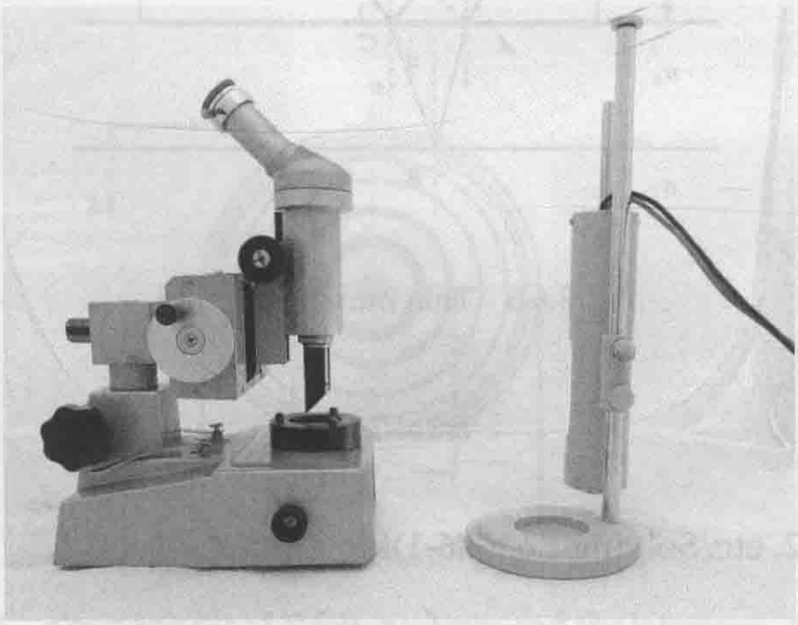


Fig. 3.6-2 Experimental setup

3.6.3 Theory

3.6.3.1 Thin film interference

Thin film interference can occur when two glass plates are placed in contact at one end and are separated by some small distance at the other (see Fig. 3.6-3). Consider a beam of light with monochromatic wavelength λ striking the glass plates at right angles to their surfaces. Part of the light wave will be reflected back at the boundary between the first glass plate and the space. Another part of the light wave will travel across the space and will be reflected back from the front surface of the second glass plate. These two reflected waves then combine at point *C* (see Fig. 3.6-3).

The difference in optical path length for the two light waves is $(2d + \lambda/2)$. The $\lambda/2$ term arises because there is a phase change of one half a wavelength when the light is reflected off the second glass plate. This effectively alters the physical path length by $\lambda/2$. The assumption is that the space between the two glass plates is filled with air, an optically less dense material than glass.

For *constructive interference* this difference in path length must equal *m* wavelengths, where *m* is an integer. Thus,

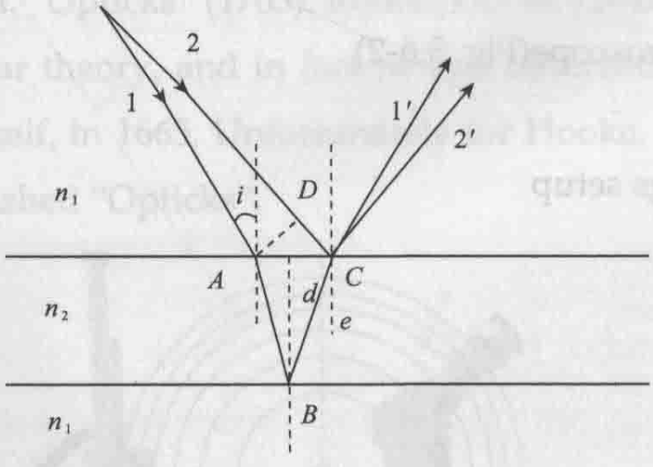


Fig. 3.6-3 Thin film interference

$$m\lambda = 2d + \frac{\lambda}{2} \tag{3.6-1}$$

where, $m = 0, 1, 2$, etc. Solving Eq. (3.6-1)for d gives

$$d = \left(m - \frac{1}{2}\right) \times \frac{\lambda}{2} \tag{3.6-2}$$

For *destructive interference* the difference in path length must equal an integral number of wavelengths more a half wavelength. Thus,

$$m\lambda + \frac{\lambda}{2} = 2d + \frac{\lambda}{2} \tag{3.6-3}$$

Solving Eq. (3.6-3) for d gives

$$d = \frac{m\lambda}{2} \tag{3.6-4}$$

3.6.3.2 Newton's rings

A typical example of film interference is Newton's rings. Newton's rings occur when monochromatic light is shone on a lens that has been placed in contact with a glass plate as shown in Fig. 3.6-4. Theoretically, at the center, where the lens is in contact with the glass plate, a dark circle is formed.

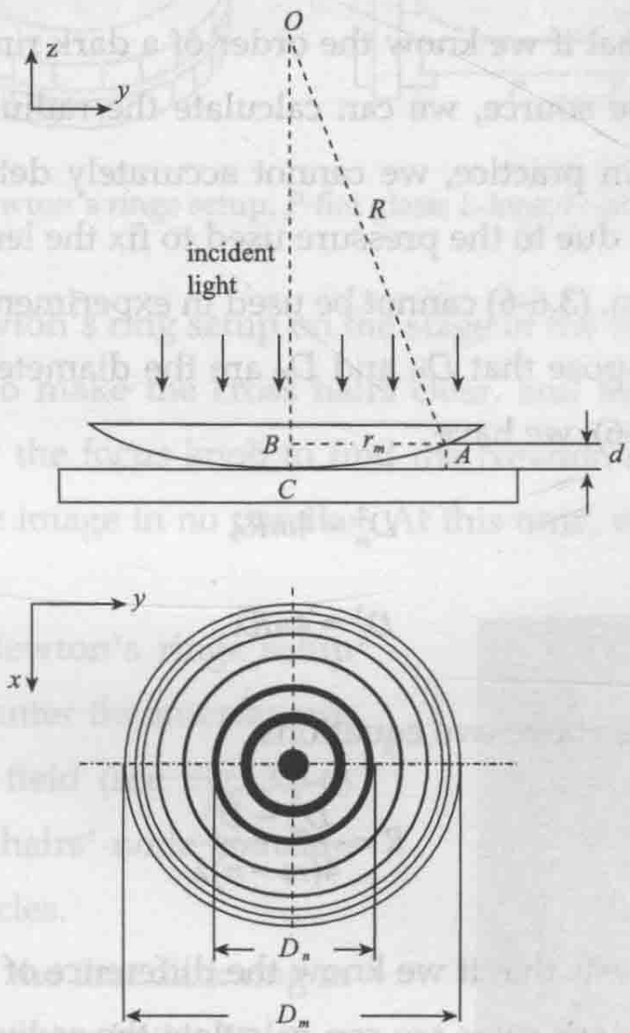


Fig. 3.6-4 Arrangement of Newton's rings and the interference pattern

In ΔAOB (see Fig. 3.6-4), it gives

$$R^2 = (R-d)^2 + r_m^2$$

where, R is the radius of curvature of lens, d is the thickness of the film, and r_m is the radius of a dark ring. Thus,

$$r_m^2 = 2Rd - d^2$$

Since d is very small as compared to R and $d^2 \ll Rd$, therefore, neglecting d^2,

$$r_m^2 = 2Rd \tag{3.6-5}$$

Suppose that we observe a dark pattern (destructive interference) at point A in Fig. 3.6-4, then substituting for d in Eq. (3.6-5) using Eq. (3.6-4)

$$r_m^2 = mR\lambda \tag{3.6-6}$$

Eq. (3.6-6) suggests that if we know the order of a dark ring and its radius, and the wavelength of the source, we can calculate the radius of curvature of the lens. Unfortunately, in practice, we cannot accurately determine the orders of the interference rings due to the pressure used to fix the lens to the glass as well as other factors. So, Eq. (3.6-6) cannot be used in experiment directly.

In Fig. 3.6-4, suppose that D_m and D_n are the diameters of two dark rings. According to Eq. (3.6-6), we have

$$D_m^2 = 4mR\lambda$$

$$D_n^2 = 4nR\lambda$$

Solving for R from the above two equations,

$$R = \frac{D_m^2 - D_n^2}{4(m-n)\lambda} \tag{3.6-7}$$

Equation (3.6-7) suggests that if we know the difference of orders instead of the exact orders for two dark rings we can calculate the radius of curvature of the lens.

Sodium light is used in this experiment. The wavelength of Sodium light

actually consists of two wavelengths called the yellow double lines. They are 588.995 0 nm and 589.592 4 nm, respectively, and have a relative intensity of about 2 : 1.

3.6.4 Experimental Procedure

In this experiment, we will measure the radius of curvature of lens utilizing the Newton's rings.

(1)Firstly, find the Newton's rings under the fluorescent lamp light. In this case, the Newton's rings are alternate colorful concentric rings around the point of contact. If the rings are not in the center of the setup you can adjust the three bolts of the setup until they are located in the center (see Fig. 3.6-5).

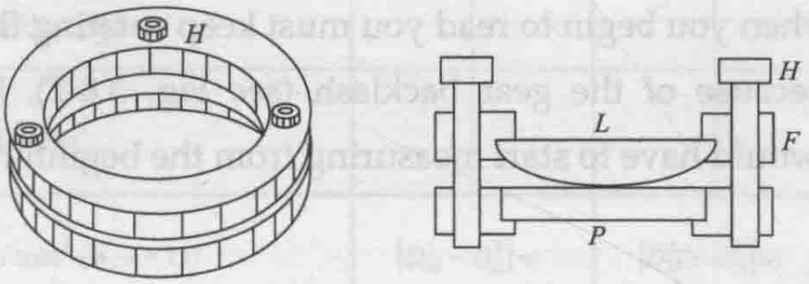


Fig. 3.6-5 The Newton's rings setup. *P*-flat glass; *L*-lens; *F*- metallic frame; *H*-bolt

(2)Place the Newton's ring setup on the stage of the microscope. Adjust the microscopic ocular to make the cross hairs clear, and then adjust microscope from down to up by the focus knob to find the Newton's rings, and make the cross hairs and circle image in no parallax. At this time, what you observe may be some fringes.

(3) Move the Newton's rings setup with your hand to center the interference pattern in the view field (see Fig. 3.6-6) and make the cross hairs' node coincide with the center of circles.

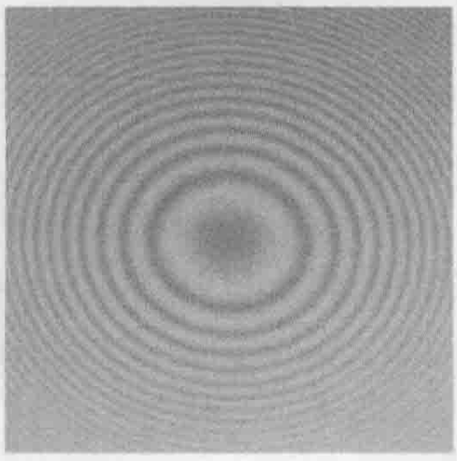


Fig. 3.6-6 The observed Newton's rings

(4) Suppose that the first dark ring in the left close to the center is the 5th ring. From this ring, you count the rings one by one until you find the 35th ring by

rotating the wheel. You don't need to read this scale. Then rotate the wheel reversely and count the rings. When you find the 30th ring place the cross hairs in the center of the ring and read the scale on the traveling microscope. It's your first data and enter it in Data Table 3.6-1.

(5) Keep rotating the screw until you find the 25th ring, and read the scale on the traveling microscope. Record the left positions of the 20th, 15th, 10th and 5th rings in Data Table 3.6-1.

(6) Keep rotating the wheel and make the cross hairs cross the center. When you find the 5th ring in the right read the position. As such, you can read the right positions of 10th, 15th, 20th, 25th, 30th rings. Record the readings in Data Table 3.6-1.

Note that when you begin to read you must keep rotating the wheel only in one direction because of the gear backlash (see Fig. 3.8-7). Do not reverse; otherwise, you would have to start measuring from the beginning again.

3.6.5 Experiment Data

Data Table 3.6-1 ***Purpose*: To measure the radius of curvature of a lens**

wavelength $\bar{\lambda} = \frac{1}{2}(\lambda_1 + \lambda_2) = \frac{1}{2}$ (_____+_____) = _____(nm)

The order of the dark rings		30th	25th	20th	15th	10th	5th
Readings /mm	Left, D_L						
	Right, D_R						
Diameter /mm $D = \lvert D_L - D_R \rvert$							
D^2 /mm^2							
$\lvert D_m^2 - D_n^2 \rvert$ / mm^2 (m–n = 15)		$\lvert D_{30}^2 - D_{15}^2 \rvert =$		$\lvert D_{25}^2 - D_{10}^2 \rvert =$		$\lvert D_{20}^2 - D_5^2 \rvert =$	
$\overline{\lvert D_m^2 - D_n^2 \rvert}$ /mm^2							
$\bar{R} = \frac{D_m^2 - D_n^2}{4(m-n)\lambda}$ /mm							

Your name and student number:_____ Instructor's initial:_____

3.6.6 Calculations

(1)Compute the radius of curvature, $\bar{R}$ of the lens. Show the following sample calculations.

(a) $D = |D_L - D_R| =$

(b) $D^2 = |D_L - D_R|^2 =$

(c) $|D_m^2 - D_n^2| =$

(d) $\overline{|D_m^2 - D_n^2|} =$

(e) $\bar{R} = \dfrac{\overline{|D_m^2 - D_n^2|}}{4|m-n|\lambda} =$

(2)Compute the uncertainty of R and give the measurement result. (Follow the steps to compute the uncertainty in R.)

(a) $\mu_{A-(D_m^2-D_n^2)} = \sqrt{\dfrac{\sum\left[\left(|D_m^2 - D_n^2|\right)_i - \overline{|D_m^2 - D_n^2|}\right]^2}{3\times(3-1)}}$

(b) $\mu_{Read-(D_m^2-D_n^2)} = \dfrac{\Delta_{Read}}{\sqrt{3}}$

(c) $\mu_{Instr.-(D_m^2-D_n^2)} = \dfrac{\Delta_{Instr.}}{\sqrt{3}}$

(d) $\sigma_{(D_m^2-D_n^2)} = \mu^2_{A-(D_m^2-D_n^2)} + \mu^2_{Read-(D_m^2-D_n^2)} + \mu^2_{Instr-(D_m^2-D_n^2)}$

(e) $\sigma_{(\bar{R})} = \dfrac{\sigma_{|D_m^2-D_n^2|}}{4|m-n|\lambda}$

3.6.7 Post Lab Questions

(1)We used sodium light lamp in this experiment. It can be treated as monochromatic light because the double lines are very close. If we use polychromatic light, what will happen?

(2)If there is a bright circle in the center for the Newton's rings, not a dark circle, what could be the reasons? Will it affect the measurement results if we keep the bright circle in the center?

(3)Is the distance between two adjacent bright ring and dark ring equal for the Newton's rings? Why?

3.7 Polarized Light

Prelab Assignment 3.7

(1)How much polarized light is passed by a combination of the first polarizer and a second that has its polarization axis at 30° with respect to the first?

(2)Explain what Brewster's angle is. Calculate Brewster's angle and its uncertainty for a piece of glass with an index of refraction of $n = 1.4 \pm 0.2$.

(3)How many types of polarized lights are involved in this experiment? Give their names.

3.7.1 Introduction and Objectives

Polarization is a fundamental property of light and a very important concept of physical optics. Not all sources of light are polarized; for instance, light from an ordinary light bulb is not polarized. In addition to unpolarized light, there is partially polarized light and totally polarized light. Light from a rainbow, reflected sunlight, and coherent laser light are examples of polarized light. There are three different types of polarization states: linear, circular and elliptical. Polaroid sunglasses are examples of polarizers. They block certain radiations such as glare from reflected sunlight. Polarizers are useful in obtaining and analyzing linear polarization.

This experiment consists of a series of basic exercises, which will introduce important techniques for analyzing the polarization of light. We will study linear and circular polarization, using linear polarizers and two types of retarders: quarter-wave and half-wave.

After performing this experiment you should be able to:

(1)Use polarizers to obtain linearly polarized light.

(2)Use polarizers to judge the types of polarization states.

(3)Understand the functions of quarter-wave and half-wave plates.

3.7.2 Required Equipment

- Diode laser
- Detector to measure the intensity of light
- Two polaroid sheets
- A microammeter to display the intensity of light
- Quarter-wave and half-wave plates (Fig. 3.7-1)

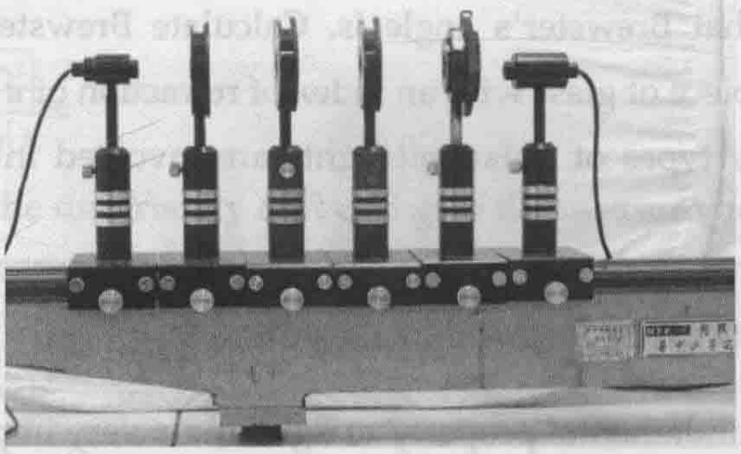


Fig. 3.7-1 Apparatus for the polarized light experiment

3.7.3 Theory

There are two ways to transmit energy between two points: by waves and by particles. Waves may have longitudinal vibrations (parallel to the propagation direction) or transverse vibrations (perpendicular to the propagation direction). Some waves, such as water waves, may have combined longitudinal and transverse vibrations. The phenomena of interference and diffraction of light show that light behaves like a wave. The fact that light may be polarized demonstrates that light waves are transverse, because longitudinal waves cannot be polarized.

Visible light is an electromagnetic wave in which the electric and magnetic fields oscillate in magnitude and direction. The electric and magnetic field vectors at any point and instant are perpendicular to each other and perpendicular to the direction of propagation (see Fig. 3.7-2). These two field vectors oscillate in phase and are proportional to each other, so we usually

represent the wave by referring only to the electric vector.

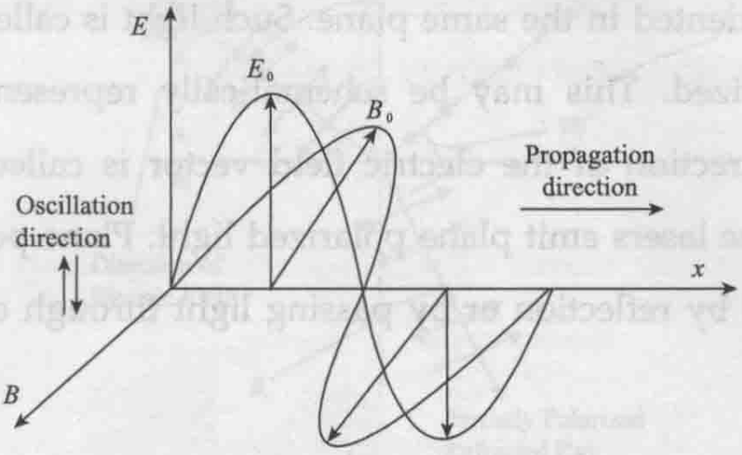


Fig. 3.7-2 An electric field vector E and a magnetic field vector B of a light wave

While much of this model of light is classical in form, one quantum aspect of light is essential for an understanding of polarization. That is the fact that light energy is not emitted in a continuous stream, but is emitted in "bursts" of energy called *photons*.

Ordinary light sources emit unpolarized light, which means that the photons they emit have their electric vectors oriented in random directions, uncorrelated with other photons. If we could keep track of the directions of the electric vectors of photons over a very large number of photons, we would find that they have no preferred orientation direction. This may be schematically represented as shown in Fig. 3.7-3, where the diagram shows that in a light beam consisting of very many photons all possible electric vector directions are represented.

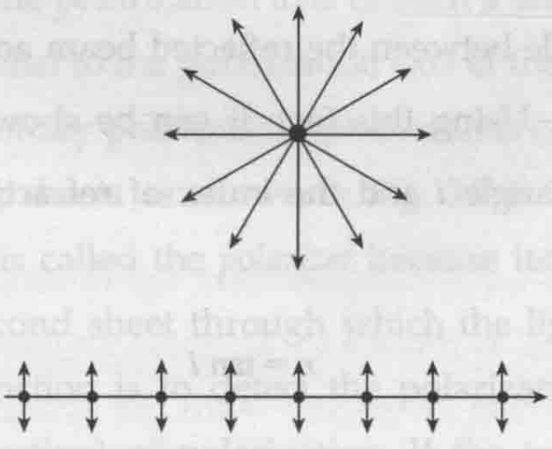


Fig. 3.7-3 Unpolarized light

Several means exist for producing light in which all photons have their electric vectors oriented in the same plane. Such light is called plane polarized or linearly polarized. This may be schematically represented as shown in Fig. 3.7-4. The direction of the electric field vector is called the direction of polarization. Some lasers emit plane polarized light. Plane polarized light may also be produced by reflection or by passing light through certain anisotropic materials.

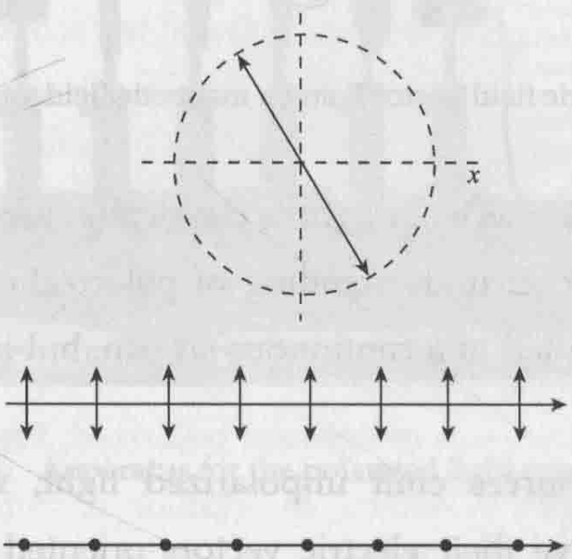


Fig. 3.7-4 Plane (or linearly) polarized light

When light is incident on a transparent medium part of the beam is reflected and part is refracted. The reflected beam will show partial polarization at all reflected angles. At one particular angle, called Brewster's angle, the reflected beam will be completely plane polarized. When light is incident at the Brewster angle the angle between the reflected beam and the refracted beam is exactly 90° (Fig. 3.7-5). Using this fact, it can be shown that the relationship between the Brewster angle i and the index of refraction of the material n is given by

$$n = \tan i \tag{3.7-1}$$

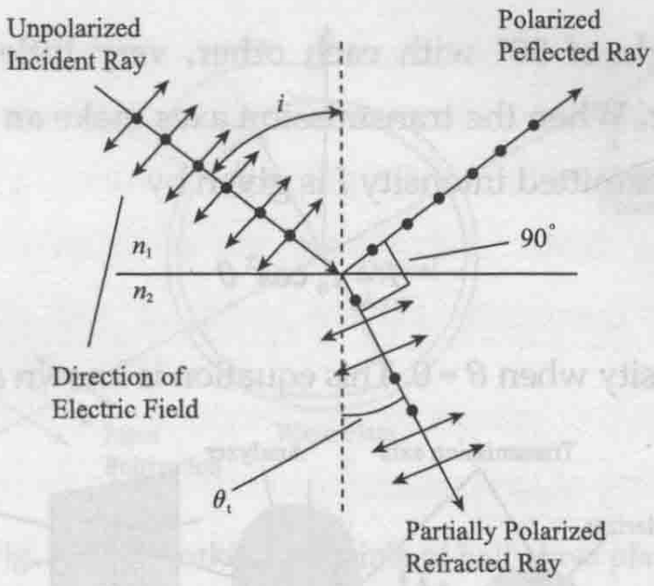


Fig. 3.7-5 Brewster angle

The second way to polarize light is to pass it through certain anisotropic materials, called *doubly refracting* or *birefringent* materials. Ordinary light passing through such materials may be thought of as consisting of two component light beams. Each component beam is plane polarized and the planes of polarization are perpendicular to each other. Many crystals found in nature, such as calcite, are doubly refracting. Some manufactured materials made up of oriented long-chain molecules also have this property. Cellophane is an example.

Some of these materials also have the property of absorbing one of the two beams, so only one plane polarized beam emerges from the material. This is useful for producing plane polarized light and for this purpose is manufactured in the form of sheets called "polarizing sheets" . Polaroid brand polarizers are a well-known example. The polarization axis of such a sheet is an imaginary line drawn on the sheet parallel to the polarization axis of the light it produces.

A common way to study polarization phenomena is to pass a beam of light through two polarizing sheets as shown in Fig. 3.7-6. The first sheet through which the light passes is called the *polarizer* because its function is to produce polarized light. The second sheet through which the light passes is called the *analyzer* because its function is to detect the polarization of the light and to determine its axis (direction) of polarization. If the transmission axes of the

sheets make an angle of 90° with each other, very little light intensity gets through the analyzer. When the transmission axes make an angle θ with respect to each other the transmitted intensity I is given by

$$I = I_0 \cos^2 \theta \tag{3.7-2}$$

where, I_0 is the intensity when $\theta = 0$. This equation is known as the law of Malus.

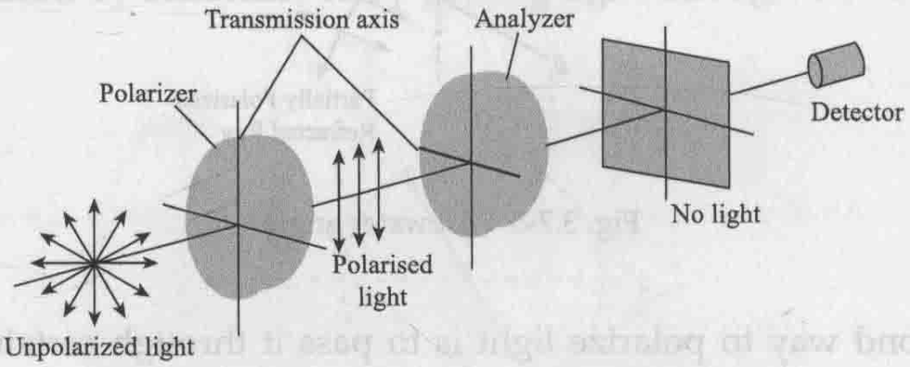


Fig. 3.7-6 Polarizer-analyzer arrangement

Wave plates are devices that delay one of the two orthogonally polarized components of the light incident upon them. The materials are asymmetric in that they have a different index of refraction in one direction (a "fast" axis) then the other (a "slow" axis). This is called *birefringence*. The "optical" or fast axis is usually indicated on any wave plate. Light polarized along this axis experiences a smaller index of refraction than light polarized perpendicular to this axis. The two orthogonal components of light, one polarized along the optical axis and one polarized perpendicular to that axis, enter the wave plate with a phase difference of zero and emerge with a phase difference of π or $\pi/2$ corresponding to either a $\lambda/2$ or a $\lambda/4$ wavelength delay — thus the name *half-wave plate* and *quarter-wave plate*. These devices are sometimes referred to as *retarders* because one of the polarization components has been slowed or retarded in its progress through the wave plate.

In the case of a half-wave plate, incident light polarized at an angle θ to the optical axis (the fast axis) is rotated by an angle 2θ (see Fig. 3.7-7). Notice how the incoming wave (making an angle θ with the optical axis) has been rotated by 2θ so that it now makes an angle θ on the other side of the optical axis.

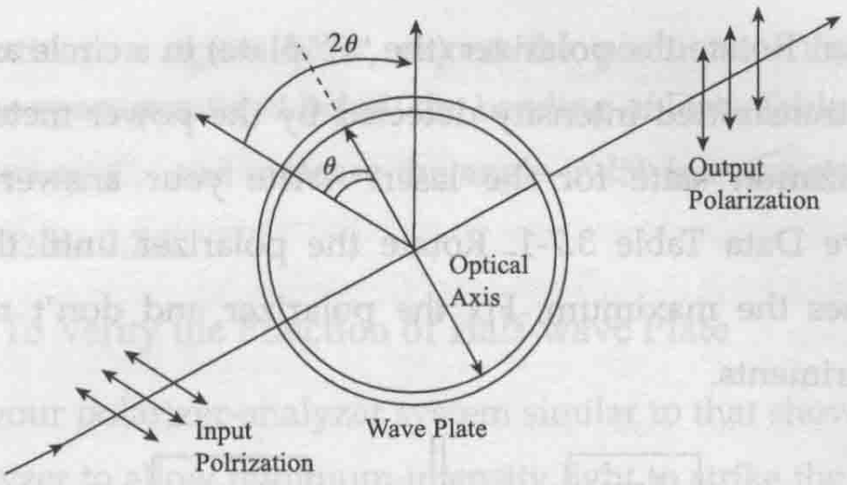


Fig. 3.7-7 Working principle of half-wave plate

A quarter-wave plate behaves in a different fashion(Fig. 3.7-8). It causes linearly polarized light to become different polarized light. It depends on the relative orientation between the directions of the incident linearly polarized light and the optical axis of a quarter-wave plate. If linearly polarized light is incident on a quarter-wave plate at 45° to the optic axis it produces circularly polarized light. If linearly polarized light is incident on a quarter-wave plate at 0° or 90° to the optic axis it produces linearly polarized light. Except the above cases, it produces elliptically polarized light. In the same way, circularly and elliptically polarized light incident on the quarter-wave plate becomes linearly polarized.

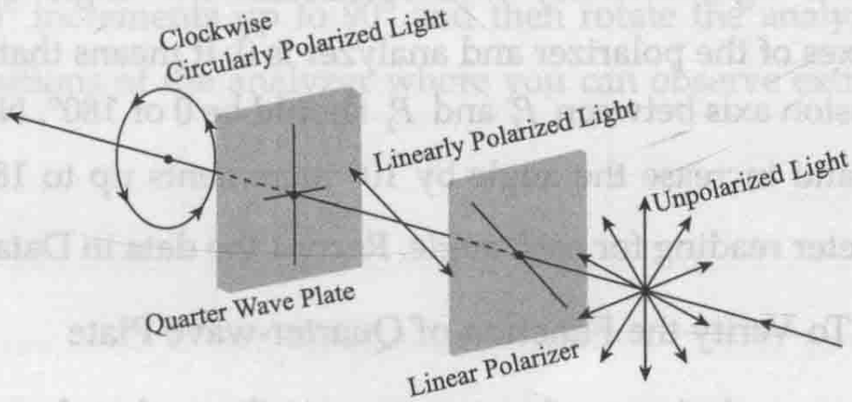


Fig. 3.7-8 Working principle of a quarter-wave plate

3.7.4 Experimental Procedure

3.7.4.1 To Verity the Law of Malus

(1)Set up the light path as shown in Fig. 3.7-9. Turn on the light source and

the power meter. Rotate the polarizer (the "P" plate) in a circle and observe the change in the transmitted intensity detected by the power meter. What is the possible polarization state for the laser? Write your answer in the space provided above Data Table 3.7-1. Rotate the polarizer until the transmitted intensity reaches the maximum. Fix the polarizer and don't rotate it in the following experiments.

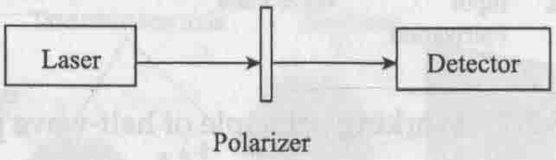


Fig. 3.7-9 Detect the polarization state of the laser source

(2)Add the analyzer (the other "P" plate) between the polarizer and the power meter and set up your polarizer-analyzer system similar to that shown in Fig. 3.7-6. Rotate the analyzer to allow maximum-intensity light to strike the photo detector. Note that the maximum intensity of light, the reading from the microammeter, should be between 200 μA and 1500 μA. Record the positions (the angles on the devices) of the polarizer and analyzer as P_1 and P_2, respectively.

(3)When the light intensity is a maximum the angle θ between the transmission axes of the polarizer and analyzer is 0. It means that the difference in the transmission axis between P_1 and P_2 should be 0 or 180°. Now start at the position of P_2 and increase the angle by 10° increments up to 180° and record the microammeter reading for each angle. Record the data in Data Table 3.7-1.

3.7.4.2 To Verity the Function of Quarter-wave Plate

(1)Set up your polarizer-analyzer system similar to that shown in Fig. 3.7-6. Rotate the analyzer to allow minimum-intensity light to strike the photodetector. The angle θ between the transmission axes of the polarizer and analyzer is 90° or 270°. Read the angles on the polarizer and analyzer as P_1 and P_2 and record them in the space provided below the heading of Data Table 3.7-2.

(2)When the light intensity is zero, insert a quarter-wave plate between the polarizer and the analyzer on the bench. Rotate the quarter-wave plate until

you observe extinction again. Now, record the position of the quarter-wave plate, C_1, in the space provided below the heading of Data Table 3.7-2.

(3)Now start at C_1 and increase the angle in 15° increments up to 90° and complete Data Table 3.7-2.

3.7.4.3 To Verity the Function of Half-wave Plate

(1)Set up your polarizer-analyzer system similar to that shown in Fig. 3.7-6. Rotate the analyzer to allow minimum-intensity light to strike the photodetector. The angle θ between the transmission axes of the polarizer and analyzer is 90° or 270°. Read the angles on the polarizer and analyzer as P_1 and P_2 and record them in the space provided below the heading of Data Table 3.7-3.

(2)When the light intensity is a minimum, insert a half-wave plate between the polarizer and the analyzer on the bench. How many times of extinction can you observe when rotating the half-wave plate from 0 to 360°?

(3)Set the half-wave at a random position where you can't observe the extinction. How many times of extinction can you observe when rotating the analyzer from 0 to 360°?

(4)Set the half-wave at a random position, C_2. Write this position down in the space below the heading of Data Table 3.7-3. Now start at C_2 and increase the angle in 15° increments up to 90° and then rotate the analyzer in a circle. Record the positions of the analyzer where you can observe extinction in Data Table 3.7-3.

3.7.5 Experimental Data

Data Table 3.7-1 *Purpose*: To verify the law of Malus

What is the possible polarization state for the laser light source? _____

P_1=_____, P_2=_____

Angle starting from the position of P_2, θ	0°	10°	20°	30°	40°	50°	60°	70°	80°	90°
Photocurrent, I_θ										
Angle starting from the position of P_2, θ	100°	110°	120°	130°	140°	150°	160°	170°	180°	
Photocurrent, I_θ										

Data Table 3.7-2 *Purpose*: To verify the function of quarter-wave plate

P_1=_____, P_2=_____, C_1=_____

Change the position of $\lambda/4$ plate starting at C_1 in steps of 15°	0°		15°		30°		45°		60°		75°		90°	
How many times of maximum and minimum intensity can you observe when you rotate the analyzer in a circle?	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min
The maximum and minimum intensities	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min
What is the polarization state of the light emerging from the $\lambda/4$ plate?														

Data Table 3.7-3 *Purpose*: To verify the function of half-wave plate

1)How many times of extinction can you observe when rotating the half-wave plate from 0 to 360°?_____

2)How many times of extinction can you observe when rotating the analyzer from 0 to 360°? _____

P_1=_____, P_2=_____, C_2=_____

Change the position of $\lambda/2$ plate starting at C_2		0°	15°	30°	45°	60°	75°	90°
Rotate the analyzer in a circle and record its positions where you can observe extinction	Position 1							
	Position 2							

Your name and student number:_____ Instructor's initial:_____

3.7.6 Data Processing

(1)Draw a graph of the microammeter reading I_θ depending on the angle θ by using the entries in Data Table 3.7-1.

(2)Draw a graph of the microammeter reading I_θ depending on $\cos^2\theta$ by using the entries in Data Table 3.7-1.

(3)Summarize the function of quarter-wave plate based on the entries in Data Table 3.7-2.

(4)Summarize the function of half-wave plate based on the entries in Data Table 3.7-3.

3.7.7 Post Lab Questions

(1)Explain how polarization phenomena indicate that light is a transverse wave.

(2)In the Step 3 of 3.7.4.3, how many times of extinction can you observe when rotating the half-wave plate from 0 to 360°? State the reasons.

(3)How can you distinguish between an unpolarized light and a circularly polarized light?

3.8 Measuring Laser wavelength and Index of Refraction of Air by Michelson Interferometer

Prelab Assignment 3.8

(1)A derivation of the path length difference of a dispersed laser beam can be described in terms of the beam angle. Show that the path length difference between paths ABE and ACD is given by $2d\cos\theta$. Note the dotted line, indicating that the lengths of DM_2 and EB are unequal(Fig. P3.8-1).

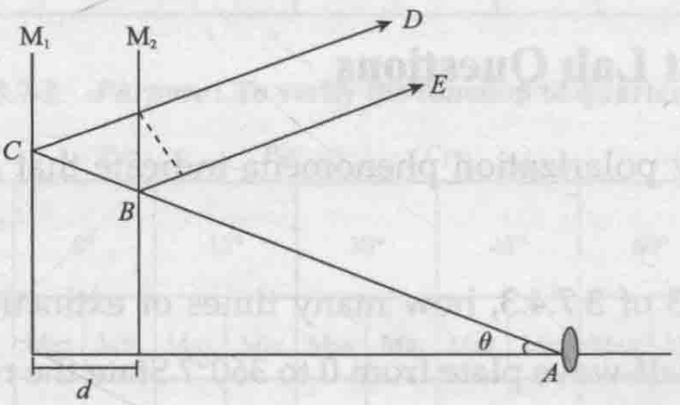


Fig. P3.8-1 The optical path difference of a dispersed beam.

(2)In Fig. 3.8-2,

a. how many times does the reflected beam pass through the beam splitter?

b. what is the purpose of the compensator plate G_2?

c. what is the characteristic of G_2 in thickness?

3.8.1 Introduction and Objectives

In 1887, Albert Abraham Michelson collaborated with colleague Edward Williams Morley in the Michelson-Morley experiment. They expected to measure the motion of the Earth relative to the aether, the hypothetical medium in which light was supposed to travel. However, their experiment resulted in a null result, which essentially disproved the existence of such an aether, leading eventually to the special relativity. A. A. Michelson received the Nobel Prize in Physics in 1907 for the Michelson-Morley experiment and the measurement of the

speed of light. He became the first American to receive the Nobel Prize in sciences.

Interferometry refers to measurements that rely on the interference of two (or more) light beams. Interferometers are widely used in science and industry for the measurement of small displacements, refractive index changes and surface irregularities.

After performing this experiment and analyzing the data, you should be able to:

(1)Understand the production of interference pattern in the Michelson interferometer.

(2)Understand how the Michelson interferometer is used to measure a small displacement.

3.8.2 Required Equipment

- Michelson interferometer(Fig. 3.8-1)
- He-Ne laser
- Pressure gauge

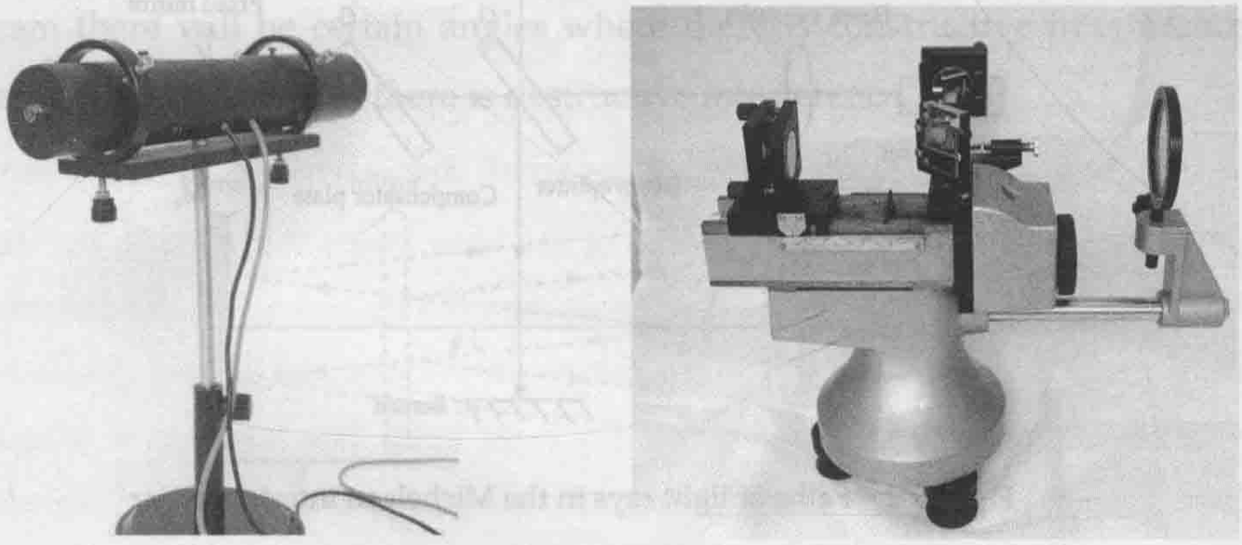


Fig. 3.8-1 He-Ne laser and Michelson interferometer

3.8.3 Theory

The Michelson interferometer permits observation of the interference between two beams of light traveling paths of about the same length, as shown in Fig. 3.8-2. Light from a laser is incident on a beam splitter (G_1) which

consists of a glass plate with a partially reflective surface, producing two perpendicular beams of equal intensity. About 50% of the light is reflected from the surface and 50% is transmitted. The reflected light hits movable mirror M_1 and is reflected back towards the beam splitter. The transmitted light as well is reflected back towards the beam splitter by a fixed mirror G_2. 50% of the intensity of each reflected beam is transmitted/reflected towards the screen for observation. There the two beams overlap and constructive and destructive interference occurs depending on the relative phase shift between the two plane waves. There is a glass plate G_2 between the G_1 and M_2. Glass plate G_2 has the same thickness with G_1 and its purpose is to ensure the reflected beam and the transmitted beam passing through the same thickness of glass. Normally, it is called compensator plate.

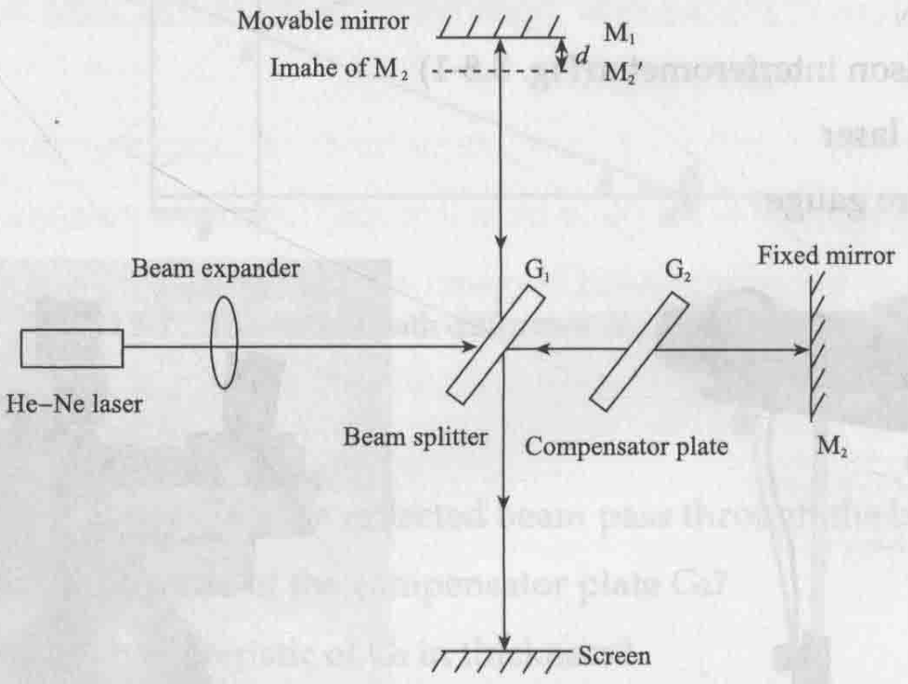


Fig. 3.8-2 Paths of light rays in the Michelson interferometer

The two beams are reflected and recombined. They will interfere constructively if their paths are of equal length, or if they differ by an integral number of wavelengths. The interference will be destructive if the paths differ by a half integral number of wavelengths.

By moving one of the mirrors, we can change the path length difference and the relative phase of the light beams. With careful alignment, it is possible

to use a laser light source to produce interference. As the path length difference changes, we would see both constructive and destructive interference. By moving Mirror M_1 slowly towards the splitter, we would see the laser point appear on the screen, reach maximum brightness, fade away, and then disappear as the path length difference is increased by one wavelength.

It is more practical to use a dispersed beam instead of a thin laser beam. In the lab, you will use a convex lens to disperse a laser light source. This lens is called beam expander. With a dispersed beam, the interferometer produces an interference pattern on the screen instead of a single point.

Figure 3.8-3 shows the path of a dispersed laser beam at an exaggerated angle θ. For convenience, the primary elements of the interferometer are shown in a linear arrangement. The parallel beams reflected towards the screen interact with each other in a constructive or destructive manner, depending on the path length difference. The path length difference, now dependent on the beam angle, θ, is $2d\cos\theta$. Since the path difference is dependent on the angle of the beam there will be certain angles where there is constructive interference and certain angles at which there is destructive interference.

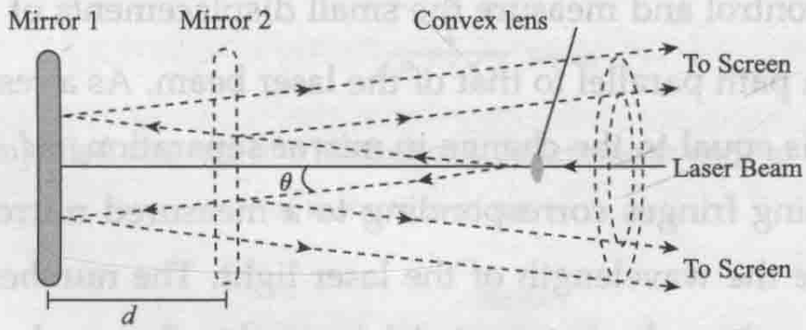


Fig. 3.8-3 The ray diagram of a dispersed laser beam at an angle θ

The condition for constructive interference is now

$$2d\cos\theta = m\lambda \qquad (3.8\text{-}1)$$

Constructive interference at specific angles gives rise to an interference pattern of bright and dark concentric circular *fringes*, as in Fig. 3.8-4. Note that Eq. (3.8-1)

still holds as a special case where $\theta = 0$, corresponding to the center of the interference pattern.

Fig. 3.8-4 Interference pattern of the Michelson interferometer

One standard use of the Michelson apparatus is to measure the wavelength of the incident light. With a dispersed beam, the path length difference can be changed by moving M_1 (see Fig. 3.8-2) to observe changes in the interference pattern. The interference fringes will move either towards or away from the center of the interference pattern, fringes are either produced or annihilated. In the center of the interference pattern, we would see the alternating light and dark spot as discussed before. A fringe will be produced or annihilated for every wavelength by which the path length difference is changed. For this reason, very fine control is required to move the mirror. The micrometer screw is designed to control and measure the small displacements of M_1. Mirror M_1 is restricted to a path parallel to that of the laser beam. As a result, the distance traveled by M_1 is equal to the change in mirror separation, Δd. By counting the number of passing fringes corresponding to a measured mirror displacement, we can calculate the wavelength of the laser light. The number of fringes that have passed in a mirror displacement, Δd, is equal to the number of wavelengths that the path length difference has changed. Therefore, Eq. (3.8-1) implies the wavelength is given by

$$\lambda = \frac{2\Delta d}{N} \tag{3.8-2}$$

where N is the number of passing fringes corresponding to the change in mirror separation.

The other use of the Michelson apparatus is to measure the index of

refraction of air. The interference pattern is sensitive to changes in the relative phase of the two split beams. So we can use the interferometer to investigate how transparent objects affect the phase of light by placing them in one of the split beam paths. Fig. 3.8-5 shows the sketch diagram of a Michelson interferometer used to measure the index of refraction of air. The apparatus includes an air cell connected to a vacuum system. The air cell is an air tight cylinder with glass windows on both faces. We can use this air cell to investigate the index of refraction of the air inside. Consider an air cell placed in the path of one of the split beams, as shown in Fig. 3.8-6.

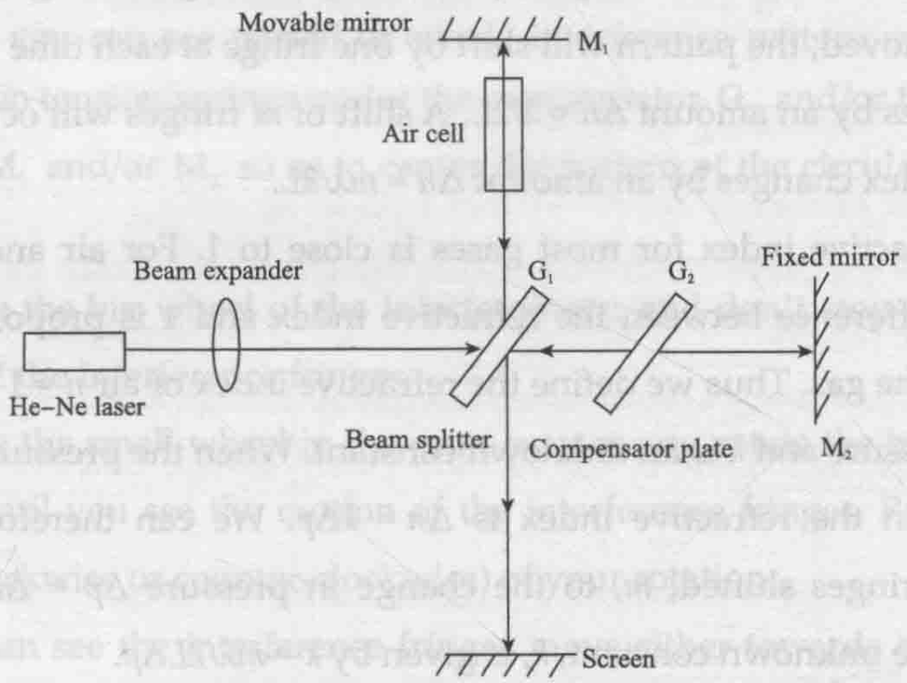


Fig. 3.8-5 A simplified diagram of a Michelson interferometer used to measure the index of refraction of air

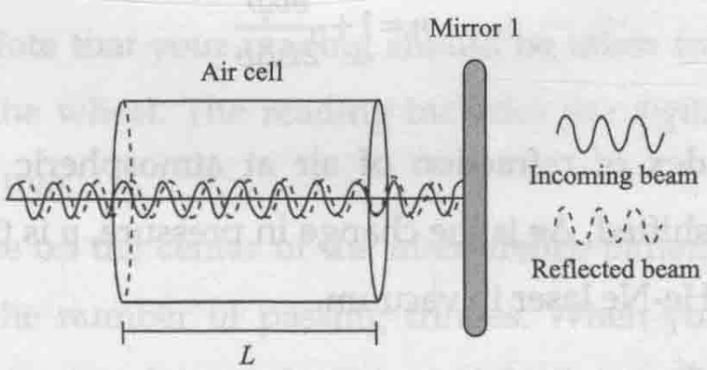


Fig. 3.8-6 Air cell placed in the path of one of the split beams

It is well known that if the optical path length of one beam changes by one

wavelength, the interference pattern is shifted by one fringe. The optical path length is equal to nL, where n is the index of refraction and L is the physical path length. The optical path length can be varied by changing either n or L.

In our experiment one of the beams passes through the cell of length L (see Fig. 3.8-6). Because the beam passes through the cell twice, the optical path length is $2nL$. The air is removed from this cell, changing the refractive index, n. The other beam passes through the same length of air, but with no cell in that beam.

If the refractive index changes by Δn, the path length changes by $2\Delta nL$. As the air is removed, the pattern will shift by one fringe at each time the refractive index changes by an amount $\Delta n = \lambda/2L$. A shift of m fringes will occur when the refractive index changes by an amount $\Delta n = m\lambda/2L$.

The refractive index for most gases is close to 1. For air and other ideal gases, the difference between the refractive index and 1 is proportional to the pressure of the gas. Thus we define the refractive index of air $n = 1 + kp$, where p is the air pressure and k is an unknown constant. When the pressure is changed, the change in the refractive index is $\Delta n = k\Delta p$. We can therefore relate the number of fringes shifted, m, to the change in pressure $\Delta p = \Delta n/k = m\lambda/2Lk$. Therefore, the unknown constant, k, is given by $k = m\lambda/2L\Delta p$.

Thus if you measure m fringes while the pressure changes by an amount Δp, you can calculate the refraction index of air at room temperature using

$$n = 1 + \frac{m\lambda p}{2L\Delta p} \tag{3.8-3}$$

where n is the index of refraction of air at atmospheric pressure, m is the number of fringes shifted, Δp is the change in pressure, p is the air pressure, λ is the wavelength of He-Ne laser in vacuum.

3.8.4 Experimental Procedure

3.8.4.1 Measuring the Wavelength of Laser

(1)Refer to Fig. 3.8-2 to set up the laser and the Michelson interferometer

without the beam expander.

(2)Align the laser and beam splitter so that their centers are at the same height. Then turn on the laser and have the laser spot just at the center of the beam splitter G_1. Ensure the reflected and the transmitted beams hit the center of the two mirrors M_1 and M_2.

(3)Several laser spots can be observed on the screen. Adjust the two screws on the back of M_1 so as to superimpose the two brightest light spots.

(4)Put the beam expander between the laser and the beam splitter and adjust its height to illuminate the beam splitter.

(5)Now, you can see partial or whole interference pattern on the screen. Adjust the two tension springs under the compensator G_2 and/or the screws on the back of M_1 and/or M_2 so as to center the pattern of the circular arcs on the screen.

(6)Rotate the big wheel of the interferometer and don't stop until you see the motion of the interference fringes.

(7)Rotate the small wheel in the same way as you rotate the big wheel and don't stop until you see the motion of the interference fringes. Remember the direction (clockwise or counter-clockwise) of your rotation.

(8)You can see the interference fringes move either towards or away from the center of the interference pattern when you rotate the small wheel. It means that you are ready for measuring.

(9)Read the scale on the interferometer as the first reading and enter it in Data Table 3.8-1. Note that your reading should be taken from the main scale, the window, and the wheel. The reading includes *five digits* after the decimal point in the unit of mm.

(10)Concentrate on the center of the interference pattern. Rotate the small wheel and count the number of passing fringes. When you have 50 passing fringes, stop rotating to read the scale and enter the data in Data Table 3.8-1.

(11)Repeat Step (10) to complete Data Table 3.8-1. Note that when you read the passing fringes, the wheel should not be reversed to avoid error caused by

gear backlash as illustrated in Fig. 3.8-7.

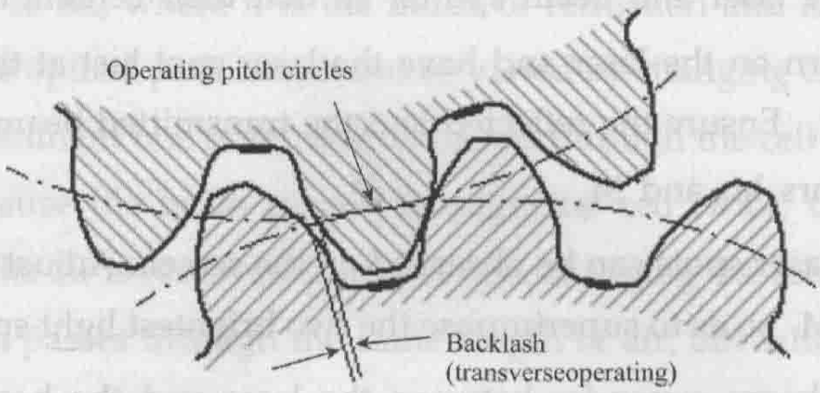


Fig. 3.8-7 Schematic diagram for gear backlash

3.8.4.2 Measuring the Index of Refraction of Air

(1)Use the Michelson interferometer with an air cell.

(2)Follow the Steps (1) through (5) in section 3.8.4.1 to see the interference pattern on the screen.

(3)Open the air valve (counter-clockwise) by a couple of turns.

(4)Squeeze the handle of the vacuum pump several times to evacuate the cell.

(5)When you have evacuated the cell to the lowest pressure you reasonably can close the valve.

(6)The reading registered on the vacuum gauge should be zero. If not, turn the knob on the panel to set it to zero. The vacuum gauge gives the difference in pressure between the air cell and the atmosphere.

(7)Squeeze the handle of the vacuum pump several times to pump up the cell. Record the reading of the vacuum gauge as p_1 and enter it in Data Table 3.8-2. Ensure p_1 is greater than 0.090 MPa.

(8)Concentrate on the center of the interference pattern and open the valve slightly until the fringes move slowly. When you have 60 passing fringes, stop evacuating to record the reading of the vacuum gauge as p_2 and enter it in Data Table 3.8-2.

(9)Repeat Steps (7) and (8) to complete Data Table 3.8-2.

(10)Record the room temperature T from the thermometer in the lab.

3.8.5 Experimental Data

Data Table 3.8-1 ***Purpose*: To measure the wavelength of He-Ne laser**

N	0	50	100	150	200	250
d/mm						
Δd_i/ mm	$\Delta d_1 = \lvert d_{150} - d_0 \rvert =$		$\Delta d_2 = \lvert d_{200} - d_{50} \rvert =$		$\Delta d_3 = \lvert d_{250} - d_{100} \rvert =$	
$\Delta\overline{d} = \dfrac{\Delta d_1 + \Delta d_2 + \Delta d_3}{3}$ / mm						
$\overline{\lambda} = \dfrac{2\Delta\overline{d}}{\Delta m} = \dfrac{2\Delta\overline{d}}{3\times 50}$ / nm						

Data Table 3.8-2 ***Purpose*: To measure the index of refraction of air**

Room temperature T = _____ ℃; Atmospheric pressure p = 1.013 25×10^5 Pa;

L = 95.0 mm; λ_0 = 633.0 nm; m = 60

Trial	1	2	3
p_1 /MPa			
p_2 /MPa			
$\Delta p_i = \lvert p_2 - p_1 \rvert$ /MPa			
$\Delta\overline{p} = \dfrac{\Delta p_1 + \Delta p_2 + \Delta p_3}{3}$ /MPa			
$n = 1 + \dfrac{\lambda_0}{2L}\dfrac{60}{\Delta p} p$			

Student's name and number:_____Instructor's initial:_____

3.8.6 Calculations

3.8.6.1 Compute the Wavelength of He-Ne Laser and its Uncertainty

(1)Compute the mirror displacements.

$$\Delta d_1 = |d_{150} - d_0|, \quad \Delta d_2 = |d_{200} - d_{50}|, \quad \Delta d_3 = |d_{250} - d_{100}|$$

(2)Compute the averaged mirror displacement.

$$\overline{\Delta d} = \frac{\Delta d_1 + \Delta d_2 + \Delta d_3}{3}$$

(3)Compute the wavelength of He-Ne laser.

$$\lambda = \frac{2\overline{\Delta d}}{\Delta m} = \frac{2\overline{\Delta d}}{3 \times 50}$$

(4)Compute relative error of the wavelength of He-Ne laser.

$$E_{\overline{\lambda}} = \frac{\overline{\lambda} - \lambda_0}{\lambda_0} \times 100\% \quad (\lambda_0 = 633.0\text{nm})$$

(5)Compute the Type A evaluation of uncertainty in $\overline{\Delta d}$.

$$\mu_{A_{\overline{\Delta d}}} = \sqrt{\frac{(\Delta d_1 - \overline{\Delta d})^2 + (\Delta d_2 - \overline{\Delta d})^2 + (\Delta d_3 - \overline{\Delta d})^2}{n(n-1)}}$$

(6)Compute the Type B evaluation of uncertainty in $\overline{\Delta d}$.

$$\mu_{B_{\overline{\Delta d}}} = \frac{\Delta_{\text{Instr.}}}{\sqrt{3}} \quad (\Delta_{\text{Instr.}} = 0.000\,05\text{mm})$$

(7)Compute the combined uncertainty in $\overline{\Delta d}$.

$$\sigma_{\overline{\Delta d}} = \sqrt{\mu_{A_{\overline{\Delta d}}^2} + \mu_{B_{\overline{\Delta d}}^2}}$$

(8)Compute the uncertainty in $\overline{\lambda}$.

$$\sigma_{\overline{\lambda}} = \frac{2\sigma_{\overline{\Delta d}}}{\Delta m} \quad (\Delta m = 3 \times 50)$$

(9)State the final result of the laser wavelength in the following form.

$$\lambda = \overline{\lambda} \pm \sigma_{\overline{\lambda}}$$

3.8.6.2 Compute the Refractive Index of Air and its Relative Error

(1)Compute changes in pressure.

$$\Delta p_1 = |p_{12} - p_{11}|, \quad \Delta p_2 = |p_{22} - p_{21}|, \quad \Delta p_3 = |p_{32} - p_{31}|$$

(2)Compute the mean value of the changes in pressure.

$$\Delta \bar{p} = \frac{\Delta p_1 + \Delta p_2 + \Delta p_3}{3}$$

(3)Compute the index of refraction of air at atmospheric pressure.

$$n = 1 + \frac{\lambda_0}{2L}\frac{60}{\Delta \bar{p}} p \quad (p = 1.013\,25 \times 10^5 \text{Pa})$$

(4)Theoretically, the index of refraction of air at atmospheric pressure can be calculated by the following equation

$$n_0 = 1 + \frac{2.8793p}{1 + 0.003\,671 \times T} \times 10^{-9}$$

where p is atmospheric pressure in Pa, T is the room temperature in Kelvin. Compute the theoretical value of the index of refraction of air.

(5)Compute the relative error of n, E_n as

$$E_n = \frac{n - n_0}{n_0} \times 100\%$$

3.8.7 Post Lab Questions

(1)Why are several light spots observed on the screen before the beam expander is placed in the light path?

(2)You are required to rotate the wheel in one direction when you count the passing fringes. Why?

(3)In the experiment of measuring the laser wavelength, we see that the interference fringes move either towards or away from the center of the interference pattern. Interpret it in terms of the changing of optical path difference?

3.9 Measurement of the Apex Angle of a Prism and the Wavelengths of Mercury Lights Using a Spectrometer

Prelab Assignment 3.9

(1)What is autocollimation? Describe the concept briefly in words with the aid of a diagram. (Hint: Autocollimation is one of methods to measure the focal length of a convex lens.)

(2)Some of the gratings used in the lab have 600 lines/mm etched on them.

Ⅰ. Find the distance, *d*, between the lines for such a grating.

Ⅱ. Suppose the wavelength of the incident light is 435.83 nm, compute the first-order diffraction angle.

3.9.1 Introduction and Objectives

A spectrometer is an instrument used to measure properties of light over a specific portion of the electromagnetic spectrum. In its simplest form, a spectrometer is nothing more than a prism or a diffraction grating and a protractor. However, because of the need for very sensitive detection and precise measurement, a real spectrometer is a bit more complicated.

After performing this experiment and analyzing the data, you should be able to:

(1)Explain the functions of various components of a spectrometer.

(2)Determine the apex angle of a glass prism.

(3)Observe the spectrum of a mercury vapor lamp and record the angle of deviation for the spectral lines.

3.9.2 Required Equipment

- A spectrometer(Fig. 3.9-1)
- A glass prism
- A diffraction grating
- A mercury vapor lamp

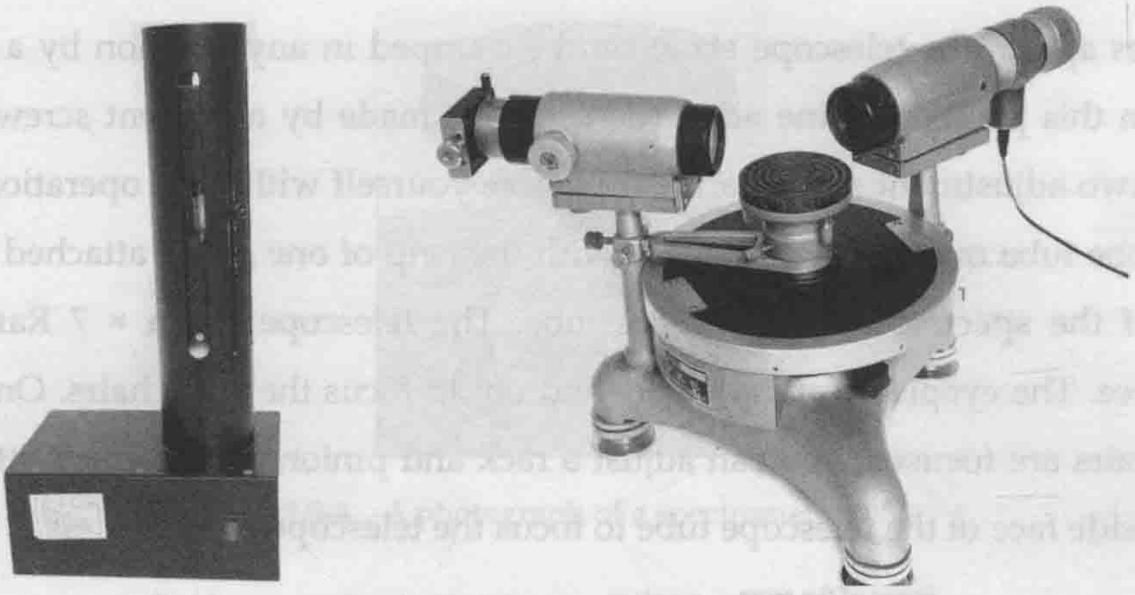


Fig. 3.9-1 A mercury vapor lamp and a spectrometer

3.9.3 Theory

3.9.3.1 The Structure of a Spectrometer

As shown in Fig. 3.9-2, a spectrometer consists of three basic components: a collimator, a diffracting element (a prism or a grating), and a telescope.

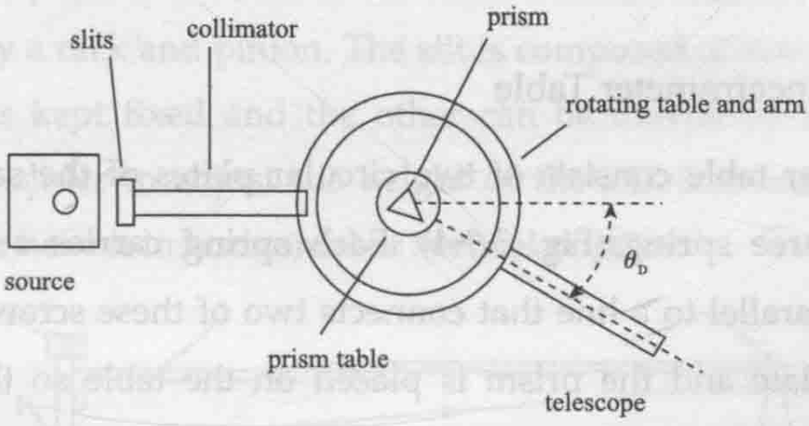


Fig. 3.9-2 Schematic diagram of an optical spectrometer

3.9.3.2 A Telescope

The schematic of a telescope is shown in Fig. 3.9-3. An astronomical (i. e. inverting) telescope with an achromatic objective and eyepiece is mounted on one arm of the spectrometer. The arm can be turned around a vertical axis passing through the center of the spectrometer. A graduated circular scale is attached to the telescope arm. This scale can be read by two verniers 180

degrees apart. The telescope stage can be clamped in any position by a screw and, in this position a fine adjustment can be made by a tangent screw. Find these two adjustment screws and familiarize yourself with their operation. The telescope tube may be slightly tilted with the help of one screw attached to the arm of the spectrometer under the tube. The telescope has a × 7 Ramsden eyepiece. The eyepiece can be slid in and out to focus the cross hairs. Once the crosshairs are focused, you can adjust a rack and pinion arrangement attached to the side face of the telescope tube to focus the telescope.

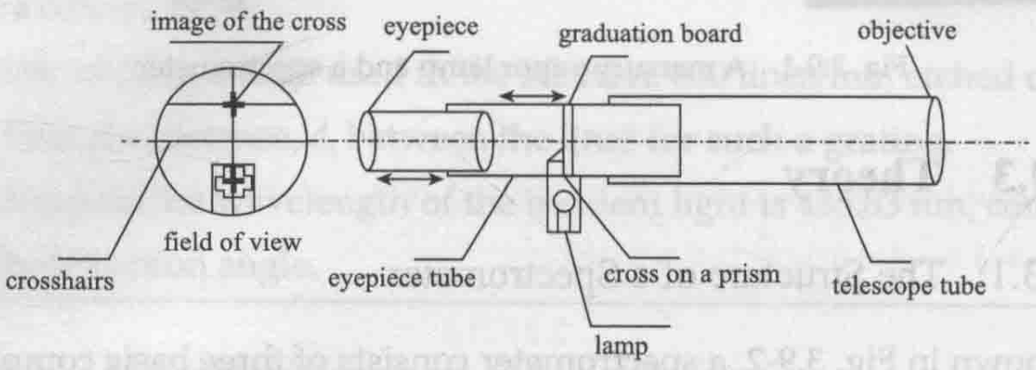


Fig. 3.9-3 Schematic diagram of a telescope and its field of view

3.9.3.3 Spectrometer Table

The spectrometer table consists of two circular plates of the same radius and separated by three springs(Fig. 3.9-4). Each spring carries a leveling screw. Straight lines, parallel to a line that connects two of these screws are engraved on the upper plate and the prism is placed on the table so that a reflecting surface is perpendicular or parallel to these lines. The vertical axis of rotation of the spectrometer passes through the center of the spectrometer table. The height of the spectrometer table can be adjusted with a clamping screw that fixes the table to the circular verniers. Thus the table can be moved around the vertical axis and its position (relative to the telescope arm scale) can be read by the verniers. Like the telescope arm it can also be fixed at any desired angle by a clamping screw and rotated very slowly using a tangent screw. Familiarize yourself with these adjustments.

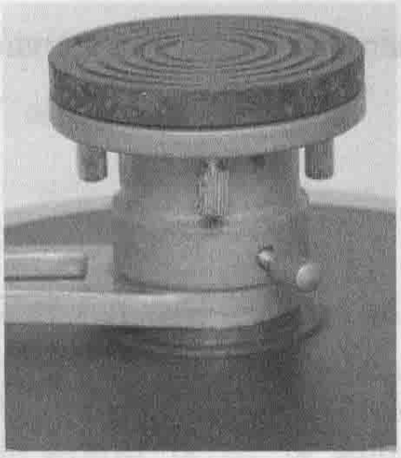


Fig. 3.9-4 A photograph of a spectrometer table

3.9.3.4 Collimator

The schematic of a collimator is shown in Fig. 3.9-5. The purpose of the collimator is to produce a parallel beam of light i. e. an object at infinity. It consists of a tube mounted horizontally on another arm of the spectrometer. This arm is fixed to the spectrometer base. The end of the tube facing the prism table has an achromatic converging lens and the other end carries a sliding tube attached to an adjustable vertical slit. The distance between the slit and the lens can be varied by a rack and pinion. The slit is composed of two sharp edges out of which one is kept fixed and the other can be moved by using the screw. There is also a slider to adjust the height of the slit. The entire "slit" can be rotated to either have it in horizontal or vertical position.

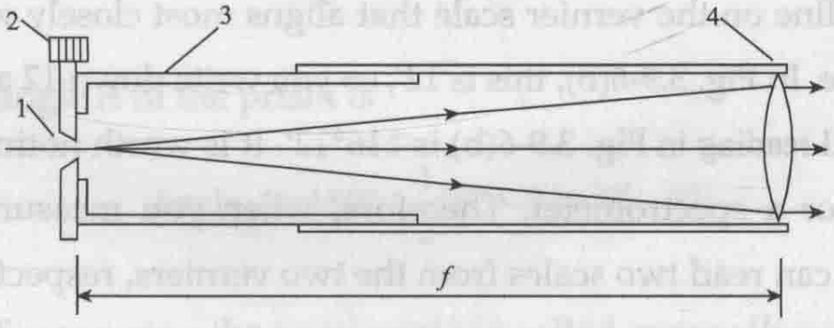


Fig. 3.9-5 A collimator: 1. slit; 2. screw; 3. tube; 4. convex lens

3.9.3.5 Scales of a Spectrometer

A spectrometer consist of a graduated circular scale attached to the telescope arm and two verniers 180 degrees apart (see Fig. 3.9-6). The primary circular scale

is in 0.5° or 30′ increment. There are thirty equal parts on the vernier. So, the smallest division is 1′.

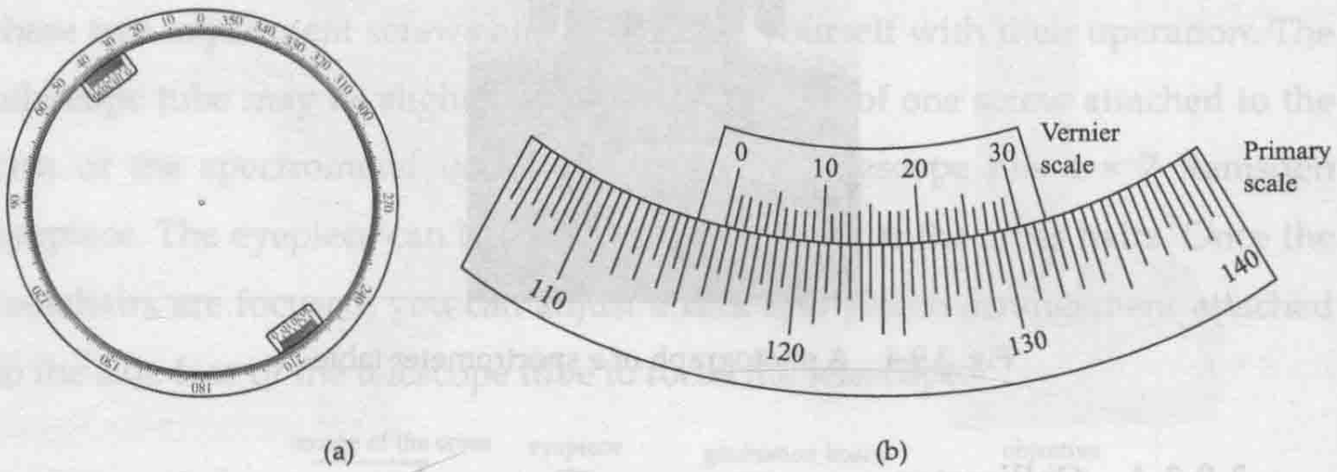


Fig. 3.9-6 Scales of a spectrometer

(a) schematic diagram of the scales; (b) vernier scales

Follow the procedure below to read the scales shown in Fig. 3.9-6 (b).

(1)Find the Zero. Locate where the zero mark of the vernier scale aligns with the degree plate.

(2)Record the degrees to the nearest, lowest 0.5°. When the zero of the top scale is between two lines on the degree plate, use the smaller value. In Fig. 3.9-6(b), the zero is between the lines 116°0′ and 116° 30′.116°0′ is the smaller of the two, so you record the degrees as 116°0′.

(3)Record the minutes. Use the magnifying glass, if there is one, to determine the line on the vernier scale that aligns most closely with any line on the degree plate. In Fig. 3.9-6(b), this is 12′, so you write down 12 as the minutes.

So, the final reading in Fig. 3.9-6(b) is 116°12′. It is worth noting that there are two verniers for a spectrometer. Therefore, when you measure the telescope arm angle you can read two scales from the two verniers, respectively.

3.9.3.6 Determination of the Apex Angle of a Prism by Autocollimation Method

Figure 3.9-7 shows a diagram to measure the apex angle of a standard triangular prism. At position 1, adjust the telescope to make its optical axis perpendicular to the side surface *AB*. Then rotate the telescope to position 2 and

make the telescope's optical axis again perpendicular to the side surface *AC*. In the quadrilateral in Fig. 3.9-7, we have

$$\text{Angle } A = 180° - \alpha \tag{3.9-1}$$

At position 1, we can read two angles, θ_1 and θ_1', from the two verniers 180 degrees apart (see Fig. 3.9-6). At position 2, we can also read two angles, θ_2 and θ_2'. The difference between θ_1 and θ_2, or θ_1' and θ_2' is the angle α. So,

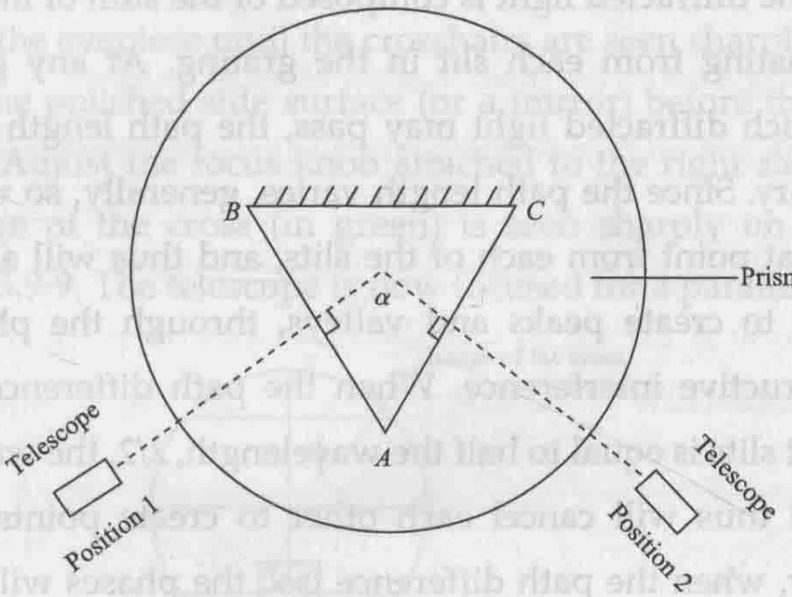


Fig. 3.9-7 Diagram for measuring the apex angle of a standard triangular prism

$$\alpha = \frac{1}{2}[|\theta_1 - \theta_2| + |\theta_1' - \theta_2'|] \tag{3.9-2}$$

Thus, the apex angle *A* of the prism is

$$\text{Angle } A = 180° - \frac{1}{2}[|\theta_1 - \theta_2| + |\theta_1' - \theta_2'|] \tag{3.9-3}$$

This method of measuring the apex angle is called *autocollimation*.

3.9.3.7 Measuring Wavelengths by a Diffraction Grating

Gratings may be of the *reflective* or *transmissive* type and they are called reflection grating or diffraction grating, respectively. We use a diffraction grating in this experiment. The term *diffraction* is a bit misleading, because the

structure in the pattern observed is dominated by interference effects. However, the interference effect is the consequence of diffraction.

In Fig. 3.9-8, there is an idealized grating made up of a set of slits of spacing d. Normally, d is wider than the wavelength of interest to cause diffraction. Assume a plane wave of wavelength λ at normal incidence (perpendicular to the grating). Each slit in the grating acts as a quasi point-source from which light propagates in all directions. After light interacts with the grating, the diffracted light is composed of the sum of interfering wave components emanating from each slit in the grating. At any given point in space through which diffracted light may pass, the path length to each slit in the grating will vary. Since the path length varies, generally, so will the phases of the waves at that point from each of the slits, and thus will add or subtract from one another to create peaks and valleys, through the phenomenon of additive and destructive interference. When the path difference between the light from adjacent slits is equal to half the wavelength, $\lambda/2$, the waves will all be out of phase, and thus will cancel each other to create points of minimum intensity. Similarly, when the path difference is λ, the phases will add together and maxima will occur. The maxima occur at angles φ, which satisfy

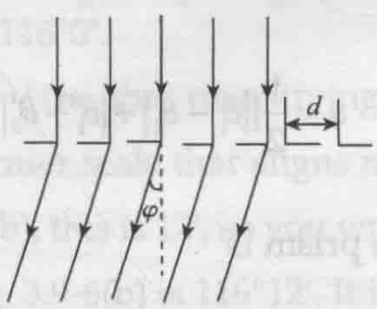


Fig. 3.9-8 An idealized grating

$$d\sin\varphi = k\lambda \quad (k=0, \pm1, \pm2, \cdots) \tag{3.9-4}$$

The above relationship between the grating spacing d and the angles of the incident and diffracted beams of light is known as the grating equation.

In this experiment a spectrometer is first carefully set up for a parallel light. It is then used to observe the emissions from a mercury vapor lamp. A diffraction grating is placed on the spectrometer table, which diffracts the green

spectral line from the lamp to a certain angle. This angle is measured and the result used to determine the spacing d of the grooves on the diffraction grating. This same grating is then used to measure the wavelengths of a number of other lines in the spectra of mercury vapor discharge lamps.

3.9.4 Experimental Procedure

3.9.4.1 Determination of the Apex Angle

(1)Turn on the power of the lamp of the telescope.

(2)Rotate the eyepiece until the crosshairs are seen sharply with your eye.

(3)Place the polished side surface (or a mirror) before the objective lens of the telescope. Adjust the focus knob attached to the right side of the telescope until the image of the cross (in green) is seen sharply on the crosshairs, as shown in Fig. 3.9-9. The telescope is now focused for a parallel incident beam.

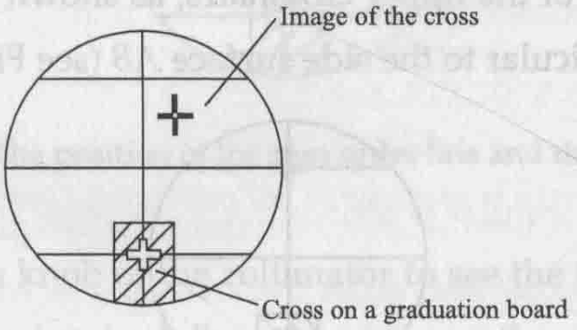


Fig. 3.9-9 The cross on a graduation board and its reflected image

(4)Place a standard triangular prism on the spectrometer table as shown in Fig. 3.9-10. In the figure, a, b, and c are three screws.

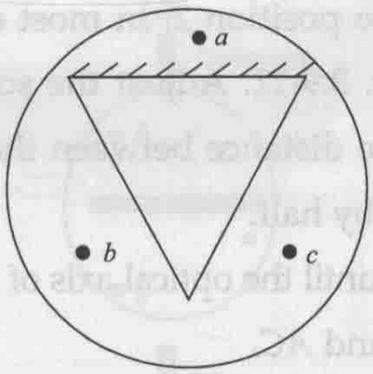
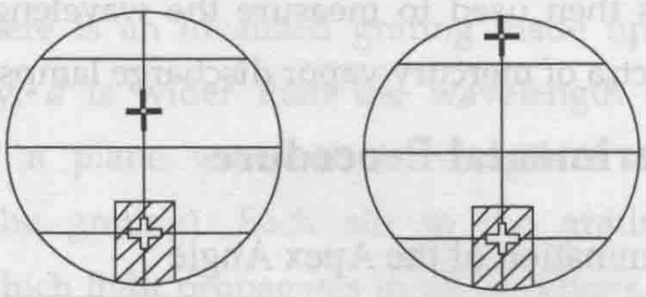


Fig. 3.9-10 The position of the prism on the table

(5)Refer to Fig. 3.9-7. Rotate the telescope to position 1 until you can see the

image of the cross as shown in Fig. 3.9-11 from the eyepiece. The image of the cross could be under or above the upper node of the crosshairs.



(a) (b)

Fig. 3.9-11 The position of the image of the cross

(a) under the upper node of the crosshairs; (b) above the upper node of the crosshairs

(6)Adjust the tilting screw under the telescope to let the image of cross overlap with the node of the upper crosshairs, as shown in Fig. 3.9-12. Now the optical axis is perpendicular to the side surface *AB* (see Fig. 3.9-7).

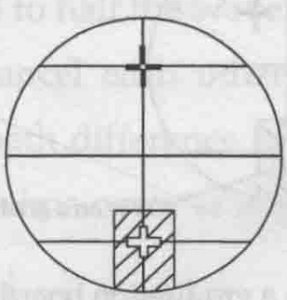


Fig. 3.9-12 The image of the cross coincides with the upper node of the crosshairs. This is autocollimation

(7)Rotate the telescope to position 2. In most cases, you can see the image of the cross as shown in Fig. 3.9-11. Adjust the screw *c* under the prism table (see Fig. 3.9-10) to reduce the distance between the image of the cross and the node of the upper crosshairs by half.

(8)Repeat Steps (5) to (7) until the optical axis of the telescope is perpendicular to both the side surfaces *AB* and *AC*.

(9)Rotate the telescope to position 1 and read two angles, θ_1 and θ_1', from the two verniers 180 degrees apart. Rotate the telescope to position 2 and read two angles, θ_2 and θ_2'. Enter these angles in Data Table 3.9-1.

(10)Repeat Step (9) to complete Data Table 3.9-1

3.9.4.2 Measuring wavelengths of the Mercury Lights by Diffraction Grating

(1)Take the prism off the spectrometer.

(2)Illuminate the slit and move the telescope so it is in-line with collimator. Loosen the screw that clamps the slit and rotate the slit to a vertical orientation so that its image will be parallel with the vertical crosshair (see Fig. 3.9-13). Retighten the screw.

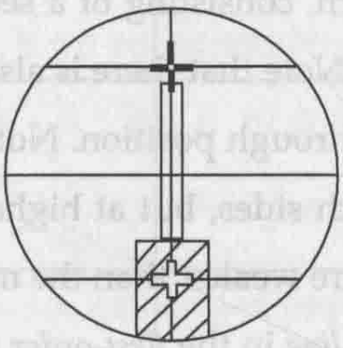


Fig. 3.9-13 The position of the zero order line and the green cross

(3)Rotate the focus knob of the collimator to see the image of the slit, i. e. the zero-order diffraction line sharply. Adjust the screw on the slit to change the width of zero-order line and set it about 0.5 mm. The collimator is now producing a parallel beam of light from the slit.

(4)Place a diffraction grating on the spectrometer table as shown in Fig. 3.9-14.

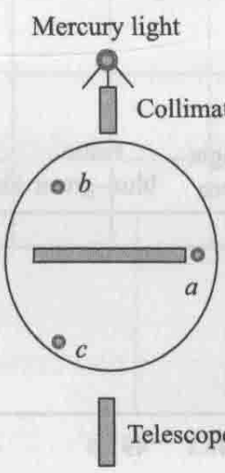


Fig. 3.9-14 Positioned diffraction grating on spectrometer table

(5)At the straight-through position, where the telescope is in line with the collimator, you can see the image of the cross from the field of view of the telescope. The image of the cross may be below or above the upper node of the crosshairs, as shown in Fig. 3.9-11. Adjust the tilting screw of the telescope or the screw under the spectrometer table to align it with the upper horizontal cross wire, as shown in Fig. 3.9-13. Now, you are ready for measurements.

(6)Move the telescope off to an angle of about 20 degrees to, say, the left side ($k = -1$), while looking through it. You should then be able to find the "first-order" diffraction pattern, consisting of a set of bright differently-colored lines, as shown in Fig. 3.9-15. Note that there is also an equivalent set of lines on the right side of the straight-through position. Note also that each set of colored lines is repeated again on both sides, but at higher angles due to higher-order diffractions. However, these are weaker than the main first-order sets.

(7)Locate the *bright green line* in the *first-order* mercury spectrum and, using the telescope's slow motion drive, carefully align the cross-wires with this line. Record the telescope's angular positions, φ_{G-L}, in Data Table 3.9-2.

(8)Rotate the telescope and align the cross-wires with the yellow doublet. Record the telescope's angular positions, φ_{Y-L1} and φ_{Y-L2}, in Data Table 3.9-2.

(9)Swing the telescope to the right-side of the straight-through position and again align the cross-wires with the first-order green and yellow doublet lines and record the telescope's angular positions, φ_{G-R}, φ_{Y-R1} and φ_{Y-R2}, in Data Table 3.9-2.

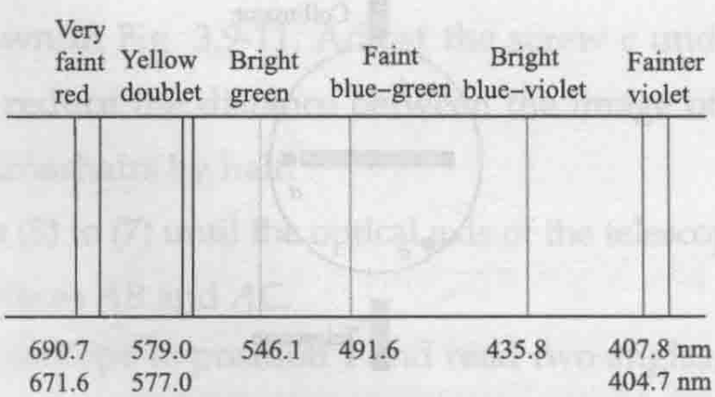


Fig. 3.9-15 Visible lines in the emission spectrum of mercury

3.9.5 Experimental Data

Data Table 3.9-1 ***Purpose*: To measure the apex angle of a prism**

Instrument error:_____

Position of telescope	Left side (position 1)		Right side (position 2)	
Trial	Vernier 1	Vernier 2	Vernier 1	Vernier 2
	θ_1 (°, ′)	θ_1′ (°, ′)	θ_2 (°, ′)	θ_2′ (°, ′)
1				
2				
3				
Average				

Data Table 3.9-2 **(*Purpose*: To measure the wavelengths of lines in the spectra of mercury**

Diffraction order	$K=-1$ (left side)						$k=+1$ (right side)					
Lines	Yellow 2		Yellow 1		Green		Green		Yellow 1		Yellow 2	
Trial	$\varphi_{Y\text{-}L21}$	$\varphi_{Y\text{-}L22}$	$\varphi_{Y\text{-}L11}$	$\varphi_{Y\text{-}L12}$	$\varphi_{G\text{-}L1}$	$\varphi_{G\text{-}L2}$	$\varphi_{G\text{-}R1}$	$\varphi_{G\text{-}R2}$	$\varphi_{Y\text{-}R11}$	$\varphi_{Y\text{-}R12}$	$\varphi_{Y\text{-}R21}$	$\varphi_{Y\text{-}R22}$
1												
2												
Average												

Student's name and number:_____Instructor's initial:_____

3.9.6 Calculations

3.9.6.1 Compute the Apex Angle of a Prism and its Uncertainty

(1)The apex angle: Angle $\overline{A}=180°-\frac{1}{2}[|\overline{\theta_1}-\overline{\theta_2}|+|\overline{\theta_1'}-\overline{\theta_2'}|]$ [Note: If the difference between the two angles is about 240 degrees, the difference should be subtracted from 360°. For example, $\overline{\theta_1}=355°45'$ and $\overline{\theta_2}=115°43'$, their difference is not $355°45'-115°43'=240°2'$ but $360°-(355°45'-115°43')=119°58'$].

(2)The Type A evaluation of uncertainty in θ_1, θ_1', θ_2, θ_2'.

(3)The Type B evaluation of uncertainty in θ_1, θ_1', θ_2, θ_2'.

(4)The combined uncertainty in θ_1, θ_1', θ_2, θ_2'.

(5)The uncertainty in apex angle A: $\sigma_A=\frac{1}{2}(\sqrt{\sigma^2_{\theta 1}+\sigma^2_{\theta 1'}+\sigma^2_{\theta 2}+\sigma^2_{\theta 2'}})$.

(6)The final result of the apex angle: $A=\overline{A}\pm\sigma_A$.

3.9.6.2 Compute the Wavelengths of Yellow Doublet Lines

(1)The diffraction angles of spectral lines of green and yellow doublets tint. For example, for green line,

$$\overline{\varphi}=\frac{1}{4}(|\varphi_{\text{G-L1}}-\varphi_{\text{G-R1}}|+|\varphi_{\text{G-L2}}-\varphi_{\text{G-R2}}|)$$

(2)The diffraction space d, $d=\frac{\lambda_{\text{green}}}{\sin\varphi_{\text{green}}}$ ($\lambda_{\text{green}}=546.07$ nm).

(3)The wavelengths of yellow doublet lines, $\lambda_{\text{Y1}}=d\times\sin\varphi_{\text{Y1}}$; $\lambda_{\text{Y2}}=d\times\sin\varphi_{\text{Y2}}$.

(4)The relative error of λ_{Y1} and λ_{Y2}. Find the accepted values in Fig. 3.9-15.

3.9.7 Post Lab Questions

(1)In Fig. 3.9-12, when the image of the cross coincides with the upper node of the crosshairs we say that the prism, the telescope, and the cross are autocollimation. Why?

(2)How can you adjust the collimator to produce a parallel beam of light?

(3)For a spectrometer, the eccentric deviation is the inconsistency between the center of the prism table and that of the dial disk. In Fig. 3.9-16, O is the center of the dial disk and O' is the center of the prism table.

i. Prove that $\varphi = \frac{1}{2}(\varphi_1 + \varphi_2)$.

ii. Use this conclusion to explain why there are two verniers on the dial disk.

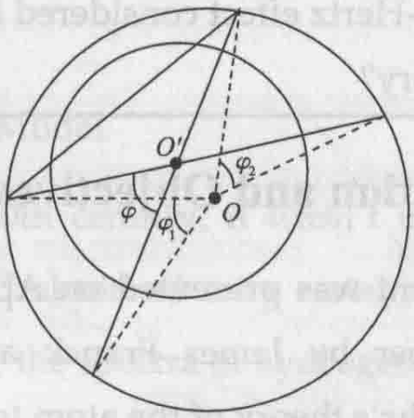


Fig. 3.9-16 The diagram of eccentric deviation

3.10 The Franck-Hertz Experiment

Prelab Assignment 3.10

(1)What do you understand by the first excitation potential of an atom?

(2)What are the basic assumptions of Bohr's theory of the hydrogen atom? Who did an experiment to provide proof for his theory?

(3)Why was the Franck-Hertz effect considered important enough to award a Nobel Prize for its discovery?

3.10.1 Introduction and Objectives

The Franck-Hertz experiment was presented on April 24, 1914 to the German Physical Society in a paper by James Franck and Gustav Hertz. It was undertaken shortly after Bohr's theory of the atom to clearly show the quantum nature of atoms. It provided one of the early indications that atoms had discrete energy levels. In the experiment, electrons are accelerated and pass through mercury vapor, where they lose energy by inelastic scattering in quantized steps as they excite mercury atoms from the ground state to an excited state. This elegant experiment yielded remarkable results that were key in the early developments of quantum theory. On December 10, 1926, Franck and Hertz were awarded the 1925 Nobel Prize in Physics "for their discovery of the laws governing the impact of an electron upon an atom".

After performing this experiment and analyzing the data, you should be able to:

(1)Understand the process of electron collision with an atom.

(2)Understand quantized excitation of argon atoms by inelastic scattering.

(3)Determine the first excitation potential of the argon atom.

3.10.2 Required Equipment

- Franck-Hertz experiment apparatus, model XD-FHW(Fig. 3.10-1)
- Wires
- Personal computer

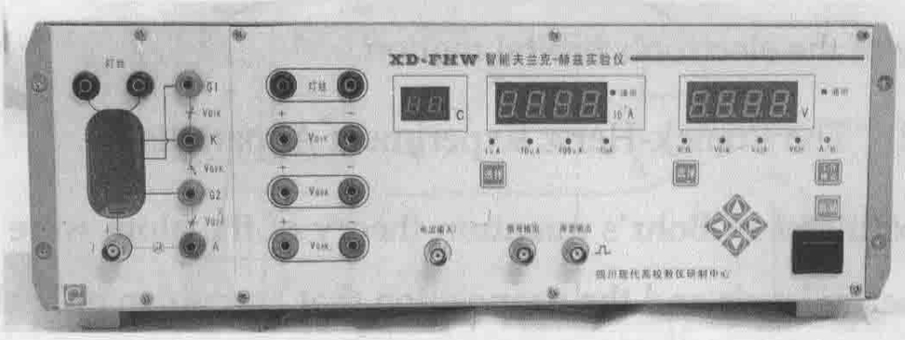


Fig. 3.10-1 Franck-Hertz apparatus

3.10.3 Theory

3.10.3.1 The Bohr Model

At the beginning of the 20th century, it wasn't understood why atoms of a given element emitted and absorbed only certain wavelengths. In 1913 Bohr provided an explanation of the spectra of hydrogen that includes some features of the currently accepted theory. His model of the hydrogen atom included the following basic assumptions:

(1)The electron moves in circular orbits about the proton under the influence of the Coulomb force of attraction. The Coulomb force produces the electron's centripetal acceleration.

(2)Only certain electron orbits are stable and allowed. In these orbits no energy in the form of electromagnetic radiation is emitted, so the total energy of the atom remains constant.

(3)Radiation is emitted by the hydrogen atom when the electron "jumps" from a more energetic initial state to a less energetic state. The "jump" can't be visualized or treated classically. The frequency ν of the radiation emitted in the jump is related to the change in the atom's energy, given by

$$E_i - E_f = h\nu \tag{3.10-1}$$

where E_i is the energy of the initial state, E_f is the energy of the final state, h is Planck's constant, and $E_i > E_f$. The frequency of the radiation is independent of

the frequency of the electron's orbital motion.

3.10.3.2 The Franck-Hertz Experiment Apparatus

Two basic postulates of Bohr's quantum theory of the atom were the existence of discrete energy levels and the proposition that atoms can only absorb or emit by amounts corresponding to the difference between two such energy levels. Bohr's theory was based on the regular characteristics of atomic line spectra (as observed in Experiment 3.9), but evidence was needed that the discrete energy levels were intrinsic to the atomic systems. One of the most direct experimental confirmations of the quantization of the energy absorbed by the atom was the experiment of J. Franck and G. Hertz. Mercury atoms were bombarded by electrons of known energy. The electrons lost only discrete amounts of energy, corresponding to differences in atomic states.

In this experiment, mercury atoms are replaced by argon atoms. The Franck-Hertz experiment apparatus, model XD-FHW, is illustrated schematically in Fig. 3.10-2. The sealed tube contains a small amount of argon atoms. A cathode K is heated by a filament F with the voltage V_1 to produce an electron beam. This process of electron emission is known as *thermionic emission*. The temperature of the filament increases with increasing voltage V_1 and a greater fraction of the electrons have kinetic energy exceeding the work function of the cathode material and thus increasing the number of electrons in the beam. The emitted electrons form a charge cloud. A grid G_1 is placed before the cathode K and the potential between the cathode K and the grid G_1 is V_2. So, the free electrons are attracted and accelerated by the electrostatic field between K and G_1. After passing through G_1, the electron beam is accelerated by another voltage V_4 and collides with argon atoms between G_1 and G_2. Finally, the electrons are retarded (decelerated) by voltage V_3 before collection at the anode A. If the electrons possess sufficient kinetic energy to overcome the retarding voltage V_3, the picoammeter I_A will give readings.

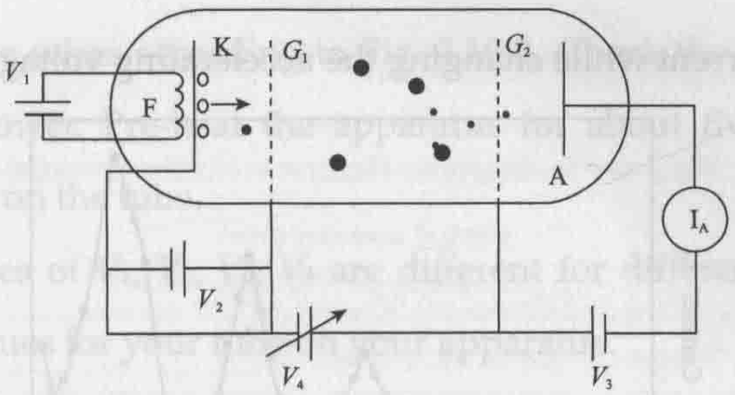


Fig. 3.10-2 Schematic diagram of the Franck-Hertz experimental apparatus

When an electron collides with an atom, several outcomes are possible:

(1)An "elastic" collision. As the argon atom is much more massive than the electron, little energy can be transferred from the electron to the atom, and the scattered electron continues with almost the same energy but on a different trajectory.

(2)An "inelastic" collision. If the bombarding electron has sufficient energy it can raise the outermost bound electron of the argon atom to a higher energy state. In this experiment 99.9997% of the argon atoms will be in their ground state, and almost all of these collisions will result in the argon electron being raised to its first excited state.

(3)A "superelastic" collision. Extremely rare will an electron encounter an already excited atom, de-excite the atom to its ground state, and thereby gain kinetic energy. This process is too rare to be considered as an explanation for any effect observed by the experimenter.

Figure 3.10-3 shows the graph of the relationship between the electron current, I_A, and the acceleration voltage V_4. In Fig. 3.10-2, the picoammeter, I_A, will measure the current of electrons, which start with energy corresponding to accelerating voltage, V_2, and pass through the interaction region to end up with more than the energy corresponding to voltage V_4. That is, it will not be reading the total electron current, but just that current of electrons which still possess some of their original energy when they finish traversing the tube. You will be

measuring this current while changing the accelerating voltage.

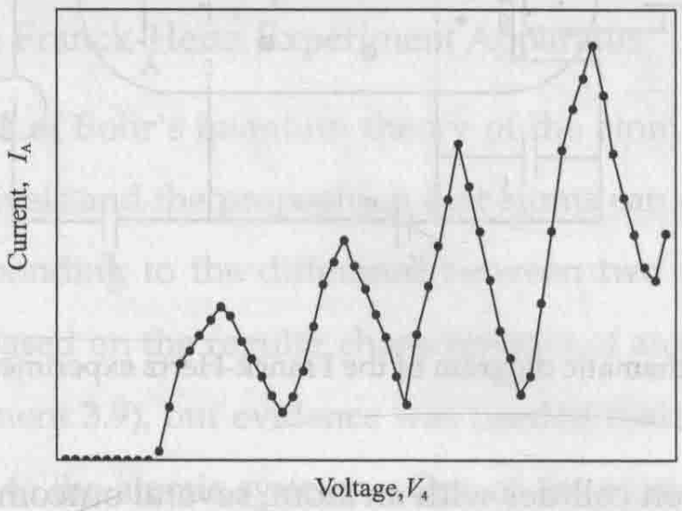


Fig. 3.10-3 Graph of the relationship between the electron current, I_A, and the acceleration voltage V_4

As the potential difference between the positive grid and the cathode is first increased from zero, no current is collected until the electrons have enough energy to overcome the slight retarding voltage V_3. The current "increases" (it is a current of negatively charged electrons so that it is actually a negative current) as the accelerating voltage is raised.

Once the accelerating potential reaches about 13.1 volts (for argon atom), inelastic collisions become possible. Those electrons which suffer such a collision will not have sufficient energy to overcome the retarding potential between grid G_2 and collector A. The current to the collector electrode will suddenly decrease. As the accelerating potential is further increased, even those electrons that have suffered inelastic collisions will again have energy to reach the collector and the current will continue to rise. Each time the accelerating potential reaches a multiple of the first excited state energy of the argon atom, the current will decrease suddenly. This gives rise to a trace as in Fig. 3.10-3.

3.10.4 Experimental Procedure

(1)Turn on the computer.

(2)Connect the wires according to Fig. 3.10-2. Check the connection before turning on the power. Pre-heat the apparatus for about five minutes before applying voltages on the tube.

(3)The voltages of V_1, V_2, V_3, V_4 are different for different tubes. Find the recommended values for your tube on your apparatus.

(4)Start the software XDPF view V2.0 on the desktop of the personal computer. Set the voltages of V_1, V_2, V_3, V_4 with the recommended values. Check these voltages again. Run the program and you will observe a curve on the screen as shown in Fig.3.10-3. The recommended voltages of V_1, V_2, V_3, V_4 are only acceptable if more than four crests or troughs would be observed on the curve. If the crests or troughs are less than four, you can increase V_1 (or V_2) or decrease V_3 in intervals of 0.1 volt.

(5)Measure the current (I_A)-accelerating voltage (V_4) relationship by manual mode. Set the values of V_1, V_2, V_3 on the panel. Increase V_4 in intervals of 1 V and read the current from the picoammeter and enter the results in Data Table 3.10-1.

(6)Measure the current (I_A)-accelerating voltage (V_4) relationship by computer. Maintain the voltages of V_2 and V_3 as in Data Table 3.10-1 and increase the value of V_1 in step of 0.1 V (or 0.2 V) from its initial value in Data Table 3.10-1 for completing Data Tables 3.10-2 and 3.10-3.

3.10.5 Experimental Data

Data Table 3.10-1 ***Purpose*: To determine the first excitation potential of argon atoms by manual measurement**

V_1=_____; V_2=_____; V_3=_____

Measurement range of the picoammeter: _____

V_4 / V	0	1	2	3	4	5	6	7	8	9
I_A(Unit:)										
V_4 / V	10	11	12	13	14	15	16	17	18	19
I_A(Unit:)										
V_4 / V	20	21	22	23	24	25	26	27	28	29
I_A(Unit:)										
V_4 / V	30	31	32	33	34	35	36	37	38	39
I_A(Unit:)										
V_4 / V	40	41	42	43	44	45	46	47	48	49
I_A(Unit:)										
V_4 / V	50	51	52	53	54	55	56	57	58	59
I_A(Unit:)										
V_4 / V	60	61	62	63	64	65	66	67	68	69
I_A(Unit:)										
V_4 / V	70	71	72	73	74	75	76	77	78	79
I_A(Unit:)										

Student's name and number:_____Instructor's initial:_____

Data Table 3.10-2 ***Purpose*: To determine the first excitation potential of argon atoms by computer**

V_1=_____; V_2=_____; V_3=_____

Measurement range of the picoammeter: _____

V_4 / V	0	1	2	3	4	5	6	7	8	9
I_A(Unit:)										
V_4 / V	10	11	12	13	14	15	16	17	18	19
I_A(Unit:)										
V_4 / V	20	21	22	23	24	25	26	27	28	29
I_A(Unit:)										
V_4 / V	30	31	32	33	34	35	36	37	38	39
I_A(Unit:)										
V_4 / V	40	41	42	43	44	45	46	47	48	49
I_A(Unit:)										
V_4/V	50	51	52	53	54	55	56	57	58	59
I_A(Unit:)										
V_4 / V	60	61	62	63	64	65	66	67	68	69
I_A(Unit:)										
V_4 / V	70	71	72	73	74	75	76	77	78	79
I_A(Unit:)										

Student's name and number:_____Instructor's initial:_____

Data Table 3.10-3 ***Purpose*: To determine the first excitation potential of argon atom by computer**

V_1=_____; V_2=_____ ; V_3=_____

Measurement range of the picoammeter: _____

V_4 / V	0	1	2	3	4	5	6	7	8	9
I_A(Unit:)										
V_4/ V	10	11	12	13	14	15	16	17	18	19
I_A(Unit:)										
V_4/ V	20	21	22	23	24	25	26	27	28	29
I_A(Unit:)										
V_4/ V	30	31	32	33	34	35	36	37	38	39
I_A(Unit:)										
V_4/ V	40	41	42	43	44	45	46	47	48	49
I_A(Unit:)										
V_4/ V	50	51	52	53	54	55	56	57	58	59
I_A(Unit:)										
V_4/ V	60	61	62	63	64	65	66	67	68	69
I_A(Unit:)										
V_4/ V	70	71	72	73	74	75	76	77	78	79
I_A(Unit:)										

Student's name and number:_____ Instructor's initial:_____

3.10.6 Plotting

Use the three sets of data in the above data tables to plot three traces in a graph.

(NOTE: Different traces should have different data symbols in the graph.)

3.10.7 Calculations

Compute the first excitation potential, U_0, from the graph.

(1)Choose one trace and read the horizontal coordinates of the crests. Enter the voltages in the following table.

Crest	1	2	3	4	5	6
Voltages, U/ V						

(2)Compute the difference of the voltages

$$U_{01} = U_2 - U_1$$

$$U_{02} = U_3 - U_1$$

$$U_{03} = U_4 - U_1$$

$$U_{04} = U_5 - U_1$$

$$U_{05} = U_6 - U_1$$

(3)Compute the first excitation potential U_0.

$$U_0 = \frac{U_{01} + U_{02} + U_{03} + U_{04} + U_{05}}{1+2+3+4+5}$$

(4)The accepted value of the first excitation potential of argon is 13.1 V. Compute the relative error of your measurement.

3.10.8 Post Lab Questions

(1)Different parameters of V_1, V_2, V_3, V_4 can be used in the experiment. How

do the voltages affect the first excitation potential of argon atom and why?

(2)What's the most important conclusion for the Franck-Hertz experiment?

(3)What happens to the excited atoms? If radiation is involved, what is its wavelength and how could it be detected?

(4)From the Data Tables 3.10-1, 3.10-2, and 3.10-3, how does the voltage V_1 affect the current through the picoammeter and why?

3.11 The Millikan Oil Drop Experiment

Prelab Assignment 3.11

(1)What forces act on the oil droplet when it is stationary in an electrostatic field? If the field is removed and it reaches its terminal speed, what forces would be acting on it?

(2)An oil droplet in a Millikan apparatus is determined to have a mass of 3.3×10^{-15} kg. It is observed to float between two parallel plates separated by a distance of 0.95 cm with 340 V of potential difference between them. Determine how many excess (extra) electrons are on the droplet.

(3)What is Stokes' law?

3.11.1 Introduction and Objectives

The oil drop experiment was performed by Robert A. Millikan and his Ph. D. student, Harvey Fletcher, from 1909 to 1913 to measure the elementary electric charge. The elementary charge is one of the fundamental physical constants and its accurate value is of great importance. The Nobel Prize in Physics 1923 was awarded to Robert A. Millikan "for his work on the elementary charge of electricity and on the photoelectric effect".

After performing this experiment and analyzing the data, you should be able to:

(1)Analyze the forces on the oil drop in the stationary and motion states.

(2)Measure the elementary charge.

3.11.2 Required Equipment

(1)Millikan oil drop apparatus, MOD-5C(Fig. 3.11-1)

(2)A CCD camera

(3)A monitor

Fig. 3.11-1 Monitor and Millikan oil drop apparatus

3.11.3 Theory

In 1907, Robert Millikan's young collaborator, Harvey Fletcher, got the idea of watching the motion of a single, charged, microscopic oil droplet under the influence of gravity and a uniform electric field between two parallel metal plates. An analysis of the forces acting on a charged droplet will allow determination of its charges.

Figure 3.11-2 shows the forces acting on the drop when it is stationary in the electric field. In Fig. 3.11-2, F_E is the electric field force, F_g is the gravity force, and F_b is the buoyant force exerted by the air. Since the density of air is only about one-thousandth that of oil, the buoyant force may be neglected and we have

$$q\frac{V}{d}=mg \tag{3.11-1}$$

where q is the electronic charge carried by the oil drop, V is voltage applied on the parallel metal plates, d is the distance between the plates, m is the mass of the oil drop, and g is the local acceleration of gravity.

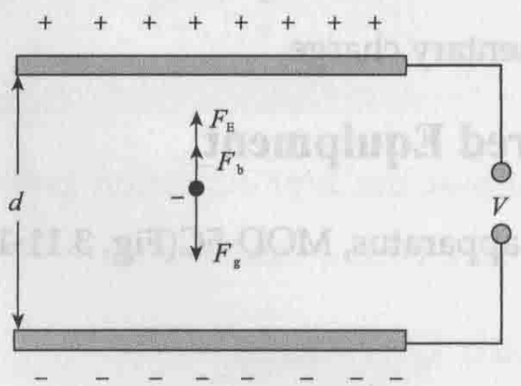


Fig. 3.11-2 The forces acting on the drop when it is stationary in the electric field

To eliminate m from Eq. (3.11-1), one uses the expression for the volume of a sphere:

$$m = \frac{4}{3}\pi r^3 \rho \tag{3.11-2}$$

where r is the radius of the oil drop, ρ is the density of the oil.

Thus, we have

$$q\frac{v}{d} = \frac{4}{3}\pi r^3 \rho g \tag{3.11-3}$$

When the voltage is removed the oil drop will fall and reach its terminal velocity in a few milliseconds. Fig. 3.11-3 shows the forces acting on the drop when it is falling in air and has reached its terminal velocity. In Fig. 3.11-3, F_v is the viscous resistance exerted by air, F_g is the gravitational force, and F_b is the buoyant force.

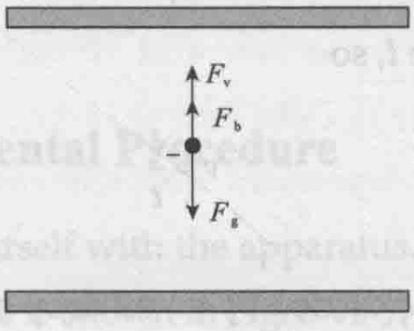


Fig. 3.11-3 Forces acting on the drop when reaching its terminal velocity

According to the Stokes' law, the viscous resistance is

$$F_v = 6\pi r \eta v \tag{3.11-4}$$

where r is the radius of the oil drop, η is the coefficient of viscosity of the air, v is the terminal velocity of the oil drop.

Neglecting the buoyant force, we have

$$\frac{4}{3}\pi r^3 \rho g = 6\pi r \eta v$$

and

$$r^2 = \frac{9\eta v}{2\rho g} \tag{3.11-5}$$

If the radius of the oil drop is as small as about 10^{-7} m, which is the order of magnitude of the mean free path of air molecules under normal laboratory conditions, a correction should be applied to the viscosity.

$$\eta' = \frac{\eta}{1+\frac{b}{pr}} \tag{3.11-6}$$

where η' is the corrected value of the viscosity, p is local atmospheric pressure, and b is a constant, 6.17×10^{-6} m·cm(Hg).

The terminal velocity of the oil drop v can be determined by measuring the falling distance l in time t, so

$$v = \frac{l}{t} \tag{3.11-7}$$

Thus, the electronic charge q carried by an oil drop can be derived from the Eqs. (3.11-3) and (3.11-5)~(3.11-7)

$$q = \frac{18\pi}{\sqrt{2\rho g}}\left[\frac{\eta l}{t\left(1+\frac{b}{pr}\right)}\right]^{\frac{3}{2}}\frac{d}{V} \tag{3.11-8}$$

where $r = \sqrt{\frac{9\eta l}{2\rho g t}}$.

The parameters used in our apparatus in Eq. (3.11-8) are depicted in Table 3.11-1.

Table 3.11-1 The parameters of density of the oil ρ, acceleration of gravity g, viscosity of air η, the distance of fall at constant speed l, correction constant b, atmospheric pressure p, the separation distance of the plates d

parameters	magnitude	unit
ρ	981	kg/m^3
g	9.79	m/s^2
η	1.83×10^{-5}	kg (m · s)
l	2.00×10^{-3}	m
b	6.17×10^{-6}	m·cm(Hg)
p	76.0	cm(Hg)
d	5.00×10^{-3}	m

Finally, we have the equation for calculating the charges of the oil drop

$$q = ne = \frac{1.43\times10^{-14}}{[t(1+0.02\sqrt{t})]^{\frac{3}{2}}}\cdot\frac{1}{V} \tag{3.11-9}$$

where n is an integer.

3.11.4 Experimental Procedure

(1)First, familiarize yourself with the apparatus. A picture of the apparatus as it should be set up for you is shown in Fig. 3.11-1.

(2)Turn on the power of the monitor and the oil drop apparatus.

(3)Introducing the droplets into the chamber. Spray droplets of oil from the atomizer through the port into the oil drop chamber. You can see hundreds of droplets on the monitor. Adjust the focusing hand wheel of the CCD so the images of the droplets are sharp.

(4)Selection of the droplet. Put the plate charging switch in the "balance" position. Tune the potentiometer to apply a voltage of about 200 volts across the parallel plates to arrest one droplet's fall. Fine tune the potentiometer to make the droplet stationary in the field. Record the balance voltage, V, registered on the voltmeter.

NOTE: The balance voltage should be in the range from 100 to 500 volts. The optimal voltage is about 200 volts.

(5)When you find an appropriately sized and charged oil droplet, fine tune the focus of the viewing scope to make it appear as a pinpoint of bright light.

(6)Put the plate charging switch in the "up" position to pull the droplet up. When it reaches the top line, put the switch in the "balance" position as soon as possible to make it stationary. In the "up" position, the voltage is greater than the balance voltage. So, the droplet goes up.

(7)Put the switch in the "down" position to measure the fall time. Keep your right hand on the timer controls and your left hand on the plate charging switch. When the droplet reaches the second horizontal line, start timing; when it reaches the sixth horizontal line, stop timing and put the switch in the "balance" position. Record the time registered on the timer.

NOTE: The fall time should be in the range from 10 to 50 seconds. The optimal time is 20 s.

(8)Pull the droplet up and repeat Steps (5) and (6) two more times to reduce the random errors of the time and voltage measurements. Record the balance voltage and fall time in Data Table 3.11-1.

(9)Select nine more independent droplets and repeat the experiment to complete Data Table 3.11-1.

3.11.5 Experimental Data

Data Table 3.11-1 ***Purpose*: To measure the electric charges carried by an oil droplet**

Oil droplets	Balance voltage *V*/V		Fall time *t*/s	
	Measurement	Average	Measurement	Average
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

Student's name and number:_____ Instructor's initial:_____

3.11.6 Calculations

(1)Compute the mean values of the balance voltage and the fall time. For the first oil droplet

$$\overline{V_1} = \frac{V_{11} + V_{12} + V_{13}}{3}$$

$$\overline{t_1} = \frac{t_{11} + t_{12} + t_{13}}{3}$$

Compute the averaged values of the balance voltages and the fall time for the ten oil droplets and enter them in Data Table 3.11-1.

(2)Use the following equation to calculate the charges carried by an oil droplet. Show the sample calculation for the first droplet and list all the charges in Table 3.11-2.

$$\overline{q} = ne = \frac{1.43\times10^{-14}}{[\overline{t_1}(1+0.02\sqrt{\overline{t}})]^{\frac{3}{2}}} \cdot \frac{1}{\overline{V}}$$

Data Table 3.11-2 Charges and the number of excess electrons on selected droplets

Oil droplets	Charges $\overline{q}$ /$\times10^{-19}$C	the number of excess electrons, n	Elementary charge e/$\times10^{-19}$C
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
		Average	

(3)Compute the number of excess electrons for every droplet. Suppose that the elementary charge $e = 1.602\times10^{-19}$ C and the number of excess electrons is

$$n = \left[\frac{q}{e}\right]$$

"[ ]" means rounding the quotient to an integer.

Show the sample calculation for the first oil droplet and enter all the results in Data Table 3.11-2.

(4)Compute the elementary charge.

$$e = \frac{\overline{q}}{n}$$

Show the sample calculation for the first oil droplet and enter all the results in Data Table 3.11-2.

(5)Compute the averaged elementary charge. Enter the final result in Data Table 3.11-2.

$$\overline{e} = \frac{1}{10}\sum_{i=1}^{10} e_i$$

(6)Compute the relative error of the elementary charge.

$$E = \frac{\overline{e} - e}{e}\times100\% \quad (e = 1.602\times10^{-19}\ \text{C})$$

3.11.7 Post Lab Questions

(1)You can observe that some oil droplets go down very fast after introducing the droplets into the chamber. Why?

(2)Some oil droplets go down very slowly after removing the potential difference applied on the two parallel plates. Are these droplets good for measurement? Why?

(3)Give some examples to show the importance of the accurate value of elementary charge.

3.12 The Wheatstone Bridge and the Prototype of Electric Balance

Prelab Assignment 3.12

(1)What is the formula for V_g in terms of voltage and resistance in the following circuit (see Fig. P3.12-1)?

(2)In Fig. P3.12-2, an unbalanced wheatstone bridge is constructed. Calculate the output voltage across points C and D and the value of resistor R_4 required to balance the bridge circuit.

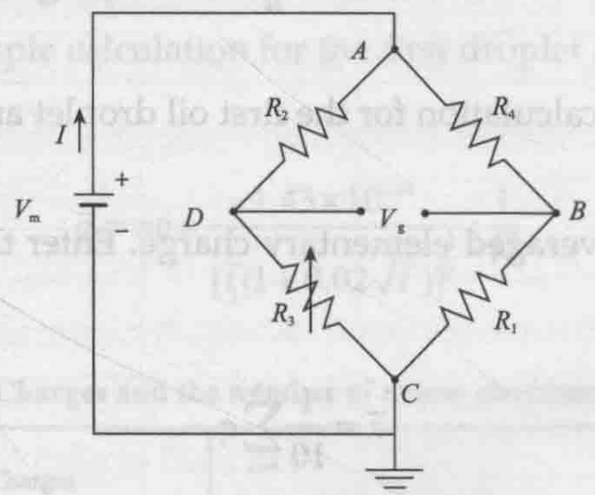


Fig. P3.12-1 An wheatstone bridge

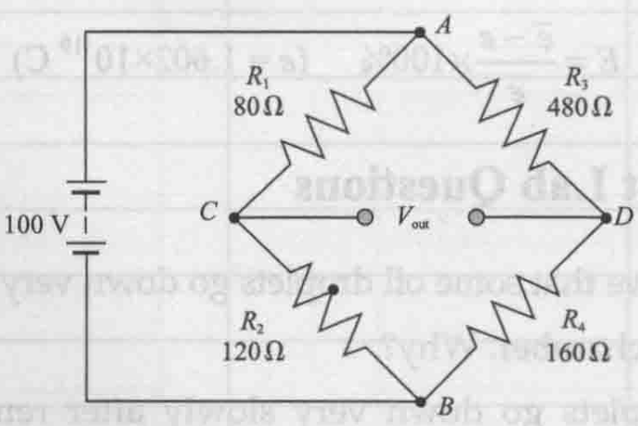


Fig. P3.12-2 An unbalanced wheatstone bridge

3.12.1 Introduction and Objectives

A wheatstone bridge is an electrical circuit design dating back to the early

1800's. It was invented by a British scientist Samuel Hunter Christie (1784~1865) and first described in 1833. It is named after its most famous user, Sir Charles Wheatstone, who did develop multiple uses for it. The purpose of this experiment is to understand a wheatstone bridge circuit and its application in the electric balance.

After performing this experiment and analyzing the data, you should be able to:

(1)Understand the operating principles of the wheatstone bridge.

(2)Understand the principle of an electric balance.

3.12.2 Required Equipment

- Two Resistance boxes(Fig. 3.12-1)
- Two fixed resistors
- Power supply
- Voltmeter
- Wires

Fig. 3.12-1 Experimental apparatus

3.12.3 Theory

3.12.3.1 The Wheatstone Bridge and Difference Voltage

1)Quarter Bridge Circuit

A Wheatstone bridge is a network of four resistive legs. It consists of four

resistors, three of which are of fixed value and a fourth which is variable, R_3 in Fig. 3.12-2. The variable resistor is the sensing element (transducer). Its design will allow its resistance to change due to a change in an environmental factor such as stress, pressure, or temperature.

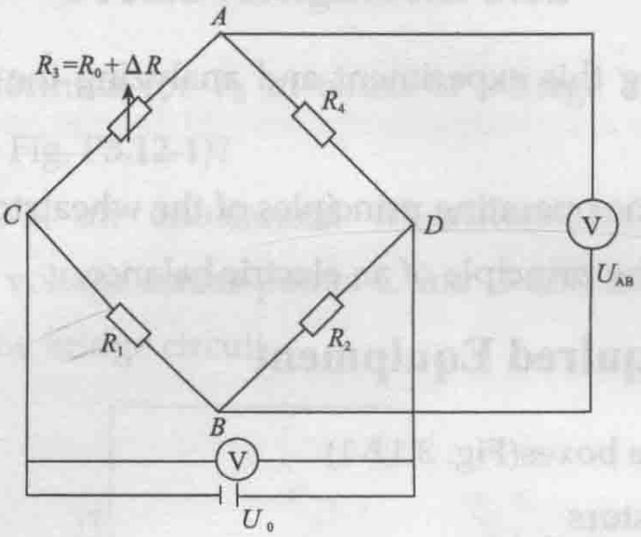


Fig. 3.12-2 Wheatstone bridge with one variable resistor

The resistor pair R_1 and R_2 is a resistive voltage divider and resistors R_3 and R_4 form another voltage divider in parallel with R_1 and R_2. The circuit is sensitive to the difference in voltage between node-A and node-B. The voltages U_A and U_B can be found by

$$U_A = \frac{R_3}{R_3 + R_4} U_0 = \frac{R_0 + \Delta R}{R_0 + \Delta R + R_4} U_0 \tag{3.12-1}$$

$$U_B = \frac{R_1}{R_1 + R_2} U_0 \tag{3.12-2}$$

where R_0 is the initial resistance value of R_3, ΔR is its resistance change caused by environmental factors.

So that the difference voltage of the wheatstone bridge is

$$U_{AB} = U_A - U_B = \left(\frac{R_3}{R_3 + R_4} - \frac{R_1}{R_1 + R_2} \right) U_0 \tag{3.12-3}$$

$$= \left(\frac{R_0 + \Delta R}{R_0 + \Delta R + R_4} - \frac{R_1}{R_1 + R_2} \right) U_0$$

$$= \frac{R_2 \Delta R}{(R_0 + \Delta R + R_4)(R_1 + R_2)} U_0$$

$$= \frac{\Delta R}{R_0 (1 + \frac{\Delta R}{R_0} + R_4 / R_0)(1 + R_1 / R_2)} U_0 \quad (3.12\text{-}4)$$

For Eq. (3.12-3), when $R_1R_4 = R_2R_3$, the circuit output is zero, $U_{AB} = 0$. The bridge is said to be balanced when $U_{AB} = 0$ volt.

Consider the ratio of the resistors $k = \frac{R_1}{R_2} = \frac{R_0}{R_4}$, the Eq. (3.12-4) can be written as

$$U_{AB} = \frac{\Delta R / R_0}{1 + k + \frac{\Delta R}{R_0} k} \cdot \frac{k}{1 + k} U_0 \quad (3.12\text{-}5)$$

The constant k is called the bridge ratio.

When $\Delta R \ll R_0$, $\Delta R/R_0$ is very small and can be neglected. Thus

$$U_{AB} = \frac{k}{(1 + k)^2} \cdot \frac{\Delta R}{R_0} \cdot U_0 \quad (3.12\text{-}6)$$

The sensitivity of the wheatstone bridge is defined as the ratio of the output voltage to the change in resistance value of the variable resistor and denoted as S_u.

$$S_u = \frac{U_{AB}}{\Delta R} = \frac{k U_0}{(1 + k)^2 R_0} \quad (3.12\text{-}7)$$

It can be seen from Eq. (3.12-6) that the output of the wheatstone bridge circuit is linearly proportional to the resistance change of the variable resistor.

When $k = 1$, the difference voltage and the sensitivity of the wheatstone bridge are at their maximum

$$U_{AB-\max} = \frac{\Delta R}{4R_0} \cdot U_0 \quad (3.12\text{-}8)$$

$$S_{u-max} = \frac{U_0}{4R_0} \tag{3.12-9}$$

2)Half Bridge Circuit

Looking carefully at the above equations and the circuit diagram, one can design a more sensitive circuit where R_3 and R_4 are both variable resistors. Such a circuit is shown in Fig. 3.12-3.

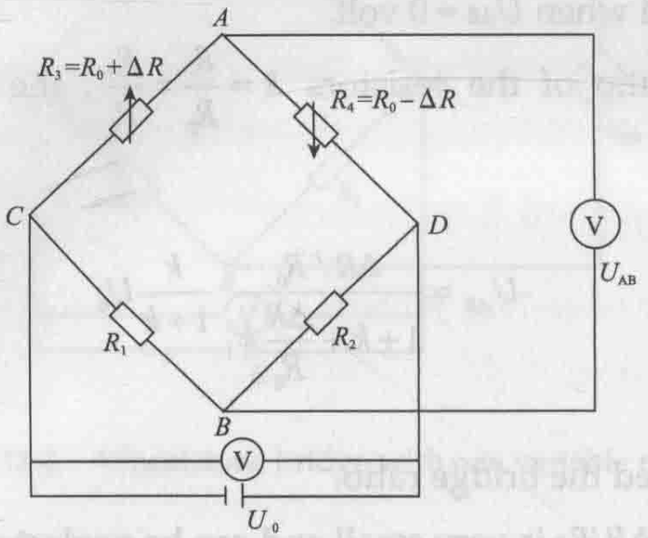


Fig. 3.12-3 Wheatstone bridge with two variable resistors

In Fig. 3.12-3, consider the case that all the four resistances are initially at R_0 each.

$$R_1 = R_2 = R_3 = R_4 = R_0$$

Suppose that R_3 increases its resistance by ΔR; reversely, R_4 decreases its resistance by ΔR. Now, Eq. (3.12-3) can be expressed as

$$U_{AB} = \frac{\Delta R}{2R_0} \cdot U_0 \tag{3.12-10}$$

The sensitivity of the wheatstone bridge with two variable resistors is

$$S_u = \frac{U_0}{2R_0} \tag{3.12-11}$$

3)Full Bridge Circuit

In Fig. 3.12-4, there are four variable resistors in all the arms. We call it full bridge

circuit. Consider the case where all the four resistances are initially at R_0 each.

$$R_1 = R_2 = R_3 = R_4 = R_0$$

Assume that R_2 and R_3 increase their resistance by ΔR; reversely, R_1 and R_4 decrease their resistances by ΔR. Eq. (3.12-3) can be expressed as

$$U_{AB} = \frac{\Delta R}{R_0} \cdot U_0 \tag{3.12-12}$$

The sensitivity of the wheatstone bridge with four variable resistors is

$$S_u = \frac{U_0}{R_0} \tag{3.12-13}$$

Therefore, the wheatstone bridge with four variable resistors is the most sensitive.

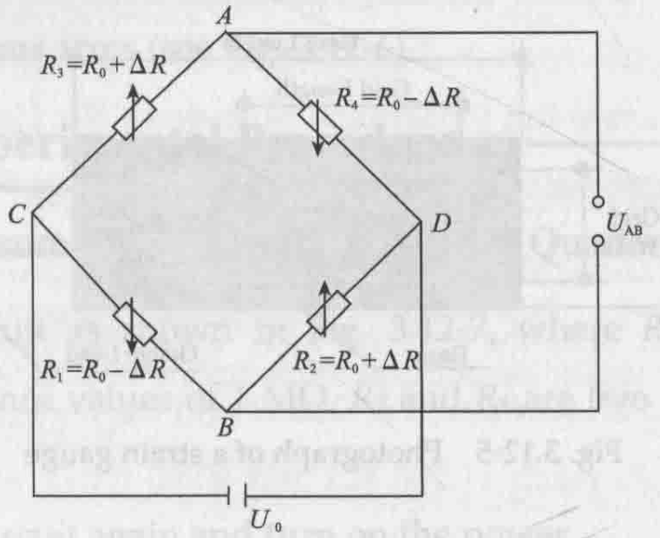


Fig.3.12-4 Wheatstone bridge with four variable resistors

3.12.3.2 The Strain Gauge and the Application of Wheatstone Bridge

Strain is the amount of deformation of a body due to an applied force. For tensile and compressive types of stress, it can be defined in terms of the fractional change in length as follows,

$$\text{strain} = \Delta L/L \tag{3.12-14}$$

where L is the original length of the material, and ΔL is the change in

length that results after the stress is applied. It is a dimensionless quantity.

Fig. 3.12-5 shows a photograph of a strain gauge. While there are several methods of measuring strain, the most common is with a strain gauge, a device whose electrical resistance varies in proportion to the amount of strain in the device. The most widely used gauge is the bonded metallic strain gauge. It consists of a very fine wire or, more commonly, metallic foil arranged in a grid pattern. The grid pattern maximizes the amount of metallic wire or foil subject to strain in the parallel direction. The cross-sectional area of the grid is minimized to reduce the effect of shear strain and Poisson strain. The grid is bonded to a thin backing, called the carrier, which is attached directly to the test specimen. Therefore, the strain experienced by the test specimen is transferred directly to the strain gauge, which responds with a linear change in electrical resistance.

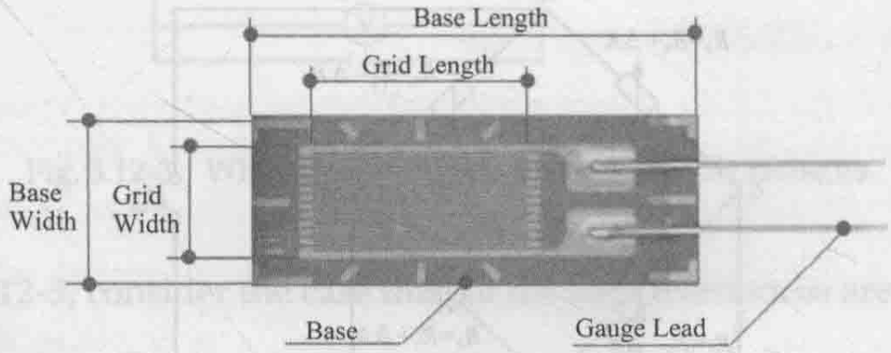


Fig. 3.12-5 Photograph of a strain gauge

A fundamental parameter of the strain gauge is its sensitivity to strain, expressed quantitatively as the gauge factor. Gauge factor is defined as the ratio of fractional change in electrical resistance to the fractional change in length (strain) and denoted as G_f.

$$G_f = \frac{\Delta R / R}{\Delta L / L} \quad \text{or} \quad \frac{\Delta R}{R} = G_f \cdot \frac{\Delta L}{L} \tag{3.12-15}$$

In general, the gauge factor for metal strain gauges is about 2.

The main fields of application of strain gauges are experimental stress analysis and the design and manufacture of transducers. In this experiment,

four strain gauges are attached to a cantilever beam to measure the applied force F, as shown in Fig. 3.12-6. This is the basic principle of electric balance. So, we can use the device to measure the mass of an object.

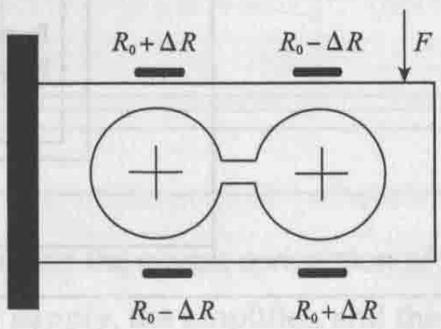


Fig. 3.12-6 Schematic diagram of a cantilever beam with four active strain gauges

Transducers generally have to comply with stringent accuracy requirement. Therefore, transducers should always have a full bridge circuit with active strain gauges in all four arms (see Fig. 3.12-6).

3.12.4 Experimental Procedure

3.12.4.1 Measure the Difference Voltage of Quarter Bridge Circuit

(1)Set up a circuit as shown in Fig. 3.12-7, where R_1 and R_2 are fixed resistors with resistance values of 1 MΩ, R_3 and R_4 are two variable resistors of resistance boxes.

(2)Check your circuit again and turn on the power.

(3)Tune the "adjustment" knob to set E = 6.00 volts (see Fig. 3.12-7). Set the resistance values of the two variable resistors to R_3= R_4=10 kΩ and the bridge is balanced and the difference voltage should be zero. However, in most cases, the difference voltage (the wheatstone bridge output) is not zero (think about the possible reasons). Tune the "bias" knob to set the difference voltage U_{AB} = 0.

(4)Increase R_3 in an increment of 200 Ω until it reaches 11.200 kΩ. Read the corresponding difference voltages registered on the voltmeter and enter them in Data Table 3.12-1.

NOTE: All the difference voltages must have more than one significant

figure. You can tune the "gain" knob to amplify the difference voltage for the first reading. However, don't tune it any more in the subsequent measurements.

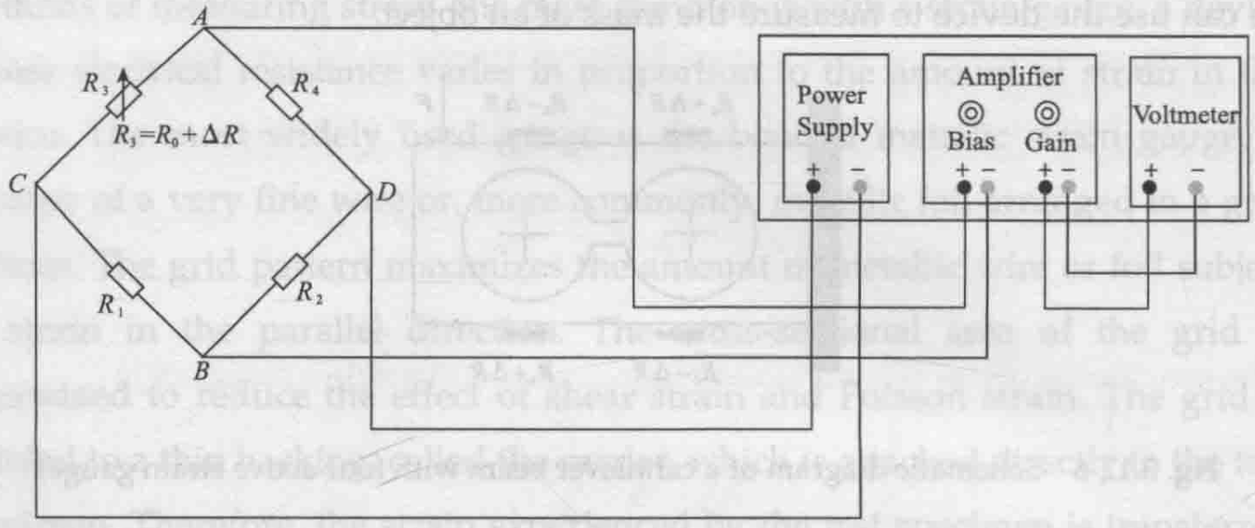


Fig. 3.12-7 Schematic diagram for the circuit connection of the resistors, power supply, amplifier, and voltmeter

3.12.4.2 Measure the Difference Voltage of the Half Bridge Circuit

(1)Use the circuit in Fig. 3.12-7. Set $E = u_{CD} = 6.00$ volts.

(2)Set the resistance values of the two variable resistors to R_3=R_4=10kΩ and the bridge is balanced. Tune the knob of "zero" to set the difference voltage U_{AB}= 0.

(3)Increase R_3 in an increment of 200 Ω and decrease R_4 in an increment of 200 Ω until R_3 and R_4 reach 11.200 kΩ and 8.800 0 kΩ, respectively. Read the difference voltages registered on the voltmeter and enter them in Data Table 3.12-2. Don't tune the "gain" knob in this step.

3.12.4.3 Determine the Sensitivity of the Prototype Electric Balance and Measure the Mass of Objects

There is a prototype electric balance available on the bench. It is made by the principles of strain gauge and wheatstone full bridge. In this section, we will determine the sensitivity of the prototype by experiment and then we will utilize this device to measure the mass of objects.

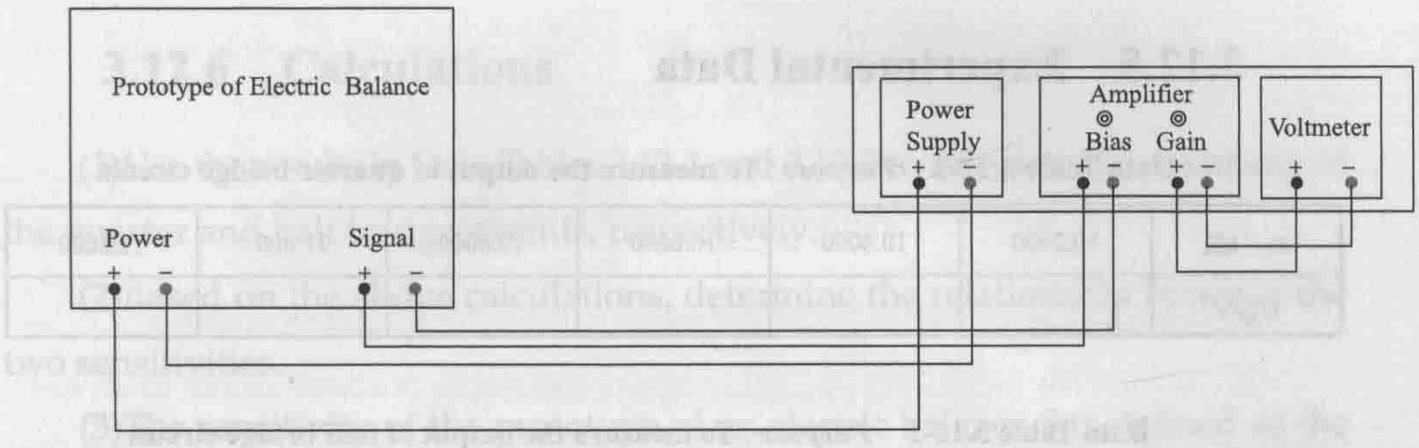


Fig. 3.12-8 Schematic diagram for the circuit connection of the prototype electric balance, the power supply, the amplifier, and the voltmeter

(1)Set up the circuit for the prototype of electric balance, as shown in Fig. 3.12-8.

(2)Turn on the power and let the prototype warm up for five minutes.

(3)Set E = 6.00 volts. Tune the "bias" knob to set the difference voltage to 0.

(4)Add a weight per time on the table of the prototype. Read the voltage registered on the voltmeter after adding each weight. Record the voltages in Data Table 3.12-3.

NOTE: All the voltages must have more than one significant figure. You can tune the "gain" knob to amplify the voltage for the first reading. However, don't tune it any more in the following measurements.

(5) Unload one weight at a time and read the voltmeter at each step. Record the readings in Data Table 3.12-3.

(6) Put one object such as a pencil box, or a cell phone on the table of the prototype. Read the voltage and enter it in Data Table 3.12-4.

(7) Take it off and put it on the table and then measure it again two more times. Record the voltages in Data Table 3.12-4.

(8) Repeat Steps (6) and (7) to measure two more objects. Record the output in Data Table 3.12-4.

(9) Put one weight on the table of the prototype. Tune the knob to set the source voltage to 2 volts. Read the output and enter it in Data Table 3.12-5. You must ensure that all the voltages have more than one digit.

(10) Increase the source voltage by 1 volt per time until it reaches 10 volts. Read the difference voltage after each increase. Record the voltages in Data Table 3.12-5.

3.12.5 Experimental Data

Data Table 3.12-1 ***Purpose*** **: To measure the output of quarter bridge circuit**

R_3 / kΩ	10.2000	10.4000	10.6000	10.8000	11.000	11.2000
U_{AB}/V						

Data Table 3.12-2 ***Purpose*** **: To measure the output of half bridge circuit**

R_3/kΩ	10.2000	10.4000	10.6000	10.8000	11.0000	11.2000
R_4/kΩ	9.8000	9.6000	9.4000	9.2000	9.0000	8.8000
U_{AB}/V						

Data Table 3.12-3 ***Purpose*** **: To measure the sensitivity of the prototype electric balance**

Weight/kg	1.000	2.000	3.000	4.000	5.000	6.000	7.000	8.000	9.000	10.000
U_+/V(Loading)										
U_-/V(Unloading)										
Average, $\bar{U}$ /V										

Data Table 3.12-4 ***Purpose*** **: To measure the mass of three random objects**

Trial	Object 1 (Name:)/V	Object 2 (Name:)/V	Object 3 (Name:)/V
1			
2			
3			
Average, $\bar{U}$			
Mass, m/kg			

Data Table 3.12-5 ***Purpose*** **: To probe the relationship between the input voltage and the output**

Input voltage, E/V	2.0	3.0	4.0	5.0	6.0	7.0	8.0	9.0	10.0
Output voltage, U_{AB} /V									

Student's name and number:_____ Instructor's initial:_____

3.12.6 Calculations

(1)Use the results in Data Tables 3.12-1 and 3.12-2 to calculate the sensitivity of the quarter and half bridge circuits, respectively.

(2)Based on the above calculations, determine the relationship between the two sensitivities.

(3)The sensitivity of the prototype of an electric balance S is defined as the ratio of change in difference voltage (ΔU) to the change in mass (Δm).

$$S = \frac{\Delta U}{\Delta m} \quad \text{(V/kg)} \qquad (3.12\text{-}16)$$

Use the results in Data Table 3.12-3 to calculate the sensitivity of your prototype by the method of least-squares fitting.

(4)After finding the sensitivity of the prototype, use Eq. (3.12-16) to calculate the mass of the measured objects in Data Table 3.12-4. Show the details of calculation for the first object.

3.12.7 Graphing

Use the data in Data Table 3.12-5 to draw a graph of prototype output (U_{AB}) depending on input voltage (E). Describe their relationship from the graph.

3.12.8 Post Lab Questions

(1)Normally, four active resistors are used in wheatstone bridge circuit. Why?

(2)For the prototype of an electric balance, what's the relationship between the output voltage and the input voltage? Can the input voltage be increased without limits in order to increase the output? Why?

3.13 Measuring the Ultrasonic Speed

Prelab Assignment 3.13

(1)Describe the physical principle of an ultrasonic sound producer.

(2)The audible frequency range for normal hearing is about 20 Hz to 20 kHz. What are the wavelengths of sound waves at these frequencies? Assume that the speed of sound in the air is 343 m/s.

(3)Diagnostic ultrasound of frequency 4.5 MHz is used to examine tumors in soft tissue. What is the wavelength in air of such a sound wave?

3.13.1 Introduction and Objectives

Ultrasonic sound waves are longitudinal with frequencies greater than 20 kHz, which exceeds the upper limit of the human hearing range. Because of their high frequency and corresponding short wavelengths, ultrasound is used in many different fields. Ultrasonic devices such as sonar are used to detect objects and measure distances. Ultrasonic imaging (sonography) is used in both veterinary medicine and human medicine. In the nondestructive testing of products and structures, ultrasound is used to detect invisible flaws. Industrially, ultrasound is used for cleaning and for mixing, and to accelerate chemical processes.

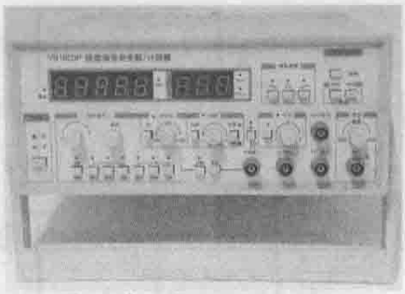
After performing this experiment and analyzing the data, you should be able to:

(1)Understand the production of ultrasonic sound by piezoelectric transducers.

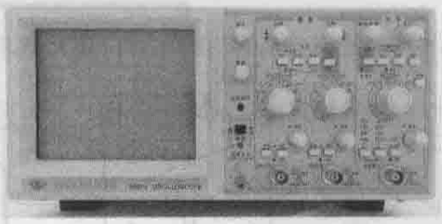
(2)Measure the speed of the ultrasonic sound waves by two methods of standing waves and phase changes.

3.13.2 Required Equipment

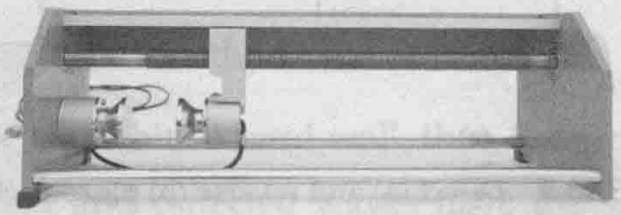
- Speed-measuring apparatus of ultrasonic sound(Fig. 3.13-1)
- Power function generator
- Dual-trace oscilloscope



Function Generator



Oscilloscope



Speed-measuring apparatus

Fig. 3.13-1 Apparatus for ultrasonic sound speed-measuring experiments

3.13.3 Theory

3.13.3.1 Piezoelectric Transducers and the Ultrasonic Sound Waves

A transducer is a device that transforms one form of energy into another, for example, a microphone (sound to electric) or loudspeaker (electric to sound). In this experiment, the ultrasonic sound waves are produced by a piezoelectric transducer.

A piezoelectric transducer is a device that transforms one type of energy to another by taking advantage of the piezoelectric effect of certain crystals or other materials. The direct piezoelectric effect is the internal generation of electrical charge resulting from an applied mechanical force, as shown in Fig. 3.13-2(a). It converts mechanical energy into electrical energy. The reverse piezoelectric effect is the internal generation of a mechanical strain resulting from an applied electrical field, as shown in Fig. 3.13-2 (b). It converts electrical energy into mechanical energy. Fig. 3.13-3 illustrates the technique that produces ultrasonic waves for various uses. Electrical contacts are made to the opposite faces of a crystal, such as quartz or strontium titanate ($SrTiO_3$). If an alternating voltage of high frequency is applied to these contacts, the crystal vibrates at the same frequency as the applied voltage, emitting a beam of ultrasonic waves.

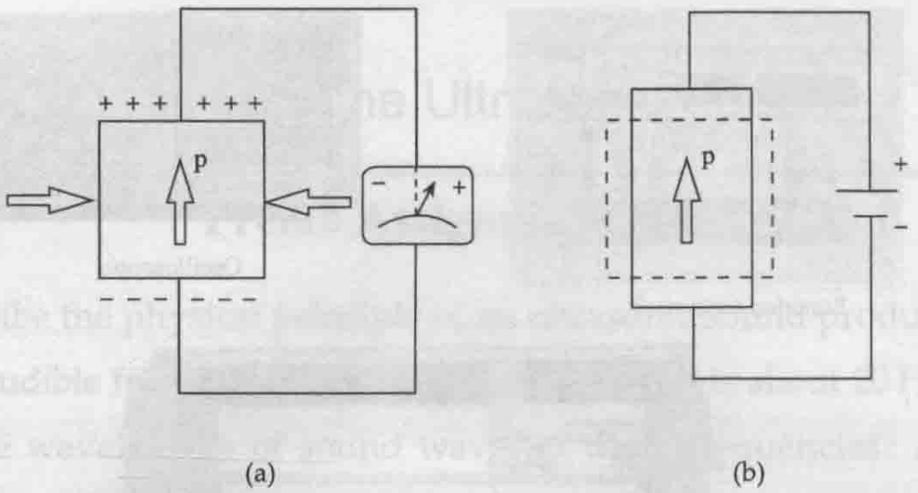


Fig. 3.13-2 Direct (a) and reverse (b) piezoelectric effect

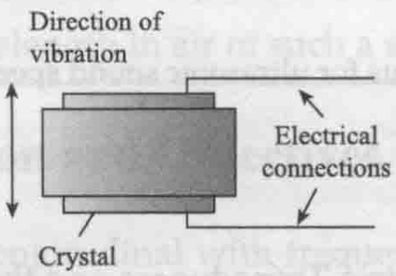


Fig. 3.13-3 An alternating voltage applied to the faces of a piezoelectric crystal causes the crystal to vibrate to produce ultrasonic sound

For a sound wave, the relationship between the frequency f, the wavelength λ, and the wave speed v is

$$v = \lambda f \tag{3.13-1}$$

It can be seen that if the frequency and wavelength are measured, the wave speed can be determined.

Systems generally have one or more natural vibrating frequencies. When a system is driven at a natural frequency, there is a maximum energy transfer and the vibrational amplitude increases to a maximum. When a system is driven at a natural frequency, we say that the system is in resonance (with the driving source) and refer to the particular frequency at which this occurs as a resonance frequency. If the frequency of the voltage applied to the piezoelectric crystal is the same as its natural frequency, the crystal will settle into steady large amplitude oscillations that produce high intensity sound waves. In this experiment, the oscillating frequency of the transducers you will use is near 40

kHz. We measure the frequency of the ultrasonic wave when the transducer system is in resonance. It equals the frequency of the applied voltage, which can be read from the function generator.

After measuring the frequency, we use two methods of standing wave and phase comparison to measure the wave length of the ultrasonic waves.

3.13.3.2 Standing Wave Method

When an ultrasonic wave hits a hard wall, then it is reflected without loss theoretically. The incident and reflected waves have the same frequency and amplitude. Superposition of the incident sound pressure wave p_1 and the reflected sound pressure wave p_2 results in the formation of a standing wave of $p = p_1 + p_2$. The function p_1 of the incident sound wave can be written as

$$p_1 = A \cdot \cos\left(\omega t - \frac{2\pi}{\lambda} x\right) \tag{3.13-2}$$

where A is the intensity, ω is angular frequency of wave, λ is the wavelength.

The function p_2 of the reflected sound wave can be written as

$$p_2 = A \cdot \cos\left(\omega t + \frac{2\pi}{\lambda} x + \pi\right) \tag{3.13-3}$$

The result of the addition of the two waves is a standing wave p

$$\begin{aligned} p &= p_1 + p_2 \\ &= A \cdot \cos\left(\omega t - \frac{2\pi}{\lambda} x\right) + A \cdot \cos\left(\omega t + \frac{2\pi}{\lambda} x + \pi\right) \\ &= 2A \sin(\omega t) \sin\left(\frac{2\pi}{\lambda} x\right) \end{aligned} \tag{3.13-4}$$

Figure 3.13-4 shows a standing wave between a transmitter and a receiver. The points of attachment to the transmitter and the receiver and all other stationary points are called nodes. The points with the maximum amplitude are called antinode. From the figure, observe that the distance d between adjacent nodes or antinodes is one-half the wavelength of the wave:

$$d = \lambda/2 \tag{3.13-5}$$

When you move the receiver, you can observe on the oscilloscope that the amplitude of the signal decreases and increases alternately. The distance between two adjacent largest signals is half wavelength $\lambda/2$.

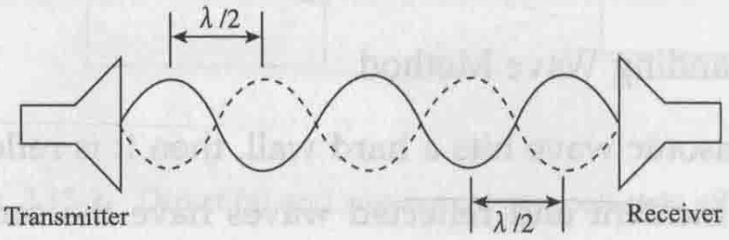


Fig. 3.13-4 A standing wave between a transmitter and a receiver

3.13.3.3 Phase Comparison Method

According to Eq. (3.13-2), the function of the wave at the point of the transmitter is

$$p_1 = A \cdot \cos \omega t \tag{3.13-6}$$

However, the function of the wave at the point of the receiver can be expressed as

$$p_2 = A \cdot \cos\left(\omega t - \frac{2\pi}{\lambda} L\right) \tag{3.13-7}$$

where L is the distance between the transmitter and the receiver.

From the Eqs. (3.13-6) and (3.13-7), we can find that the phase difference $\Delta\varphi$ between the two waves is $\Delta\varphi = 2\pi L/\lambda$. When the distance between the transmitter and the receiver is changed by a wavelength, the phase difference is changed 2π. Therefore, we can measure the phase difference of the two waveforms by Lissajous patterns to determine the wavelength.

The two waveforms applied to the transmitter and received by the receiver are both sinusoidal with the same frequency. A Lissajous pattern can be observed on the oscilloscope by connecting them to the vertical and horizontal deflection plates through the *X*-*Y* mode. The shape of the Lissajous pattern will

change with the phase difference. Fig. 3.13-5 shows the Lissajous patterns and the phase difference between the two input waveforms.

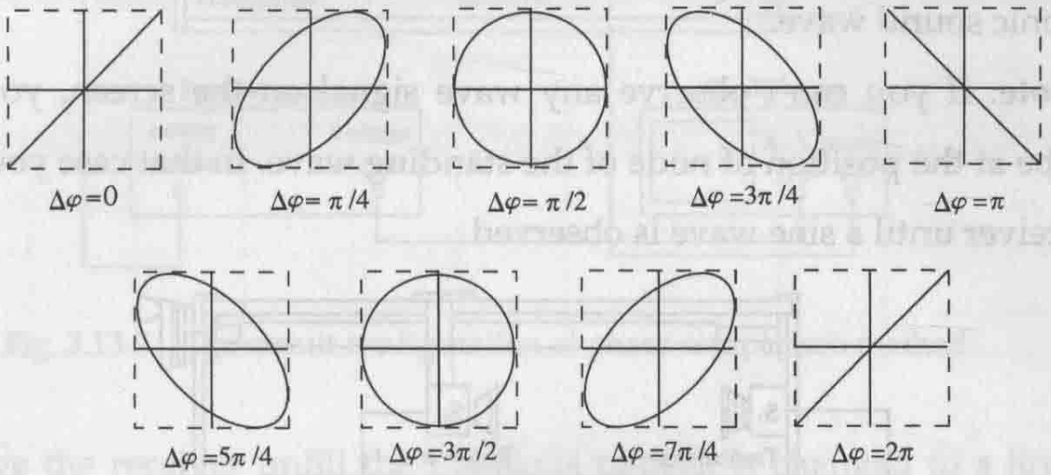


Fig. 3.13-5 Lissajous patterns and the phase difference between the two input waveforms

As shown in Fig. 3.13-5, when the position of the receiver is moved it can be observed that the shape of the Lissajous pattern changes. When the Lissajous pattern changes in correspondence with the change in phase difference from $\Delta\varphi = 0$ to $\Delta\varphi = 2\pi$, the traveling distance is λ.

3.13.4 Experimental Procedure

3.13.4.1 Determination of Resonance Frequency

(1)Familiarize yourself with the function generator, the ultrasonic speed-measuring set-up, and the dual-trace oscilloscope. Set up the circuit as shown in Fig. 3.13-6.

(2)Turn on the oscilloscope and the function generator. Choose sinusoidal wave with frequency 35.000 kHz from the function generator. Adjust the distance between the transmitter and the receiver to about 5 mm.

(3)Adjust the knobs in the panel to make the sine wave stand still on the oscilloscope screen.

(4)Tune the frequency of the sine wave from the function generator from 35.000 kHz to 45.000 kHz. You will find that the sine wave on the screen is changing in frequency and amplitude. When the amplitude of the received sine

wave reaches the maximum, the frequency of the ultrasonic wave is found. Record the frequency on the function generator as the frequency of the ultrasonic sound wave.

Note: If you can't observe any wave signal on the screen, your receiver could be at the position of node of the standing wave. In that case you can move the receiver until a sine wave is observed.

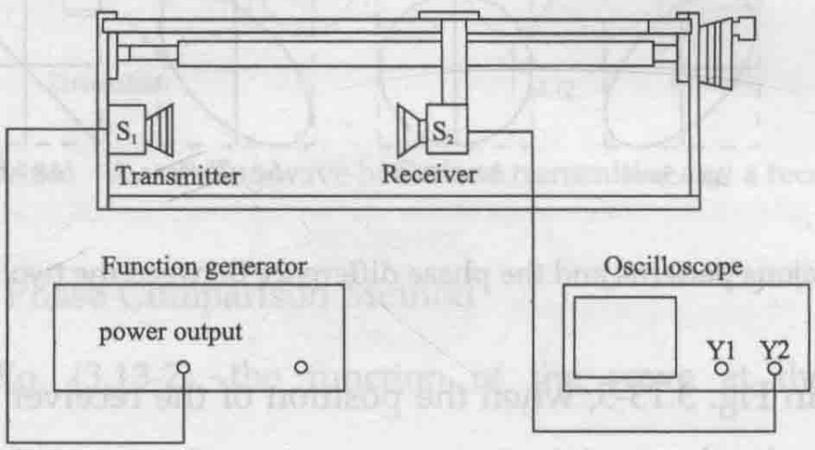


Fig. 3.13-6 Circuit configuration of the standing wave method

3.13.4.2 Using Standing Wave Method to Measure the Wavelength

(1) Move the receiver slowly to increase the distance between the transmitter and the receiver. When the amplitude of the sine wave reaches the maximum, record the position and record it in Data Table 3.13-1 as l_1.

(2) Move the receiver to further increase the distance. When the amplitude reaches the next maximum again, record the position and continue the process to complete Data Table 3.13-1.

3.13.4.3 Using Phase Comparison Method to Measure the Wavelength

(1)Set up the circuit as shown in Fig. 3.13-7.

(2)Turn the TIME/DIV knob of the oscilloscope fully anticlockwise to *X*-*Y* mode. Now, you can observe a Lissajous pattern on the screen.

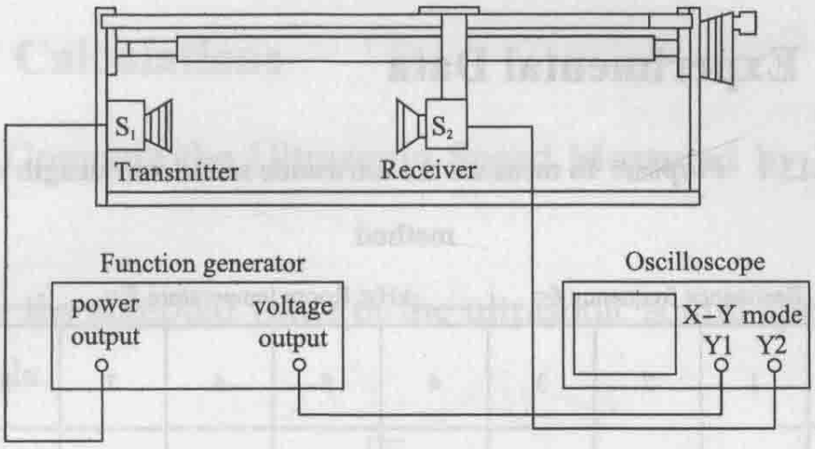


Fig. 3.13-7 The circuit configuration of phase comparison method

(3)Move the receiver untill the Lissajous pattern is changed to a line with positive slope as shown in Fig. 3.13-5. Record the position of the receiver as L_1 and enter it in Data Table 3.13-2.

(4)Move the receiver untill the line with positive slope is visible on the screen again. Record the positions of the receiver to complete Data Table 3.13-2.

(5)Record the room temperature from the thermometer on the wall near the entrance.

3.13.5 Experimental Data

Data Table 3.13-1 ***Purpose*: To measure the ultrasonic sound wavelength with standing wave method**

Resonance frequency f = _____kHz; Room temperature T =_____℃

Trial	1	2	3	4	5	6	7	8	9	10
l_i/mm										

Data Table 3.13-2 ***Purpose*: To measure the ultrasonic sound wavelength with phase comparison method**

Trial	1	2	3	4	5	6	7	8	9	10
L_i /mm										

Student's name and number:_____Instructor's initial:_____

3.13.6 Calculations

3.13.6.1 Compute the Ultrasonic Speed Measured by Standing Wave Method

(1)Compute the accepted value of the ultrasonic sound speed in air by the following formula.

$$V_0 = 331.30 \times \sqrt{\frac{T}{T_0}} \quad (T_0=273.15\ \mathrm{K})$$

(2)Compute the wavelength measured by standing wave method.

$$\bar{\lambda}_s = \frac{2}{25}[|l_{10} - l_5| + |l_9 - l_4| + |l_8 - l_3| + \cdots + |l_6 - l_1|]$$

(3)Compute the ultrasonic sound speed.

$$\bar{V}_s = \bar{\lambda}_s \times f$$

(4)Compute the relative error.

$$E_s = \frac{\bar{V}_s - V_0}{V_0} \times 100\%$$

3.13.6.2 Compute the Ultrasonic Speed Measured by Phase Comparison Method

(1)Compute the wavelength measured by phase comparison method.

$$\bar{\lambda}_p = \frac{1}{25}\left[|L_{10} - L_5| + |L_9 - L_4| + |L_8 - L_3| + \cdots + |L_6 - L_1|\right]$$

(2)Compute the ultrasonic sound speed.

$$\bar{V}_p = \bar{\lambda}_p \times f$$

(3)Compute the relative error.

$$E_p = \frac{\bar{V}_p - V_0}{V_0} \times 100\%$$

3.13.6.3 Compute the Uncertainty in the Ultrasonic Speed Measured by Standing Wave Method

(1)Compute the following difference in positions.

$$L_1=|l_6-l_1|,\quad L_2=|l_7-l_2|,\quad L_3=|l_8-l_3|,\quad L_4=|l_9-l_4|,\quad L_5=|l_{10}-l_5|,$$

$$\bar{L}=\frac{1}{5}(L_1+L_2+L_3+L_4+L_5)$$

(Note: In the following calculations, L_1, L_2, L_3, L_4, and L_5 are treated as direct measurement results.)

(2)Compute the Type A uncertainty in $\bar{L}$.

$$\mu_{A_{\bar{L}}}=\sqrt{\frac{\sum_{i=1}^{5}(L_i-\bar{L})^2}{5\times(5-1)}}$$

(3)Compute the Type B uncertainty in $\bar{L}$.

$$\mu_{B_{\bar{L}}}=\frac{\Delta_{\text{Instr.}}}{\sqrt{3}}$$

(4)Compute the combined uncertainty in L.

$$\sigma_L=\sqrt{\mu_{A_{\bar{L}}}{}^2+\mu_{B_{\bar{L}}}{}^2}$$

(5)Compute the uncertainty in λ.

$$\sigma_\lambda=\frac{2}{5}\sigma_l$$

(6)Compute the uncertainty in f.

$$\sigma_f=\mu_{B_f}=\frac{\Delta_{\text{Instr.}}}{\sqrt{3}}$$

(Note: Single measurement was performed for the frequency. The unit of the last digit is the instrument error for the digital cymometer.)

(7)Compute the relative uncertainty in the ultrasonic speed V_s.

$$\frac{\sigma_{V_s}}{V_s}=\sqrt{\left(\frac{\sigma_\lambda}{\lambda}\right)^2+\left(\frac{\sigma_f}{f}\right)^2}$$

(8)Compute the uncertainty in V.

$$\sigma_{V_s}=\overline{V}_s\times\frac{\sigma_V}{V}$$

(9)Write the final result of ultrasonic speed in the form,

$$V_s=\overline{V}_s\pm\sigma_{V_s}$$

3.13.7 Post Lab Questions

(1)Two methods are used to measure the ultrasonic sound wavelength in this experiment. Which one is more accurate? Why?

(2)How is it determined that the transmitter is in resonance?

3.14 The Photoelectric Effect

Prelab Assignment 3.14

(1) What is photoelectric effect threshold frequency for a given element?

(2) What is work function for a given element?

(3) A sodium surface is illuminated with light of wavelength 0.300 μm. The work function for sodium is 2.46 eV. Calculate,

a. The energy of each photon in electron volts;

b. The maximum kinetic energy of the ejected photoelectrons;

c. The cutoff wavelength for sodium.

3.14.1 Introduction and Objectives

In the latter part of the 19th century, experiments showed that light incident on certain metallic surfaces caused the emission of electrons from the surfaces. This phenomenon is known as the *photoelectric effect*, and the emitted electrons are called *photoelectrons*. The first discovery of this phenomenon was made by Hertz in 1887. This photoelectric effect was unexplained until Einstein connected this experimental curiosity with Planck's idea that radiation comes in small packets, or quanta. In 1905, he proposed that the energy of the ejected electrons is proportional to the energy of the incident light with a constant offset that is unique to the metal, referred to as the work function. This phenomenon was a crucial precursor to the formulation of quantum mechanics as it was one of the first to show the wave-particle duality of light. Albert Einstein received the 1921 Nobel Prize in Physics for his services to theoretical physics and especially for his discovery of the law of the photoelectric effect.

After performing this experiment and analyzing the data, you should be able to:

(1)Test the Einstein's law of the photoelectric effect.

(2)Extract values for Planck's constant.

3.14.2 Required Equipment

- Mercury-arc lamp(Fig. 3.14-1)
- Light filter
- Photoelectric effect measurement apparatus

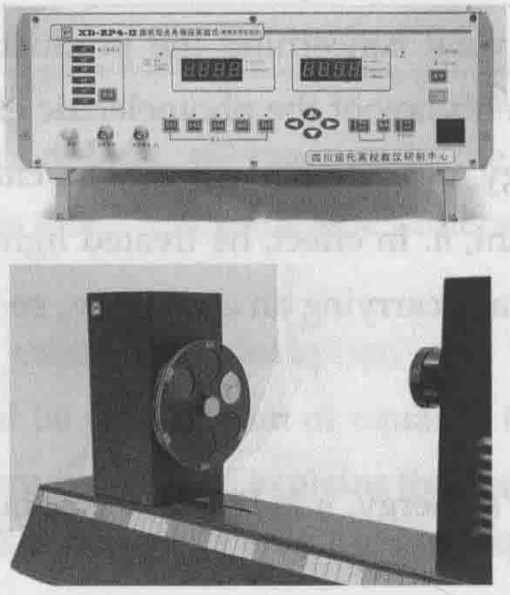


Fig. 3.14-1 Photoelectric effect apparatus

3.14.3 Theory

There are several features of the photoelectric effect that can't be explained with classical physics or with the wave theory of light:

(1)No electrons are emitted if the incident light frequency falls below some cutoff frequency f_c, also called the threshold frequency, which is characteristic of the material being illuminated. This fact is inconsistent with the wave theory, which predicts that the photoelectric effect should occur at any frequency, provided the light intensity is sufficiently high.

(2)The maximum kinetic energy of the photoelectrons is independent of light intensity. According to wave theory, light of higher intensity should carry more energy into the metal per unit time and therefore eject photoelectrons having higher kinetic energies.

(3)The maximum kinetic energy of the photoelectrons increases with increasing light frequency. The wave theory predicts no relationship between

photoelectron energy and incident light frequency.

(4)Electrons are emitted from the surface almost instantaneously (less than 10^{-9} s after the surface is illuminated), even at low light intensities. Classically, we expect the photoelectrons to require some time to absorb the incident radiation before they acquire enough kinetic energy to escape from the metal.

Motivated by these, Einstein produced one of his seminal papers in 1905, "A heuristic point of view concerning the production and transformation of light", which put forth his law of the photoelectric effect. He extended Planck's notion that the energy of radiation comes in chunks of the frequency, v multiplied by a constant, h. In effect, he treated light as if it were composed of particles, or photons, each carrying an energy hv, so that

$$E = hv \tag{3.14-1}$$

where E is the photon's energy, h is Planck's constant and v is the frequency of the light.

The key point here is that the light energy lost by the emitter, hv, stays sharply localized in a tiny packet or particle called a photon. In Einstein's model a photon is so localized that it can give *all* its energy hv to a single electron in the metal. According to Einstein, the maximum kinetic energy for these liberated photoelectrons is

$$E_{\mathrm{K\,max}} = hv - w \tag{3.14-2}$$

where E_{Kmax} is the maximum kinetic energy for the photoelectron, w is called the work function of the metal. The work function, which represents the minimum energy with which an electron is bound in the metal, is on the order of a few electron volts. Table3.14-1 lists work functions for various metals.

Table 3.14-1 Work functions (w) for some selected metals

Metal	w/eV
Ag	4.73
Al	4.08
Cu	4.70

Continued

Metal	w/eV
Fe	4.50
Na	2.46
Pb	4.14
Pt	6.35
Zn	4.31

With the photon theory of light, we can explain the previously mentioned features of the photoelectric effect that cannot be understood using concepts of classical physics:

(1)Photoelectrons are created by absorption of a single photon, so the energy of that photon must be greater than or equal to the work function, else no photoelectrons will be produced. This explains the cutoff frequency.

(2)From Eq. (3.14-2), E_{Kmax} depends only on the frequency of the light and the value of the work function. Light intensity is immaterial because absorption of a single photon is responsible for the electron's change in kinetic energy.

(3)Eq. (3.14-2) is linear in the frequency, so E_{Kmax} increases with increasing frequency.

(4)Electrons are emitted almost instantaneously, regardless of intensity, because the light energy is concentrated in packets rather than spread out in waves. If the frequency is high enough, no time is needed for the electron to gradually acquire sufficient energy to escape from the metal.

Figure 3.14-2 is a schematic diagram of a photoelectric effect apparatus. An evacuated glass tube known as a photocell contains a metal plate K (the emitter) connected to the negative terminal of a variable power supply. Another metal plate, A (the collector), is maintained at a positive potential by the power supply. When the tube is kept in the dark, the ammeter reads zero, indicating that there is no current in the circuit. When plate K is illuminated by light having a wavelength shorter than some particular wavelength that depends on the material used to make plate K, however, a current is detected by the ammeter, indicating a flow of charges across the space between K and A. This

current arises from photoelectrons emitted from the negative plate K and collected at the positive plate A.

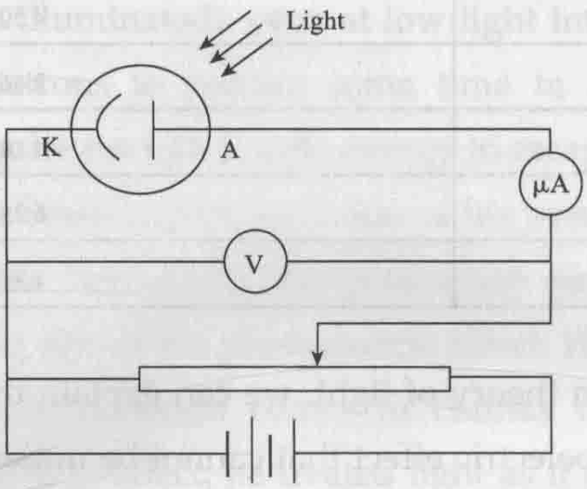


Fig. 3.14-2 A schematic circuit diagram for studying the photoelectric effect

Figure 3.14-3 is a plot of the photoelectric current versus the potential difference ΔV between K and A for two light intensities. At large values of ΔV, the current reaches a maximum value. In addition, the current increases as the incident light intensity increases, as you might expect. Finally, when ΔV is negative—that is, when the power supply in the circuit is reversed to make K positive and A negative—the current drops to a low value because most of the emitted photoelectrons are repelled by the now negative plate A. In this situation only those electrons having a kinetic energy greater than the magnitude of $e\Delta V$ reach A, where e is the charge on the electron.

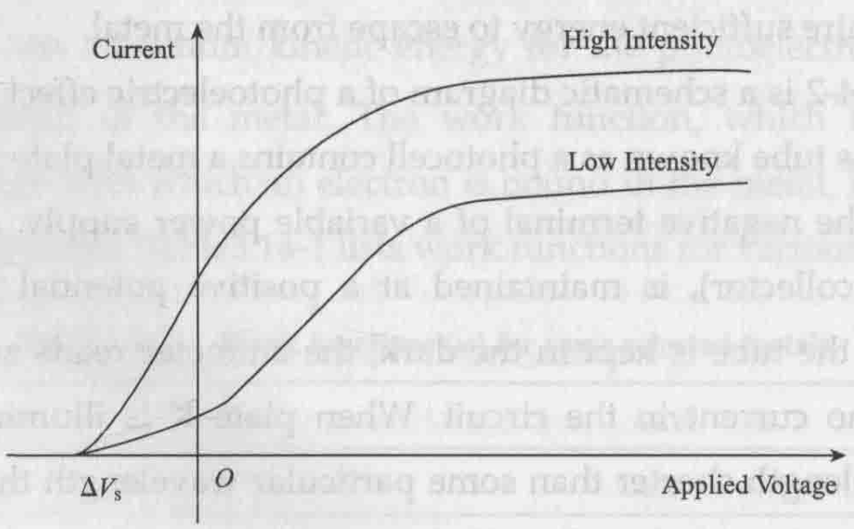


Fig. 3.14-3 Photoelectric current versus applied potential difference for two light intensities

When ΔV is equal to or more negative than ΔV_s, the stopping potential, no electrons reach A and the current is zero. The stopping potential is independent of the radiation intensity. The maximum kinetic energy of the photoelectrons is related to the stopping potential through the relationship

$$E_{\mathrm{K\,max}} = e\Delta V_s \tag{3.14-3}$$

Combining Eqs. (3.14-2) and (3.14-3), we have

$$E_{\mathrm{K\,max}} = e\Delta V_s = h\nu - w \tag{3.14-4}$$

Thus,

$$\Delta V_s = \frac{h}{e}\nu - \frac{w}{e} \tag{3.14-5}$$

If ΔV_s is plotted as a function of ν, Einstein's theory predicts a straight line with slope h/e and y-intercept $-w/e$. Thus we can find an empirical value for h and the work function w of our photocathode simply by measuring stopping voltages.

3.14.4 Experimental Procedure

3.14.4.1 Measuring the Stopping Potential

(1)First, familiarize yourself with the apparatus.

(2)Rotate the filter disk to cover the window of the photoelectric tube. Cover the window of the light source with a cap. Turn on the power of the light source and the photoelectric effect measurement apparatus. Preheat the light source for about five minutes.

(3)Select the ranges of 10^{-13}A and –2~0V for the ammeter and voltmeter, respectively. After preheating, tune the "zero" knob to set the current to zero and then press the "Calibration" button.

(4)Move the photoelectric tube to adjust the distance between the light source and photoelectric tube to about 40 cm. Uncover the light source.

(5)Rotate the filter disk to mount the 365 nm filter on the window of the photoelectric tube.

(6)Adjust the magnitude of the voltage to find a value which makes the current zero. That's the stopping potential. Record the voltage in Data Table 3.14-1.

(7)Select the filters of 405 nm, 436 nm, 546 nm, 577 nm. Repeat Steps (5) and (6) to find the stopping potentials for the different lights. Record the voltages in DataTable 3.14-1.

3.14.4.2 Measuring the Current-voltage Characteristics of the Photoelectric Tube

(1)Move the photoelectric tube to set the distance d between the light source and photoelectric tube to about 30cm.

(2)Select the ranges of 10^{-11}(or 10^{-12}) A and $-2\sim50$V for the ammeter and voltmeter, respectively.

Note: If you change the scale of the ammeter you need to tune the "zero" knob to set the current to zero when the photoelectric tube is covered.

Rotate the filter disk to mount the 577nm filter on the window of the photoelectric tube.

(3)Adjust the magnitude of the voltage to read the current registered on the ammeter. Record the current in Data Table 3.14-2.

(4)Move the photoelectric tube to adjust the distance d between the light source and photoelectric tube to about 40 cm. Repeat Step (4) to complete Data Table 3.14-2.

3.14.5 Experimental Data

Data table 3.14-1 ***Purpose*: To measure the stopping potentials for different lights**

Wavelength /nm	Frequency /Hz	Stopping potential /V
365		
405		
436		
546		
577		

Data Table 3.14-2 ***Purpose*: To measure current-voltage characteristics of the photoelectric tube**

ΔV/V	–2.0	0.0	2.0	4.0	6.0	8.0	10.0	12.0
d = 30cm, I/×10^{-}( )A								
d = 35cm, I/×10^{-}( )A								
d = 40cm, I/×10^{-}( )A								
ΔV/V	14.0	16.0	18.0	20.0	22.0	24.0	26.0	28.0
d = 30cm, I/×10^{-}( )A								
d = 35cm, I/×10^{-}( )A								
d = 40cm, I/×10^{-}( )A								
ΔV/V	30.0	32.0	34.0	36.0	38.0	40.0	42.0	44.0
d = 30cm, I/×10^{-}( )A								
d = 35cm, I/×10^{-}( )A								
d = 40cm, I/×10^{-}( )A								
ΔV/V	46.0	48.0	50.0					
d = 30cm, I/×10^{-}( )A								
d = 35cm, I/×10^{-}( )A								
d = 40cm, I/×10^{-}( )A								

Student's name and number:_____ Instructor's initial:_____

3.14.6 Calculations

(1)Use the following formula to compute the frequencies of the five lights. Show one sample calculation and enter the data in Data Table 3.14-1.

$$\nu = \frac{c}{\lambda} \tag{3.14-6}$$

Where c is the speed of light ($c = 3.000\times10^8$ m/s), and λ is the wavelength of light.

(2)Suppose that the frequency ν and the stopping potential ΔV_s satisfy the relationship $\Delta V_s = k\nu + b$. Use the data in Data Table 3.14-1 and the least-squares fitting method to determine the values of the Planck's constant and the work function of the metal in the photoelectric tube. Calculate the relation coefficient [see Eq. (2.28)] to interpret how well the line fits the data.

3.14.7 Graphing

(1)Use the entries in Data Table 3.14-1 to draw a smooth line described by the data points.

(2)Use the entries in Data Table 3.14-2 to draw a graph of the voltage dependent current for the photoelectric tube illuminated by different intensities.

3.14.8 Post Lab Questions

(1)For a given light intensity, when the accelerating voltage increases, the photoelectric current increases and then reaches the saturation. Why?

(2)State what property of light is demonstrated by the photoelectric effect.

(3)How would you claim that the Einstein's equation is verified by the photoelectric effect experiment?

3.15 The Potentiometer

Prelab Assignment 3.15

(1)Can we use a voltmeter to measure the emf of a cell? Why?

(2)What is the null balance measuring method in electricity?

(3)In Fig. 3.15-4, consider E_s = 2.000V. What's the resistance value of R in order to obtain a working current of 5.000mA in the source circuit? In the measuring circuit, R_x is 201.5Ω when the galvanometer reads zero. What's the measured emf?

3.15.1 Introduction and Objectives

We cannot simply attach a voltmeter to the battery and say that the voltage we read is the electromotive force (emf) of the battery. The reason for this is because the battery has its own internal resistance and when the voltmeter sends current through the battery to measure the voltage, the internal resistance of the battery affects the voltage measured. Therefore, in order to accurately measure the emf of a battery, we have to use a device that does not draw any current through the battery. The potentiometer is one such device that will accomplish this goal. This method was proposed by Johann Christian Poggendorff around 1841 and became a standard laboratory technique for measuring emf.

After performing this experiment and analyzing the data, you should be able to:

(1)Understand the principle of a box potentiometer.

(2)Measure a thermocouple's emf.

3.15.2 Required Equipment

- A box potentiometer(Fig. 3.15-1)
- Standard power supply
- Sensitive galvanometer

- A heating vessel
- Thermocouples

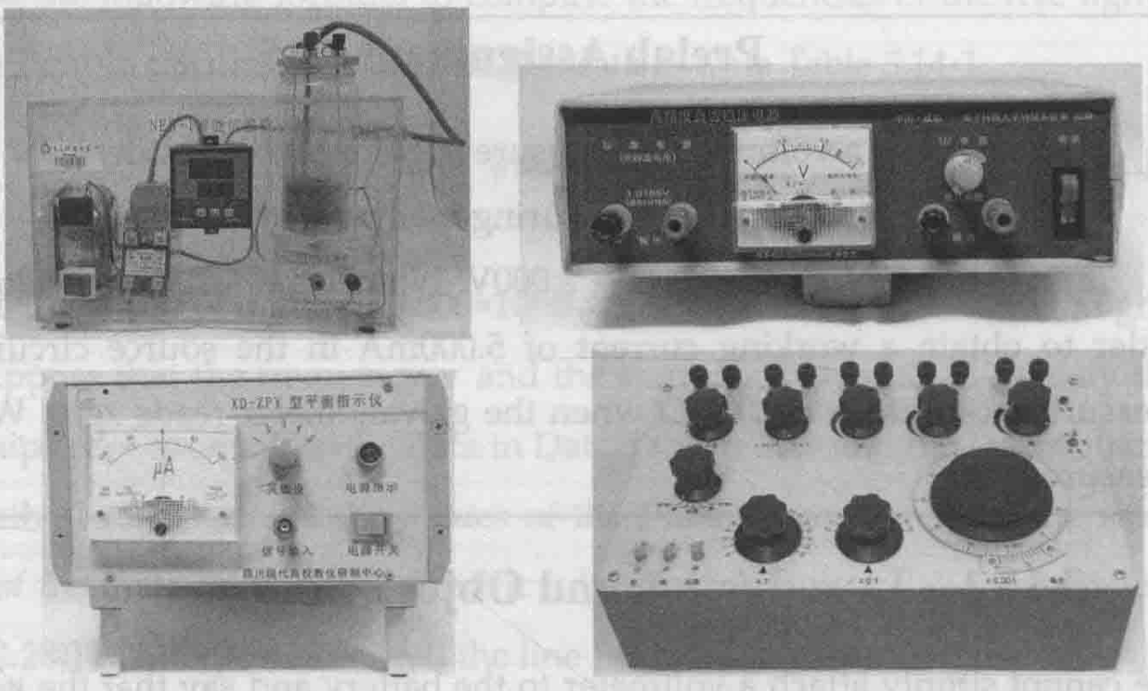


Fig. 3.15-1 Experimental apparatus

3.15.3 Theory

3.15.3.1 The Principle of the Potentiometer

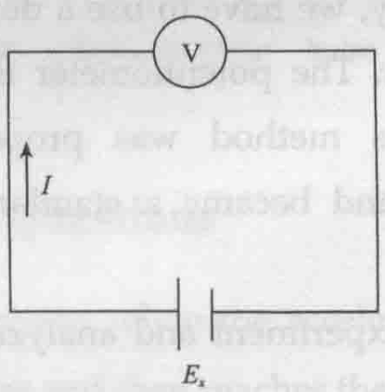


Fig. 3.15-2 Measurement of the terminal voltage of a cell by a voltmeter

The terminal voltage of a cell is the potential difference between its terminals. The emf of a cell may be defined as the terminal voltage of the cell when not under load, that is, delivering no current. Fig. 3.15-2 shows a circuit to measure the terminal voltage of a cell by a voltmeter. Since a voltmeter is always connected in parallel with the component or components under test, it draws

some current in the tested circuit. The measured terminal voltage is always lower than the emf due to the current I through the circuit and internal resistance of the battery.

Figure 3.15-3 shows the schematic circuit diagram of the null balance measuring method. The galvanometer G is a sensitive device capable of indicating the presence of very small current. Its purpose is to accurately indicate a condition of zero current, rather than to indicate any specific (nonzero) quantity as a normal ammeter would. E_0 is a standard cell, whose emf is adjustable. When the deflection of the galvanometer is observed, adjust the emf of the standard cell until the galvanometer no longer deflects from zero. It means the galvanometer draws no current from the unknown source, and the magnitude of the unknown emf E_x is equal to that of the standard cell. The key point of null balance measuring method is to measure electrical potentials without any current draw.

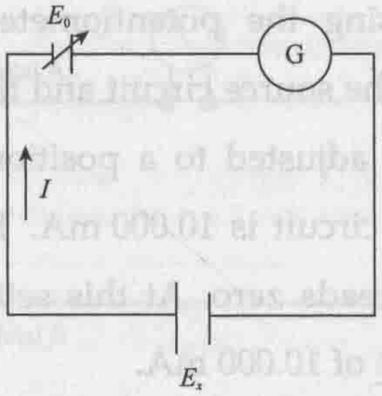


Fig. 3.15-3 Schematic circuit diagram of the null balance measuring method

The box potentiometer is constructed on the principle of null balance. A schematic diagram of the potentiometer is shown in Fig. 3.15-4, where E is a stable current source with a constant voltage greater than any voltage to be measured, E_s is a standard cell with emf of 1.0185 V, E_x is an unknown emf produced by thermocouple in this experiment, R_0 is a variable resistor, R_A and R_B are two resistors, C and C' are two sliding contacts which varies the resistances in the calibration and measurement circuits, R and R_x are the

resistances in the calibration and measurement circuits, G is a sensitive galvanometer which indicates zero current when the pointer deflects neither to the left nor to the right, K_E, K_P and K are three switches.

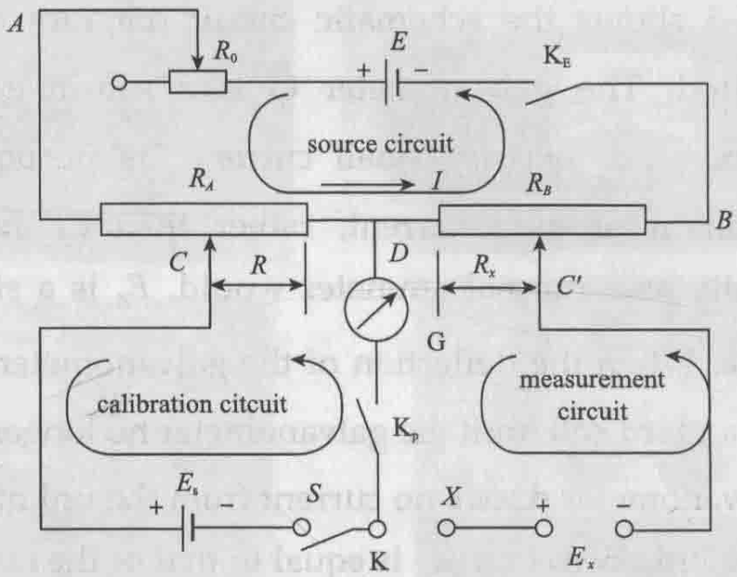


Fig. 3.15-4 A schematic circuit diagram of the potentiometer

The first step in using the potentiometer is to calibrate it with the calibration circuit. When the source circuit and the calibration circuit are closed the movable contact C is adjusted to a position to set R = 101.85 Ω. So, the current in the calibration circuit is 10.000 mA. Then adjust R_0 to a position at which the galvanometer reads zero. At this setting, the current in the source circuit is equal to the value of 10.000 mA.

Next, the switch K is turned to connect with X, that is, the standard cell is replaced by the source of unknown emf E_x. The movable contact C' is adjusted to find a point where the galvanometer reads zero. It indicates that the resistance R_x doesn't draw any current in the measurement circuit. The terminal voltage across R_x is the emf E_x. Thus, we have

$$E_x = IR_x \tag{3.15-1}$$

where I = 10.000 mA, R_x can be read directly. For the box potentiometer, the current through R_B (or R_x) is fixed to 10.000 mA and the measured R_x is transferred to the value of E_x. So, we can read the emf directly from the

potentiometer.

3.15.3.2 The thermocouple

If two dissimilar metals, such as copper and iron, are joined together, as in Fig. 3.15-5, and if the two junctions are at different temperatures, a thermal emf will be generated. This is now known as the *thermoelectric effect* or *Seebeck effect*. This is because electrons in these metals are free to move, and those electrons at a hotter junction of the thermocouple will move more quickly than those electrons at the colder end. This will result in a net effect of some charge moving from one end of the thermocouple to the other, which is what current is. The corresponding potential difference is then called the thermal emf. This effect was discovered by the German–Estonian physicist Thomas Johann Seebeck in 1821. It is utilized in making a thermocouple, an instrument for the accurate and rapid measurement of temperatures.

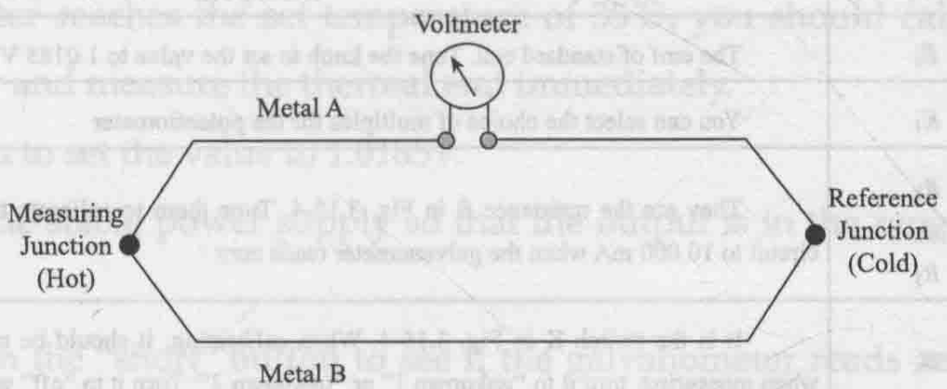


Fig. 3.15-5 A schematic circuit diagram of the thermocouple

Thermocouples are widely used in science and industry. Applications include temperature measurement for kilns, gas turbine exhaust, diesel engines, and other industrial processes. In this experiment, the unknown emf is produced by a thermocouple.

3.15.4 Experimental Procedure

Fig. 3.15-6 shows the panel of a box potentiometer. The knobs on the panel are introduced in Table 3.15-1.

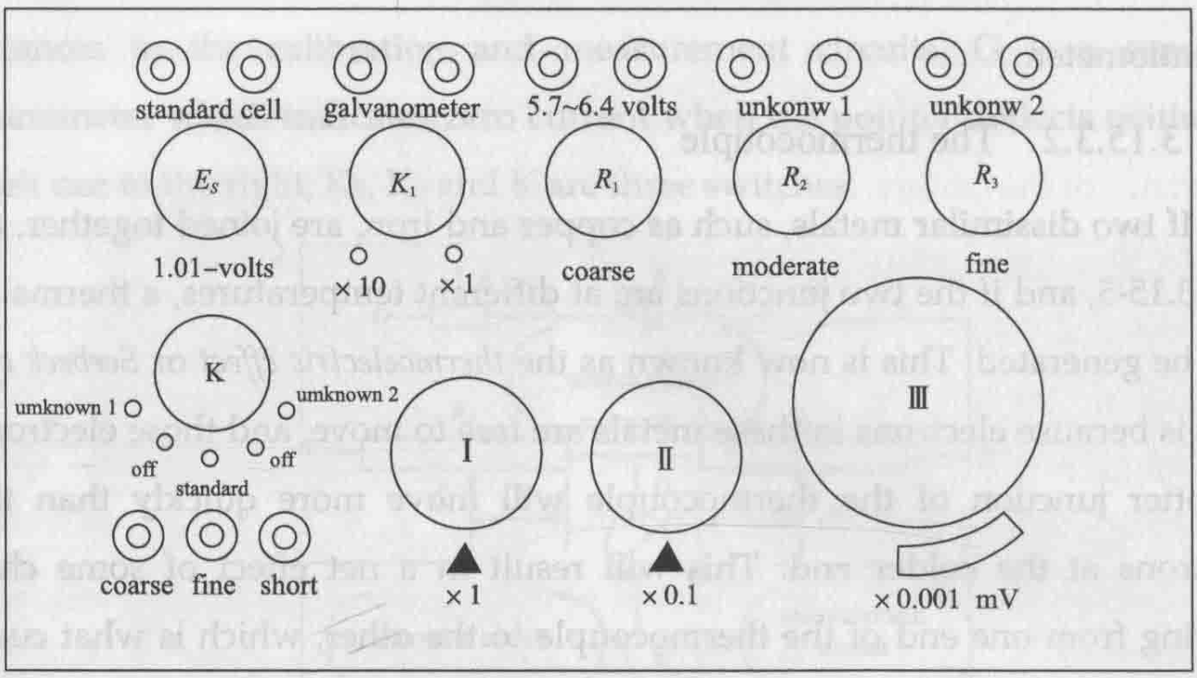


Fig. 3.15-6 Panel of the box potentiometer

Table 3.15-1 The knobs and their description on the panel of the potentiometer

Symbol (knob)	Description
E_s	The emf of standard cell. Tune the knob to set the value to 1.0185 V in this experiment
K_1	You can select the choice of multiples for the potentiometer
R_1 R_2 R_3	They are the resistance R in Fig. 3.15-4. Tune them to calibrate the current in source circuit to 10.000 mA when the galvanometer reads zero
K	It is the switch K in Fig. 3.15-4. When calibrating, it should be turned to "standard"; when measuring, turn it to "unknown 1" or "unknown 2"; Turn it to "off" when it is not in use
Coarse Fine Short	They are the switch K_p in Fig. 3.15-4. You will use it when calibrating and measuring. At the beginning of calibration or measuring, press down the "coarse" button and it will be observed that the galvanometer deflects far from zero. After the galvanometer deflects close to zero you can lock the "coarse" button and push down the "fine" button. The "short" button is used to calibrate the galvanometer
I II III	The resistance R_B (or R_x) in Fig. 3.15-4. Tune them to make the measurement circuit null balance when the galvanometer reads zero. The values are transferred to millivolt due to the fixed current of 10.000mA through R_B in the source circuit

3.15.4.1 Heating the Water to a Setting Temperature

(1)Familiarize yourself with the potentiometer, the stable power supply, the galvanometer, and the heating vessel.

(2)Connect the potentiometer, the stable power supply, the galvanometer,

and the heating vessel correctly.

(3)Check your circuit and turn on the power of the potentiometer, the stable supply, and the galvanometer to preheat them for at least five minute.

(4)Rotate K_1 to choose the multiple of "×1".

(5)Turn on the power of the heating vessel. Turn the switch on the panel to choose "auto control" mode of heating. There are two displays on the panel. The upper reads the actual temperature of water and the lower reads the setting temperature.

(6)Press the "set" button on the panel and the digits on the lower display will flash. Press the "▼" and/or "▲" button to set the temperature to the first set value of 35 ℃. Finally, press the "set" button again and water will be heated automatically to the setting temperature.

3.15.4.2 Calibrating the Potentiometer

When the water reaches the set temperature of 35℃, you should calibrate the potentiometer and measure the thermal emf immediately.

(7)Tune E_s to set the value to 1.0185V.

(8)Tune the stable power supply so that the output is in the range of 5.7～6.4 V.

(9)Press in the "short" button to see if the galvanometer reads zero. If not, adjust it mechanically by a screwdriver. There are five sensitivity grades for the galvanometer. The greater the number, the more sensitive the galvanometer is. You can choose the sensitivity as high as possible but the pointer of the galvanometer should not oscillate when current goes through it.

(10)Rotate the switch K to "standard" and set grade "1" for the galvanometer. Push in the "coarse" knob at the lower left corner. If the galvanometer deflects far from zero, adjust R_1 and R_2 (see Fig. 3.15-6) until the galvanometer deflects little from zero. Rotate the "coarse" knob to lock it. Push in the "fine" knob and increase the sensitivity of the galvanometer. Tune R_3 until the galvanometer reads zero. Choose the sensitivity as high as possible so that

the pointer of the galvanometer can read zero and doesn't oscillate. Now, the current in the source circuit is calibrated to 10.000 mA.

(11)Rotate and pull out both the "fine" and "coarse" knobs.

3.15.4.3 Measuring the emf Produced by the Thermocouple

(12)Turn the switch K to "unknown 1" or "unknown 2" to measure the unknown emf.

(13)Push in the "coarse" knob. If the galvanometer deflects far from zero, adjust knobs "Ⅰ" and "Ⅱ" (see Fig. 3.15-6) until the galvanometer deflects little from zero. Rotate the "coarse" knob to lock it. Push in the "fine" knob. Tune knob "Ⅲ" until the galvanometer reads zero. Now, read the scales on the knobs "Ⅰ" "Ⅱ" and "Ⅲ" and add them up. The result is the emf produced by the thermocouple at temperature of 35℃.

(14)Repeat Steps (6) through (13) to complete Data Table 3.15-1 for the temperatures given in the table.

3.15.5 Experimental Data

Data Table 3.15-1 *Purpose*: To measure the emf produced by a thermocouple

Room temperature _____ ; Multiple of the potentiometer_____

Temperature, T/℃	35	45	55	65	75	85
Thermal emf, E_x/mV						

Student's name and number:_____Instructor's initial:_____

3.15.6 Calculations

We consider the thermocouple emf E_x and the temperature T are connected by a linear relation of the form $E_x = a + bT$. Use the data in Data Table 3.15-1 and least-squares fitting method to find the best estimates for the constants a and b.

3.15.7 Graphing

Use the data in Data Table 3.15-1 to draw a smooth line described by the data points.

3.15.8 Post Lab Questions

(1)The output of the stable power supply is in the range of 5.7～6.4 V. We call it the working voltage of the potentiometer. Can we measure an emf by a potentiometer which is greater than the working voltage? Why?

(2)In Fig. 3.15-4, the purpose of the calibration circuit is to get a current of 10.000 mA in the source circuit. Can the calibration circuit be replaced by an ammeter? Why?

APPENDIX I
Laboratory Report Writing Format

A1-1 Abstract

The abstract should be able to stand by itself, and it should be brief. Generally, it consists of three parts which answers these questions:

(1)What is the aim of the experiment? – A statement of the purpose of the experiment, a concise description of the experiment and physics principles investigated.

(2)What were your results? – Highlight the most significant results of the experiment.

(3)What do these results tell you? – Depending on the type of experiment, this is conclusions and implications of the results or it may be lessons learned from the experiment.

Write the abstract after all the other sections are completed. (You need to know everything in the report before you can write a summary of it.)

A1-2 Introduction

Write on the background, the physical principles, and the objectives of the experiment (what it seeks to achieve).

A1-3 Experimental Procedure

(1)Indicate lab equipment used (could be a simple list or include diagrams).

(2)Describe the experimental process (steps) in a chronological order. Using paragraph structure, be concise on all the steps.

A1-4 Results

(1)State all the results obtained here in a well-organized manner.

(2)Graphics should be easily read and well labeled.

A1-5 Discussion

This is where all your analytical skills come to play. Critical analysis on your results is done here.

(1)Explain why you got the results you have based on your theoretical knowledge and understanding of the experiment.

(2)Compare expected results with those obtained and make specific and definite conclusions on disparities or agreements (DO NOT talk about human error as that will suggest you are not competent—in other words reduce human error to the bearers possible minimum during the experiment).

A1-6 Conclusion

In a brief statement, put down what you now know for sure from your discussion. It is a statement of the final outcome of your experiment.

A1-7 References

(1) Include laboratory manual and other materials you might have read in preparation for the experiment or related to the experiment.

(2) The recommended referencing style is the following.

i. If a reference is a book

Author name, book title, publisher, city, year.

For example:

John R. Taylor, An Introduction to Error Analysis, University Science Books, CA, 1982.

ii. If a reference is a journal article

Author name(s), article title, Journal name, year (volume): start page-end page (or article ID).

For example:

H. Fu, Z. Ma, X. J. Zhang, D. H. Wang, B. H. Teng, E. Agurgo Balfour, Table-like magnetocaloric effect in the Gd-Co-Al alloys with multi-phase structure, Applied Physics Letters, 2014(104):072401.

A1-8 Appendices

(1)Include raw data, calculations, and tables that will distract a reader if included in the report.

(2) Each item should be contained in a separate appendix.

APPENDIX II
Materials Properties and Physical Constants

Table AII-1 Young's Modulus (E) for Some Metals

Metals	$E/(\times 10^{10}\ \mathrm{N/m^2})$
Aluminum	6.5
Brass	9.0
Copper	12.0
Iron (cast)	9.0
Iron (wrought)	19.0
Steel	19.2

Table AII-2 Coefficient of Linear Thermal Expansion

Substances	$\times 10^{-6}$ /℃
Aluminum	24.0
Brass	18.8
Copper	16.8
Glass (window)	8.5
Glass (Pyrex)	3.3
Iron	11.4
Lead	29.4
Nickel	12.8
Silver	18.8
Steel	13.4
Tin	26.9
Zinc	26.4

Table AII-3 Major Visible Spectral Lines of Some Elements

Element	Wavelength/nm	Color	Relative Intensity
Helium	388.9	Violet	1000
	396.5	Violet	50
	402.6	Violet	70
	438.8	Blue-violet	30
	447.1	Dark blue	100
	471.3	Blue	40
	492.2	Blue-green	50
	501.5	Green	100
	587.6	Yellow	1000
	667.8	Red	100
	706.5	Red	70
Mercury	404.7	Violet	300
	407.8	Violet	150
	435.8	Blue	500
	491.6	Blue-green	50
	546.1	Green	2000
	577.0	Yellow	200
	579.0	Yellow	1000
	690.7	Red	125
Sodium	449.4	Blue	60
	449.8	Blue	70
	466.5	Blue	80
	466.9	Blue	200
	498.3	Green	200
	514.9	Green	400
	515.3	Green	600
	567.0	Green	100
	567.5	Green	150
	568.3	Green	80
	568.8	Green	300
	589.0	Yellow-orange	9000
	589.6	Yellow-orange	5000
	615.4	Orange	500
	616.1	Orange	500

Table AII-4 Physical Constants

Physical quantity	Symbol	Magnitude
Acceleration due to gravity	g	$9.8\ m/s^2$
Universal gravitational constant	G	$6.67 \times 100^{-11}\ N \cdot m^2/kg^2$
Electron charge	e	$1.60 \times 10^{-19}\ C$
Speed of light	c	$3.0 \times 10^{-8}\ m/s$
Boltzmann's constant	k	$1.38 \times 10^{-23}\ J/K$
Planck's constant	h	$6.63 \times 10^{-34}\ J \cdot s = 4.14 \times 10^{-15}\ eV \cdot s$
	$\hbar$	$h/2\pi = 1.05 \times 10^{-34}\ J \cdot s = 6.58 \times 10^{-15}\ eV \cdot s$
Permittivity of free space	ε_0	$8.85 \times 10^{-12}\ C^2/(N \cdot m^2)$
Permeability of free space	μ_0	$4\pi \times 10^{-7} = 1.26 \times 10^{-6}\ T \cdot m/A$

APPENDIX III
Physical Experiments I
Sample Examination Paper

Score

Part A Multiple Choice ($2 \times 20 = 40$ points)

Please choose only one answer to each question and write your answer in the following table.

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer																				

1. Which of the following processes would you refer to as a measurement?

A. In a supermarket, a customer was told that the apples were 1.1 kilograms.

B. Jack and John are brothers. Their mother wants to know who is taller by observing their heights.

C. The thickness d of an object was determined with a vernier caliper and the result is $d = (10.24 \pm 0.01)$ mm.

D. A carpenter estimated that the height of a doorway was about 210 cm.

2. The measured acceleration due to gravity is 9.78 m/s^2 and its accepted value is 9.80 m/s^2. What are the absolute error and the relative error, respectively?

A. -0.02 m/s^2, -0.2%.

B. 0.02 m/s^2, 0.2%.

C. 0.02 m/s^2, -0.2%.

D. -0.02 m/s^2, 0.2%.

3. For all measured quantities, the last digit is.

A. Doubtful and significant.

B. Reliable and significant .

C. Doubtful but insignificant.

D. Reliable but insignificant.

4. A quantity was measured only once. Which of the following statements is correct?

A. The Type A evaluation of uncertainty can be calculated.

B. Both the Type A and Type B evaluation of uncertainty can be calculated.

C. The Type A evaluation of uncertainty is zero and the Type B evaluation of uncertainty can be estimated.

D. There is no Type B evaluation of uncertainty in a single measurement.

5. Which is an indirect measurement quantity?

A. The current I registered on the ammeter is 15.28 mA.

B. The local acceleration due to gravity g is measured using a pendulum by measuring its period T by a stopwatch and the corresponding length L by a tape measure.

C. The mass m of a sample on an electric scale is 1.325g.

D. The velocity v measured by a speedometer is 80km/h.

6. Which statement is correct on systematic and random errors?

A. Both the systematic and random errors can be eliminated.

B. Both the systematic and random errors can be reduced.

C. The systematic error can be reduced with repeated measurements and the random error can be eliminated with special method.

D. The systematic error can be eliminated with special method and the random error can be reduced with repeated measurements.

7. A student measures the density of a liquid five times and gets the results, 1.8, 2.0, 2.0, 1.9, 1.8 (all in g/cm^3). Calculate the standard deviation and standard deviation of the mean, respectively.

A. 0.09 g/cm^3, 0.1 g/cm^3.

B. 0.1 g/cm^3, 0.09 g/cm^3.

C. 0.09 g/cm^3, 0.04 g/cm^3.

D. 0.1 g/cm^3, 0.1 g/cm^3.

8. A quantity x is measured N times and the average, the standard deviation, and the standard deviation of the mean are denoted $\overline{x}$, σ_x, and $\sigma_{\overline{x}}$. Which statement is true?

A. If we make one more measurement, there is a 70% probability that the

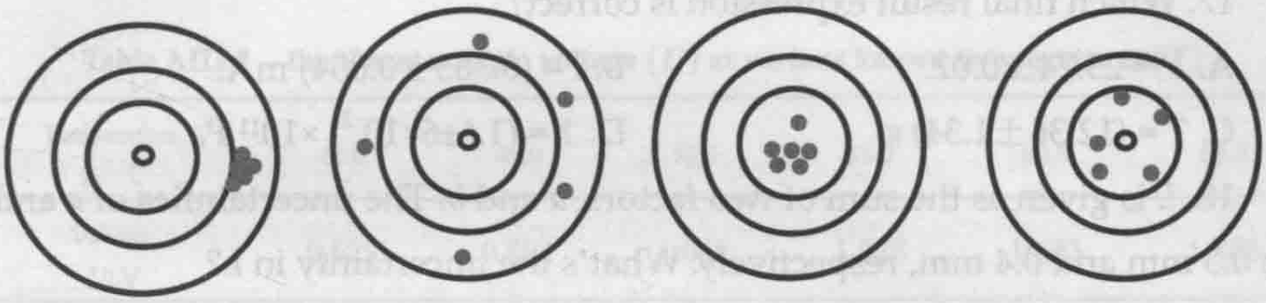
new measurement will be in the range from $\overline{x}-\sigma_{\overline{x}}$ to $\overline{x}+\sigma_{\overline{x}}$.

B. If we make one more measurement, there is a 70% probability that the new measurement will be in the range from $\overline{x}-\sigma_x$ to $\overline{x}+\sigma_x$.

C. If we make more measurements, there is a 70% probability that the true value will be in the range from $\overline{x}-\sigma_{\overline{x}}$ to $\overline{x}+\sigma_{\overline{x}}$.

D. If we make more measurements, there is a 70% probability that the true value will be in the range from $\overline{x}-\sigma_x$ to $\overline{x}+\sigma_x$.

9. In the following graphs, the bull's eye is assumed to be the true value. Which one signifies good accuracy but poor precision?



A. B. C. D.

10. In the defined relationship, 1 foot = 12 inches, how many significant figures for the number 12?

A. 2.

B. 0

C. 1.

D. It is certain and with infinite significant figures.

11. Which answer contains four significant figures?

A. 0.0013140. B. 1.00314.

C. 135000.0. D. 1.300×10^5.

12. All the zeros are significant in.

A. 3700.0. B. $1\overline{2}00000$.

C. 0.3042. D. 0.0255.

13. How many significant figures for the product of $2\times\pi\times r$ when r is measured as 3.569 mm?

A.1. B.2. C.3. D.4.

14. What's the most appropriate result of lg3869 considering that 3869 is a measured number?

A. 3.5876. B. 3.588.

C. 3.59. D. 3.587 598 729 7……

15. What is the result of $\frac{10.0^2 \times 2.000}{27.3211 - 27.311} + 350$? All the numbers are assumed to be obtained by measurements.

A. 2.0×10^4. B. 2×10^4. C. 20 350. D. 20 152.

16. The Type A and Type B uncertainties can be combined by.

A. Sum. B. Quadratic sum.

C. The average. D. The root mean square.

17. Which final result expression is correct?

A. $V = 23.74 \pm 0.02$. B. $I = (6.365 \pm 0.004)$ mA.

C. $T = (12.36 \pm 1.34)$ s. D. $Y = (1.6\pm6\times10^{-2}) \times10^{11}$ Pa.

18. L is given as the sum of two factors, a and b. The uncertainties of a and b are 0.3 mm and 0.4 mm, respectively. What's the uncertainty in L?

A. 0.7 mm. B. 0.1 mm. C. 0.5 mm. D. 0.6 mm.

19. L is given as the product of two factors, a and b. The uncertainties of a and b are 0.3 mm and 0.4 mm, respectively. What's the uncertainty in L?

A. 0.7 mm.

B. 0.12 mm.

C. 0.5 mm.

D. It depends on the values of a and b and their uncertainties.

20. Consider an analog ammeter with an accuracy class of 1.0 used to measure current. If its scale is 30 mA and it reads the current 13.2 mA. What is the instrument error?

A. 0.3 mA. B. 1 mA. C. 0.132 mA. D. 0.1 mA.

Score

Part B Long Questions (2 × 30 = 60 points)

21. The thermocouples must be calibrated before it is used in any experiment. To calibrate, we measure the thermocouple voltage at various known temperatures, so as to develop a correlation between thermocouple voltage and thermocouple temperature. The measured results are listed in the following table.

Table AIII-1 the thermocouple voltage (U) at various known temperatures (T)

Temperature, T / ℃	40.0	45.0	50.0	55.0	60.0	65.0
Voltage, U/ V	0.675	0.884	1.068	1.276	1.693	1.862

(a)Use the data in Table AIII-1 to draw a graph of thermocouple voltage (U) depending on temperature (T). (16 points)

(b)Choose two points on the above graph and write down their coordinates. Use these two data to find the relationship of the current and the voltage. (14 points) (**DO NOT WRITE THE ANSWERS ONLY**)

22. A student measures two lengths L_1 and L_2. L_1 is measured once with a vernier caliper and the result is (13.26 ± 0.01)mm. L_2 is measured five times with a micrometer whose instrument error is 0.01 mm and the values are shown in Table AIII-2.

Table AIII-2 the measured values of L_2

Trial	1	2	3	4	5
L_2/mm	3.215	3.218	3.214	3.213	3.216

(a)Compute the average of L_2. (2 points) (**DO NOT WRITE THE ANSWERS ONLY**)

(b)Compute the Type A uncertainty σ_{AL_2} in L_2. (4 points) (**DO NOT WRITE THE ANSWERS ONLY**)

(c)Compute the Type B uncertainty σ_{BL_2} in L_2. (6 points) (**DO NOT WRITE THE ANSWERS ONLY**)

(d)Compute the combined uncertainty σ_{L_2} in L_2. (4 points) (**DO NOT WRITE THE ANSWERS ONLY**)

(e)If q is the sum of L_1 and L_2, $q = L_1 + L_2$. Find q and its uncertainty. (10 points) **(DO NOT WRITE THE ANSWERS ONLY)**

(f)Write the final result of q. (4 points)

The End

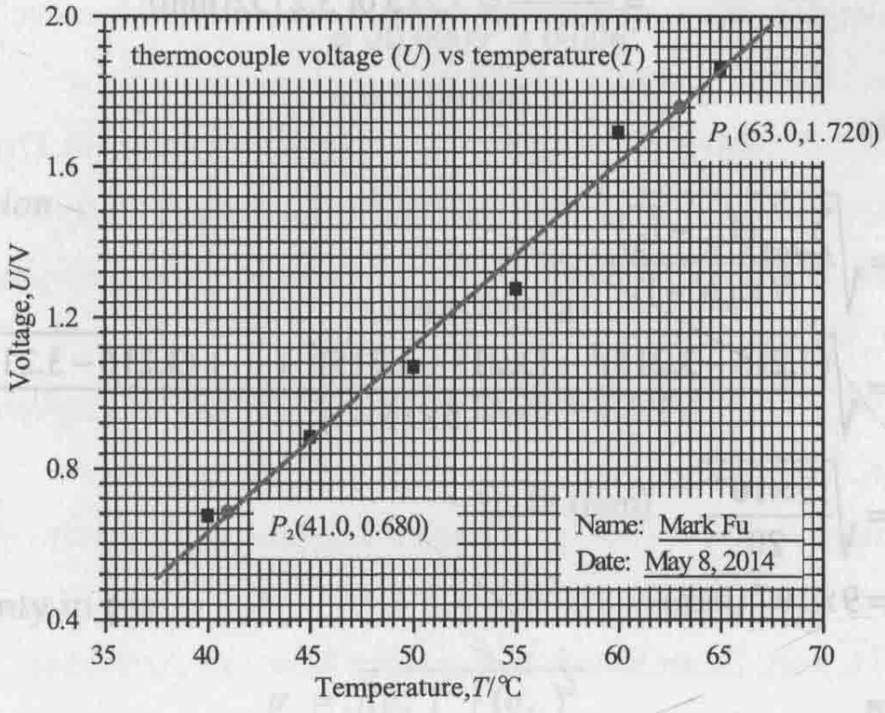
Answers to Sample Questions

Part A Multiple Choice (2×20=40 points)

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	C	A	A	C	B	D	C	B	D	D	D	A	D	A	A	B	B	C	D	A

Part B Long Questions (2×30=60 points)

(21) (a)



(b)

Solution

Suppose that the thermocouple voltage (U) and temperature (T) have the relationship, $U = aT+b$. Choose two far apart points, say P_1(63.0, 1.720), and P_2(41.0, 0.680), to calculate a and b.

$$a = \frac{U_2 - U_1}{T_2 - T_1} = \frac{0.680 - 1.720}{41.0 - 63.0} = 0.473(\text{V/℃})$$

Then, use the calculated a and the first point P_1 to calculate b

$$B = U_1 - aT_1 = 1.720 - 0.0473 \times 63.0 = -1.260 \text{ (V)}$$

Thus, the relationship between resistance and temperature is

$$U = 0.0473T - 1.260 \text{ (V)}$$

(22) (a)***Solution***

$$\overline{L_2} = \frac{1}{N}\sum_{i=1}^{5} L_{2i}$$

$$= \frac{3.215 + 3.218 + \cdots + 3.216}{5}$$

$$= \frac{16.076}{5} = 3.215 \text{ or } 3.2152(\text{mm})$$

(b)***Solution***

$$\mu_{AL_2} = \sqrt{\frac{\sum_{i=1}^{N}(L_{2i} - \overline{L_2})^2}{N \times (N-1)}}$$

$$= \sqrt{\frac{(3.215-3.215)^2 + (3.218-3.215)^2 + \cdots + (3.216-3.215)^2}{5 \times (5-1)}}$$

$$= \sqrt{\frac{15 \times 10^{-6}}{20}}$$

$$= 9 \times 10^{-4}(\text{mm})$$

(c)***Solution***

The instrument error of a micrometer and its component of the Type B uncertainty are

$$\Delta_{\text{Instru.}} = 0.01 \text{ mm}$$

$$\mu_{\Delta\text{Instru.}} = \frac{\Delta_{\text{Instr.}}}{\sqrt{3}} = \frac{0.01}{\sqrt{3}} = 0.006(\text{mm})$$

The reading error and its Type B uncertainty component are

$$\Delta_{\text{Read.}} = 0.2 \times \text{least count} = 0.2 \times 0.01 = 0.002 \text{ (mm)}$$

$$\mu_{\Delta \text{Read}} = \frac{\Delta_{\text{Read}}}{\sqrt{3}} = \frac{0.002}{\sqrt{3}} = 0.001(\text{mm})$$

So, the combined Type B uncertainty is

$$\begin{aligned}\mu_{BL_2} &= \sqrt{\mu^2_{\Delta \text{Instru.}} + \mu^2_{\Delta \text{Read}}} \\ &= \sqrt{0.006^2 + 0.001^2} \\ &= 0.006(\text{mm})\end{aligned}$$

(d)***Solution***

$$\begin{aligned}\sigma_{L_2} &= \sqrt{\mu^2_{AL_2} + \mu^2_{BL_2}} \\ &= \sqrt{0.0009^2 + 0.006^2} \\ &= 0.006(\text{mm})\end{aligned}$$

(e)***Solution***

$$q = L_1 + L_2$$

$$= 13.26 + 3.215$$

$$= 16.48 \text{ (mm)}$$

The uncertainty in q is

$$\begin{aligned}\sigma_q &= \sqrt{(\sigma_{L_1})^2 + (\sigma_{L_2})^2} \\ &= \sqrt{(0.01)^2 + (0.006)^2} \\ &= 0.01(\text{mm})\end{aligned}$$

(f)Write the final result of q (4 marks)

$$q = 16.48 \pm 0.01 \text{ (mm)}$$

THE END

References

1. David C. Baird. Experimentation: An Introduction to Measurement Theory and Experiment Design, 2nd ed., Prentice Hall, 1988.

2. Daryl W. Preston. Experiments in Physics: A Laboratory Manual for Scientists and Engineers, John Wiley and Sons, 1985.

3. N. C. Barford. Experimental Measurements: Precision, Error, and Truth, AddisonWesley, 1967.

4. Arthur Mittler, Walter Schier David Pul. Physics Laboratory Experiments, 8th ed., The McGraw-hill Companies, 2010.

5. Yaakov Kraftmakher. Experiments and Demonstrations in Physics: Bar-ilan Physics Laboratory, World Scientific Publishing Company, 2006

6. Cicero H. Bernard, Chirold D. Epp. Laboratory Experiments in College Physics, 7th ed., Wiley, 1994.

7. J. R. Green, D. Margerison. Statistical Treatment of Experimental Data, 1st ed., Elsevier, 1977.

8. Holt Rinehart and Winston. Physics Laboratory Experiments Teacher's Edition, Random House Value Publishing, 1998.

9. Paul G. Hewitt, Dean Baird. Laboratory Manual: Activities, Experiments, Demonstrations and Tech Labs for Conceptual Physics, 12th ed., Pearson, 2014.

10. G. L. Squires. Practical Physics, 4th ed., Cambridge, 2001.

11. P. R. Bevington, K. D. Robinson. Data Reduction and Error Analysis for the Physical Sciences, McGraw-Hill, 1992.

12. Edward L. Robinson. Data Analysis for Scientists and Engineers, Princeton University, 2016.

13. E. M. Pugh, G. H. Winslow. The Analysis of Physical Measurements, AddisonWesley, 1966.

[General Information]
书名=14423491
SS号=14423491